University of Waterloo

Stochastic Processes II

STAT 433

Lecture Notes

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Preface to the Reader

These notes offer an in-depth and rigorous treatment of probability theory, martingale theory, Brownian motion, and stochastic calculus. They begin with the foundations of measure-theoretic probability and proceed until expressions such as "a continuous, nowhere differentiable path with finite quadratic variation" start to sound reasonable.

The guiding philosophy is simple: no important detail should be left unexplained merely because someone, somewhere, called it obvious. Definitions are unpacked, proofs are developed carefully, and limits are not permitted to pass through expectations without presenting the appropriate documentation. Almost-sure convergence has already tried this and can no longer be trusted unsupervised.

Along the way, we encounter Kolmogorov's foundations, conditional expectations, filtrations, and martingales: processes whose expected future value, given everything currently known, is their present value. This is also approximately the reader's assessment of their understanding halfway through a difficult proof.

Brownian motion then arrives to demonstrate that continuity does not imply good behaviour. Stochastic calculus follows to explain how to integrate against something that cannot be differentiated, and why the ordinary chain rule now requires a correction term. Applications to sampling and neural networks provide reassurance that all this machinery occasionally leaves the mathematics department.

The reader is encouraged to proceed slowly, work through the algebra, and question every apparently harmless interchange of operations. If a proof requires several readings, this is normal. If it requires only one, check that you have read the assumptions.

Understanding is the goal. Almost surely would be nice; everywhere may be asking too much.

> "Let me have a moment. Assuming it exists."
>
> — Kenneth Zhang, almost surely.

1 Foundations of Probability Theory

1.1 Samples Spaces & Probability Spaces and Measures

Definition 1.1 (Sample Space)**.** The *sample space* Ω is the set of all possible outcomes of a random experiment. Each element $\omega \in \Omega$ represents one possible outcome.

Remark 1.1. Think of Ω as the set of all possible worlds, with each $\omega \in \Omega$ representing one complete scenario. We do *not* observe ω directly but instead observe quantities determined by it. For example, Lehman's stock price at time t is represented by a function,

$$S_t : \Omega \to \mathbb{R}, \qquad \omega \mapsto S_t(\omega).$$

Thus, each possible world determines a stock price, and we observe that value.

Example 1.1 (A finite sample space)**.** Let $\Omega = \{\omega_1, \omega_2\}$ with equally likely outcomes.

Define $C : \Omega \to \mathbb{R}$ by,

$$C(\omega_1) = 0, \quad C(\omega_2) = 1.$$

Then,

$$\mathbb{P}(\{\omega_1\}) = \mathbb{P}(\{\omega_2\}) = \frac{1}{2}.$$

On a finite sample space, individual outcome probabilities determine the probability of every event by summation. This approach does not generally extend to *uncountable* sample spaces.

Definition 1.2 (σ-algebra)**.** A collection $\mathcal{F}$ of subsets of Ω is a σ-algebra if,

(i) $\Omega \in \mathcal{F}$.
(ii) If $A \in \mathcal{F}$, then $A^c = \Omega \backslash A \in \mathcal{F}$.
(iii) For every sequence $A_1, A_2, \cdots \in \mathcal{F}$,

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

The sets in $\mathcal{F}$ are called *events*.

Definition 1.3 (Probability Measure)**.** A *probability measure* on $(\Omega, \mathcal{F})$ is a function,

$$\mathbb{P} : \mathcal{F} \to [0, 1], \qquad \mathbb{P}(\Omega) = 1$$

and countable additivity. That is, for every sequence of pairwise disjoint events $A_1, A_2, \ldots$,

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

$\mathbb{P}(A)$ represents the probability that event A occurs.

Definition 1.4 (Probability space)**.** A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is a sample space, $\mathcal{F}$ is a σ-algebra of subsets of Ω, and $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$.

Example 1.2. The probability axioms reflect three natural requirements. Probabilities lie in $[0, 1]$, the entire sample space is certain, so $\mathbb{P}(\Omega) = 1$, and probabilities add when events are mutually exclusive. In particular, for pairwise disjoint events, $A_1, \ldots, A_m$,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Countable additivity extends this rule to countably infinite sequences of events.

Notice, however, we need not require uncountable additivity. Consider the uniform distribution on $\Omega = [0, 1]$. Every singleton has probability zero, yet the whole interval has probability one,

$$\mathbb{P}(\{x\}) = 0 \quad \forall x \in [0, 1], \quad \mathbb{P}([0, 1]) = 1.$$

Since $[0, 1]$ is the disjoint union of its singletons, uncountable additivity would imply,

$$1 = \mathbb{P}([0, 1]) = \mathbb{P}\left(\bigcup_{x\in[0,1]} \{x\}\right) = \sum_{x\in[0,1]} \mathbb{P}(\{x\}) = 0,$$

a contradiction. Thus, we require countable additivity, requiring uncountable additivity would exclude distributions such as the uniform distribution.

Proposition 1.1 (Basic Properties of Probability)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For any events $A, B \in \mathcal{F}$,*

(i) ***Empty Event:*** $\mathbb{P}(\emptyset) = 0$.
(ii) ***Complement:*** $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
(iii) ***Monotonicity:*** *If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.*
(iv) ***Inclusion-Exclusion:***

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof. We prove **(i)** – **(iv)** of **Proposition 1.1**.

Proof of (i): Applying countable additivity to the pairwise disjoint sequence $\Omega, \emptyset, \emptyset, \ldots$,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(\Omega) + \sum_{n=1}^{\infty} \mathbb{P}(\emptyset).$$

Since all terms are non-negative, this equality forces $\mathbb{P}(\emptyset) = 0$. Consequently, countable additivity also gives finite additivity: for pairwise disjoint events, $E_1, \ldots, E_m$, append empty events to so that we shall obtain,

$$\mathbb{P}\left(\bigcup_{i=1}^{m} E_i\right) = \sum_{i=1}^{m} \mathbb{P}(E_i).$$

Proof of (ii): Since A and A^c are disjoint and their union is Ω, finite additivity gives,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c).$$

Re-arranging proves the claim.

Proof of (iii): The event $B\backslash A = B \cap A^c$ belongs to $\mathcal{F}$. Since,

$$B = A \cup (B\backslash A)$$

is a disjoint uniont, finite additivity gives,

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B\backslash A) \geq \mathbb{P}(A).$$

Inequality of the above follows from non-negativity.

Proof of (iv): Using the disjoint decompositions,

$$A \cup B = A \cup (B\backslash A), \quad B = (A \cap B) \cup (B\backslash A).$$

Finite additivity gives,

$$\begin{aligned} \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B\backslash A) \\ \mathbb{P}(B) &= \mathbb{P}(A \cap B) + \mathbb{P}(B\backslash A). \end{aligned}$$

Solving the second equality for $\mathbb{P}(B\backslash A)$ and substituting into the first yields,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

□

Proposition 1.2 (Probability over a countable partition)**.** *Let $A_1, A_2, \cdots \in \mathcal{F}$ be pairwise disjoint events satisfying the following union,*

$$\bigcup_{i=1}^{\infty} A_i = \Omega.$$

Then, for every event $B \in \mathcal{F}$,

$$\mathbb{P}(B) = \sum_{i=1}^{\infty} \mathbb{P}(B \cap A_i).$$

Proof. Each $B \cap A_i$ belongs to $\mathcal{F}$, since a σ-algebra is closed under intersections. These events are pairwise disjoint: for $i \neq j$,

$$(B \cap A_i) \cap (B \cap A_j) = B \cap (A_i \cap A_j) = \emptyset.$$

Moreover, they cover the set B,

$$\bigcup_{i=1}^{\infty} (B \cap A_i) = B \cap \left(\bigcup_{i=1}^{\infty} A_i \right) = B \cap \Omega = B.$$

Countable additivity therefore gives,

$$\mathbb{P}(B) = \mathbb{P}\left(\bigcup_{i=1}^{\infty}(B \cap A_i)\right) = \sum_{i=1}^{\infty}\mathbb{P}(B \cap A_i).$$

□

Example 1.3 (Stock prices and Pre-images)**.** Let $X : \Omega \to \mathbb{R}_+$ assign a stock price $X(\omega)$ to each outcome ω. To find the probability that the price lies in $[10, 11]$, we consider the corresponding set of outcomes to the preimage given as,

$$\{\omega \in \Omega : 10 \leq X(\omega) \leq 11\} = X^{-1}([10, 11]).$$

This set is the *pre-image* of $[10, 11]$ under X. If it belongs to $\mathcal{F}$, its probability is defined as,

$$\mathbb{P}(10 \leq X \leq 11) = \mathbb{P}(X^{-1}([10, 11])).$$

A *random variable* is a function whose pre-image of every Borel set is an event in $\mathcal{F}$, ensuring that these probabilities are well-defined.

1.2 Random Variables

Definition 1.5 (Preimage)**.** Let $X : \Omega \to \mathbb{R}$ be a function and $B \subseteq \mathbb{R}$.

The *preimage* of B under X,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}.$$

It consists of all outcomes whose values under X lie in B. The notation $X^{-1}(B)$ does not require X to have an inverse function. A function X is not automatically a random variable: we must additionally require that $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \subseteq \mathbb{R}$.

The probability that X takes a value in $B \subseteq \mathbb{R}$ is the probability of the corresponding preimage,

$$\mathbb{P}(X \in B) := \mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\}) = \mathbb{P}(X^{-1}(B)).$$

This probability is defined only when $X^{-1}(B) \in \mathcal{F}$, since $\mathbb{P}$ assigns probabilities to events in $\mathcal{F}$.

We must therefore specific a collection of sets $B \subseteq \mathbb{R}$ whose preimages are required to be events. This collection should include intervals, so that expressions such as $\mathbb{P}(a \leq X \leq b)$ are well-defined. The *Borel σ-algebra* provides this collection.

This motivates the question:

What is the smallest σ-algebra on $\mathbb{R}$ containing every interval?

Intervals allows us to ask basic questions about observed values. For example, whether a stock price lies between \$10 and \$11. However, intervals alone do not form a σ-algebra: the complement of countable unions of intervals need not be an interval.

We therefore include all intervals together with the additional sets needed for closure under complements and countable unions.

The resulting collection is the Borel σ-algebra, denoted $\mathcal{B}(\mathbb{R})$. Requiring,

$$X^{-1}(B) \in \mathcal{F} \qquad \forall B \in \mathcal{B}(\mathbb{R}),$$

then ensure that probabilities involving these events of observed values are well-defined.

The Vitali theorem motivates why we do not simply use every subset: it prevents extending the usual notion of length to a countably additive, translation-invariant measure on all subsets of $\mathbb{R}$. It does not rule out every probability measure on all subsets of $\mathbb{R}$.

Definition 1.6 (Borel σ-algebra)**.** The *Borel σ-algebra* on $\mathbb{R}$, denoted by $\mathcal{B}(\mathbb{R})$, is the smallest σ-algebra containing every interval in $\mathbb{R}$. Equivalently,

$$\mathcal{B}(\mathbb{R}) = \sigma\left(\{I \subseteq \mathbb{R} : I \text{ is an interval}\}\right).$$

Its members are called *Borel sets*.

Definition 1.7 (Random variable)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

A function $X : \Omega \to \mathbb{R}$ is a random variable if,

$$X^{-1}(B) \in \mathcal{F} \qquad \forall B \in \mathcal{B}(\mathbb{R}).$$

The preimage of every Borel set is an event, so $\mathbb{P}(X \in B)$ is well-defined.

Definition 1.8 (σ-algebra generated by a random variable)**.** Let $X : \Omega \to \mathbb{R}$ be a random variable. The σ-algebra generated by X is,

$$\sigma(X) = \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}.$$

This is the smallest sub-σ-algebra of $\mathcal{F}$ with respect to which X is measurable, and represents the information revealed by observing X. Its events are expressed as $\{X \in B\}$ for a Borel set B.

Definition 1.9 (Measurability)**.** Let $\mathcal{G}$ be a σ-algebra. A function $X : \Omega \to \mathbb{R}$ is $\mathcal{G}$-measurable if, $X^{-1}(B) \in \mathcal{G}$ for all $B \in \mathcal{B}(\mathbb{R})$. Equivalently, $\sigma(X) \subseteq \mathcal{G}$.

Hence, a real-valued random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ is precisely an $\mathcal{F}$-measurable function.

Example 1.4 (Measurability of a stock price)**.** Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$ and define the stock price by, $X(\omega) = 50 + 100\omega$. For the Borel set of prices $B = [90, 110]$, its preimage is,

$$\begin{aligned} X^{-1}([90, 110]) &= \{\omega \in [0, 1] : 90 \leq 50 + 100\omega \leq 100\} \\ &= \{\omega \in [0, 1] : 0.4 \leq \omega \leq 0.6\} \\ &= [0.4, 0.6]. \end{aligned}$$

Since $[0.4, 0.6] \in \mathcal{F}$, the condition that the stock prices lies between \$90 and \$110 defines an event. Checking this one preimage does not establish measurability of X; the condition must hold for every Borel set. Here, X is continuous, so the preimage of every Borel set is Borel. Hence, X is $\mathcal{F}$-measurable and therefore a random variable.

A σ-algebra represents the information available about a random outcome. To illustrate, consider a sequence of stock prices, $S_0, S_1, S_2, \ldots$ where each $S_t : \Omega \to \mathbb{R}$ is a random variable.

A single outcome ω determines the entire price sequence $S_0(\omega), S_1(\omega), S_2(\omega)$. Observing S_0 reveals the information represented by $\sigma(S_0)$. Observing both S_0 and S_1 reveals the information represented by $\sigma(S_0, S_1)$, the smallest σ-algebra with respect to which both variables are measurable. As we observe more prices, the available information increases.

If S_2 is completely determined by the information in S_0 and S_1, then S_2 is $\sigma(S_0, S_1)$-measurable,

$$S_2^{-1}(B) \in \sigma(S_0, S_1) \quad \forall B \in \mathcal{B}(\mathbb{R}).$$

This is an additional assumption: a future price need not be determined by past observations.

Theorem 1.1 (Natural Filtration)**.** *Let $(S_t)_{t\geq 0}$ be a discrete-time stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$. Define, $\mathcal{F}_t := \sigma(S_0, \ldots, S_t)$ for $t = 0, 1, 2, \ldots$. Then, $(\mathcal{F}_t)_{t\geq 0}$ is a filtration,*

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq F_2 \subseteq \cdots \subseteq \mathcal{F}.$$

Such a sequence is called the natural filtration of the process. Here $\mathcal{F}_t$ represents the information revealed by all observations up to time t. In particular, for every $s \leq t$ and $B \in \mathcal{B}(\mathbb{R})$,

$$\{S_s \in B\} = S_s^{-1}(B) \in \mathcal{F}_t.$$

Proof. By defintion $\mathcal{F}_{t+1}$ makes $S_0, \ldots, S_{t+1}$ measurable, so it also makes $S_0, \ldots, S_t$ measurable. Since $\mathcal{F}_t$ is the smallest σ-algebra with this latter property,

$$\mathcal{F}_t \subseteq \mathcal{F}_{t+1}.$$

Moreover, every S_t is $\mathcal{F}$-measurable, so $\mathcal{F}_t \subseteq \mathcal{F}$. □

Instead of treating the prices $S_0, \ldots, S_T$ separately, we can collect them into a single vector,

$$S : \Omega \to \mathbb{R}^{T+1}, \quad S(\omega) = (S_0(\omega), \ldots, S_T(\omega)).$$

Each outcome ω therefore determines an entire finite price path. To describe events involving this vector, we use the Borel σ-algebra on $\mathbb{R}^{T+1}$.

Definition 1.10 (Borel σ-algebra on $\mathbb{R}^{T+1}$)**.** The Borel σ-algebra on $\mathbb{R}^{T+1}$ is,

$$\mathcal{B}(\mathbb{R}^{T+1}) := \sigma\left(\{I_0 \times \cdots \times I_T : I_0, \ldots, I_T \text{ are intervals in } \mathbb{R}\}\right).$$

This is the smallest σ-algebra containing all such rectangles.

Definition 1.11 (Finite-time process as a random vector)**.** Let $S_0, \ldots, S_T$ be real-valued random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. The process can be represented by the random vector,

$$S : \Omega \to \mathbb{R}^{T+1}, \quad S(\omega) = (S_0(\omega), \ldots, S_T(\omega)).$$

This map is measurable with respect to $\mathcal{F}$ and $\mathcal{B}(\mathbb{R}^{T+1})$.

Indeed, for every generating rectangle,

$$S^{-1}(I_0 \times \cdots \times I_T) = \bigcap_{t=0}^{T} S_t^{-1}(I_t) \in \mathcal{F}.$$

Since these rectangles generate $\mathcal{B}(\mathbb{R}^{T+1})$, this suffices to establish measurability.

Remark 1.2 (Product σ-algebra). Equivalently,

$$\mathcal{B}(\mathbb{R}^{T+1}) = \underbrace{\mathcal{B}(\mathbb{R}) \otimes \cdots \otimes \mathcal{B}(\mathbb{R})}_{T+1 \text{ factors}},$$

where the product σ-algebra is generated by sets $B_0 \times \cdots \times B_T$ with each $B_t \in \mathcal{B}(\mathbb{R})$.

In two dimensions, a rectangle is a product of two intervals. More generally, in $\mathbb{R}^{T+1}$,

$$I_0 \times \cdots \times I_T = \left\{(x_0, \ldots, x_T) \in \mathbb{R}^{T+1} : x_t \in I_t \quad \forall t = 0, \ldots, T\right\}.$$

A process $S : \Omega \to \mathbb{R}^{T+1}$ is measurable if $S^{-1}(B) \in \mathcal{F}$. for every $B \in \mathcal{B}(\mathbb{R}^{T+1})$.

We can similarly represent a continuous-time process with continuous paths as a function $S : \Omega \to \mathcal{C}([0,T])$ where $\mathcal{C}([0,T])$ is the space of continuous real-valued functions on $[0,T]$. Each outcome ω determines a path $S(\omega)$ whose value at time t is,

$$S(\omega)(t) = S_t(\omega).$$

To describe sets of paths, we begin with conditions imposed at finitely many times.

Definition 1.12 (Cylinder sets on $\mathcal{C}([0,T])$)**.** A *cylinder set* is a set of paths of the form

$$C = \{f \in \mathcal{C}([0,T]) : (f(t_1), \ldots, f(t_k)) \in B\},$$

where $t_1, \ldots, t_k \in [0,T]$ and $B \in \mathcal{B}(\mathbb{R}^k)$. In particular, a *rectanglar cylinder set* has the form,

$$C = \{f \in \mathcal{C}([0,T]) : f(t_1) \in I_1, \ldots, f(t_k) \in I_k\}$$

where each $I_i \subseteq \mathbb{R}$ is an interval.

Such a set restricts the path at the specified times and imposes no additional conditions at other times. Its preimage under the process S is the event,

$$S^{-1}(C) = \bigcap_{i=1}^{k} \{\omega \in \Omega : S_{t_i}(\omega) \in I_i\}.$$

These rectangular cylinder sets generate the cylinder σ-algebra on $\mathcal{C}([0,T])$. On this space, it coincides with the Borel σ-algebra induced by the uniform norm,

$$\|f\|_\infty = \sup_{t \in [0,T]} |f(t)|.$$

1.3 Distributions, Laws and Measure

Definition 1.13 (Distribution of a random variable)**.** Let $X : \Omega \to \mathbb{R}$ be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. The *distribution*, or *law*, of X is the probability measure μ_X on $\mathcal{B}(\mathbb{R})$ defined by,

$$\mu_X(B) := \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)), \quad B \in \mathcal{B}(R).$$

Hence, $\mu_X(B)$ is the probability that the observed value of X lies in B.

The measure μ_X is also called the *pushforward* of $\mathbb{P}$ under X, written $\mu_X = \mathbb{P} \odot X^{-1}$. It transfers the probability measure from the original sample space to the space of observed values,

$$(\Omega, \mathcal{F}, \mathbb{P}) \xrightarrow{X} (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_X).$$

In particular, $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_X)$ is itself a probability space.

Example 1.5. Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$ and let $\mathbb{P}$ be the uniform probability measure. Define $X(\omega) = 50 + 100\omega$. The distribution μ_X assigns probabilities directly to sets of stock prices by taking their preimages under X. For example, since $X^{-1}([90, 110]) = [0.4, 0.6]$,

$$\begin{aligned} \mu_X([90, 110]) &= \mathbb{P}(X^{-1}([90, 110])) \\ &= \mathbb{P}([0.4, 0.6]) \\ &= 0.6 - 0.4 = 0.2. \end{aligned}$$

Thus, the probability that the stock price lies between \$90 and \$110 is 0.2.

More generally, for $50 \leq a \leq b \leq 150$,

$$X^{-1}([a, b]) = \left[\frac{a - 50}{100}, \frac{b - 50}{100}\right] \iff \mu_X([a, b]) = \frac{b - a}{100}.$$

Hence, X is uniformly distributed on $[50, 150]$ written, $X \sim \text{Unif}([50, 150])$.

The pushforward construction transfers probability from outcomes to observed prices,

$$([0, 1], \mathcal{B}([0, 1]), \mathbb{P}) \xrightarrow{X} (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_X), \quad \mu_X([50, 150]) = 1.$$

To compute this new measure, we pull a set of prices back to the sample space and apply $\mathbb{P}$,

$$B \mapsto X^{-1}(B) \mapsto \mathbb{P}(X^{-1}(B)) = \mu_X(B).$$

The *law* of a random variable X is the probability measure,

$$\mu_X(B) = \mathbb{P}(X \in B), \quad B \in \mathcal{B}(\mathbb{R}).$$

It describes how probability is distributed over the possible observed values of X. Probability mass functions and probability density functions are two ways of describing this measure. We begin with point masses, the building blocks of discrete distributions.

In this course, we restrict attention to laws that can be represented by point masses, a density, or a mixture of the two. General probability measures need not have this form.

Definition 1.14 (Dirac measure)**.** For $a \in \mathbb{R}$, the *Dirac measure* at a is the probability measure,

$$\delta_a : \mathcal{B}(\mathbb{R}) \to [0, 1]$$

defined by,

$$\delta_a(B) = \begin{cases} 1, & a \in B \\ 0, & a \notin B. \end{cases}$$

Thus, δ_a places all probability mass at the single point a and assigns probability one to any Borel set containing a, and probability zero to any Borel set that does not.

For $p \geq 0$, the measure $p\delta_a$ places mass p at a,

$$(p\delta_a)(B) = p\delta_a(B) = \begin{cases} p, & a \in B \\ 0, & a \notin B. \end{cases}$$

Hence, its total mass is p, so $p\delta_a$ is itself a probability measure only when $p = 1$.

Remark 1.3 (A measure acts on sets). The *Dirac measure* is a function of a *Borel set* B, rather than an ordinary density function of a real number x.

The expression $\delta_a(B)$ asks whether the set B contains the point a.

Example 1.6 (Bernoulli distribution)**.** Let $X \sim \text{Bernoulli}(p)$ where $0 \leq p \leq 1$. Then,

$$\mathbb{P}(X = 0) = 1 - p, \qquad \mathbb{P}(X = 1) = p.$$

Since X takes only the values 0 and 1, its law places mass $1 - p$ at 0 and mass p at 1.

$$\mu_X = (1 - p)\delta_0 + p\delta_1.$$

This equality of measures also implies that for every $B \in \mathcal{B}(\mathbb{R})$,

$$\mu_X(B) = (1 - p)\delta_0(B) + p\delta_1(B).$$

Each term contributes probability mass precisely when its corresponding point belongs to a Borel set B. For example, $\mu_X(\{0\}) = 1 - p$, $\mu_X(\{1\}) = p$, and henceforth. The special case is when $\mu_X((0, 1)) = 0$, since the open interval contains neither possible value of X.

Remark 1.4 (Integration against a Dirac measure). For any measure μ, the notation,

$$\int_B \mu(dx) = \int_{\mathbb{R}} \mathbb{I}_B(x)\mu(dx) = \mu(B),$$

expresses the mass that μ assigns to B. In particular,

$$\int_B \delta_a(dx) = \delta_a(B), \qquad \int_B p\delta_a(dx) = \begin{cases} p, & a \in B \\ 0, & a \notin B. \end{cases}$$

More generally, integrating a Borel-measurable function $\phi : \mathbb{R} \to \mathbb{R}$ against δ_a evaluates it at a,

$$\int_{\mathbb{R}} \phi(x)\delta_a(dx) = \phi(a).$$

All the mass is concentrated at a, so only the value $\phi(a)$ contributes. Consequently, for the Bernoulli law derived previously,

$$\int_{\mathbb{R}} \phi(x)\mu_X(dx) = (1-p)\phi(0) + p\phi(1).$$

Proposition 1.3 (Representation of a discrete law)**.** *Suppose X takes values in a finite or countable set of distinct points, represented as $\{x_j : j \in J\}$ with,*

$$p_j = \mathbb{P}(X = x_j), \qquad p_j \geq 0, \qquad \sum_{j\in J} p_j = 1.$$

Then its law is,

$$\mu_X = \sum_{j\in J} p_j \delta_{x_j}.$$

Equivalently, for every Borel set B,

$$\mu_X(B) = \sum_{j\in J} p_j \delta_{x_j}(B) = \sum_{j\in J; x_j \in B} p_j.$$

Proof. Apart from an event of probability zero, $X \in B$ occurs precisely when $X = x_j$ for one of the values $x_j \in B$. These events are pairwise disjoint, so countable additivity gives,

$$\mathbb{P}(X \in B) = \sum_{j\in J; x_j \in B} \mathbb{P}(X = x_j) = \sum_{j\in J; x_j \in B} p_j.$$

This is precisely the mass assigned to B by $\sum_{j\in J} p_j \delta_{x_j}$. □

Example 1.7 (Computing probabilities from a discrete law)**.** Suppose X takes values $\{1, 3, 7\}$ with corresponding probabilities,

$$\mathbb{P}(X = 1) = 0.2, \qquad \mathbb{P}(X = 3) = 0.5, \qquad \mathbb{P}(X = 7) = 0.3.$$

Its law is hence defined as,

$$\mu_X = 0.2\delta_1 + 0.5\delta_3 + 0.3\delta_7.$$

To find the probability that $X \in [2, 5]$, evaluate this measure on $[2, 5]$,

$$\begin{aligned}\mu_X([2,5]) &= 0.2\delta_1([2,5]) + 0.5\delta_3([2,5]) + 0.3\delta_7([2,5]) \\ &= 0.2(0) + 0.5(1) + 0.3(0) \\ &= 0.5.\end{aligned}$$

Only 3 belongs on $[2, 5]$, so only its probability mass contributes. Thus,

$$\mu_X([2, 5]) = \mathbb{P}(2 \leq X \leq 5) = \mathbb{P}(X = 3) = 0.5.$$

For discrete laws, we compute probabilities by summing point masses. For laws with a density, we compute probabilities by integrating that density against a *Lebesgue measure*, which assigns each intervals its length. We will introduce this measure next.

Definition 1.15 (Lebesgue measure)**.** The *Lebesgue measure* on $\mathbb{R}$ is its measure, $\lambda : \mathcal{B}(\mathbb{R}) \to [0, \infty]$ that assigns each bounded intervals its length,

$$\lambda([a, b]) = b - a, \qquad a \leq b.$$

It is countably additive, hence for every sequence of pairwise disjoint Borel sets $A_1, A_2, \ldots,$

$$\lambda\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \lambda(A_i).$$

Lebesgue measure extends the notion of length from intervals to Borel sets, In particular, we specify that $\lambda(\{a\}) = 0$ and $\lambda(\mathbb{R}) = \infty$. Thus, λ is not a probability measure since its total mass is infinite. The follow remark explores integration with respect to length.

Remark 1.5 (Integration with respect to length). Integrating 1 over a Borel set gives its *Lebesgue measure* defined as,

$$\int_B 1\lambda(dx) = \lambda(B).$$

We usually write dx instead of $\lambda(dx)$ so,

$$\int_B \lambda(dx) = \int_B dx = \lambda(B).$$

For example, the integral $\displaystyle\int_{[a,b]} dx = b - a$. More generally, the notations,

$$\int_B f(x)\lambda(dx), \qquad \int_B f(x)d\lambda(x), \qquad \int_B f(x)dx$$

all denote the same Lebesgue integral.

For a Riemann-integrable function on a bounded interval, in particular, a continuous or piecewise continuous function, the Lebesgue and Riemann integrals agree. Integration with respect to a measure also lets us treat densities and point masses within the same framework.

Recall that the *law*, or *distribution*, of a random variable X is the probability measure,

$$\mu_X(B) = \mathbb{P}(X \in B), \qquad B \in \mathcal{B}(\mathbb{R}).$$

For a discrete random variable, this measure is a sum of weighted Dirac measures. We now describe laws that have a density, followed by laws that combine a density with point masses.

Definition 1.16 (Probability density function)**.** Let $X : \Omega \to \mathbb{R}$ be a random variable. A nonnegative Borel-measurable function $f : \mathbb{R} \to [0, \infty)$ is a *probability density function* of X if,

$$\mu_X(B) = \mathbb{P}(X \in B) = \int_B f(x)dx \qquad \forall B \in \mathcal{B}(\mathbb{R}).$$

Since μ_X is a probability measure, necessarily $\int_{\mathbb{R}} f(x)dx = 1$. Hence, we express this relationship compactly as $\mu_X(dx) = f(x)dx$. In particular, we may write,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)dx.$$

The density weights length, that is, probabilities are obtained by integrating f over sets, rather than evaluating f at individual points. A law with this representation is called absolutely continuous with respect to Lebesgue measure,

$$\int_{\mathbb{R}} f(x)dx = \mu_X(\mathbb{R}) = 1.$$

We write this relationship precisely as,

$$\mu_X(dx) = f(x)dx.$$

Intuitively, f describes how heavily probability is concentrrated near each value. Integrating f over a set B gives the probability that $X \in B$. The value $f(x)$ itself is not the probability of observing x. Indeed, it follows that,

$$\mathbb{P}(X = x) = \int_{\{x\}} f(u)du = 0.$$

Definition 1.17 (Laws with point masses and a density)**.** Suppose the law of X has the form,

$$\mu_X(dx) = \sum_{i \in J} p_i \delta_{a_i}(dx) + f(x)dx,$$

where J is finite or countable, the points a_i are distinct, $p_i \geq 0$, and f is nonnegative and Borel-measurable. The components must satisfy,

$$\sum_{i \in J} p_i + \int_{\mathbb{R}} f(x)dx = 1.$$

Equivalently, for every Borel set B,

$$\begin{aligned} \mu_X(B) &= \sum_{i \in J} p_i \delta_{a_i}(B) + \int_B f(x)dx \\ &= \sum_{i \in J; a_i \in B} p_i + \int_B f(x)dx. \end{aligned}$$

When both components have positive total mass, this is a mixed discrete–continuous law. Purely discrete laws and laws with a density arise by setting the other component to zero. We restrict attention to these forms in this course; not every probability law admits this representation.

Remark 1.6 (Interpreting components of mixed law). The sum counts the point masses whose locations lie in B, while the integral gives the contribution from the density over B. The notation $\mu_X(dx)$ summarizes this decomposition. That is, to evaluate the measure on a set, we integrate,

$$\mu_X(B) = \int_B \mu_X(dx).$$

In a mixed law, f is the density of only the continuous component so it does not integrate to one. If, some continuous function q that satisfies,

$$q := \int_{\mathbb{R}} f(x)dx > 0,$$

we can write $f = qg$ where $g = \frac{f}{q}$ is a probability density satisfying $\int_{\mathbb{R}} g(x)dx = 1$.

Example 1.8 (Point mass at zero)**.** Suppose $X = 0$ with probability 0.2 while conditional on $X \neq 0$, its distribution has density g. Then,

$$\mu_X(dx) = 0.2\delta_0(dx) + 0.8g(x)dx, \qquad \int_{\mathbb{R}} g(x)dx = 1.$$

Thus, 20% of the probability is concentrated at zero, and the remaining 80% is distributed according to g. For $B = [-1, 2]$, the interval contains zero so,

$$\begin{aligned}\mu_X([-1, 2]) &= 0.2\delta_0([-1, 2]) + 0.8 \int_{-1}^{2} g(x)dx \\ &= 0.2 + 0.8 \int_{-1}^{2} g(x)dx.\end{aligned}$$

For $B = [1, 2]$, the interval excludes zero so,

$$\mu_X([1, 2]) = 0.8 \int_{-1}^{2} g(x)dx.$$

In both cases, the continuous contribution carries the weight 0.8.

Example 1.9 (Bankruptcy risk)**.** Fix a time t. Suppose a stock price S_t is zero with probability p, representing bankruptcy, and otherwise has density g on $[0, \infty)$ with,

$$\int_0^\infty g(x)dx = 1.$$

Extending g by zero outside $[0, \infty)$, the law of S_t is,

$$\mu_{S_t}(dx) = p\delta_0(dx) + (1 - p)g(x)dx.$$

For any Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}(S_t \in B) = p\mathbb{I}_{\{0 \in B\}} + (1-p)\int_{B\cap[0,\infty)} g(x)dx.$$

The bankruptcy mass contributes precisely when $0 \in B$. For example, for $b > 0$,

$$\begin{aligned}\mathbb{P}(0 \le S_t \le b) &= \mathbb{P}(0 < S_t \le b)\\ &= p + (1-p)\int_0^b g(x)dx\\ &= (1-p)\int_0^b g(x)dx.\end{aligned}$$

Hence, the difference between these probabilities is exactly $\mathbb{P}(S_t = 0) = p$.

The law of a single random variable describes its possible values and their probabilities. For a stochastic process, however, knowing the law of S_t at each time t does not tell us how values at different times vary together. To describe this dependence, we need joint distributions.

Definition 1.18. Let $X_1, \ldots, X_n$ be real-valued random variables on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Their joint distribution is the law of the random vector,

$$\mathbf{X} : \Omega \to \mathbb{R}^n, \qquad \mathbf{X}(\omega) = \begin{bmatrix} X_1(\omega) \\ \vdots \\ X_n(\omega) \end{bmatrix}.$$

It is the probability measure,

$$\mu_{\mathbf{X}} : \mathcal{B}(\mathbb{R}^n) \to [0,1]$$

defined by $\mu_{\mathbf{X}}(B) = \mathbb{P}(X \in B)$ for $B \in \mathcal{B}(\mathbb{R}^n)$.

For a rectangular set $B = B_1 \times \cdots \times B_n$, this becomes,

$$\mu_{\mathbf{X}}(B_1 \times \cdots \times B_n) = \mathbb{P}(X_1 \in B_1, \ldots, X_n \in B_n).$$

The right-hand side is the probability that all the stated conditions hold simultaneously. *Dirac measure* extend this setting as well. For $a \in \mathbb{R}^n$, the point mass at a is,

$$\delta_{\mathbf{a}}(B) = \begin{cases} 1, & \mathbf{a} \in B, \\ 0, & \mathbf{a} \notin B. \end{cases}$$

Definition 1.19 (Marginal distribution)**.** The law of an individual coordinate X_i is called a *marginal distribution* of X. It is obtained from the *joint law* by leaving every other coordinate unrestricted,

$$\mu_{X_i}(B) = \mu_{\mathbf{X}}\left(\mathbb{R}^{i-1} \times B \times \mathbb{R}^{n-i}\right), \qquad B \in \mathcal{B}(\mathbb{R}).$$

For example, for a pair of random variables (X, Y),

$$\mu_X(B) = \mu_{(X,Y)}(B \times \mathbb{R}), \qquad \mu_Y(B) = \mu_{(X,Y)}(\mathbb{R} \times B).$$

The joint law determines all its marginal laws, but the marginal laws generally do not determine the joint law, that is, they do not specify the dependence between coordinates.

Example 1.10 (Identical marginals, different joint laws)**.** Let $U \sim \text{Bernoulli}(1/2)$.

Consider the two random vectors (U, U) and $(U, 1-U)$. Every coordinate has the same Bernoulli(1/2) distribution. Their joint laws, however, are,

$$\mu_{(U,U)} = \frac{1}{2}\delta_{(0,0)} + \frac{1}{2}\delta_{(1,1)},$$
$$\mu_{(U,1-U)} = \frac{1}{2}\delta_{(0,1)} + \frac{1}{2}\delta_{(1,0)}.$$

In the first model, the coordinates always agree while in the second, they always differ. In particular, observe that their probabilities,

$$\mathbb{P}(U = 1, U = 1) = \frac{1}{2}, \qquad \mathbb{P}(U = 1, 1 - U = 1) = 0.$$

Thus, identical coordinate distributions can produce different probabilities for joint events.

Definition 1.20 (Joint density)**.** A random vector $\mathbf{X} \in \mathbb{R}^n$ has joint density if there exists a nonnegative Borel-measurable function $f_{\mathbf{X}} : \mathbb{R}^n \to [0, \infty)$ such that,

$$\mu_{\mathbf{X}}(B) = \int_B f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}, \qquad \forall B \in \mathcal{B}(\mathbb{R}^n).$$

Here, $d\mathbf{x} = dx_1 \cdots dx_n$ denotes integration with respect to n-dimensional Lebesgue measure. Necessarily, we have that, $\int_{\mathbb{R}^n} f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} = 1$. In two dimensions,

$$\mathbb{P}\left((X, Y) \in B\right) = \int \int_B f_{X,Y}(x, y) dx dy.$$

Note that, here $B = I_1 \times I_2$, the rectangular Borel set. We can express the marginal law of X on Borel set $B = I_1 \times I_2$ as,

$$\mu_X(B) = \mathbb{P}((X, Y) \in I_1 \times I_2) = \mathbb{P}(X \in I_1, Y \in I_2).$$

More generally, for a rectangle $B = [a, b] \times [c, d]$,

$$\mathbb{P}(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y) dy dx.$$

Geometrically, this is the volume under the density surface above the region B.

Proposition 1.4 (Marginal densities)**.** *If (X, Y) has joint density $f_{X,Y}$, then marginal densities given by,*

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y)dy, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)dx.$$

These identities hold almost everywhere.

Proof. For any Borel set $A \subseteq \mathbb{R}$, leaving Y unrestricted gives,

$$\begin{aligned}\mathbb{P}(X \in A) &= \mathbb{P}((X, Y) \in A \times \mathbb{R}) \\ &= \int_A \left(\int_{\mathbb{R}} f_{X,Y}(x, y)dy \right) dx.\end{aligned}$$

The inner integral therefore defines a density for X. The argument for Y is identical. □

Example 1.11 (Joint law without a joint density)**.** Let $U \sim \text{Unif}([0, 1])$ and set,

$$X = U, \qquad Y = U.$$

Both coordinates have uniform densities, but the vector (X, Y) always lies on the diagonal $D = \{(u, u) : u \in [0, 1]\}$. Thus, $\mathbb{P}((X, Y) \in D) = 1$. The diagonal has two-dimensional Lebesgue measure zero. If the joint law had density f, we would instead obtain,

$$\mathbb{P}((X, Y) \in D) = \int \int_D f(x, y)dxdy = 0.$$

Hence, a contradiction. Moreover, every individual point has probability zero,

$$\mathbb{P}((X, Y) = (u, u)) = \mathbb{P}(U = u) = 0.$$

This law is therefore neither a collection of point masses nor a law with a two dimensional density. In particular, even when each coordinate has a density, the joint law need not have one.

1.4 Gaussian Processes

Definition 1.21 (Multivariate Gaussian distribution)**.** A random vector $\mathbf{X} \in \mathbb{R}^n$ is jointly Gaussian if every linear combination,

$$\mathbf{c}^\top \mathbf{X} = \sum_{i=1}^{n} c_i X_i, \qquad \mathbf{c} \in \mathbb{R}^n$$

has a one-dimensional normal distribution, allowing variance zero.

The law of a Multivariate Gaussian (MVN) is determined by its mean vector $\mathbf{m}$ and covariance matrix $\mathbf{\Sigma}$, whose entries are,

$$m_i = \mathbb{E}X_i, \qquad \Sigma_{ij} = \text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - m_i)(X_j - m_j)].$$

We write, $\mathbf{X} \sim \mathcal{N}(\mathbf{m}, \Sigma)$. The covariance matrix is symmetric and *positive semi-definite*,

$$\mathbf{c}^\top \Sigma \mathbf{c} = \mathrm{Var}(\mathbf{c}^\top \mathbf{X}) \geq 0.$$

If Σ is positive definite, the joint law has density,

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2} (\mathbf{x} - \mathbf{m})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{m}) \right).$$

If Σ is singular, the Gaussian law is still well defined, but it has no density with respect to the n-dimensional Lebesgue measure. The inverse-and-determinant formula above does not apply in that case. Consider the 2-dimensional case for some vector (X, Y).

Example 1.12 (Bivariate Gaussian law)**.** Suppose $\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}(\mathbf{m}, \Sigma)$ where $\mathbf{m} \in \mathbb{R}^2$ and Σ is a positive definite 2×2 covariance matrix. Writing $\mathbf{z} = (x, y)^\top$, the joint density is,

$$f_{X,Y}(x, y) = \frac{1}{2\pi \sqrt{\det(\Sigma)}} \exp\left(-\frac{1}{2} (\mathbf{z} - \ \mathbf{m})^\top \Sigma^{-1} (\mathbf{z} - \mathbf{m}) \right).$$

For example,

$$\mathbb{P}(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y) dy dx.$$

The density is evaluated at the numerical coordinates x, y where the random variables X, Y describe the vector whose law it represents. We now apply joint distributons to stochastic processes. A process may contain infinitely many random variables, but selecting finitely many times produces an ordinary random vector.

Definition 1.22 (Finite-dimensional distributions)**.** Let $(S_t)_{t \in [0,T]}$ be a stochastic process.

For $k \geq 1$ and times $t_1, \ldots, t_k \in [0, T]$, its finite-dimensional distribution at these times is the joint law of,

$$\mathbf{S_t} = \begin{pmatrix} S_{t_1} \\ \vdots \\ S_{t_k} \end{pmatrix}, \qquad \mathbf{t} = (t_1, \ldots, t_k).$$

In particular,

$$\mu_{\mathbf{S_t}}(B_1 \times \cdots \times B_k) = \mathbb{P}(S_{t_1} \in B_1, \ldots, S_{t_k} \in B_k).$$

If the process has continuous sample paths, we can also regard it as a random element,

$$S : \Omega \to \mathcal{C}([0, T]), \qquad S(\omega)(t) = S_t(\omega).$$

For the cylinder set,

$$C = \{ f \in \mathcal{C}([0, T]) : f(t_i) \in B_i : i = 1, \ldots, k \}$$

its path law satisfies,

$$\begin{aligned}\mu_S(C) &= \mathbb{P}(S_{t_1} \in B_1, \ldots, S_{t_k} \in B_k) \\ &= \mu_{\mathbf{S_t}}(B_1 \times \cdots \times B_k) \\ &= \int_{B_1 \times \cdots \times B_k} \mathbf{f_t}(x_1, \ldots, x_k) dx_1 \cdots dx_k,\end{aligned}$$

where the last line follows if the finite-dimensional law has a density $\mathbf{f_t}$. Thus, finite-dimensional distributions give probabilities to cylinder events. Since these events generate the Borel σ-algebra on $\mathcal{C}([0,T])$, they determine the law of a continuous-path process.

Definition 1.23 (Gaussian process)**.** A stochastic process $(S_t)_{t\in[0,T]}$ is a *Gaussian process* if every finite collection of its values is jointly Gaussian. That is, for every $k \geq 1$ and every choice $t_1, \ldots, t_k \in [0,T]$ the vector is follows the law,

$$\begin{pmatrix} S_{t_1} \\ \vdots \\ S_{t_k} \end{pmatrix} \sim \mathcal{N}(\mathbf{m_t}, \mathbf{\Sigma_t})$$

where,

$$\mathbf{m_t} = \begin{pmatrix} m(t_1) \\ \vdots \\ m(t_k) \end{pmatrix}, \qquad \Sigma_t = \begin{bmatrix} K(t_1,t_1) & \cdots & K(t_1,t_k) \\ \vdots & \ddots & \vdots \\ K(t_k,t_1) & \cdots & K(t_k,t_k) \end{bmatrix}.$$

Here, $m(t) = \mathbb{E}[S_t]$ is the mean function and $K(s,t) = \mathrm{Cov}(S_s, S_t)$ is the covariance function, also called the covariance kernel. We write $S \sim \mathrm{GP}(m, K)$.

The functions m and K are defined over the time index. Choosing finitely many times converts them into an ordinary mean vector and covariance matrix. These determine the corresponding finite-dimensional Gaussian law. The validity of such covariance functions is specified below.

Remark 1.7 (Valid covariance functions). A covariance function must be symmetric,

$$K(s,t) = K(t,s)$$

and must produce a positive semidefinite matrix for every finite choice of times. Equivalently,

$$\sum_{i=1}^{k} \sum_{j=1}^{k} c_i c_j K(t_i, t_j) \geq 0$$

for all $k \geq 1$, all times $t_1, \ldots, t_k$, and all real coefficients $c_1, \ldots, c_k$. Consequently, an arbitrary function of two times cannot be used as a covariance function. For example, $K(s,t) = e^{-|t-s|}$ is a valid covariance function, whereas $e^{|t-s|}$ is not because its diagonal values are one, but its off-diagonal values exceed one, violating,

$$|K(s,t)| \leq \sqrt{K(s,s)K(t,t)}.$$

Remark 1.8 (Centered Gaussian processes). A Gaussian process is centered, or zero-mean, if $m(t) = 0$ for every t. Then, $\mathbf{S_t} \sim \mathcal{N}(0, \Sigma_t)$.

All finite-dimensional distributions are determined by the covariance function alone. Gaussianity requires joint Gaussianity at every finite collection of times. Having a normal marginal distribution each individual time is therefore not sufficient.

Also, indexing a process by continuous time does not imply continuous sample paths, and Gaussianity alone does not guarantee path continuity.

Example 1.13 (Brownian motion)**.** Standard Brownian motion $(W_t)_{t\geq 0}$ is a centered Gaussian process with continuous sample paths and covariance function,

$$K(s,t) = \min\{s,t\}.$$

Thus, $\mathbb{E}[W_t] = 0$ and $\mathrm{Var}(W_t) = t$. For $0 < t_1 < \cdots < t_k$,

$$\begin{pmatrix} W_{t_1} \\ \vdots \\ W_{t_k} \end{pmatrix} \sim \mathcal{N}(0, \Sigma_t), \qquad (\Sigma_t)_{ij} = \min\{t_i, t_j\}.$$

For example, at two times $0 < s < t$,

$$\begin{pmatrix} W_s \\ W_t \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} s & s \\ s & t \end{pmatrix} \right)$$

For distinct, strictly positive times, Σ_t is positive definite, and the finite-dimensional density,

$$f_t(\mathbf{x}) = \frac{1}{(2\pi)^{k/2} \det(\Sigma_t)^{1/2}} \exp\left(-\frac{1}{2} \mathbf{x}^\top \Sigma_t^{-1} \mathbf{x} \right).$$

If time zero is included, the vector has deterministic coordinate $W_0 = 0$, so its covariance matrix is singular. Its joint law remains Gaussian, but the displayed density formula no longer applies.

1.5 Expectations and Integrability

The expectation of a random variable is a probability-weighted average of its possible values. We express this average as an integral with respect to its law. Before defining a finite expectation, we require a condition ensuring that the positive and negative contributions are both finite.

Definition 1.24 (Integrability)**.** Let $X : \Omega \to \mathbb{R}$ be a random variable with law μ_X.

We say that X is *integrable* if,

$$\int_{\mathbb{R}} |x| \mu_X(dx) < \infty.$$

Equivalently, $\mathbb{E}[|X|] < \infty$.

For a law consistent of point masses and a density,

$$\mu_X(dx) = \sum_{j\in J} p_j \delta_{a_j}(dx) + f(x)dx,$$

the integrability condition becomes,

$$\sum_{j\in J} p_j|a_j| + \int_{\mathbb{R}} |x| f(x)dx < \infty.$$

Thus, for a discrete law we check the sum, for a law with a density, we check the integral, and for a mixed law we check both. We now define expectation.

Definition 1.25 (Expectation)**.** If X is integrable, its expectation is,

$$\mathbb{E}[X] := \int_{\mathbb{R}} x\mu_X(dx).$$

In particular, for the mixed law,

$$\mathbb{E}[X] = \sum_{j\in J} p_j a_j + \int_{\mathbb{R}} xf(x)dx.$$

The pont masses contribute a weighted sum and the density contributes a weighted integral. The notations, $\int_{\mathbb{R}} x\mu_X(dx)$ and $\int_{\mathbb{R}} xd\mu_X(dx)$ represents the same integral.

Remark 1.9 (Why require integrability?). Define the positive and negative parts of X by,

$$X^+ := \max\{X, 0\}, \qquad X^- := \max\{-X, 0\}.$$

Then, $X = X^+ - X^-$ and $|X| = X^+ + X^-$. Consequently, integrability ensure that,

$$\mathbb{E}[X^+] < \infty, \quad \mathbb{E}[X^-] < \infty,$$

that is, expectations of both parts remain finite. So,

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$$

is a well-defined number.

For a nonnegative random variable, its expectation is always defined as an extended value in $[0, \infty)$. More generally, if exactly one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is infinite, an extended expectation can be defined. If both are infinite, the expectation is undefined since the expression $\infty - \infty$ has no meaning. Hence, we use integrability to ensure that expectations are finite.

Example 1.14. Let X have the standard Cauchy density,

$$f(x) = \frac{1}{\pi(1+x^2)}, \qquad x \in \mathbb{R}.$$

Note, $f(x)$ is a valid density because,

$$\int_{-\infty}^{\infty} f(x)dx = \frac{1}{\pi}\left[\arctan x\right]_{-\infty}^{\infty} = 1.$$

To test integrability, first integrate over $[-R, R]$,

$$\begin{aligned}\int_{\mathbb{R}} |x| d\mu(x) &= \int_{-R}^{R} |x| f(x)dx \\ &= \frac{2}{\pi}\int_{0}^{R} \frac{x}{1+x^2}dx \\ &= \frac{2}{\pi}\frac{1}{2}\left[\log(1+x^2)\right]_0^R \\ &= \frac{1}{\pi}\log(1+R^2).\end{aligned}$$

Letting $R \to \infty$, $\log(1+R^2) \to \infty$ hence,

$$\int_{\mathbb{R}} |x| d\mu(x) = \infty.$$

Hence, X is not integrable. Moreover, symmetry gives

$$\mathbb{E}[X^+] = \mathbb{E}[X^-] = \infty$$

so its expectation is undefined. Although,

$$\int_{-R}^{R} xf(x)dx = 0 \qquad \forall R > 0$$

this does not establish that $\mathbb{E}[X] = 0$. Symmetric trunction forces the positive and negative contributions to cancel but does not make either contribution finite.

Example 1.15 (Absolute moments of Cauchy random variables)**.** For $p > 0$, the p-th absolute moment is computed as,

$$\mathbb{E}[|X|^p] = \frac{2}{\pi}\int_0^{\infty} \frac{x^p}{1+x^2}dx.$$

Near zero, the integrand is bounded by x^p, which is integrable on $[0, 1]$.

For $x \geq 1$, $x^2 \leq 1 + x^2 \leq 2x^2$ so,

$$\frac{1}{2}x^{p-2} \leq \frac{x^p}{1+x^2} \leq x^{p-2}.$$

Therefore, the tail integral converges exactly when $\int_1^{\infty} x^{p-2}dx < \infty$ which holds precisely when $p < 1$. Thus, for positive orders,

$$\mathbb{E}[|X|^p] < \infty \iff 0 < p < 1.$$

Hence, the Cauchy density decays sufficiently quickly to integrate to one, but not sufficiently quickly for its first absolute moment to be finite.

Proposition 1.5 (Expectation of a function of a random variable)**.** *Let X have law μ_X and let,*

$$\phi : \mathbb{R} \to \mathbb{R}$$

be Borel-measurable. If,

$$\int_{\mathbb{R}} |\phi(x)| \mu_X(dx) < \infty$$

then $\phi(X)$ is integrable and,

$$\mathbb{E}\phi(X) = \int_{\mathbb{R}} \phi(x) \mu_X(dx) = \int_{\Omega} \phi(X(\omega)) \mathbb{P}(d\omega).$$

For a nonnegative measurable ϕ, the same identity holds with $+\infty$ being permitted. These integrals compute the same expectation over different spaces. The first integrates over the possible values x weighted by the law of X. The second integrates over the underlying outcomes ω weighted by $\mathbb{P}$. Their equality follows from the pushforward relation,

$$\mu_X = \mathbb{P} \odot X^{-1}.$$

For a law with point masses and density, this gives,

$$\mathbb{E}[\phi(X)] = \sum_{j \in J} p_j \phi(x_j) + \int_{\mathbb{R}} \phi(x) f(x) dx$$

provided,

$$\sum_{j \in J} p_j |\phi(a_j)| + \int_{\mathbb{R}} |\phi(x)| f(x) dx < \infty.$$

In particular, we can compute $\mathbb{E}[\phi(X)]$ from the law of X, without first finding the law of $\phi(X)$.

Definition 1.26 (Sure and Almost sure events)**.** An event $A \in \mathcal{F}$ occurs surely in $A = \Omega$.

It occurs *almost surely* if $\mathbb{P}(A) = 1$ or equivalently $\mathbb{P}(A^c) = 0$.

A sure event contains every possible outcome. An almost sure event may exclude outcomes, but the excluded set has probability zero. A property holds almost surely if the event on which it holds has probability one. Consider the following example.

Example 1.16 (Probability zero does not imply empty)**.** Let $\Omega = [0, 1]$ and $U(\omega) = \omega$.

Equip such a probability space with uniform probability measure. The event,

$$A = \{U > 0\} = (0, 1],$$

which has probability one since $\mathbb{P}(A^c) = \mathbb{P}(U = 0) = 0$. Thus, $U > 0$ almost surely, but not surely. It fails at the outcome $\omega = 0$. More generally, if $D \subseteq [0, 1]$ is finite or countable then,

$$\mathbb{P}(U \in D) = \sum_{x \in D} \mathbb{P}(U = x) = 0.$$

For example, a uniform random variable is irrational almost surely.

Proposition 1.6 (Null sets do not contribute to expectations)**.** *Let $N \in \mathcal{F}$ satisfy $\mathbb{P}(N) = 0$. For any real-valued measurable random variable X,*

$$\int_N |X(\omega)|\mathbb{P}(d\omega) = 0$$

and therefore,

$$\int_N X(\omega)\mathbb{P}(d\omega) = 0.$$

Consequently, if X is integrable and A occurs almost surely then,

$$\mathbb{E}[X] = \int_A X(\omega)\mathbb{P}(d\omega).$$

In particular, if $X = Y$ almost surely, then $\mathbb{E}[|X - Y|] = 0$. Hence X is integrable if and only if Y is integrable, and in that case, $\mathbb{E}[X] = \mathbb{E}[Y]$. Changing a random variable on a measurable set of probability zero does not change its law or its expectation.

Finally, countably many null events can be discarded together.

If $\mathbb{P}(N_j) = 0$ for every j, countable subadditivity gives,

$$\mathbb{P}\left(\bigcup_{j=1}^{\infty} N_j\right) \leq \sum_{j=1}^{\infty} \mathbb{P}(N_j) = 0.$$

This conclusion does not extend to arbitrary uncountable unions since each singleton in $[0, 1]$ has uniform probability zero, but their union is the entire sample space.

For a Borel-measurable function $\phi : \mathbb{R} \to \mathbb{R}$ satisfying,

$$\int_{\mathbb{R}} |\phi(x)|\mu_X(dx) < \infty$$

we can compute its expectation in either of two independent ways,

$$\mathbb{E}[\phi(X)] = \int_{\mathbb{R}} \phi(X)\mu_X(dx) = \int_{\Omega} \phi(X(\omega))\mathbb{P}(d\omega).$$

The first integral averages over the possible values of X using law μ_X whereas the second averages over the underlying outcomes ω using $\mathbb{P}$. For discrete laws, these calculations rely on the following property of Dirac measures.

Proposition 1.7 (Integration against a Dirac measure)**.** *Let $a \in \mathbb{R}$ and let $\phi : \mathbb{R} \to \mathbb{R}$ be Borel-measurable. Then,*

$$\int_{\mathbb{R}} \phi(x)\delta_a(dx) = \phi(a).$$

Since δ_a places all its mass at a, only the value $\phi(a)$ contributes to the integral.

Example 1.17 (Second moment of a Bernoulli random variable)**.** Let $X \sim \text{Bernoulli}(p)$ and take $\phi(x) = x^2$. Since the law of X, denoted μ_X is,

$$\mu_X = (1-p)\delta_0 + p\delta_1.$$

Taking expectations, we obtain,

$$\begin{aligned}
\mathbb{E}[X^2] &= \int_{\mathbb{R}} x^2 \mu_X(dx) \\
&= (1-p)\int_{\mathbb{R}} x^2 \delta_0(dx) + p\int_{\mathbb{R}} x^2 \delta_1(dx) \\
&= (1-p)0^2 + p1^2 \\
&= p.
\end{aligned}$$

This follows directly from $X^2 = X$ almost surely, since X takes only values 0 and 1.

Example 1.18 (Expected stock price with bankruptcy)**.** Suppose a stock S_t has law,

$$\mu_{S_t}(dx) = p\delta_0(dx) + (1-p)g(x)dx$$

where p is the probability of bankruptcy and g is a probability density on $(0, \infty)$.

Assume,

$$\int_0^\infty xg(x)dx < \infty$$

so that S_t is integrable. Computing its expectation directly from its law gives,

$$\begin{aligned}
\mathbb{E}[S_t] &= \int_{\mathbb{R}} x\mu_{S_t}(dx) \\
&= \int_{\mathbb{R}} x\left[p\delta_0(dx) + (1-p)g(x)dx\right] \\
&= p\int_{\mathbb{R}} x\delta_0(dx) + (1-p)\int_0^\infty xg(x)dx \\
&= p\cdot 0 + (1-p)\int_0^\infty xg(x)dx \\
&= (1-p)\int_0^\infty xg(x)dx.
\end{aligned}$$

The Diract integral evaluates the function $x \mapsto x$ at zero,

$$\int_{\mathbb{R}} x\delta_0(dx) = 0.$$

Thus, the bankruptcy outcome contributes zero to the expected price. If Y has density g, then $\mathbb{E}[Y] = \int_0^\infty xg(x)dx$ and therefore, $\mathbb{E}[S_t] = (1-p)\mathbb{E}[Y]$. The overall expected price is the mean price in the non-bankruptcy case, weighted by the probability $1-p$ of that case occurring.

Example 1.19 (Expected outcome of a fair die)**.** Let X be the result of rolling a fair six-sided die such that,

$$\mathbb{P}(X = n) = \frac{1}{6}, \qquad n = 1, \ldots, 6.$$

Its law places probability mass $\frac{1}{6}$ at each possible value,

$$\mu_X(dx) = \sum_{n=1}^{6} \frac{1}{6} \delta_n(dx).$$

Since X is bounded, it is integrable. Computing its expectation directly from its law,

$$\begin{aligned}
\mathbb{E}[X] &= \int_{\mathbb{R}} x \mu_X(dx) \\
&= \int_{\mathbb{R}} x \left(\sum_{n=1}^{6} \frac{1}{6} \delta_n(dx) \right) \\
&= \sum_{n=1}^{2} \frac{1}{6} \int_{\mathbb{R}} x \delta_n(dx) \\
&= \sum_{n=1}^{6} \frac{n}{6} \\
&= \frac{1+2+3+4+5+6}{6} \\
&= \frac{7}{2}.
\end{aligned}$$

Here, each Dirac integral evaluates $x \mapsto x$ at n hence,

$$\int_{\mathbb{R}} x \delta_n(dx) = n.$$

Thus, the expected outcome is 3.5. Evidently, the expectation is a probability-weighted average and need not be a value that X can actually take.

An event probability can be expressed as an expectation by using a random variable that records whether the event occurs. Consider the following random variable definition.

Definition 1.27 (Indicator random variable)**.** For an event $A \in \mathcal{F}$, its *indicator* is the random variable,

$$\mathbb{I}_A : \Omega \to \{0, 1\}, \qquad \mathbb{I}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

An *Indicator random variable* records whether A occurs. Its distribution is Bernoulli with parameter $\mathbb{P}(A)$, that is, it is specified by,

$$\mathbb{P}(\mathbb{I}_A = 1) = \mathbb{P}(A), \qquad \mathbb{P}(\mathbb{I}_A = 0) = 1 - \mathbb{P}(A).$$

Theorem 1.2 (Expectation of an indicator)**.** *For every event $A \in \mathcal{F}$,*

$$\mathbb{E}[\mathbb{I}_A] = \mathbb{P}(A).$$

Proof. Since $0 \leq \mathbb{I}_A \leq 1$, the indicator is integrable. Splitting the integral over A and A^c,

$$\begin{aligned}
\mathbb{E}[\mathbb{I}_A] &= \int_\Omega \mathbb{I}_A(\omega)\mathbb{P}(d\omega) \\
&= \int_A \mathbb{I}_A \mathbb{P}(d\omega) + \int_{A^c} \mathbb{I}_A(\omega)\mathbb{P}(d\omega) \\
&= \int_A 1\mathbb{P}(d\omega) + \int_{A^c} 0\mathbb{P}(d\omega) \\
&= \mathbb{P}(A).
\end{aligned}$$

The third equality uses the fact that $\mathbb{I}_A = 1$ on A and $\mathbb{I}_A = 0$ on A^c.

Thus, event probabilities are a special case of expectation, that is, averaging a variable that records occurrence gives the probability of occurrence. □

Example 1.20 (Indicator of a stock-price event)**.** Let S_t be a stock price at time t, and define,

$$A = \{\omega \in \Omega : S_t(\omega) < 100\}.$$

Since S_t is a random variable, $A \in \mathcal{F}$. Its indicator is,

$$\mathbb{I}_A(\omega) = \begin{cases} 1, & S_t(\omega) > 100, \\ 0, & S_t(\omega) \leq 100. \end{cases}$$

Suppose $\mathbb{P}(S_t > 100) = 0.35$. Then, $\mathbb{P}(\mathbb{I}_A = 1) = 0.35$ and $\mathbb{P}(\mathbb{I}_A = 0) = 0.65$. Hence, its law,

$$\mu_{\mathbb{I}_A} := 0.65\delta_0 + 0.35\delta_1.$$

Computing its expectation,

$$\begin{aligned}
\mathbb{E}\mathbb{I}_A &= \int_{\mathbb{R}} x\mu_{\mathbb{I}_A}(dx) \\
&= 0.65 \int_{\mathbb{R}} x\delta_0(dx) + 0.35 \int_{\mathbb{R}} x\delta_1(dx) \\
&= 0.65 \cdot 0 + 0.35 \cdot 1 \\
&= 0.35 \\
&= \mathbb{P}(A).
\end{aligned}$$

Indicators therefore also let us separate a random variable into its contributions on different events. The product $X\mathbb{I}_A$ equals X on A and zero elsewhere so for integrable X,

$$\mathbb{E}X\mathbb{I}_A = \int_A X(\omega)\mathbb{P}(d\omega).$$

Proposition 1.8 (Expectation decomposition over a finite partition)**.** *Let $A_1, \ldots, A_n \in \mathcal{F}$ be pairwise disjoint events satisfying,*

$$\bigcup_{i=1}^{n} A_i = \Omega.$$

If X is integrable then,

$$\mathbb{E}X = \sum_{i=1}^{n} \mathbb{E}X\mathbb{I}_A.$$

Proof. Each outcome ω belongs to exactly one of the events A_i. Therefore,

$$\sum_{i=1}^{n} \mathbb{I}_{A_i}(\omega) = 1, \qquad \forall \omega \in \Omega.$$

Multiplying $X(\omega)$ gives the pointwise identity,

$$X(\omega) = \sum_{i=1}^{n} X(\omega)\mathbb{I}_{A_i}(\omega).$$

Moreover, each term is integrable because $|X\mathbb{I}_{A_i}| \leq |X|$ for $\mathbb{E}[|X|] < \infty$.

Taking expectations and by linearity,

$$\begin{aligned}\mathbb{E}X &= \mathbb{E}\left[X \sum_{i=1}^{n} \mathbb{I}_{A_i}\right] = \mathbb{E}\left[\sum_{i=1}^{n} X\mathbb{I}_{A_i}\right] \\ &= \sum_{i=1}^{n} \mathbb{E}X\mathbb{I}_{A_i}.\end{aligned}$$

Each term is the contribution to the overall expectation from outcomes in A_i so,

$$\int_{\Omega} X d\mathbb{P} = \sum_{i=1}^{n} \int_{A_i} X d\mathbb{P}.$$

□

Expectation inherits basic algebraic and order properties of integration. That is, it respects linear combinations and almost-sure inequalities while taking absolute values before averaging prevents cancellation between positive and negative values.

Before stating these properties, recall that $X \leq Y$ almost surely means that,

$$\mathbb{P}\left(\{\omega \in \Omega : X(\omega) \leq Y(\omega)\}\right) = 1.$$

The inequality may fail at some outcomes, but those outcomes form an event of probability zero. This comparison event is measurable. Indeed, the joint map $(X, Y) : \Omega \to \mathbb{R}^2$ is measurable and $B = \{(x, y) \in \mathbb{R}^2 : x \leq y\}$ is closed, hence Borel. Therefore,

$$\{X \leq Y\} = (X, Y)^{-1}(B) \in \mathcal{F}.$$

Proposition 1.9 (Basic properties of expectation.)**.** *Let X, Y be integrable random variables on the same probability space and let $a, b \in \mathbb{R}$. Then,*

(i) ***Linearity.*** *The random variable $aX + bY$ is integrable, and*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

(ii) ***Monotonicity.*** *If $X \leq Y$ almost surely, then*

$$\mathbb{E}[X] \leq \mathbb{E}[Y].$$

(iii) ***Absolute-value inequality.***

$$|\mathbb{E}[X]| \leq \mathbb{E}[|X|].$$

(iv) ***Zero expectation for nonnegative random variable.*** *If $X \geq 0$ almost surely, and $\mathbb{E}[X] = 0$, then*

$$X = 0 \quad \textit{almost surely.}$$

Proof. We aim to prove properties **(i)** − **(iv)**.

Proof of (i). The ordinary triangle inequality gives,

$$|aX(\omega) + bY(\omega)| \leq |a||X(\omega)| + |b||Y(\omega)|, \qquad \forall \omega \in \Omega.$$

Integrating, we obtain

$$\begin{aligned}\int_\Omega |aX + bY| d\mathbb{P} &\leq |a| \int_\Omega |X| d\mathbb{P} + |b| \int_\Omega |Y| d\mathbb{P} \\ &= |a|\mathbb{E}[|X|] + |b|\mathbb{E}[|Y|] \\ &< \infty.\end{aligned}$$

Hence, $aX + bY$ is integrable. Therefore, we split its integral and factor out the constants,

$$\begin{aligned}\mathbb{E}[aX + bY] &= \int_\Omega (aX + bY)\, d\mathbb{P} \\ &= a \int_\Omega X\, d\mathbb{P} + b \int_\Omega Y\, d\mathbb{P} \\ &= a\mathbb{E}[X] + b\mathbb{E}[Y].\end{aligned}$$

No independence assumption is needed.

Proof of (ii). If $X \leq Y$ almost surely, then $Y - X \geq 0$ almost surely. A nonnegative function has a nonnegative integral, and a null event does not affect that integral. Hence,

$$0 \leq \mathbb{E}[Y - X] = \mathbb{E}[Y] - \mathbb{E}[X].$$

Re-arranging, we obtain,

$$\mathbb{E}[X] \leq \mathbb{E}[Y].$$

Proof of (iii). Split X according to its sign,

$$X = X\mathbb{I}_{\{X\geq 0\}} + X\mathbb{I}_{\{X<0\}}.$$

Observe that both terms are integrable because their absolute values are bounded by $|X|$. By linearity and the triangle inequality for real numbers,

$$\begin{aligned}|\mathbb{E}[X]| &= |\mathbb{E}[X\mathbb{I}_{\{X\geq 0\}}] + \mathbb{E}[X\mathbb{I}_{\{X<0\}}]| \\ &\leq |\mathbb{E}[X\mathbb{I}_{\{X\geq 0\}}]| + |\mathbb{E}[X\mathbb{I}_{\{X<0\}}]|.\end{aligned}$$

Since $X\mathbb{I}_{\{X\geq 0\}} > 0$ and $X\mathbb{I}_{\{X<0\}} \leq 0$, by monotonicity, both expectations have the same respective signs. Therefore, re-writing,

$$\begin{aligned}|\mathbb{E}[X]| &\leq \mathbb{E}[X\mathbb{I}_{\{X\geq 0\}}] - \mathbb{E}[X\mathbb{I}_{\{X<0\}}] \\ &= \mathbb{E}[X\mathbb{I}_{\{X\geq 0\}} - X\mathbb{I}_{\{X<0\}}] \\ &= \mathbb{E}[|X|].\end{aligned}$$

We obtain the last line via the identity $X\mathbb{I}_{\{X\geq 0\}} - X\mathbb{I}_{\{X<0\}} = |X|$. Hence, positive and negative values can cancel when computing $\mathbb{E}[X]$ but contribute positively when computing $\mathbb{E}[|X|]$.

An equivalent shorter proof follows from $-|X| \leq X \leq |X|$. Taking expectations we obtain,

$$-\mathbb{E}[|X|] \leq \mathbb{E}[X] \leq \mathbb{E}[|X|]$$

which is precisely the desired inequality.

Proof of (iv). Suppose $X \geq 0$ almost surely and $\mathbb{E}[X] = 0$. For each integer $n \geq 1$ define,

$$A_n = \left\{\omega \in \Omega : X(\omega) \geq \frac{1}{n}\right\}.$$

This is an event because $A_n = X^{-1}([1/n, \infty]) \in \mathcal{F}$. On A_n, the value of X is at least $1/n$. Outside A_n, it is nonnegative almost surely. Thus,

$$X \geq \frac{1}{n}\mathbb{I}_{A_n} \qquad \text{almost surely.}$$

By monotonicity and linearity, and the expectation of an indicator,

$$0 = \mathbb{E}[X] \geq \frac{1}{n}\mathbb{E}[\mathbb{I}_{A_n}] = \frac{1}{n}\mathbb{P}(A_n) \geq 0.$$

It follows that $\mathbb{P}(A_n) = 0$ for every $n \geq 1$. Every strictly positive number is at least $1/n$ for some sufficiently large n. Consequently,

$$\{X > 0\} = \bigcup_{n=1}^{\infty} A_n.$$

Countable subadditivity now gives,

$$\mathbb{P}(X > 0) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n) = 0.$$

Since $X \geq 0$ almost surely implies $\mathbb{P}(X < 0) = 0$. Hence,

$$\mathbb{P}(X \neq 0) = \mathbb{P}(X > 0) + \mathbb{P}(X < 0) = 0.$$

Hence, $X = 0$ almost surely. □

The nonnegativity assumption in part (iv) is essential as positive and negative contributions can cancel, that is, a random variable taking 1 and -1 with equal probability has expectation zero but is never zero. Linearity also explains the partition identity established earlier.

If $A_1, \ldots, A_m$ are pairwise disjoint events covering Ω then,

$$\sum_{i=1}^{m} \mathbb{I}_{A_i} = 1, \qquad X = \sum_{i=1}^{n} X\mathbb{I}_{A_i}.$$

Therefore,

$$\mathbb{E}[X] = \sum_{i=1}^{m} \mathbb{E}[X\mathbb{I}_{A_i}].$$

If $X = x_i$ on each A_I, this simplifies to,

$$\mathbb{E}X\mathbb{I}_{A_i} = x_i\mathbb{E}[\mathbb{I}_{A_i}] = x_i\mathbb{P}(A_i).$$

Hence, we obtain our familiar weighted-average formulation,

$$\mathbb{E}[X] = \sum_{i=1}^{m} x_i\mathbb{P}(A_i).$$

1.6 Sub-σ-Algebras and Independence

Knowing the individual laws of X and Y does not generally determine their joint law, that is, the marginal distributions do not describe how the variables depend on one another. Independence, hence, specifies a relationship under which probabilities of simultaneous events factor into products. Since a σ-algebra represents information, we first define independence between σ-algebras, then apply it to random variables.

Definition 1.28 (Independent σ-algebras)**.** Let $\mathcal{G}_1, \ldots, \mathcal{G}_n$ be a sub-σ-algebra of $\mathcal{F}$.

They are *independent* if for every choice of events $A_i \in \mathcal{G}$,

$$\mathbb{P}\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} \mathbb{P}(A_i).$$

An arbitrary family of sub-σ-algebras is independent if every finite subfamily is independent. In particular, two sub-σ-algebras $\mathcal{G}$ and $\mathcal{H}$ are independent if,

$$\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C), \qquad \forall A \in \mathcal{G}, C \in \mathcal{H}.$$

This requirement applies to every choice of events from the respective σ-algebras. Verifying the equality for one selected pair is not sufficient. If $\mathbb{P}(C) > 0$, the equality can also be written as,

$$\mathbb{P}(A \mid C) = \frac{\mathbb{P}(A \cap C)}{\mathbb{P}(C)} = \mathbb{P}(A).$$

Definition 1.29 (Independent random variables)**.** Random variables $X_1, \ldots, X_n$ are independent if their generated σ-algebras, $\sigma(X_1), \ldots, \sigma(X_n)$ are independent.

Equivalently, for every choice of Borel sets $B_1, \ldots, B_n \subseteq \mathbb{R}$,

$$\mathbb{P}(X_1 \in B_1, \ldots, X_n \in B_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i).$$

In terms of their laws, we may write this as,

$$\mu_{(X_1,\ldots,X_n)}(B_1 \times \cdots \times B_n) = \prod_{i=1} 6n\mu_{X_i}(B_i).$$

A sequence $(X_i)_{i\geq 1}$ is independent if every finite sub-collection is independent. This equivalence follows because the events in $\sigma(X_i)$ are precisely those of the form,

$$A_i = X_i^{-1}(B_i) = \{X_i \in B_i\}, \qquad B_i \in \mathcal{B}(\mathbb{R}).$$

Substituting these events into *Definition 1.29* gives the displayed factorization. For two independent random variables, the joint law assigns a rectangle $B_1 \times B_2$ the probability,

$$\mu_{(X,Y)}(B_1 \times B_2) = \mu_X(B_1)\mu_Y(B_2).$$

This resembles length times width, except the factors are marginal probabilities. The rectangle's "shape" alone does not justify multiplication as independence is the assumption that makes the factorization valid. Consider the simple fair coin example below.

Example 1.21. Consider $\Omega = \{HH, HT, TH, TT\}$ with $\mathcal{F} = 2^{\Omega}$ and assign probability $1/4$ to each outcome. Define the events,

$$X_1(\omega) = \begin{cases} 1, & \text{the first toss is heads,} \\ 0, & \text{the first toss is tails,} \end{cases} \qquad X_2(\omega) = \begin{cases} 1, & \text{the second toss is heads,} \\ 0, & \text{the second toss is tails.} \end{cases}$$

Observing X_1 tells us whether the first toss was heads or tails but does not allow us to distinguish HH from HT nor TH from TT. Hence the corresponding generated σ-algebras,

$$\sigma(X_1) = \{\emptyset, \Omega, \{HH, HT\}, \{TH, TT\}\},$$
$$\sigma(X_1) = \{\emptyset, \Omega, \{HH, TH\}, \{HT, TT\}\}.$$

Observing both variables distinguishes every outcome, consequently,

$$\sigma(X_1, X_2) = 2^{\Omega} = \mathcal{F}.$$

To verify independence, suppose we have arbitrary events, $A \in \sigma(X_1)$ and $C \in \sigma(X_2)$. If either event is $\emptyset$ or Ω, then $\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C)$ holds immediately. Otherwise, A specifies the result of the first toss and C specifies the result of the second.

Each has probability $\frac{1}{2}$ and their intersection contains exactly one outcome with probability $\frac{1}{4}$. Hence, we may compute the probability of the intersection as,

$$\mathbb{P}(A \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(A)\mathbb{P}(C).$$

This covers every possible pair of events, so $\sigma(X_1)$ and $\sigma(X_2)$ are independent and therefore, X_1 and X_2 are independent. For instance,

$$\{X_1 = 1\} \cap \{X_2 = 1\} = \{HH\},$$

and the corresponding joint probability is,

$$\mathbb{P}(X_1 = 1, X_2 = 1) = \frac{1}{4} = \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 1).$$

The uniform assignment on the four outcomes is essential as merely stating that each individual toss is fair would specify the marginals and not by itself establish independence.

Definition 1.30 (Independent events)**.** Events $A, C \in \mathcal{F}$ are independent if,

$$\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C).$$

Equivalently, their generated σ-algebras,

$$\sigma(A) = \{\emptyset, \Omega, A, A^c\}, \qquad \sigma(C) = \{\emptyset, \Omega, C, C^c\}$$

are independent. To see why the single equality suffices, observe that it also gives,

$$\begin{aligned} \mathbb{P}(A \cap C^c) &= \mathbb{P}(A) - \mathbb{P}(A \cap C) \\ &= \mathbb{P}(A)(1 - \mathbb{P}(C)) \\ &= \mathbb{P}(A)\mathbb{P}(C^c). \end{aligned}$$

The same argument gives,

$$\mathbb{P}(A^c \cap C) = \mathbb{P}(A^c)\mathbb{P}(C),$$

and equivalently,

$$\begin{aligned} \mathbb{P}(A^c \cap C^c) &= 1 - \mathbb{P}(A) - \mathbb{P}(C) + \mathbb{P}(A \cap C) \\ &= (1 - \mathbb{P}(A))(1 - \mathbb{P}(C)) \\ &= \mathbb{P}(A^c)\mathbb{P}(C^c). \end{aligned}$$

The remaining cases involve $\emptyset$ and Ω and hold automatically. Since $\sigma(\mathbb{I}_A) = \sigma(A)$ and $\sigma(\mathbb{I}_C) = \sigma(C)$, we also obtain that A and C are independent if and only if $\mathbb{I}_A$ and $\mathbb{I}_C$ are independent.

Finally, independence should be distinguished from disjointness. Disjoint events cannot occur together. If both have positive probability, then,

$$\mathbb{P}(A \cap C) = 0 \quad \text{but} \quad \mathbb{P}(A)\mathbb{P}(C) > 0$$

so they are hence not independent.

Independence also has two useful consequences: expectations of products factor into products of expectations, and applying separate measurable functions preserves independence.

Proposition 1.10 (Expectation of a product)**.** *Let X and Y be independent, integrable random variables. Then XY is integrable and,*

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Proof. Independence means that the joint law is the product of the marginal laws,

$$\mu_{(X,Y)} = \mu_X \otimes \mu_Y.$$

First establish integrability. Since $|x||y|$ is nonnegative, we may evalluate its integral one coordinate at a time. That is, under absolute expectations,

$$\begin{aligned}
\mathbb{E}[|XY|] &= \int_{\mathbb{R}} \int_{\mathbb{R}} |x||y| \mu_Y(dy) \mu_X(dx) \\
&= \int_{\mathbb{R}} |x| \left(\int_{\mathbb{R}} |y| \mu_Y(dy) \right) \mu_X(dx) \\
&= \left(\int_{\mathbb{R}} |x| \mu_X(dx) \right) \left(\int_{\mathbb{R}} |y| \mu_Y(dy) \right) \\
&= \mathbb{E}[|X|]\mathbb{E}[|Y|] \\
&< \infty.
\end{aligned}$$

Thus, XY is integrable.

We can now perform the same calculation with the signed integrand xy,

$$\begin{aligned}
\mathbb{E}[XY] &= \int_{\mathbb{R}} \int_{\mathbb{R}} xy \mu_Y(dy) \mu_X(dx) \\
&= \int_{\mathbb{R}} x \left(\int_{\mathbb{R}} y \mu_Y(dy) \right) \mu_X(dx) \\
&= \left(\int_{\mathbb{R}} x \mu_X(dx) \right) \left(\int_{\mathbb{R}} y \mu_Y(dy) \right) \\
&= \mathbb{E}[X]\mathbb{E}[Y],
\end{aligned}$$

as required. □

For discrete laws, this calculation is the separation of a double sum. For laws with densities, it is the separate of an iterated integral. Independence provides the product structure that

makes this possible, that is, it is unnecessary to assume separately that XY is integrable as independence and integrability of X, Y already imply it.

The converse, however, is false. The single equality $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ does not establish the factorization of probabilities required for independence.

Example 1.22 (Factorization of expectations without independence)**.** Suppose X takes the values $-1, 0, 1$ each with probability $1/3$ and let $Y = X^2$. Then,

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[Y] = \frac{2}{3}, \quad \mathbb{E}[XY] = \mathbb{E}[X^3] = 0.$$

Hence, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. Nevertheless, X and Y are not independent because,

$$\mathbb{P}(X = 0, Y = 0) = \frac{1}{3} \neq \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9} = \mathbb{P}(X = 0)\mathbb{P}(Y = 0).$$

We next consider what happens when independent variables are transformed separately. A measurable function of X uses only information already contained in X and cannot introduce dependence on an independent variable Y.

Proposition 1.11 (Functions of independent random variables)**.** *Let X, Y be independent random variables and let $g, h : \mathbb{R} \to \mathbb{R}$ be Borel-measurable. Then,*

$$g(X) \perp h(Y).$$

No integrability assumptions are needed.

We denote independence using $\perp$ as shorthand notation in this course.

Proof. Let $A, B \in \mathcal{B}(\mathbb{R})$ be arbitrary.

The events involving the transformed variables can be written using the preimages,

$$\{g(X) \in A\} = \{X \in g^{-1}(A)\}, \qquad \{h(Y) \in B\} = \{Y \in h^{-1}(B)\}.$$

Because g, h are Borel-measurable functions,

$$g^{-1}(A) \in \mathcal{B}(\mathbb{R}), \quad h^{-1}(B) \in \mathcal{B}(\mathbb{R}).$$

By independence of X, Y, we obtain,

$$\begin{aligned}\mathbb{P}(g(X) \in A,\ h(Y) \in B) &= \mathbb{P}(X \in g^{-1}(A),\ Y \in h^{-1}(B)) \\ &= \mathbb{P}(X \in g^{-1}(A))\mathbb{P}(Y \in h^{-1}(B)) \\ &= \mathbb{P}(g(X) \in A)\mathbb{P}(h(Y) \in B).\end{aligned}$$

Since this holds for every pair of Borel sets A, B, the transformed variables are independent. □

Remark 1.10 (Transformations preserve independence). The same argument can be expressed in terms of information,

$$\sigma(g(X)) \subseteq \sigma(X), \quad \sigma(h(Y)) \subseteq \sigma(Y).$$

For example, an event determined by $g(X)$ has the form,

$$\{g(X) \in A\} = \{X \in g^{-1}(A)\}$$

so it is already an event in $\sigma(X)$. Passing to smaller sub-σ algebras preserves independence because the required factorisation already holds for every event in the large ones. Thus, transforming X and Y separately may discard information but preserves their independence.

Combining the two propositions, whenever $g(X)$ and $h(Y)$ are integrable,

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)].$$

Integrability hence must be checked for the transformed variables. Integrability of X alone does not, for example, guarantee integrability of X^2.

Definition 1.31 (Equality in distribution)**.** Random variables X, Y are equal in distribution, written,

$$X \overset{d}{=} Y$$

if they have the same law,

$$\mathbb{P}(X \in B) = \mathbb{P}(Y \in B), \qquad \forall B \in \mathcal{B}(\mathbb{R}).$$

Note, they need not be defined in the same probability space. Equality in distribution describes the probabilities of possible values, rather than equality of the realized values. If X and Y are defined on the same probability space, then

$$X = Y \quad \text{a.s.} \implies X \overset{d}{=} Y.$$

However, the converse need not hold.

Definition 1.32 (Independent and Identically distributed random variables)**.** A sequence $(X_i)_{i\geq 1}$ is independent and identically distributed, abbreviated *i.i.d.* if,

(i) The random variables are independent: every finite subcollection is independent, and
(ii) They have common law μ,

$$\mu_{X_i} = \mu, \qquad \forall i \geq 1.$$

We often write $X_1, X_2, \ldots \overset{i.i.d.}{\sim} \mu$. Equivalently for any distinct indices $i_1, \ldots, i_k$ and Borel sets $B_1, \ldots, B_k$, the joint probability is equal to the product over independent Borel sets,

$$\mathbb{P}(X_{i_1} \in B_1, \ldots, X_{i_k} \in B_k) = \prod_{j=1}^{k} \mu(B_j).$$

Example 1.23 (Same distribution, different dependence)**.** Let $U \sim \text{Bernoulli}(1/2)$. If $X = U$ and $Y = 1 - U$, then $X \overset{d}{=} Y$ because both take 0 and 1 with equal probability.

Nevertheless, they always differ and are dependent,

$$\mathbb{P}(X = 1, Y = 1) = 0 \neq \frac{1}{4} = \mathbb{P}(X = 1)\mathbb{P}(Y = 1).$$

If instead X and Y record two independent fair coin tosses, they are i.i.d., yet they are still not equal almost surely, $\mathbb{P}(X = Y) = \frac{1}{2}$. Thus, identically distributed implies the same law, not the same outcome. The i.i.d. setting models repeated independent observations from one distribution. Going forward, it will be central to the laws of large numbers and Central Limit Theorem, with appropriate moment assumptions.

1.7 Variance, Probability & Tail Bounds

Expectation describes the average value of a random variable. Variance measures its average squared deviation from that mean, while covariance measures how the deviations of two variables vary together. We work with random variables having finite second moments,

$$\mathbb{E}X^2 < \infty, \qquad \mathbb{E}Y^2 < \infty.$$

These assumptions ensure that the expectations below are finite. Indeed,

$$|X| \leq \frac{1 + X^2}{2}, \qquad |Y| \leq \frac{X^2 + Y^2}{2},$$

so X, Y and XY are integrable.

Definition 1.33 (Variance and Covariance)**.** Let X, Y have finite second moments.

The *variance* of X is,

$$\text{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

The *covariance* of X and Y is,

$$\text{Cov}(X, Y) = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right].$$

Variance is nonnegative because it is the expectation of a square. Moreover, $\text{Var}(X) = 0$ if and only if $X = \mathbb{E}[X]$ almost surely. Indeed, a nonnegative random variable has expectation zero exactly when it is zero almost surely. Applying this to $(X - \mathbb{E}[X])^2$ gives the equivalence.

For covariance, the product of the centered values is positive when both variables lie on the same side of their respective means, and negative when they lie on opposite sides. Covariance averages these signed products. In particular,

$$\text{Cov}(X, X) = \text{Var}(X), \qquad \text{Cov}(X, Y) = \text{Cov}(Y, X).$$

Proposition 1.12 (Basic Variance and Covariance identities)**.** *Let $X_1, \ldots, X_n$ have finite second moments and let $a, b \in \mathbb{R}$. Then,*

(i) ***Variance identity.***

$$\mathrm{Var}(X_1) = \mathbb{E}[X_1^2] - (\mathbb{E}[X_1])^2 .$$

(ii) ***Covariance identity.***

$$\mathrm{Cov}(X_1, X_2) = \mathbb{E}[X_1 X_2] - \mathbb{E}[X_1]\mathbb{E}[X_2].$$

(iii) ***Variance under scaling and translation.***

$$\mathrm{Var}(aX_1 + b) = a^2 \,\mathrm{Var}(X_1).$$

(iv) ***Variance of a sum.***

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2 \sum_{1 \le i < j \le n} \mathrm{Cov}(X_i, X_j).$$

Proof. We aim to prove properties **(i)** − **(iv)**. Let $m_i = \mathbb{E}[X_i]$.

Proof of (i). Expanding the square and using linearity of expectation,

$$\begin{aligned}
\mathrm{Var}(X_1) &= \mathbb{E}[(X_1 - m)^2] \\
&= \mathbb{E}[X_1^2 - 2m_1 X_1 - m_1^2] \\
&= \mathbb{E}[X_1^2] - 2m_1 \mathbb{E}[X_1] + m_1^2 \\
&= \mathbb{E}[X_1^2] - m_1^2.
\end{aligned}$$

Proof of (ii). Similarly,

$$\begin{aligned}
\mathrm{Cov}(X_1, X_2) &= \mathbb{E}[(X_1 - m_1)(X_2 - m_2)] \\
&= \mathbb{E}[X_1 X_2 - m_2 X_1 - m_1 X_2 + m_1 m_2] \\
&= \mathbb{E}[X_1 X_2] - m_2 m_1 m_1 m_2 + m_1 m_2 \\
&= \mathbb{E}[X_1 X_2] - m_1 m_2.
\end{aligned}$$

Proof of (iii). Centering the transformed variable gives,

$$\begin{aligned}
aX_1 + b - \mathbb{E}[aX_1 + b] &= aX_1 + b - (am_1 + b) \\
&= a(X_1 - m_1).
\end{aligned}$$

Therefore, we obtain,

$$\begin{aligned}
\mathrm{Var}(aX_1 + b) &= \mathbb{E}[a^2 (X_1 - m_1)^2] \\
&= a^2 \,\mathrm{Var}(X_1).
\end{aligned}$$

Hence, adding constant shifts mean without changing deviations from it. Multiplying by a multiplies each deviation by a, and hence each squared deviation by a^2.

Proof of (iv). By linearity,

$$\sum_{i=1}^{n} X_i - \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n}(X_i - m_i).$$

Square of a finite sum consists of its squared terms and twice each distinct cross-product,

$$\left(\sum_{i=1}^{n}(X_i - m_i)\right)^2 = \sum_{i=1}^{n}(X_i - m_i)^2 + 2\sum_{1\le i<j\le m}(X_i - m_i)(X_j - m_j).$$

Taking expectations,

$$\begin{aligned}\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) &= \sum_{i=1}^{n}\mathbb{E}[(X_i - m_i)^2] \\ &\quad + 2\sum_{1\le i<j\le n}\mathbb{E}[(X_i - m_i)(X_j - m_j)] \\ &= \sum_{i=1}^{n}\operatorname{Var}(X_i) + 2\sum_{1\le i<j\le n}\operatorname{Cov}(X_i, X_j).\end{aligned}$$

All terms are integrable by the finite-second moment assumptions.

□

For two variables, the last identity reads,

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y).$$

Thus, the variance of a sum depends on both the individual variances and the dependence between the summands. Positive covariance increases the variance of the sum while negative covariance reduces it. Under independence, covariance dependence vanishes.

Corollary 1.1 (Variance of an independent sum)**.** *If $X_1, \ldots, X_n$ are independent and have finite second moments, then*

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n}\operatorname{Var}(X_i).$$

Proof. For $i \neq j$, independence gives,

$$\mathbb{E}[X_i X_j] = \mathbb{E}[X_i]\mathbb{E}[X_j].$$

Hence,

$$\operatorname{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] = 0.$$

Every covariance term in the variance-of-a-sum formula therefore vanishes.

□

Definition 1.34 (Uncorrelated random variables)**.** Random variables X, Y with finite second moments are uncorrelated if,

$$\mathrm{Cov}(X, Y) = 0.$$

Independence implies that variables with finite second moments are uncorrelated. However, the converse is false. For example, if X is uniform on $\{-1, 0, 1\}$ and $Y = X^2$, then

$$\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0.$$

Nevertheless, X and Y are dependent as observing X determines Y.

The variance-of-an-independent-sum formula therefore holds under the weaker assumption that the summands are pairwise uncorrelated which makes every cross-covariance term zero.

Definition 1.35 (Correlation coefficient)**.** Let X, Y have finite, strictly positive variances. Their *correlation coefficient* is,

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}.$$

Correlation, hence, normalizes covariance by the standard deviations producing a quantity without units. It is undefined when either variance is zero.

Consider the standardized variables,

$$U = \frac{X - \mathbb{E}[X]}{\sqrt{\mathrm{Var}(X)}}, \qquad V = \frac{Y - \mathbb{E}[Y]}{\sqrt{\mathrm{Var}(Y)}}.$$

Then $\mathbb{E}[U^2] = \mathbb{E}[V^2] = 1$ and $\mathbb{E}[UV] = \rho(X, Y)$. Since squares have nonnegative expectations,

$$0 \le \mathbb{E}[(U - V)^2] = 2 - 2\rho(X, Y)$$
$$0 \le \mathbb{E}[(U + V)^2] = 2 + 2\rho(X, Y).$$

Together, these imply,

$$-1 \le \rho(X, Y) \le 1.$$

Zero correlation means zero covariance, but generally does not imply independence. Variance becomes especially useful when it controls the probability that a random variable lies far from its mean. We begin with a simpler principle: a nonnegative random variable cannot frequently take large values unless its expectation is also large. This observation leads to Markov's inequality, and applying it to $(X - \mathbb{E}[X])^2$ will give a probability bound in terms of variance.

Proposition 1.13 (Markov's Inequality)**.** *Let $X \ge 0$ almost surely. Then, for every $a > 0$,*

$$\mathbb{P}(X \ge a) \le \frac{\mathbb{E}[X]}{a}.$$

Proof. Fix $a > 0$. Let $A := \{X \leq a\}$. We compare X with $a\mathbb{I}_A$.

(i) On A, $\mathbb{I}_A = 1$, so $X \geq a\mathbb{I}_A = a$,
(ii) On A^c, $\mathbb{I}_{A^c} = 0$, so $X \geq a\mathbb{I}_A = 0$.

Hence,

$$X \geq a\mathbb{I}_A \quad \text{almost surely.}$$

Taking expectations,

$$\begin{aligned}\mathbb{E}[X] &\geq \mathbb{E}[a\mathbb{I}_A] \\ &= a\mathbb{E}[\mathbb{I}_A] \\ &= a\mathbb{P}(A) \\ &= a\mathbb{P}(X \geq a).\end{aligned}$$

Re-arranging we obtain,

$$\mathbb{E}[X] \geq a\mathbb{P}(X \geq a) \iff \mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

This is precisely *Markov's inequality*, as required. □

The indicator isolates the event $\{X \geq a\}$. Every outcome in this event contributes at least a, so this event alone contributes at least $a\mathbb{P}(X \geq a)$ to the expectation. Since X is. nonnegative elsewhere, the remaining outcomes cannot reduce that contribution. The nonnegative assumption is essential. If negative values were allowed, they could offset large positive values and make the expectation small without making the upper-tail probability small.

Proposition 1.14 (Chebyshev's inequality)**.** *Let X have a finite second moment, $\mathbb{E}[X^2] < \infty$. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left(|X - \mathbb{E}[X]| \geq \varepsilon\right) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

Proof. Define $Y := (X - \mathbb{E}[X])^2$. Since a square is always nonnegative, $Y \geq 0$.

Since $\varepsilon > 0$ then,

$$|X - \mathbb{E}[X]| \geq \varepsilon \iff (X - \mathbb{E}[X])^2 \geq \varepsilon^2.$$

Applying Markov's inequality,

$$\begin{aligned}\mathbb{P}\left(|X - \mathbb{E}[X]| \geq \varepsilon\right) &= \mathbb{P}(Y \geq \varepsilon^2) \\ &= \frac{\mathbb{E}[Y]}{\varepsilon^2} \\ &= \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{\varepsilon^2} \\ &= \frac{\mathrm{Var}(X)}{\varepsilon^2}.\end{aligned}$$

□

Remark 1.11 (Controlling deviations from the mean). Chebyshev's inequality controls both tails because,

$$\{|X - \mathbb{E}[X]| \geq \varepsilon\} = \{X \leq \mathbb{E}[X] - \varepsilon\} \cup \{X \geq \mathbb{E}[X] + \varepsilon\}.$$

These events are disjoint so,

$$\mathbb{P}(X \leq \mathbb{E}[X] - \varepsilon) + \mathbb{P}(X \geq \mathbb{E}[X] + \varepsilon) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

For a fixed threshold ε, a smaller variance gives a smaller upper bound on the probability of a deviation of that size. For a fixed variance, the bound decreases as $1/\varepsilon^2$ which describes the rate of the bound, not necessarily the actual tail probability.

Equivalently,

$$\mathbb{P}(|X - \mathbb{E}[X]| < \varepsilon) \geq 1 - \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

Thus, small variance forces most of the probability to lie near the mean.

Suppose $\sigma = \sqrt{\mathrm{Var}(X)} > 0$, choosing $\varepsilon = k\sigma$ gives,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq k\sigma) \leq \frac{1}{k^2}, \qquad k > 0.$$

For example, the probability of being at least two standard deviations from the mean is at most $1/4$. This bound requires no particular distributional shape and only a finite second moment.

Averaging independent observations preserves their common mean while reduce their variance. Chebyshev's inequality then turns this shrinking variance into a bound on the probability of an estimation error. The following theorem specifies this bound.

Theorem 1.3 (Weak Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with,*

$$\mathbb{E}[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2 < \infty.$$

Define the sample mean $\bar{X}_n := \dfrac{1}{n}\sum_{i=1}^{n} X_i$. Then, for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

We say that $\bar{X}_n$ converges in probability to μ and write $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. First, linearity of expectation gives,

$$\begin{aligned}
\mathbb{E}[\bar{X}_n] &= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] \\
&= \frac{1}{n}\sum_{i=1}^{n} \mu \\
&= \mu.
\end{aligned}$$

Thus, the sample mean has the same expectation as each observation.

Next, scaling a random variable by $\frac{1}{n}$ scales it variance by $\frac{1}{n^2}$. Independence makes the variance of the sum equal to the sum of the variance. Therefore,

$$\begin{aligned}\operatorname{Var}(\bar{X}_n) &= \operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\operatorname{Var}(X_i) \\ &= \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n}.\end{aligned}$$

Fix $\varepsilon > 0$. Applying Chebyshev's inequality,

$$\begin{aligned}\mathbb{P}\left(|\bar{X}_n - \mu| > \varepsilon\right) &\leq \mathbb{P}\left(|\bar{X}_n - \mathbb{E}[\bar{X}_n]| \geq \varepsilon\right) \\ &\leq \frac{\operatorname{Var}(\bar{X}_n)}{\varepsilon^2} \\ &= \frac{\sigma^2}{n\varepsilon^2}.\end{aligned}$$

Consequently, we have established the bound,

$$0 \leq \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \to 0 \qquad n \to \infty.$$

Hence, we have proven convergence in probability. □

Convergence in probability means that for any fixed error tolerance $\varepsilon > 0$, the probability of exceeding that tolerance tends to zero. It does not assert that the error decreases at every step or that every individual sequence of observations converges. We supply the bound,

$$\mathbb{P}\left(|\bar{X}_n - \mu| > \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

For example, to make this probability at most $\delta \in (0, 1)$ it suffices to chose $n \geq \frac{\sigma^2}{n\varepsilon^2}$. This is a sufficient sample size obtained from Chebyshev's inequality since the actual error probability may be much smaller.

1.8 Modes of Convergence

For a sequence of real numbers, $x_n \to x$ means that for every $\varepsilon > 0$ there exists N such that $|x_n - x| < \varepsilon$ for all $n \geq N$. Eventually, the sequence enters any prescribed neighbourhood of x and remains there. For random variables, convergence requires additional assumptions because each X_n is a function,

$$X_n : \Omega \to \mathbb{R}.$$

Throughout this section, $X_1, X_2, \ldots$ and X are defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Fixing an outcome ω produces a numerical sequence, $X_1(\omega), X_2(\omega), \ldots$. Different notions of convergence specify how these sequences approach $X(\omega)$ and which outcomes are allowed.

Definition 1.36 (Sure convergence)**.** A sequence $(X_n)_{n\geq 0}$ converges *surely*, or *pointwise*, to X if,

$$\lim_{n\to\infty} X_n(\omega) = X(\omega) \qquad \forall \omega \in \Omega.$$

Equivalently, for every $\omega \in \Omega$ and every $\varepsilon > 0$, there exists an integer $N = N(\omega, \varepsilon)$ such that,

$$|X_n(\omega) - X(\omega)| < \varepsilon \quad \forall n \geq N.$$

Hence, convergence is checked outcome by outcome, that is, fix ω the examine the numerical sequence $X_n(\omega)$. The index N may depend on both the outcome and the desired accuracy. Sure convergence does not require one common N to work for every outcome.

Example 1.24 (Sure convergence to zero)**.** Let $\Omega = [0, 1]$ and define,

$$X_n(\omega) = \frac{\omega}{n}, \quad X(\omega) = 0.$$

For every $\omega \in [0, 1]$,

$$|X_n(\omega) - X(\omega)| = \frac{\omega}{n} \leq \frac{1}{n} \to 0.$$

Thus, $X_n \to 0$ surely. The bound $\frac{1}{n}$ even allows the same N to work for every outcome.

Sure convergence permits no exceptions, including outcomes having probability zero. Since null events do not affect probabilities or expectations, it is natural to allow convergence to fail on such an event. However, such a notion of convergence is often too *strong* to be useful in practice, hence we loosen the notion of convergence.

Definition 1.37 (Almost sure convergence)**.** A sequence $(X_n)_{n\geq 0}$ converges *almost surely* to X if,

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

We write this as $X_n \overset{a.s.}{\to} X$ or $X_n \to X$ almost surely. Equivalently, there exists an event $A \in \mathcal{F}$ with $\mathbb{P}(A) = 1$ such that for every $\omega \in A$ and every $\varepsilon > 0$ there is an $N = N(\omega, \varepsilon)$ satisfying,

$$|X_n(\omega) - X(\omega)| < \varepsilon, \qquad \forall n \geq N.$$

The exceptional event A^c has probability zero. Almost sure convergence therefore means that on one event of probability one, the entire numerical sequence convergences. The event A is fixed such that the index N can vary with ω and ε.

Example 1.25 (Almost sure convergence without sure convergence)**.** Let $\Omega = [0,1]$ and $\mathcal{F} = \mathcal{B}([0,1])$ with uniform probability measure, and define $X_n(\omega) = \omega^n$.

For every $0 \leq \omega \leq 1$, $\omega^n \to 0$. At $\omega = 1$ however,

$$X_n(1) = 1 \qquad \forall n$$

hence convergence to zero fails there. Consequently, the event on which $X_n \to 0$ is $A = [0,1)$ and $\mathbb{P}(A) = 1$ and $\mathbb{P}(A^c) = \mathbb{P}(\{1\}) = 0$. Thus, $X_n \overset{a.s.}{\to} 0$ but X_n does not converge surely to zero.

In fact, its pointwise limit is,

$$X(\omega) = \mathbb{I}_{\{1\}}(\omega) = \begin{cases} 0, & 0 \leq \omega < 1, \\ 1, & \omega = 1. \end{cases}$$

Hence, $X_n \to \mathbb{I}_{\{1\}}$ surely, while $\mathbb{I}_{\{1\}} = 0$ almost surely.

Hence, almost sure statements disregard differences on null events.

Remark 1.12 (Exceptional set must have probability zero). It is not enough to say that each outcome where convergence fails has probability zero. The whole set of such outcomes must have probability zero,

$$\mathbb{P}(\{\omega : X_n(\omega) \not\to X(\omega)\}) = 0.$$

Under the uniform measure, every singleton in $[0,1]$ has probability zero, but their uncountable unions is the entire sample space and has probability one.

Remark 1.13 (Convergence event is measurable). The event in the definition can be expressed using only countable unions and intersections,

$$\{X_n \to X\} = \bigcap_{m=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{ |X_n - X| < \frac{1}{m} \right\}.$$

Each event $\{|X_n - X| < 1/m\}$ is measurable so the entire expression belongs to $\mathcal{F}$. Reading from the outside inward, the event states for every accuracy threshold $\frac{1}{m}$ there is some starting index N after which all errors are smaller than $\frac{1}{m}$. These countably many accuracy levels suffice to express ordinary convergence. Sure convergence always implies almost sure convergence, but *Example 1.25* shows that the converse fails. Both notions concern the eventual behaviour of the sequence along fixed outcomes. The next notion, *convergence in probability*, instead asks whether the probability of a large error at index n tends to zero.

Almost sure convergence examines the entire sequence along each fixed outcome. Convergence in probability instead examines the probability of a large error at each index n.

Definition 1.38 (Convergence in probability)**.** Let $X_1, \ldots, X_n$ and X be random variables on the same probability space. We say that X_n *converges in probability* to X, written

$$X_n \xrightarrow{\mathbb{P}} X$$

if, for every $\varepsilon > 0$,

$$\mathbb{P}\left(|X_n - X| > \varepsilon\right) \to 0 \qquad \text{as } n \to \infty.$$

For a fixed tolerance $\varepsilon > 0$, we define the error event as,

$$A_n(\varepsilon) = \{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}.$$

Convergence in probability, hence, means that the probability of such an event goes to zero, that is $\mathbb{P}(A_n(\varepsilon)) \to 0$. Thus, for sufficiently large n, an observation of X_n is unlikely to differ from X more than a fixed tolerance ε.

Note, the events $A_n(\varepsilon)$ may change with n. Their probabilities can thus shrink even if each fixed outcome belongs to infinitely many of them. This is the distinction that allows convergence in probability without almost sure convergence.

Example 1.26 (A rare nonzero value)**.** Suppose $(X_n)_{n\geq 1}$ is a sequence of random variables satisfying,

$$\mathbb{P}(X_n = 1) = \frac{1}{n}, \qquad \mathbb{P}(X_n = 0) = 1 - \frac{1}{n}.$$

For $0 < \varepsilon < 1$, the event $\{|X_n| > \varepsilon\}$ occurs exactly when $X_n = 1$. For $\varepsilon \geq 1$, it is empty.

Therefore,

$$\mathbb{P}(|X_n| > \varepsilon) = \begin{cases} \frac{1}{n}, & 0 < \varepsilon < 1, \\ 0, & \varepsilon \geq 1. \end{cases}$$

In either case, the probability tends to zero, so $X_n \xrightarrow{\mathbb{P}} 0$.

Observe in the example above, no independence assumption is required. These marginal probabilities suffice to establish convergence in probability to zero, but they do not specify whether the sequence converges almost surely. For that, the relationship between the variables across different indices matters.

Example 1.27 (Convergence in probability without pointwise convergence)**.** Consider $\Omega = [0, 1]$ and $\mathcal{F} = \mathcal{B}([0, 1])$ with uniform probability measure. For each integer $m \geq 0$, partition $(0, 1]$ into 2^m intervals and define such intervals,

$$I_{m,k} = \left(\frac{k}{2^m}, \frac{k+1}{2^m}\right], \quad k = 0, \ldots, 2^m - 1.$$

Define $X_{2^m+k}(\omega) := \mathbb{I}_{I_{m,k}}(\omega)$. Half-open intervals ensure that each $\omega \in (0,1]$ belongs to exactly one interval at every level m. The sequence begins by scanning the whole interval, then its two halves then its four quarters. Hence,

$$\begin{aligned} X_1 &= \mathbb{I}_{(0,1]}, \\ X_2 &= \mathbb{I}_{(0,1/2]}, & X_3 &= \mathbb{I}_{(1/2,1]}, \\ X_4 &= \mathbb{I}_{(0,1/4]}, & X_5 &= \mathbb{I}_{(1/4,1/2]}, \\ X_6 &= \mathbb{I}_{(1/2,3/4]}, & X_7 &= \mathbb{I}_{(3/4,1]}. \end{aligned}$$

At level m, the indices run from 2^m to $2^{m+1}-1$ and each indicator is supported on an interval of length 2^{-m}. For $0 < \varepsilon < 1$ and $n = 2^m + k$,

$$\begin{aligned} \mathbb{P}(|X_n| > \varepsilon) &= \mathbb{P}(X_n = 1) \\ &= \mathbb{P}(I_{m,k}) \\ &= \frac{1}{2^m}. \end{aligned}$$

As $n \to \infty$, its corresponding level m also tends to infinity. Hence, this probability tends to zero. For $\varepsilon \geq 1$ it is already zero, so we write $X_n \xrightarrow{\mathbb{P}} 0$.

Suppose we now fix any outcome $\omega \in (0,1]$. At each level m, exactly one interval contains ω. Consequently, among $X_{2^m}(\omega), \ldots, X_{2^{m+1}-1}(\omega)$ exactly one value is 1. For every $m \geq 1$. there is also at least one value equal to 0. Since there are infinitely many levels, the numerical sequence $X_n(\omega)$ takes both values 0 and 1 infinitely often.

Therefore,

$$\liminf_{n\to\infty} X_n(\omega) = 0, \qquad \limsup_{n\to\infty} X_n(\omega) = 1.$$

Hence, it does not converge for any $\omega \in (0,1]$. In particular,

$$\mathbb{P}(\{\omega : X_n(\omega) \to 0\}) = 0.$$

So, convergence to zero is not almost sure. At each index, the set where $X_n = 1$ is small. Nevertheless, these sets move through the sample space and revisit every outcome infinitely often. Convergence in probability controls the size of the error event at each index but does **not** require individual outcomes eventually to stop encountering errors.

Often, our goal is to approximate the *law* of a random variable X, rather than its value on each individual outcome. For example, a numerical method may produce approximations X_n whose distributions approach the distribution of X.

This motivates a *weaker* notion of convergence that compares only the laws. Unlike almost sure convergence and convergence in probability, it does not require that the random variables to be defined on the same probability space.

Definition 1.39 (Convergence in distribution)**.** Let $X_1, X_2, \ldots$ and X be real-valued random variables possibly defined on different probability spaces.

We say that X_n *converges in distribution* to X written,

$$X_n \xrightarrow{d} X$$

if,

$$\mathbb{E}[\phi(X_n)] \to \mathbb{E}\phi(X) \quad \text{for every } \phi \in C_b(\mathbb{R}),$$

where $C_b(\mathbb{R})$ is the collection of bounded, continuous functions $\phi : \mathbb{R} \to \mathbb{R}$.

Writing $\mu_n = \mathcal{L}(X_n)$ and $\mu = \mathcal{L}(X)$ the definition is equivalent to,

$$\int_{\mathbb{R}} \phi(x)\mu_n(dx) \to \int_{\mathbb{R}} \phi(x)\mu(dx) \quad \forall \phi \in C_b(\mathbb{R}).$$

This is also referred to as *weak convergence of the probability measures* μ_n to μ.

The functions ϕ are called test functions. Each one measures a feature of the distribution through its average. Continuity prevents abrupt changes under small changes in the input, while boundedness ensures that these expectations are finite without imposing moment assumptions.

Convergence in distribution therefore depends only on μ_n, μ and not on how the random variables are related outcome by outcome. However, it remains to determine which event probabilities converge.

Remark 1.14. Convergence in distribution does not guarantee that $\mathbb{P}(X_n \in A) \to \mathbb{P}(X \in A)$ for every Borel set A. An indicator $\mathbb{I}_A$ is generally discontinuous, so it need not be an admissible test function. However, this convergence does hold whenever,

$$\mathbb{P}(X \in \delta A) = 0.$$

Here, δA is the boundary of A. Such a set is called a continuity set of the limiting law. Intuitively, the limit places no probability on the boundary where small changes could move a value into or out of A.

Proposition 1.15 (CDF characterization)**.** *Let* $F_n(x) = \mathbb{P}(X_n \leq x)$ *and* $F(x) = \mathbb{P}(X \leq x)$*. Then,*

$$X_n \xrightarrow{d} X \iff F_n(x) \to F(x) \quad \forall \textit{ continuity points } x \textit{ of } F.$$

The continuity requirement concerns the limiting CDF F. Convergence, hence, need not hold at jumps. The test-function definition is especially useful because it extends directly to random vectors by replacing $C_b(\mathbb{R})$ with $C_b(\mathbb{R}^k)$ and to random elements in more general metric spaces, including spaces of continuous paths.

Example 1.28 (Convergence of Gaussian laws)**.** Suppose,

$$X_n \sim \mathcal{N}\left(0, 1 + \frac{1}{n}\right), \quad X \sim \mathcal{N}(0, 1).$$

Let Φ denote the standard normal CDF. For every $x \in \mathbb{R}$,

$$F_n(x) = \Phi\left(\frac{x}{\sqrt{1+\frac{1}{n}}}\right) \iff \frac{x}{\sqrt{1+\frac{1}{n}}} \to x.$$

Since Φ is continuous, $F_n(x) \to \Phi(x) = F(x)$. Consequently, $X_n \xrightarrow{d} X$.

Hence, convergence of the variance establishes convergence of the laws because all the variables belong to the specified Gaussian family and their means are fixed. For arbitrary distributions, convergence of means and variances alone would *not* suffice.

Example 1.29 (Continuous laws can converge to a point mass)**.** Let $X_n \sim \text{Unif}([0, 1/n])$ with $X = 0$. For any bounded continuous function ϕ,

$$\begin{aligned}
|\mathbb{E}[\phi(X_n)] - \mathbb{E}[\phi(X)]| &= \left| n \int_0^{\frac{1}{n}} \phi(x)dx - \phi(0) \right| \\
&= \left| n \int_0^{\frac{1}{n}} (\phi(x) - \phi(0))\, dx \right| \\
&\leq \sup_{0 \leq x \leq \frac{1}{n}} |\phi(x) - \phi(0)| \\
&\to 0
\end{aligned}$$

by continuity of ϕ at zero. Therefore, $X_n \xrightarrow{d} 0$.

Each X_n has a continuous distribution, but the limiting law is the point mass δ_0. Their CDFs do not converge at zero, that is, $F_n(0) = 0$ but $F(0) = 1$.

However, this is permitted because zero is a discontinuity point of F.

The example above distinguishes convergence of laws from equality of laws. That is, if two probability measures satisfy,

$$\int_{\mathbb{R}} \phi d\mu_1 = \int_{\mathbb{R}} \phi d\mu_2, \quad \forall \phi \in C_b(\mathbb{R})$$

then it must be the case that $\mu_1 = \mu_2$. In particular, a law with an atom and an atomless law cannot agree on every bounded continuous test function. The equality is ordinary equality of measures where we denote it as $X \overset{d}{=} Y$ for random variables with the same law.

Proposition 1.16 (Implications between modes of convergence)**.** *Let $X_1, X_2, \ldots$ be random variables on the same probability space. Then,*

$$X_n \overset{a.s.}{X} \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X.$$

Neither reverse implication holds in general.

Proof. We prove the two implications separately.

Almost surely convergence implies convergence in probability.

Suppose $X_n \to X$ almost surely. Fix tolerance $\varepsilon > 0$ and define,

$$A_n := \bigcup_{n=N}^{\infty} \{|X_n - X| > \varepsilon\}.$$

The event A_N occurs when an error greater than ε happens at least once at or after index N. As N increases, there are fewer indices included in the union, so $A_{N+1} \subseteq A_N$.

Moreover,

$$\bigcap_{N=1}^{\infty} A_N = \{\omega : |X_n(\omega) - X(\omega)| > \varepsilon \text{ for infinitely many } n\}.$$

Instead of belonging to every A_N means that no matter ho far along the sequence we start, another error greater than ε occurs. On any outcome where $X_n(\omega) \to X(\omega)$ such errors eventually stop. Therefore, intersection is $\subseteq \{\omega : X_n(\omega) \not\to X(\omega)\}$. Almost sure convergence implies that the event on the right has probability zero. Hence,

$$\mathbb{P}(A_N) \to \mathbb{P}\left(\bigcap_{N=1}^{\infty} A_N\right) = 0.$$

Finally, $\{|X_N - X| > \varepsilon\} \subseteq A_N$ so,

$$0 \leq \mathbb{P}(|X_N - X| > \varepsilon) \leq \mathbb{P}(A_N) \to 0.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $X_n \xrightarrow{\mathbb{P}} X$.

Convergence in probability implies convergence in distribution.

Suppose $X_n \to X$ in probability. Let $F_n(x) = \mathbb{P}(X_n \leq x)$ and $F(x) = \mathbb{P}(X \leq x)$. By *Proposition 1.15* of convergence in distribution, it suffices to show that $F_n(x) \to F(x)$ for every continuity point x of F. Fix such a point x and let $\delta > 0$.

Define the error event,

$$E_n := \{|X_n - X| > \delta\}.$$

Outside E_n, the variables differ by at most δ,

$$X - \delta \leq X_n \leq X + \delta.$$

If $X \leq x - \delta$ and E_n does not occur then,

$$X_n \leq X + \delta \leq x.$$

Hence, the event $\{X \leq x - \delta\} \subseteq \{X_n \leq x\} \cup E_n$. Taking probabilities and by subadditivity,

$$F(x - \delta) \leq F_n(x) + \mathbb{P}(E_n) \iff F_n(x) \geq F(x - \delta) - \mathbb{P}(E_n).$$

Conversely, if $X_n \leq x$ and E_n does not occur then,

$$X \leq X_n + \delta \leq x + \delta.$$

Hence, $\{X_n \leq x\} \subseteq \{X \leq x + \delta\} \cup E_n$. Hence,

$$F_n(x) \leq F(x + \delta) + \mathbb{P}(E_n).$$

Combining the bounds from both cases,

$$F(x - \delta) - \mathbb{P}(|X_n - X| > \delta) \leq F_n(x) \leq F(x + \delta) + \mathbb{P}(|X_n - X| > \delta).$$

For each fixed $\delta > 0$, convergence in probability gives,

$$\mathbb{P}(|X_n - X| > \delta) \to 0.$$

Taking lower and upper limits as $n \to \infty$,

$$\begin{aligned} F(x - \delta) &\leq \lim_{n \to \infty} \sup F_n(x) \\ &\leq \lim_{n \to \infty} \sup F_n(x) \\ &\leq F(x + \delta). \end{aligned}$$

Let $\delta \to 0$. Since F is continuous at x, then $F(x - \delta) \to F(x)$ and $F(x + \delta) \to F(x)$. The lower and upper limits are therefore both $F(x)$ proving that $F_n(x) \to F(x)$.

Hence, we conclude that $X_n \xrightarrow{d} X$.

□

For the reverse implications, the moving-interval sequence example converges to zero in probability but fails to converge at every outcome. Thus, convergence in probability does not imply almost sure convergence. To see that convergence in distribution does not imply convergence in probability, suppose that $X \sim \text{Bernoulli}(1/2)$ and $X_n = 1 - X$ for every $n \in \mathbb{N}$.

Since $X_n \stackrel{d}{=} X$, we have that $X_n \xrightarrow{d} X$. However,

$$\mathbb{P}\left(|X_n - X| > \frac{1}{2}\right) = 1.$$

Hence, for every n, convergence in probability fails.

The weak law of large numbers tells us that the sample mean approaches the population mean. The central limit theorem describes the fluctuations that remain. That is, after multiplying the error by $\sqrt{n}$, its distribution approaches a Gaussian law.

Theorem 1.4 (Central limit theorem)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables satisfying,*

$$\mathbb{E}[X_i] = \mu, \quad \mathrm{Var}(X_i) = \sigma^2, \quad 0 < \sigma^2 < \infty.$$

Then,

$$\frac{1}{\sigma\sqrt{n}} \sum_{i=1}^{n} (X_i - \mu) \xrightarrow{d} Z, \quad Z \sim \mathcal{N}(0,1).$$

We can equivalently write,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

so that standardizing gives,

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} Z.$$

The standard normal distribution has density $f_Z(z) = \dfrac{1}{\sqrt{2\pi}} e^{-z^2/2}$ for all $z \in \mathbb{R}$. The normalization follows from the mean and variance of the sum,

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = n\mu, \quad \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = n\sigma^2.$$

Subtracting $n\mu$ centers the sum and dividing by its standard deviation $\sigma\sqrt{n}$ gives a variable with mean zero and variance one. The theorem says that its entire distribution not just these two moments, approaches the standard normal law.

Remark 1.15. The weak law gives $\bar{X}_n - \mu \xrightarrow{\mathbb{P}} 0$. The Central Limit Theorem adds that the error has approximately Gaussian distribution on the scale $\frac{1}{\sqrt{n}}$.

Informally, for large n, $\bar{X}_n - \mu$ has approximately the law of $\frac{\sigma}{\sqrt{n}} Z$. Equivalently,

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

These are distributional approximations, not pointwise identities involving a fixed Z. More precisely, if Φ is the standard normal CDF, then for ever $z \in \mathbb{R}$,

$$\mathbb{P}\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z\right) \to \Phi(z).$$

Note that the original distribution need not be Gaussian, but the independence, identical distribution, and finite-variance assumptions are essential to this version of the theorem.

Proof of CLT. Assume additionally that $\mathbb{E}|X_1|^3 < \infty$.

We first prove the convergence against smooth, compactly supported test functions. The central idea is to replace the summads by Gaussian variables one at a time. Matching means and variances makes the first two Taylor terms canceld, leaving a small remainder.

By replacing X_i with $\frac{X_i - \mu}{\sigma}$ we may assume that $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[X_i^2] = 1$.

Let $Z_1, Z_2, \ldots$ be i.i.d. standard normal random variables, independent of the entire sequence (X_i). Such variables can be introduced on an enlarged probability space without changing the laws of the X_i. Fix a smooth, compactly supported function $\phi : \mathbb{R} \to \mathbb{R}$.

For $k = 0, \ldots, n$ define,

$$W_{n,k} := \frac{1}{\sqrt{n}} \left(\sum_{j=1}^{k} X_j + \sum_{j=k+1}^{n} Z_j \right)$$

with empty sums interpreted as zero. At the two endpoints,

$$W_{n,n} = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} X_j, \qquad X_{n,0} = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} Z_j \sim \mathcal{N}(0,1).$$

Thus, we wish to show that,

$$\mathbb{E}[\phi(W_{n,n})] - \mathbb{E}[\phi(W_{n,0})] \to 0.$$

We can write this difference as a telescoping sum,

$$\mathbb{E}\phi(W_{n,n}) - \mathbb{E}\phi(W_{n,0}) = \sum_{k=1}^{n} \left(\mathbb{E}\phi(W_{n,k}) - \mathbb{E}\phi(W_{n,k-1}) \right).$$

Each term measures the effect of replacing one Gaussian summand Z_k by W_k. Next, we can expand around the common part of two consecutive sums. Define,

$$V_{n,k} := \frac{1}{\sqrt{n}} \left(\sum_{j=1}^{k+1} X_j + \sum_{j=k+1}^{n} Z_j \right).$$

Then, we have that,

$$W_{n,k} = V_{n,k} + \frac{X_k}{\sqrt{n}}, \qquad W_{n,k-1} = V_{n,k} + \frac{Z_k}{\sqrt{n}}.$$

The common part $V_{n,k}$ contains neither X_k nor Z_k and is independent of both. Since ϕ is smooth with compact support, its third derivative is bounded. Taylor's theorem gives the expansion,

$$\phi\left(w + \frac{x}{\sqrt{n}}\right) = \phi(w) + \frac{x}{\sqrt{n}}\phi'(w) + \frac{x^2}{2n}\phi''(w) + R_n(w,x)$$

where $|R_n(w,x)| \le \dfrac{C|x|^3}{n^{3/2}}$ with $C = \dfrac{\|\phi'''\|_\infty}{6}$. Applying the expansion with $w = V_{n,k}$, we first use $x = X_k$ then let $x = Z_k$. Subtracting and taking expectations we obtain,

$$\begin{aligned}
\mathbb{E}[\phi(W_{n,k})] - \mathbb{E}[\phi(W_{n,k-1})] &= \frac{1}{\sqrt{n}} \underbrace{\mathbb{E}[(X_k - Z_k)\phi'(V_{n,k})]}_{=(\mathbb{E}[X_k]-\mathbb{E}[Z_K])\mathbb{E}[\phi'(V_{n,k})]=0} + \frac{1}{2n} \underbrace{\mathbb{E}[(X_k^2 - Z_k^2)\phi''(V_{n,k})]}_{=(\mathbb{E}[X_k^2]-\mathbb{E}[Z_k^2])\mathbb{E}[\phi''(V_{n,k})]=0} \\
&\quad + \mathbb{E}[R_N(V_{n,k}, X_k) - R_n(V_{n,k}, Z_k)] \\
&\le \mathbb{E}[|R_n(V_{n,k}, X_k)|] + \mathbb{E}[|R_n(V_{n,k}, Z_k)|], \qquad \text{constant terms,} \\
&\le \frac{C}{n^{3/2}} \left(\mathbb{E}[|X_1|^3] + \mathbb{E}[|Z_1|^3]\right).
\end{aligned}$$

Using the telescoping sum and the triangle inequality,

$$\begin{aligned}
|\mathbb{E}[\phi(W_{n,n})] - \mathbb{E}[\phi(W_{n,0})]| &\le \sum_{k=1}^{n} |\mathbb{E}\phi(W_{n,k}) - \mathbb{E}\phi(W_{n,k-1})| \\
&\le \frac{C}{\sqrt{n}} \left(\mathbb{E}[|X_1|^3] + \mathbb{E}[|Z_1|^3]\right) \\
&\longrightarrow 0.
\end{aligned}$$

Since $W_{n,0}$ is a standard normal, we have showed that,

$$\mathbb{E}\phi(W_{n,n}) \to \mathbb{E}\phi(Z).$$

□

Remark 1.16. The definition of convergence in distribution requires that every $\phi \in C_b(\mathbb{R})$. To pass from smooth test functions to these more general functions, we control the tails and then approximate on a bounded interval. Write,

$$S_n = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} X_j.$$

Both S_n and Z has mean zero and variance one, so Chebyshev's inequality gives,

$$\mathbb{P}(|S_j| > R) \le \frac{1}{R^2}, \qquad \mathbb{P}(|Z| > R) \le \frac{1}{R^2}.$$

The first bound holds uniformly in n. Let $M = \|\phi\|_\infty$. For any $R > 0$ and $\eta > 0$, choose a smooth and compactly supported function ψ such that,

$$\sup_{|x| \le R} |\phi(x) - \psi(x)| \le \eta, \qquad \|\psi\|_\infty \le M + \eta.$$

For $U = S_n$ or $U = Z$, splitting according to $|U| \le R$ gives,

$$\begin{aligned}
\mathbb{E}[|\phi(U) - \psi(U)|] &\le \eta + (2M + \eta)\mathbb{P}(|U| > R) \\
&\le \eta + \frac{2M + \eta}{R^2}.
\end{aligned}$$

Consequently,

$$|\mathbb{E}[\psi(S_n)] - \mathbb{E}[\phi(Z)]| \leq |\mathbb{E}[\psi(S_n)] - \mathbb{E}[\psi(Z)]| + 2\eta + \frac{2(2M+\eta)}{R^2}.$$

Hence, for fixed R, η, ψ, the first term tends to zero by the preceding argument. Taking $n \to \infty$ then $R \to \infty$ and finally $\eta \to 0$ proves convergence for every bounded continuous ϕ.

Remark 1.17. The finite third moment was used only to control the Taylor remainder. It is not an assumption of the central limit theorem. For a smooth, compactly supported ϕ, the same remainder satisfies both a cubic and quadratic bound,

$$|R_n(w,x)| \leq C_3 \frac{|x|^3}{n^{3/2}}, \qquad |R_n(w,x)| \leq C_2 \frac{x^2}{n}$$

for constants depending only on ϕ. The quadratic bound follows by writing the second-order remainder in terms of difference of the bounded function ϕ''. Suppose we fix $\delta > 0$. Then, for a random variable U with finite second moments, we can split according to whether $|U| \leq \delta\sqrt{n}$. On such an event,

$$\frac{|U|^3}{n^{3/2}} \leq \frac{\delta U^3}{n}.$$

On its complement, we use the quadratic bound. Hence, uniformly in w,

$$\mathbb{E}\left[|R_n(w,U)|\right] \leq \frac{C_3\delta}{n}\mathbb{E}[U^2] + \frac{C_2}{n}\mathbb{E}\left[U^2 \mathbb{I}_{\{|U|>\delta\sqrt{n}\}}\right].$$

Applying this to X_k and Z_k, and summing the n replacement errors gives,

$$|\mathbb{E}\phi(S_n) - \mathbb{E}\phi(Z)| \leq 2C_3\delta + C_2\mathbb{E}\left[X_1^2 \mathbb{I}_{\{|X_1|>\delta\sqrt{n}\}}\right] + C_2\mathbb{E}\left[Z_1^2 \mathbb{I}_{\{|Z_1|>\delta\sqrt{n}\}}\right].$$

For fixed $\delta > 0$, both tail expectations tend to zero because the second moments are finite. Hence,

$$\lim_{n\to\infty} \sup |\mathbb{E}\phi(S_n) - \mathbb{E}\phi(Z)| \leq 2C_3\delta.$$

Letting $\delta \to 0$ proves the required convergence without a third-moment assumption. The bounded continuous approximation argument then completes the proof under the stated finite-variance assumptions.

1.9 Chebyshev's Inequality versus the Central Limit Theorem

Chebyshev's inequality gives a bound that holds for every sample size, using only the mean and varaince. The central limit theorem gives a limiting probability by identifying the asymptotic shape of the standardized distribution. These serve different purposes: one provides a finite-sample guarantee while the other provides an asymptotic description.

Example 1.30 (Fluctuations of a Bernoulli random variable)**.** Let $X_1, X_2, \cdots \sim \text{Bernoulli}(p)$ with $0 < p < 1$ and define the sample mean,

$$S_n := \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Since $\mathbb{E}[X_i] = p$ and $\text{Var}(X_i) = p(1-p)$, by linearity and independent, the expectation and variance of the sample mean are $\mathbb{E}[S_n] = p$ and $\text{Var}(S_n) = \frac{p(1-p)}{n}$. Consequently,

$$\text{SD}(S_n) = \frac{\sqrt{p(1-p)}}{\sqrt{n}}.$$

We examine deviations on the same scale as this standard deviation.

Suppose we fix $c > 0$ and consider,

$$\mathbb{P}\left(S_n - p \leq \frac{c}{\sqrt{n}} \right).$$

This is precisely the probability that the sample mean does not exceed its population mean by more than $\frac{c}{\sqrt{n}}$. By Chebyshev's inequality letting $\varepsilon_n = \dfrac{c}{\sqrt{n}}$ and $\varepsilon_n^2 = \dfrac{c^2}{n}$ then,

$$\begin{aligned} \mathbb{P}\left(|S_n - p| \geq \frac{c}{\sqrt{n}} \right) &\leq \frac{\text{Var}(S_n)}{c^2/n} \\ &= \frac{p(1-p)/n}{c^2/n} \\ &= \frac{p(1-p)}{c^2}. \end{aligned}$$

Taking complements,

$$\mathbb{P}\left(|S_n - p| < \frac{c}{\sqrt{n}} \right) \geq 1 - \frac{p(1-p)}{c^2}.$$

Moreover, the event $\left\{ |S_n - p| < \frac{c}{\sqrt{n}} \right\} \subseteq \left\{ S_n - p \leq \frac{c}{\sqrt{n}} \right\}$. Hence the probability,

$$\mathbb{P}\left(S_n - p \leq \frac{c}{\sqrt{n}} \right) \geq 1 - \frac{p(1-p)}{c^2}.$$

The right-hand side is a lower bound, not an exact probability. If it is negative, the trivial lower bound zero is stronger. For $p = \frac{1}{2}$ and $c = 1$, $p(1-p) = \frac{1}{4}$ hence,

$$\mathbb{P}\left(S_n - \frac{1}{2} \leq \frac{1}{\sqrt{n}} \right) \geq \frac{3}{4} \qquad \forall n \geq 1.$$

The bound does not improve with n because the threshold shrinks at the same rate as the standard deviation. Indeed,

$$\frac{c/\sqrt{n}}{\text{SD}(S_n)} = \frac{c}{\sqrt{p(1-p)}}$$

which is independent of n. We are keeping the threshold at a fixed number of standard deviations above the mean. Now, using the central limit theorem, we obtain,

$$\frac{\sqrt{n}(S_n - p)}{\sqrt{p(1-p)}} \xrightarrow{d} Z, \quad Z \sim \mathcal{N}(0,1).$$

Re-writing the probability in terms of this standardized variable,

$$\begin{aligned}\mathbb{P}\left(S_n - p \leq \frac{c}{\sqrt{n}}\right) &= \mathbb{P}(\sqrt{n}(S_n - p) \leq c) \\ &= \mathbb{P}\left(\frac{\sqrt{n}(S_n - p)}{\sqrt{p(1-p)}} \leq \frac{c}{\sqrt{p(1-p)}}\right).\end{aligned}$$

The threshold on the right is fixed. Since the standard normal CDF Φ is continuous everywhere, from convergence in distribution we obtain,

$$\mathbb{P}\left(S_n - p \leq \frac{c}{\sqrt{n}}\right) \to \Phi\left(\frac{c}{\sqrt{p(1-p)}}\right).$$

For $p = \frac{1}{2}$ and $c = 1$, the standardized threshold is $\frac{1}{\sqrt{1/4}} = 2$. Hence,

$$\mathbb{P}\left(S_n - \frac{1}{2} \leq \frac{1}{\sqrt{n}}\right) \to \Phi(2) \approx 0.97725.$$

Hence, under Chebyshev, the probability is at least 0.75 for every n whereas under the central limit theorem, the probability tends to approximately 0.97725 as $n \to \infty$. There is no contradiction. Chebyshev gives a conservative guarantee while the CLT identifies the limiting value. The CLT alone does not specify the approximation error at a particular finite sample space.

For comparison, the corresponding two-sided probability satisfies,

$$\mathbb{P}\left(|S_n - \frac{1}{2}| < \frac{1}{\sqrt{n}}\right) \to \mathbb{P}(|Z| < 2) = 2\Phi(2) - 1 \approx 0.95450.$$

Chebyshev also gives 0.75 as a lower bound for this event. Thus, part of the conservatism in our original calculation comes from using a two-sided bound to control a one-sided event.

Remark 1.18 (Information used by Chebyshev). Chebyshev's inequality uses only the variance of the random variable being bounded. In this example, independence was used to calculate $\mathrm{Var}(S_n)$ but Chebyshev does not otherwise exploit the Bernoulli structure, boundedness, or higher moments. Its bound must therefore remain valid across a broad class of distributions.

Indeed, the two-sided Chebyshev bound can be attained for $a \geq \sigma > 0$ let,

$$\mathbb{P}(Y = a) = \mathbb{P}(Y = -a) = \frac{\sigma^2}{2a^2}, \qquad \mathbb{P}(Y = 0) = 1 - \frac{\sigma^2}{a^2}.$$

Then, $\mathbb{E}[Y] = 0$ and $\mathrm{Var}(Y) = \sigma^2$ and $\mathbb{P}(|Y| \geq a) = \frac{\sigma^2}{a^2}$.

Thus, the general two-sided bound cannot be uniformly improved using only the variance. Additional structure, or a specifically one-sided inequality, can yield stronger bounds.

Remark 1.19 (Exponential moments give stronger tail bounds). When more information is available, Markov's inequality can be applied to a different nonnegative function of the variable. For example, if $Z \sim \mathcal{N}(0,1)$, then every positive absolute moment is finite,

$$\mathbb{E}[|Z|^p] < \infty \qquad \forall q > 0.$$

Its moment-generating function is also finite for every $\theta \in \mathbb{R}$ $\mathbb{E}[e^{\theta Z}] = e^{\theta^2/2}$. For $t > 0$ and $\theta > 0$ applying Markov's inequality to $e^{\theta Z}$ gives,

$$\begin{aligned}\mathbb{P}(Z \geq t) &= \mathbb{P}(e^{\theta Z} \geq e^{\theta t}) \\ &\leq e^{-\theta t}\mathbb{E}[e^{\theta Z}] \\ &= \exp\left(\frac{\theta^2}{2} - \theta t\right).\end{aligned}$$

Choosing $\theta = t$ minimizes this expression yielding,

$$\mathbb{P}(Z \geq t) \leq e^{-t^2/2}.$$

By symmetry, $\mathbb{P}(|Z| \geq t) \leq 2e^{-t^2/2}$.

For large t, this decays much faster than Chebyshev's bound $\frac{1}{t^2}$. The improvement comes from using the Gaussian exponential moments. Convergence to a Gaussian law alone does not automatically transfer this bound to a finite-sample standardized sum.

2 Conditional Expectation, Martingales, and Convergence

In introductory probability, conditioning on an event B with $\mathbb{P}(B) > 0$ is defined by,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

We now extend conditioning from a single event to an entire collection of information, represented by a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$. The conditional expectation of X given $\mathcal{G}$ describes the average value of X using the information available in $\mathcal{G}$.

Because that information can differ from one outcome to another, the conditional expectation is itself a random variable, rather than a single number.

2.1 Conditional Expectation

Definition 2.1 (Conditional expectation)**.** Let X be an integrable random variable on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra.

An integrable random variable Z is a conditional expectation of X given $\mathcal{G}$ if,

(i) Z is $\mathcal{G}$-measurable,

(ii) For every $A \in \mathcal{G}$,

$$\mathbb{E}[Z\mathbb{I}_A] = \mathbb{E}[X\mathbb{I}_A].$$

Equivalently, the second condition is,

$$\int_A Z d\mathbb{P} = \int_A X d\mathbb{P}, \qquad \forall A \in \mathcal{G}.$$

We write $Z := \mathbb{E}[X \mid \mathcal{G}]$ to represent this conditional expectation.

Remark 2.1 (Interpreting the definition conditions). Measurability requires that the random variable $Z := \mathbb{E}[X \mid \mathcal{G}]$ use only the information available in $\mathcal{G}$. That is,

$$Z^{-1}(B) \in \mathcal{G} \qquad \forall B \in \mathcal{B}(\mathbb{R}).$$

Thus, the available information must suffice to determine Z, even when it does not suffice to determine X. The second condition, called the *testing identity*, requires Z to preserve the expected contribution of X on every event distinguishable using that information,

$$\mathbb{E}[(X - Z)\mathbb{I}_A] = 0, \qquad \forall A \in \mathcal{G}.$$

In particular, taking $A = \Omega$ gives,

$$\mathbb{E}Z = \mathbb{E}X.$$

This definition requires more than this overall equality, that is, the averages must agree on every event $\mathcal{G}$. For $A \in \mathcal{G}$ with $\mathbb{P}(A) > 0$, dividing the testing identity by $\mathbb{P}(A)$ gives

$$\frac{\mathbb{E}[Z\mathbb{I}_A]}{\mathbb{P}(A)} = \frac{\mathbb{E}[X\mathbb{I}_A]}{\mathbb{P}(A)}.$$

These are the averages of Z and X within the event A. Conditional expectation therefore provides a $\mathcal{G}$-measurable description of X that preserves all such averages.

The definition, however, does not require $Z(\omega) = X(\omega)$ for all $\omega \in \Omega$ but rather requires that Z retain the averages that can be tested using $\mathcal{G}$.

Lemma 2.1 (Almost sure uniqueness of Conditional expectation)**.** *Let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra. If Z and Z' both satisfy the definition of $\mathbb{E}[X \mid \mathcal{G}]$ then,*

$$Z = Z' \quad \textit{almost surely.}$$

Proof. Since both Z and Z' satisfy the testing identity,

$$\mathbb{E}[Z\mathbb{I}_A] = \mathbb{E}[X\mathbb{I}_A] = \mathbb{E}[Z'\mathbb{I}_A] \qquad \forall A \in \mathcal{G}.$$

Subtracting we obtain that,

$$\mathbb{E}[(Z - Z')\mathbb{I}_A] = 0, \qquad \forall A \in \mathcal{G}.$$

Our goal is to show that $Z - Z'$ cannot be positive or negative on a set of positive probability. We thus test separately on the events where it has each sign, preventing positive and negative differences from cancelling. First define,

$$A_+ := \{Z > Z'\}.$$

Because Z, Z' are $\mathcal{G}$-measurable, their difference is $\mathcal{G}$-measurable. Hence,

$$A_+ := (Z - Z')^{-1}((0, \infty)) \in \mathcal{G}.$$

This is precisely where measurability matters as it allows us to use A_+ in the testing identity. Taking $A = A_+$ we obtain,

$$\mathbb{E}[(Z - Z')\mathbb{I}_{A_+}] = 0.$$

Hence, the random variable $Y_+ := (Z - Z')\mathbb{I}_{A_+}$ is nonnegative everywhere. It is also integrable since $|Y_+| \leq |Z| + |Z'|$ and both Z, Z' are integrable. A nonnegative random variable with expectation zero must be zero almost surely. Therefore, $Y_+ = 0$ almost surely.

But Y_+ is strictly positive exactly on A_+ so,

$$\mathbb{P}(Z > Z') = \mathbb{P}(Y_+ > 0) = 0.$$

Similarly, define, $A_- := \{Z < Z'\} \in \mathcal{G}$. Testing on A_- gives,

$$\mathbb{E}[(Z - Z')\mathbb{I}_{A_-}] = 0 = \mathbb{E}[(Z' - Z)\mathbb{I}_{A_-}].$$

Hence, the integrable random variable $(Z' - Z)\mathbb{I}_{A_-}$ is nonnegative and strictly positive exactly on A_-. The same argument yields $\mathbb{P}(Z < Z') = 0$. Finally,

$$\{Z \neq Z'\} = \{Z > Z'\} \cup \{Z < Z'\}$$

so taking probabilities under $\mathbb{P}$,

$$\mathbb{P}(Z \neq Z') \leq \mathbb{P}(Z > Z') + \mathbb{P}(Z < Z') = 0.$$

Therefore, $\mathbb{P}(Z = Z') = 1$, as required.

□

Different random variables satisfying the definition are called versions of the conditional expectation. The lemma shows that any two versions agree outside a null event, so the notation $\mathbb{E}[X \mid \mathcal{G}]$ is unambiguous up to almost sure equality.

This proves uniqueness, existence is a separate result. Finally, conditional probability fits into the same framework by taking an indicator,

$$\mathbb{P}(A \mid \mathcal{G}) := \mathbb{E}[\mathbb{I}_A \mid \mathcal{G}].$$

We will see that conditioning on the information generated by an event B recovers the familiar ratio $\mathbb{P}(A \cap B)/\mathbb{P}(B)$ on B whenever $\mathbb{P}(B) > 0$.

2.2 Conditioning and σ-Algebras

Theorem 2.1. *Let X be an integrable random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.*

(i) If X is $\mathcal{G}$-measurable, equivalently $\sigma(X) \subseteq \mathcal{G}$ then,

$$\mathbb{E}[X \mid \mathcal{G}] = X \quad \textit{almost surely.}$$

(ii) If $\mathcal{G} = \{\emptyset, \Omega\}$ then,

$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X] \quad \textit{almost surely}$$

where $\mathbb{E}[X]$ is regarded as a constant random variable.

(iii) Let $A_1, \ldots, A_m$ be pairwise disjoint events satisfying,

$$\bigcup_{i=1}^{m} A_i = \Omega, \quad \mathbb{P}(A_i) > 0 \quad \forall i.$$

If $\mathcal{G} = \sigma(A_1, \ldots, A_m)$ then,

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{i=1}^{m} \frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)} \mathbb{I}_{A_i} \quad \textit{almost surely.}$$

Proof. We directly verify each of the follow cases.

Proof of (i). We verify directly that X satisfies the definition of conditional expectation of X given $\mathcal{G}$. First, X is $\mathcal{G}$-measurable by assumption. Second, X is integrable that is $\mathbb{E}[|X|] < \infty$. Finally choosing $Z = X$ for every $A \in \mathcal{G}$,

$$\mathbb{E}[Z\mathbb{I}_A] = \mathbb{E}[X\mathbb{I}_A].$$

Thus, X itself satisfies all the defining conditions. By *Lemma 2.1*,

$$\mathbb{E}[X \mid \mathcal{G}] = X \quad \text{almost surely.}$$

Conditioning on $\mathcal{G}$ restricts us to the information contained in $\mathcal{G}$. Hence, if X is already $\mathcal{G}$-measurable, that information determines X, so conditioning leaves it unchanged. In particular,

$$\mathbb{E}[X \mid \sigma(X)] = X \quad \text{almost surely.}$$

Proof of (ii). Let $Z := \mathbb{E}[X \mid \mathcal{G}]$ be a version of the conditional expectation of X given $\mathcal{G}$. Since Z is $\mathcal{G}$-measurable and $\mathcal{G} = \{\emptyset, \Omega\}$, the preimage of every Borel set must be either $\emptyset$ or Ω. This forces Z to be constant. Indeed, if there were outcomes ω_1, ω_2 with,

$$Z(\omega_1) < Z(\omega_2)$$

we could choose r strictly between these values, that is $Z(\omega_1) < r < Z(\omega_2)$. The event $\{Z \leq r\} \in \mathcal{G}$ would contain ω_1 but not ω_2 making it neither empty nor all of Ω.

This is precisely a contradiction. Therefore, $Z = c$ for some $c \in \mathbb{R}$.

Taking $A = \Omega$ in the testing identity,

$$c = \mathbb{E}[c] = \mathbb{E}[Z] = \mathbb{E}[Z\mathbb{I}_\Omega] = \mathbb{E}[X\mathbb{I}_\Omega] = \mathbb{E}[X].$$

Conversely, $\mathbb{E}[X]$ is a constant, integrable, and $\mathcal{G}$-measurable, and satisfies the testing identity on the only two events $\emptyset$ and Ω. Hence,

$$\mathbb{E}[X \mid \{\emptyset, \Omega\}] = \mathbb{E}[X] \quad \text{almost surely.}$$

Proof of (iii). Define,

$$Z := \sum_{i=1}^{m} \frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)} \mathbb{I}_{A_i}.$$

We verify that Z is $\mathcal{G}$-measurable, integrable, and satisfies,

$$\mathbb{E}[Z\mathbb{I}_B] = \mathbb{E}[X\mathbb{I}_B] \qquad \forall B \in \mathcal{G}.$$

First, Z is $\mathcal{G}$-measurable. Since the A_i's are pairwise disjoint and cover Ω such that $\bigcup_{i=1}^m A_i = \Omega$, every outcome $\omega \in \Omega$ belongs to exactly one cell. If $\omega \in A_i$ then,

$$\begin{aligned} Z(\omega) &= \sum_{j=1}^{m} \frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}\mathbb{I}_{A_j}(\omega) \\ &= \frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)}\mathbb{I}_{A_i} + \sum_{j\neq i} \frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}\mathbb{I}_{A_j} \\ &= \frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)}(1) + \sum_{j\neq i} \frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}(0) \\ &= \frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)}, \end{aligned}$$

Thus, Z is constant on each partition cell A_i since its value depends only on which cell occurred. More formally, for every Borel set $D \subseteq \mathbb{R}$, its preimage

$$Z^{-1}(D) = \bigcup_{\left\{i:\frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)}\in D\right\}} A_i \in \mathcal{G}.$$

Hence, Z is $\mathcal{G}$-measurable. Second, we verify that Z is integrable.

Since exactly one indicator is nonzero at each outcome,

$$\begin{aligned} \mathbb{E}[|Z|] &= \mathbb{E}\left[\sum_{i=1}^{m} \frac{|\mathbb{E}[X\mathbb{I}_{A_i}]|}{\mathbb{P}(A_i)}\mathbb{I}_{A_i}\right] \\ &= \sum_{i=1}^{m} \frac{|\mathbb{E}[X\mathbb{I}_{A_i}]|}{\mathbb{P}(A_i)}\mathbb{P}(A_i) \\ &= \sum_{i=1}^{m} |\mathbb{E}[X\mathbb{I}_{A_i}]| \\ &\leq \sum_{i=1}^{m} \mathbb{E}[|X|\mathbb{I}_{A_i}] = \mathbb{E}\left[|X|\sum_{i=1}^{m}\mathbb{I}_{A_i}\right] = \mathbb{E}[|X|] < \infty. \end{aligned}$$

Third, we verify the testing identity. Let $B \in \mathcal{G}$. Because $\mathcal{G}$ is generated by the finite partition, every event in $\mathcal{G}$ is a union of partition cells. Thus for some $I \subseteq \{1, \ldots, m\}$,

$$B = \bigcup_{i\in I} A_i.$$

Disjointedness allows us to write this as,

$$\mathbb{I}_B = \sum_{i\in I} \mathbb{I}_{A_i}.$$

Substituting the formulation previously obtained for Z,

$$\begin{aligned}
\mathbb{E}[Z\mathbb{I}_B] &= \mathbb{E}\left[\left(\sum_{j=1}^{m}\frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}\mathbb{I}_{A_j}\right)\mathbb{I}_B\right] \\
&= \mathbb{E}\left[\sum_{j=1}^{m}\frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}\mathbb{I}_{A_j}\mathbb{I}_B\right] \\
&= \sum_{j=1}^{m}\frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}\mathbb{E}[\mathbb{I}_{A_j}\mathbb{I}_B] \\
&= \sum_{j=1}^{m}\frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}\mathbb{E}[\mathbb{I}_{A_j\cap B}], \qquad \mathbb{I}_{A_j}\mathbb{I}_B = \mathbb{I}_{A_j\cap B} \\
&= \sum_{j=1}^{m}\frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}\mathbb{P}(A_j\cap B), \qquad \text{by expectation of indicators.}
\end{aligned}$$

The second last line follows since the product of the two indicators $\mathbb{I}_{A_j}\mathbb{I}_B$ is equivalent to an indicator over the intersection of A_j and B. Since B is a union of the selected cells,

$$A_j \cap B = A_j \cap \left(\bigcup_{i\in I} A_i\right) = \begin{cases} A_i, & j \in I, \\ \emptyset, & j \notin I. \end{cases}$$

Hence, by expectation of indicators,

$$\mathbb{E}[\mathbb{I}_{A_j}\mathbb{I}_B] = \mathbb{P}(A_j\cap B) = \begin{cases} \mathbb{P}(A_j), & j \in I, \\ 0, & j \notin I. \end{cases}$$

Since only the indices in I contribute to the sum,

$$\begin{aligned}
\mathbb{E}[Z\mathbb{I}_B] &= \sum_{j\in I}\frac{\mathbb{E}[X\mathbb{I}_{A_j}]}{\mathbb{P}(A_j)}\mathbb{P}(A_j) = \sum_{j\in I}\mathbb{E}[X\mathbb{I}_{A_j}] \\
&= \mathbb{E}\left[\sum_{j\in I} X\mathbb{I}_{A_j}\right] \\
&= \mathbb{E}\left[X\sum_{j\in I}\mathbb{I}_{A_j}\right] \\
&= \mathbb{E}[X\mathbb{I}_B].
\end{aligned}$$

Hence, this establishes the testing identity for every $B \in \mathcal{G}$. Consequently, Z satisfies all the conditions for conditional expectation. By *Lemma 2.1*,

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{i=1}^{m}\frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)}\mathbb{I}_{A_i} \quad \text{almost surely.}$$

□

Remark 2.2. The numerator,

$$\mathbb{E}[X\mathbb{I}_{A_i}] = \int_{A_i} X \in d\mathbb{P}$$

is the contribution of A_i to the overall expectation. Dividing by $\mathbb{P}(A_i)$ normalizes the probability within that cell and produces its averages,

$$\mathbb{E}[X \mid A_i] := \frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)}.$$

Thus, the formulation may be read as,

$$\mathbb{E}[X \mid \mathcal{G}] := \sum_{i=1}^{m} \mathbb{E}[X \mid A_i]\mathbb{I}_{A_i}.$$

Observe that $\mathbb{E}[X \mid A_i]$ is a number so the conditional expectation given $\mathcal{G}$ is a random variable that selects this number when A_i occurs. The denominator is also forced by the testing identity, that is, assigning the displayed average to all A_i gives,

$$\left(\frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)}\right)\mathbb{P}(A_i) = \mathbb{E}[X\mathbb{I}_{A_i}]$$

so the total expected contribution on that cell is preserved.

Example 2.1 (Observing the parity of a die roll)**.** Let X be a fair die roll and suppose we observe only whether it is odd or even. Set $A_1 = \{X \in \{1,3,5\}\}$, $A_2 = \{X \in \{2,4,6\}\}$ and $\mathcal{G} = \sigma(A_1, A_2)$. Since, $\mathbb{P}(A_1) = \mathbb{P}(A_2) = 1/2$,

$$\frac{\mathbb{E}[X\mathbb{I}_{A_i}1]}{\mathbb{P}(A_1)} = \frac{(1+3+5)/6}{1/2} = 3,$$
$$\frac{\mathbb{E}[X\mathbb{I}_{A_2}]}{\mathbb{P}(A_2)} = \frac{(2+4+6)/6}{1/2} = 4.$$

Therefore, $\mathbb{E}[X \mid \mathcal{G}] = 3\mathbb{I}_{A_1} + 4\mathbb{I}_{A_2}$.

An odd rolls gives conditional expectation 3 and an even roll gives conditional expectation 4. The information does not reveal the exact roll, but it determines which group average to use.

Remark 2.3 (Recovering the conditional probability). For an event $C \in \mathcal{F}$, apply *Theorem 2.1* to $X = \mathbb{I}_C$. Since we previously saw,

$$\mathbb{E}[I_C I_{A_i}] = \mathbb{P}(C \cap A_i)$$

we obtain the conditional probability,

$$\mathbb{P}(C \mid \mathcal{G}) := \mathbb{E}[\mathbb{I}_C \mid \mathcal{G}] = \sum_{i=1}^{m} \frac{\mathbb{P}(C \cap A_i)}{\mathbb{P}(A_i)}\mathbb{I}_{A_i} \quad \text{almost surely.}$$

On each partition cell, this is precisely the introductory conditional probability,

$$\mathbb{P}(C \mid A_i) = \frac{\mathbb{P}(C \cap A_i)}{\mathbb{P}(A_i)}.$$

Remark 2.4 (Cells of probability zero). If a partition cell satisfies $\mathbb{P}(A_i) = 0$, the ratio is undefined. We may instead assign any finite constant on that cell. Both,

$$\mathbb{E}[X\mathbb{I}_{A_i}] = 0 = \mathbb{E}[Z\mathbb{I}_{A_i}]$$

so the testing identity imposes no restriction there. This freedom is consistent with conditional expectation being unique only up to almost sure equality.

For a finite partition, recall that conditional expectation assigns an average to each cell. If the realized outcome ω belongs to A_i, the information in $\mathcal{G} = \sigma(A_1, \ldots, A_m)$ tells us that A_i occurred. Our conditional estimate of X is then,

$$\mathbb{E}[X \mid A_i] = \frac{\mathbb{E}[X\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)}.$$

Consequently,

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{i=1}^{m} \mathbb{E}[X \mid A_i]\mathbb{I}_{A_i} \quad \text{almost surely.}$$

The coefficients are fixed numbers as the indicators select the coefficient corresponding to the observed cell. This interpretation also explains conditional probability, that is, to estimate the probability of an event B, we condition its indicator.

Definition 2.2 (Conditional probability given a σ-algebra)**.** For $B \in \mathcal{F}$ and a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ define,

$$\mathbb{P}(B \mid \mathcal{G}) := \mathbb{E}[\mathbb{I}_B \mid \mathcal{G}].$$

Hence, $\mathbb{P}(B \mid \mathcal{G})$ is a random variable representing the probability of B given the information available in $\mathcal{G}$. When $\mathcal{G}$ is generated by a finite partition $A_1, \ldots, A_m$ with positive cell probabilities, the partition formula gives,

$$\mathbb{E}[\mathbb{I}_B \mid \mathcal{G}] = \sum_{i=1}^{m} \mathbb{E}[\mathbb{I}_B \mid A_i]\mathbb{I}_{A_i}.$$

For each cell,

$$\begin{aligned}
\mathbb{E}[\mathbb{I}_B \mid A_i] &= \frac{\mathbb{E}[\mathbb{I}_B\mathbb{I}_{A_i}]}{\mathbb{P}(A_i)} = \frac{\mathbb{E}[\mathbb{I}_{B\cap A_I}]}{\mathbb{P}(A_i)} \\
&= \frac{\mathbb{P}(B \cap A_i)}{\mathbb{P}(A_i)} \\
&= \mathbb{P}(B \mid A_i).
\end{aligned}$$

Therefore,

$$\mathbb{P}(B \mid \mathcal{G}) = \sum_{i=1}^{m} \mathbb{P}(B \mid A_i)\mathbb{I}_{A_i} \quad \text{almost surely.}$$

The familiar conditional probabilities $\mathbb{P}(B \mid A_i)$ are the numerical values take by this random variable on the respective cells. We can similarly condition on an observed random variable Y. Observing Y reveals precisely the information represented by $\sigma(Y)$.

Definition 2.3 (Conditioning on a random variable)**.** Let X be integrable and let Y be a random variable. We define,

$$\mathbb{E}[X \mid Y] := \mathbb{E}[X \mid \sigma(Y)].$$

Likewise, for $B \in \mathcal{F}$,

$$\mathbb{P}(B \mid Y) := \mathbb{E}[\mathbb{I}_B \mid Y].$$

Example 2.2 (Conditioning on a finite-valued random variable)**.** Suppose Y takes distinct values $y_1, \ldots, y_m$ with $\mathbb{P}(Y = y_i) > 0$ almost surely. Then, the events,

$$\{Y = y_1\}, \ldots, \{Y = y_m\}$$

form a partition Ω and,

$$\sigma(Y) = \sigma\left(\{Y = y_1\}, \ldots, \{Y = y_m\}\right).$$

Hence, knowing Y therefore identifies the level set $\{Y = y_i\}$ contains the outcome.

Applying the finite partition formula,

$$\begin{aligned}
\mathbb{E}[X \mid Y] &= \mathbb{E}[X \mid \sigma(Y)] \\
&= \sum_{i=1}^{m} \frac{\mathbb{E}[X\mathbb{I}_{\{Y=y_i\}}]}{\mathbb{P}(Y = y_i)}\mathbb{I}_{\{Y=y_i\}} \\
&= \sum_{i=1}^{m} \underbrace{\mathbb{E}[X \mid Y = y_i]}_{:=\frac{\mathbb{E}[X\mathbb{I}_{\{Y=y_i\}}]}{\mathbb{P}(Y=y_i)}}\mathbb{I}_{\{Y=y_i\}} \quad \text{almost surely.}
\end{aligned}$$

If $Y(\omega) = y_i$ then,

$$\begin{aligned}
\mathbb{E}[X \mid Y](\omega) &= \sum_{j=1}^{m} \mathbb{E}[X \mid Y = y_i]\mathbb{I}_{\{Y=y_i\}}(\omega) \\
&= \mathbb{E}[X \mid Y = y_i].
\end{aligned}$$

Only the indicator corresponding to the observed value is one with all others zero.

Conditional expectation retains the linearity and order properties of ordinary expectation. The different is that its outputs are random variables, so the resulting identities and inequalities hold almost surely.

Proposition 2.1 (Basic properties of conditional expectation)**.** *Let $X, Y \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra and let $a, b \in \mathbb{R}$. Then,*

(i) ***Linearity.***

$$\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[X \mid \mathcal{G}] \quad \textit{almost surely.}$$

(ii) ***Positivity.*** *If $X \geq 0$ almost surely then,*

$$\mathbb{E}[X \mid \mathcal{G}] \geq 0 \quad \textit{almost surely.}$$

(iii) ***Monotonicity.*** *If $X \leq Y$ almost surely then,*

$$\mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}] \quad \textit{almost surely.}$$

Proof. We prove the properties of conditional expectation.

Proof of (i). Define,

$$Z := a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}].$$

Both conditional expectations at $\mathcal{G}$-measurable so their linear combination is also $\mathcal{G}$-measurable. Moreover,

$$\mathbb{E}[|Z|] \leq |a|\mathbb{E}[|\mathbb{E}[X \mid \mathcal{G}]|] + |b|\mathbb{E}[|\mathbb{E}[Y \mid \mathcal{G}]|] < \infty$$

because conditional expectations are integrable by definition.

For every $A \in \mathcal{G}$. linearity of ordinary expectation and the testing identities gives,

$$\begin{aligned}\mathbb{E}[Z\mathbb{I}_A] &= a\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbb{I}_A] + b\mathbb{E}[\mathbb{E}[Y \mid \mathcal{G}]\mathbb{I}_A] \\ &= a\mathbb{E}[X\mathbb{I}_A] + b\mathbb{E}[Y\mathbb{I}_A] \\ &= \mathbb{E}[(aX + bY)\mathbb{I}_A].\end{aligned}$$

Thus, Z satisfies the definition of $\mathbb{E}[aX + bY \mid \mathcal{G}]$. Almost sure uniqueness proves the identity.

Proof of (ii). Suppose $X \geq 0$ almost surely, and write,

$$Z := \mathbb{E}[X \mid \mathcal{G}].$$

We must show that the event on which Z is negative has probability zero. Define the event $A := \{Z < 0\}$. Since Z is $\mathcal{G}$-measurable, $A \in \mathcal{G}$, so we may use it in the testing identity. On A, Z is negative. Outside of A, $Z\mathbb{I}_A = 0$. Hence, $Z\mathbb{I}_A \leq 0$.

On the contrary, $X\mathbb{I}_A \geq 0$ almost surely. Therefore,

$$0 \geq \mathbb{E}[Z\mathbb{I}_A] = \mathbb{E}[X\mathbb{I}_A] \geq 0.$$

It follows that $\mathbb{E}[-Z\mathbb{I}_A] = 0$. The random variable $-Z\mathbb{I}_A$ is nonnegative and integrable, so it must equal zero almost surely. It is strictly positive exactly on A hence $\mathbb{P}(Z < 0) = \mathbb{P}(A) = 0$.

Proof of (iii). If $X \leq Y$ almost surely, then $Y - X \geq 0$ almost surely.

Applying positivity then linearity,

$$\begin{aligned} 0 &\leq \mathbb{E}[Y - X \mid \mathcal{G}] \\ &= \mathbb{E}[Y \mid \mathcal{G}] - \mathbb{E}[X \mid \mathcal{G}] \quad \text{almost surely.} \end{aligned}$$

Re-arranging gives the result.

□

In constructing proofs, we ideally wish to construct a candidate and verify its testing identity or test on a $\mathcal{G}$-measurable event where the desired property would fail. If the available information is independent of X, conditioning does not change its mean. The information may reveal something about other random quantities, but it provides no information about X.

Lemma 2.2 (Conditioning on independent information)**.** *Let X be integrable and suppose X and Y are independent. Then,*

$$\mathbb{E}[X \mid \sigma(Y)] = \mathbb{E}[X] \quad \textit{almost surely.}$$

Equivalently, we can say that $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$ almost surely.

Proof. Set $Z(\omega) := \mathbb{E}[X]$ for every $\omega \in \Omega$. Since Z is constant, it is $\sigma(Y)$-measurable.

It is also integrable so,

$$\mathbb{E}[|Z|] = |\mathbb{E}[X]| \leq \mathbb{E}[|X|] < \infty.$$

Now take any $A \in \sigma(Y)$. By independence of X, Y implies that $\sigma(X)$ and $\sigma(Y)$ are independent. Since $\sigma(\mathbb{I}_A) \subseteq \sigma(Y)$, the random variables X and $\mathbb{I}_A$ are independent. Both are integrable so,

$$\mathbb{E}[X\mathbb{I}_A] = \mathbb{E}[X]\mathbb{E}[\mathbb{I}_A] = \mathbb{E}[X]\mathbb{P}(A).$$

For the candidate Z,

$$\begin{aligned} \mathbb{E}[Z\mathbb{I}_A] &= \mathbb{E}[\mathbb{E}[X]\mathbb{I}_A] \\ &= \mathbb{E}[X]\mathbb{P}(A) \\ &= \mathbb{E}[X\mathbb{I}_A]. \end{aligned}$$

Thus, Z satisfies the testing identity for every $A \in \sigma(Y)$. By *Lemma 2.1*,

$$\mathbb{E}[X \mid \sigma(Y)] = \mathbb{E}[X] \quad \text{almost surely.}$$

□

Note that the same proof shows that $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$ almost surely.

2.3 Conditional Expectation as an Orthogonal Projection

We now develop a geometric interpretation of conditional expectation. In Euclidean space, projecting a vector onto a subspace produces its closest approximation within that subspace. For square-integrable random variables, conditional expectation has exactly this interpretation. That is, it is the closest approximation to X among random variables determined by $\mathcal{G}$, when distance is measured by mean-squared error.

Definition 2.4 ($\mathcal{L}^p$ spaces)**.** For $1 \leq p < \infty$, define

$$\mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P}) = \{X : \Omega \to \mathbb{R} : X \in \mathcal{F}, \mathbb{E}[|X|^p] < \infty\}.$$

$X \in \mathcal{F}$ in this setting implies that X is $\mathcal{F}$-measurable. Its norm is,

$$\|X\|_{\mathcal{L}^p} := \left(\mathbb{E}[|X|^P]\right)^{1/p}.$$

Thus, $X \in \mathcal{L}^1$ if and only if X is integrable while $X \in \mathcal{L}^2$ if and only if X is square-integrable. We will abbreviate $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ as $\mathcal{L}^2(\mathcal{F})$. For $\mathcal{G} \subseteq \mathcal{F}$ we can define,

$$\mathcal{L}^2(\mathcal{G}) := \{X \in \mathcal{L}^2(\mathcal{F}) : X \in \mathcal{G}\}.$$

This is precisely the subspace of square-integrable random variables that can be determined using the information in $\mathcal{G}$. Identifying almost surely equal variables is necessary for this geometry. If $X = Y$ almost surely, their distance should be zero and they should represent the same point.

Definition 2.5 (Inner product on $\mathcal{L}^2$)**.** For $X, Y \in \mathcal{L}^2(\mathcal{F})$ define,

$$\langle X, Y \rangle := \mathbb{E}[XY].$$

This is finite because $2|XY| \leq X^2 + Y^2$.

From MATH 235, we know that the inner product satisfies,

(i) **Linearity.**

$$\langle aX + bY, W \rangle = a\langle X, Y \rangle + b\langle Y, W \rangle.$$

(ii) **Symmetry.**

$$\langle X, Y \rangle = \langle Y, X \rangle.$$

(iii) **Nonnegativity.**

$$\langle X, X \rangle = \mathbb{E}[X^2] \geq 0.$$

(iv) **Positive definiteness up to almost sure equality.**

$$\langle X, X \rangle = 0 \iff X = 0 \quad \text{almost surely.}$$

The induced norm and distance respectively are,

$$\|X\|_{\mathcal{L}^2} := \sqrt{\mathbb{E}[X^2]}, \qquad \|X - Y\|_{\mathcal{L}^2} := \sqrt{\mathbb{E}[(X-Y)^2]}.$$

Two random variables are *orthogonal* if,

$$\langle X, Y \rangle = \mathbb{E}[XY] = 0.$$

The standard $\mathcal{L}^2(\mathcal{F})$ is complete under this norm making it a Hilbert space. We will use this completeness statement as a standard fact without proof.

Remark 2.5 ($\mathcal{L}^2(\mathcal{G})$ is a closed subspace). Linear combinations of $\mathcal{G}$-measurable, square-integrable variables remain in $\mathcal{L}^2(\mathcal{G})$, hence it is a linear subspace.

It is also closed as an $\mathcal{L}^2$ limit of elements of $\mathcal{L}^2(\mathcal{G})$ has a $\mathcal{G}$-measurable representative. Indeed, $\mathcal{L}^2$ convergence implies convergence in probability, from which one can extract an almost surely convergent subsequence. Taking the pointwise limit of its $\mathcal{G}$-measurable representatives where that limit exists and assigning zero elsewhere, produces the required representative.

Closedness ensures that the closest approximation remains inside $\mathcal{L}^2(\mathcal{G})$ rather than merely being approached by elements of its. For a closed linear subspace H of a Hilbert space, the orthogonal projection theorem gives a unique closest point $M \in H$.

This point is precisely characterized by,

$$\langle X - M, W \rangle = 0 \quad \forall W \in H.$$

Thus, the error is orthogonal to every direction available within the subspace.

Lemma 2.3 (Conditional expectation as an orthogonal projection)**.** *Let $X \in \mathcal{L}^2(\mathcal{F})$ and let $\mathcal{G} \subseteq \mathcal{F}$. Then,*

$$\mathbb{E}[X \mid \mathcal{G}] = \mathrm{proj}_{\mathcal{L}^2(\mathcal{G})} X.$$

In particular, $\mathbb{E}[X \mid \mathcal{G}] \in \mathcal{L}^2(\mathcal{G})$ and,

$$\mathbb{E}\left[(X - \mathbb{E}[X \mid \mathcal{G}])\, W\right] = 0 \quad \forall W \in \mathcal{L}^2(\mathcal{G}).$$

Proof. Since $\mathcal{L}^2(\mathcal{G})$ is a closed linear subspace of $\mathcal{L}^2(\mathcal{F})$, the orthogonal projection theorem gives a unique point,

$$M := \mathrm{proj}_{\mathcal{L}^2(\mathcal{G})} X \in \mathcal{L}^2(\mathcal{G}),$$

such that $\langle X - M, W \rangle = 0$ for every $W \in \mathcal{L}^2(\mathcal{G})$. It remains for us to show that M satisfies the defining conditions of $\mathbb{E}[X \mid \mathcal{G}]$. First choose a $\mathcal{G}$-measurable representative of M.

Second, M is integrable because it is square-integrable. By the *Cauchy-Schwarz* inequality,

$$\begin{aligned} \mathbb{E}[|M|] &= \mathbb{E}[|M| \cdot 1] \\ &\leq \sqrt{\mathbb{E}[M^2]}\sqrt{\mathbb{E}[1^2]} \\ &= \sqrt{\mathbb{E}[M^2]} < \infty. \end{aligned}$$

Finally, take any $A \in \mathcal{G}$. Its indicator is $\mathcal{G}$-measurable and satisfies,

$$\mathbb{E}[\mathbb{I}_A^2] = \mathbb{E}[\mathbb{I}_A] = \mathbb{P}(A) \leq 1.$$

Therefore, $\mathbb{I}_A \in \mathcal{L}^2(\mathcal{G})$. Consequently, we may choose $W = \mathbb{I}_A$ in the orthogonality condition,

$$\begin{aligned} 0 &= \langle X - M, \mathbb{I}_A \rangle \\ &= \mathbb{E}[(X - M)\mathbb{I}_A] \\ &= \mathbb{E}[X\mathbb{I}_A] - \mathbb{E}[M\mathbb{I}_A]. \end{aligned}$$

Re-arranging, we obtain the testing identity $\mathbb{E}[X\mathbb{I}_A] = \mathbb{E}[M\mathbb{I}_A]$ for all $A \in \mathcal{G}$.

Hence, M is $\mathcal{G}$-measurable, integrable, and satisfies the testing identity. By almost sure uniqueness, $M = \mathbb{E}[X \mid \mathcal{G}]$ almost surely. Since M is the orthogonal projection, its residual is orthogonal to the whole subspace. Hence,

$$X - \mathbb{E}[X \mid \mathcal{G}] \perp \mathcal{L}^2(\mathcal{G}).$$

□

Remark 2.6 (Best approximation in mean-squared error). The projection interpretation gives,

$$\mathbb{E}[X \mid \mathcal{G}] = \arg_{Y \in \mathcal{L}^2(\mathcal{G})} \min \mathbb{E}[(X - Y)^2]$$

with uniqueness by *Lemma 2.1* understood up to almost sure equality.

To see this, take any $Y \in \mathcal{L}^2(\mathcal{G})$ and decompose,

$$X - Y = (X - \mathbb{E}[X \mid]) + (\mathbb{E}[X \mid \mathcal{G}] - Y)\,.$$

The second term belongs to $\mathcal{L}^2(\mathcal{G})$ so it is orthogonal to the first. Expanding the square,

$$\begin{aligned} \mathbb{E}[(X - Y)^2] = \mathbb{E}\left[(X - \mathbb{E}[X \mid \mathcal{G}])^2\right] + 2\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])(\mathbb{E}[X \mid \mathcal{G}] - Y)] \\ \mathbb{E}\left[(\mathbb{E}[X \mid \mathcal{G}] - Y)^2\right] \\ = \mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])^2] + \mathbb{E}\left[(\mathbb{E}[X \mid \mathcal{G}] - Y)^2\right]. \end{aligned}$$

This is precisely the Pythagorean identity in $\mathcal{L}^2$-space. The final term is nonnegative so we establish the inequality,

$$\mathbb{E}[(X - Y)^2] \geq \mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])^2].$$

Note that equality holds precisely when $Y = \mathbb{E}[X \mid \mathcal{G}]$ almost surely. Hence, every other square-integrable estimate based on $\mathcal{G}$ has an additional mean-squared error equal to its squared $\mathcal{L}^2$ distance from the conditional expectation.

The testing identity is the event-based form of this orthogonality,

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])\mathbb{I}_A] = 0 \qquad \forall A \in \mathcal{G}.$$

In $\mathcal{L}^2$, it extends to every square-integrable $\mathcal{G}$-measurable test variable. This means that the residual has zero inner product with every such variable but does not generally imply that the residual is independent of $\mathcal{G}$. Conditional expectation itself only requires $X \in \mathcal{L}^1$. The projection and mean-squared error interpretation requires the stronger assumption $X \in \mathcal{L}^2$.

Example 2.3 (Conditional expectation of a Gaussian random variable)**.** Let Y and Z be independent standard normal random variables and let $\rho \in (-1, 1)$. Define,

$$X := \rho Y + \sqrt{1-\rho^2} Z, \qquad Y, Z \sim \mathcal{N}(0,1).$$

We wish to compute $\mathbb{E}[X \mid \sigma(Y)]$. Because Y is $\sigma(Y)$-measurable, conditioning on $\sigma(Y)$ leaves it unchanged. Hence, $\mathbb{E}[Y \mid \sigma(Y)] = Y$. On the other hand, $Z \perp Y$ so,

$$\mathbb{E}[Z \mid \sigma(Y)] = \mathbb{E}[Z] = 0.$$

By linearity of conditional expectation,

$$\begin{aligned}
\mathbb{E}[X \mid \sigma(Y)] &= \mathbb{E}\left[\rho Y + \sqrt{1-\rho^2} Z \mid \sigma(Y)\right] \\
&= \rho \mathbb{E}[Y \mid \sigma(Y)] + \sqrt{1-\rho^2}\mathbb{E}[Z \mid \sigma(Y)] \\
&= \rho Y + \sqrt{1-\rho^2}(0) \\
&= \rho Y.
\end{aligned}$$

Thus, observing Y determines the component ρY while the independent, centered component $\sqrt{1-\rho^2}Z$ has conditional mean zero. Since X is square-integrable,

$$X = \rho Y + \sqrt{1-\rho^2} Z$$

is precisely the sum of a conditional expectation and a residual.

The first term belongs to $\mathcal{L}^2(\sigma(Y))$ and the residual is orthogonal to every random variable in $\mathcal{L}^2(\sigma(Y))$. Indeed, if $W \in \mathcal{L}^2(\sigma(Y))$ then $Z \perp W$ so it follows,

$$\begin{aligned}
\mathbb{E}[(X - \rho Y)W] &= \sqrt{1-\rho^2}\mathbb{E}[ZW] \\
&= \sqrt{1-\rho^2}\mathbb{E}[Z]\mathbb{E}[W] \\
&= 0.
\end{aligned}$$

Consequently, $\operatorname{proj}_{\mathcal{L}^2(\sigma(Y))} X = \rho Y$.

Remark 2.7 (Independence and orthogonality). Independence and orthogonality are different concepts. In $\mathcal{L}^2$, orthogonality meansd that $\langle Y, Z\rangle = \mathbb{E}[YZ] = 0$. Independence gives $\mathbb{E}[YZ] = \mathbb{E}[Y]\mathbb{E}[Z]$ so independent square-integrable random variables are orthogonal when at least one has mean zero. Orthogonality alone does not generally imply independence.

In *Example 2.3*, Y, Z are independent and centered so they are orthogonal. However, conditional expectation projects onto the entire space $\mathcal{L}^2(\sigma(Y))$ which contains all square-integrable functions of Y, not merely the multiples of Y.

The projection happens to be a linear function ρY in this Gaussian example. Once we condition on $\mathcal{G}$, any $\mathcal{G}$-measurable random variable is already determined by the available information. This leads to the following conditional analogue of pulling a constant out of an expectation.

Proposition 2.2 (Taking Out What Is Known)**.** *Let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and let Y be a real-valued, $\mathcal{G}$-measurable random variable. If $XY \in \mathcal{L}^1$ then $Y\mathbb{E}[X \mid \mathcal{G}]$ is integrable and,*

$$\mathbb{E}[XY \mid \mathcal{G}] = Y\mathbb{E}[X \mid \mathcal{G}] \quad \textit{almost surely.}$$

Proof. We first prove the result for bounded Y then extend it to the stated assumptions.

Suppose $Y := \mathbb{I}_B$ where $B \in \mathcal{G}$. The random variable $\mathbb{I}_B\mathbb{E}[X \mid \mathcal{G}]$ is $\mathcal{G}$-measurable and integrable. For every $A \in \mathcal{G}$, $A \cap B \in \mathcal{G}$ hence,

$$\begin{aligned}\mathbb{E}[\mathbb{I}_B\mathbb{E}[X \mid \mathcal{G}]] &= \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbb{I}_{A\cap B}] \\ &= \mathbb{E}[X\mathbb{I}_{A\cap B}] = \mathbb{E}[X\mathbb{I}_B\mathbb{I}_A].\end{aligned}$$

Thus, $\mathbb{E}[X\mathbb{I}_B] = \mathbb{I}_B\mathbb{E}[X \mid \mathcal{G}]$ almost surely. By linearity, the same conclusion holds whenever Y is a $\mathcal{G}$-measurable simple random variable. Suppose $|Y| \leq M$.

Choose $\mathcal{G}$-measurable simple random variables Y_n such that $Y_n \to Y$ pointwise for $|Y_n| \leq M$.

For each $A \in \mathcal{G}$, recall previously,

$$\mathbb{E}[Y_n\mathbb{E}[X \mid \mathcal{G}]\mathbb{I}_A] = \mathbb{E}[XY_n\mathbb{I}_A].$$

The integrands are bounded in absolute value by the integrable random variables $M|\mathbb{E}[X \mid \mathcal{G}]|$ and $M|X|$ respectively. Dominated convergence yields,

$$\mathbb{E}[Y\mathbb{E}[X \mid \mathcal{G}]\mathbb{I}_A] = \mathbb{E}[XY\mathbb{I}_A].$$

Since $Y\mathbb{E}[X \mid \mathcal{G}]$ is also $\mathcal{G}$-measurable and integrable, it is precisely a version of $\mathbb{E}[X \mid \mathcal{G}]$. Now we will generalize this to Y with $XY \in \mathcal{L}^1$. First, conditional monotonicity applied to,

$$-|X| \leq X \leq |X|$$

gives $|\mathbb{E}[X \mid \mathcal{G}]| \leq \mathbb{E}[|X| \mid \mathcal{G}]$ almost surely. For each n, the random variable $|Y| \wedge n$ is bounded and $\mathcal{G}$-measurable. By the bounded case, taking ordinary expectations yields,

$$\mathbb{E}[(|Y| \wedge n)\mathbb{E}[|X| \mid \mathcal{G}]] = \mathbb{E}[(|Y| \wedge n) \mid X].$$

Letting $n \to \infty$ and using monotone convergence theorem,

$$\mathbb{E}[|Y| \mid \mathbb{E}[X \mid \mathcal{G}]] \leq \mathbb{E}[|Y|\mathbb{E}[|X| \mid \mathcal{G}]] = \mathbb{E}[|XY|] < \infty.$$

Thus, the proposed conditional expectation is integrable.

Now define the bounded truncations, $Y_n := Y\mathbb{I}_{\{|Y|\leq n\}}$. For every $A \in \mathcal{G}$, the bounded case gives,

$$\mathbb{E}[Y_n\mathbb{E}[X \mid \mathcal{G}]\mathbb{I}_A] = \mathbb{E}[XY_n\mathbb{I}_A].$$

Because,

$$|Y_n\mathbb{E}[X \mid \mathcal{G}]\mathbb{I}_A| \leq |Y||\mathbb{E}[X \mid \mathcal{G}]|, \quad |XY\mathbb{I}_A| \leq |XY|$$

dominated convergence applies to both sides. Therefore,

$$\mathbb{E}[Y\mathbb{E}[X \mid \mathcal{G}]\mathbb{I}_A] = \mathbb{E}[XY\mathbb{I}_A] \quad \forall A \in \mathcal{G}.$$

Together with measurability and integrability this proves the result.

□

When $\mathcal{L}^2(\mathcal{F})$ and Y is bounded and $\mathcal{G}$-measurable, there is a shorter proof using orthogonal projections. These assumptions ensure that $XY \in \mathcal{L}^2(\mathcal{F})$ with $Y\mathbb{E}[X \mid \mathcal{G}] \in \mathcal{L}^2(\mathcal{G})$.

For any $W \in \mathcal{L}^2(\mathcal{G})$ boundedness of Y also gives $YW \in \mathcal{L}^2(\mathcal{G})$. Hence,

$$\begin{aligned} 0 &= \langle X - \mathbb{E}[X \mid \mathcal{G}], YW\rangle \\ &= \mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])YW] \\ &= \mathbb{E}[(XY - Y\mathbb{E}[X \mid \mathcal{G}])W] \\ &= \langle XY - Y\mathbb{E}[X \mid \mathcal{G}], W\rangle. \end{aligned}$$

Thus, $Y\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of XY onto $\mathcal{L}^2(\mathcal{G})$.

The assumption $X \in \mathcal{L}^2$ is needed for this projection argument as boundedness of Y and $X \in \mathcal{L}^1$ alone only guarantee that $XY \in \mathcal{L}^1$.

We next consider conditioning on two nested collections of information. If $\mathcal{H} \subseteq \mathcal{G}$, then $\mathcal{H}$ contains no more information that $\mathcal{G}$. The tower property describes how successive condition interacts with the inclusion.

2.4 Successive & Iterated Conditioning

Proposition 2.3 (The Tower Property)**.** *Let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$ and let $H \subseteq \mathcal{G} \subseteq \mathcal{F}$ be sub-σ-algebras. Then,*

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] \quad \textit{almost surely.}$$

Proof. We prove the two identities separately. First consider the nest conditional expectation,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}].$$

By definition, this random variable is $\mathcal{H}$-measurable and integrable. To show that it equals $\mathbb{E}[X \mid \mathcal{H}]$, it remains to verify the defining integral identity.

Let $A \in \mathcal{H}$. Since $H \subseteq \mathcal{G}$, then $A \in \mathcal{G}$, too. Consequently,

$$\begin{aligned} \mathbb{E}[\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}]\mathbb{I}_A] &= \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbb{I}_A] \\ &= \mathbb{E}[X\mathbb{I}_A]. \end{aligned}$$

The first equality uses conditioning on $\mathcal{H}$ while the second uses conditioning on $\mathcal{G}$.

Thus, almost sure uniqueness of conditional expectation gives,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}].$$

For the other identity, $\mathbb{E}[X \mid \mathcal{H}]$ is $\mathcal{H}$-measurable and therefore also $\mathcal{G}$-measurable. Conditioning an integrable, $\mathcal{G}$-measurable random variable on $\mathcal{G}$ leaves it unchanged. Hence,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{H}].$$

This proves both identities, as required.

□

Remark 2.8 (Successive conditioning). The two identities have complementary interpretations. The identity,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$$

states that first estimating X using information contained in $\mathcal{H}$ then retaining only the information in $\mathcal{H}$ gives the same result as conditioning directly on $\mathcal{H}$. The identity,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{H}]$$

states that the estimate based on $\mathcal{H}$ is already known given $\mathcal{G}$.

Conditioning that estimate on $\mathcal{G}$ therefore leaves it unchanged. In particular, taking the trivial σ-algebra $\mathcal{H} = \{\emptyset, \Omega\}$ gives the law of total expectation,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X].$$

Remark 2.9 (The Geometry of the Tower Property). When $X \in \mathcal{L}^2(\mathcal{F})$, the inclusion of information corresponds to nested closed subspaces,

$$\mathcal{L}^2(\mathcal{H}) \subseteq \mathcal{L}^2(\mathcal{G}) \subseteq \mathcal{L}^2(\mathcal{F}).$$

Hence, the tower property becomes,

$$\begin{aligned}\operatorname{proj}_{\mathcal{L}^2(\mathcal{H})}\left(\operatorname{proj}_{\mathcal{L}^2(\mathcal{G})} X\right) &= \operatorname{proj}_{\mathcal{L}^2(\mathcal{H})} X,\\ \operatorname{proj}_{\mathcal{L}^2(\mathcal{H})}\left(\operatorname{proj}_{\mathcal{L}^2(\mathcal{H})} X\right) &= \operatorname{proj}_{\mathcal{L}^2(\mathcal{H})} X.\end{aligned}$$

Thus, projecting the first onto the larger subspace and the onto the smaller one gives the same point as projecting directly onto the smaller subspace.

Conversely, once a point lies in the smaller subspace, it also lies in the larger one, so projecting it onto the larger subspace does nothing.

3 Martingales, Stopping Times, and Stopped Processes

Before we define the notion of a martingale, it is important to define a structure that properly captures the evolution of information. Hence, a filtration describes how information accumulates over time. Once we specify this information, we can distinguish processes whose current values are observable, decisions made before the next observation, and processes whose conditional expected changes are zero.

3.1 Filtrations, Adaptedness and Predictability

Definition 3.1 (Filtration)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A filtration is a sequence of sub-σ-algebras $(\mathcal{F}_n)_{n\geq 0}$ satisfying,

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}.$$

The collection specified by $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\geq 0}, \mathbb{P})$ is called a filtered probability space.

For a stochastic process $(X_n)_{n\geq 0}$, its natural filtration is,

$$\mathcal{F}_n^X := \sigma(X_1, \ldots, X_n), \quad n \geq 0.$$

Remark 3.1 (Information available over time). We interpret $\mathcal{F}_n$ as the information available at time n. If $A \in \mathcal{F}_n$, then this information determines whether A has occurred. The inclusion,

$$\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$$

means that information available at time n remains available at time $n+1$.

Hence, the natural filtration contains precisely the information needed revealed by observing the process up to the current time. In particular, $\mathcal{F}_n^X$ is the smallest σ-algebra making all $X_0, \ldots, X_n$ possible. A filtration specifies the available information. Adaptedness and predictability describes why the value of a process can be determined from that information.

Definition 3.2 (Adapted and Predictable processes)**.** Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration.

(i) A process $(X_n)_{n\geq 0}$ is *adapted* to $(\mathcal{F}_n)_{n\geq 0}$ if X_n is $\mathcal{F}_n$-measurable for every $n \geq 0$,
(ii) A process $(H_n)_{n\geq 1}$ is *predictable* wrt $(\mathcal{F}_n)_{n\geq 0}$ if H_n is $\mathcal{F}_{n-1}$-measurable for every $n \geq 1$.

Thus, an adapted process has a value at time n that is known from the information available at time n. On the contrary, a predictable process has its time n value determined one step earlier using information available at time $n-1$.

Remark 3.2 (Adaptedness to the Natural filtration). Every process is adapted to its natural filtration. Indeed, $\mathcal{F}_n^X := \sigma(X_0, \ldots, X_n)$ makes X_n measurable by construction.

Also, every predictable process is adapted at times $n \geq 1$ since $\mathcal{F}_{n-1} \subseteq \mathcal{F}_n$.

The converse, however, need not hold.

That is, a value that is observable at time n may not have been known at time $n-1$.

Example 3.1 (A Predictable Trading Strategy)**.** Let $(S_n)_{n\geq 0}$ be a stock-price process with natural filtration $\mathcal{F}_n^S = \sigma(S_0, \ldots, S_n)$. Suppose H_n represents the number of shares held during the price movement from time $n-1$ to n. The position must be chosen using information available at time $n-1$ before observing S_n. For example, define

$$H_n = \mathbb{I}_{\{S_{n-1}>100\}} = \begin{cases} 1, & S_{n-1} > 100, \\ 0, & S_{n-1} \leq 100. \end{cases}$$

Because S_{n-1} is $\mathcal{F}_{n-1}^S$-measurable, $\{S_{n-1} > 100\} \in \mathcal{F}_{n-1}^S$ so H_n is $\mathcal{F}_{n-1}^S$-measurable.

Therefore, $(H_n)_{n\geq 1}$ is predictable. By contrast, the rule $H_n = \mathbb{I}_{\{S_n>100\}}$ is adapted but generally not predictable. It uses the price at the end of the movement to choose the position held during that movement. The gain over this period is,

$$H_n(S_n - S_{n-1}).$$

Predictability expresses the requirement that the position H_n be selected before the price change quantified as $S_n - S_{n-1}$ is observed.

Example 3.2. Let $\xi_1, \xi_2, \ldots$ be independent Rademacher random variables satisfying,

$$\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2}, \quad \text{denoted } \xi_k \overset{i.i.d.}{\sim} \text{Rademacher}.$$

In the future, we will simply refer to these as Rademacher random variables. Define the simple symmetric random walk by $S_0 = 0$ and $S_n = \sum_{i=1}^n \xi_i$ for $n \geq 1$.

Its natural filtration is $\mathcal{F}_n = \sigma(S_0, \ldots, S_n)$. For each $n \geq 1$, $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ since observing the increments determines the entire path up to time n since,

$$S_k = \sum_{i=1}^k \xi_i, \quad k \leq n.$$

Conversely, observing the path determines every increment up to time n because $\xi_k = S_k - S_{k-1}$ for $k \leq n$. Thus, the observed path and the observed increments generate exactly the same information. The process $(S_n)_{n\geq 0}$ is adapted to $(F_n)_{n\geq 0}$.

Now define,

$$H_n = \mathbb{I}_{\{S_{n-1}>0\}} = \begin{cases} 1, & S_{n-1} > 0, \\ 0, & S_{n-1} \leq 0, \end{cases} \quad n \geq 1.$$

Since $\{S_{n-1} > 0\} \in \mathcal{F}_{n-1}$, the random variable H_n is $\mathcal{F}_{n-1}$-measurable. Hence $(H_n)_{n\geq 1}$ is predictable since its value is determined before the next increment ξ_n is observed.

Given these examples, we can now formulate the idea of a *fair process*. Given everything currently known, its expected value at the next time equals its present value.

3.2 Martingales, Submartingales and Supermartingales

Definition 3.3 (Martingale, Submartingale, and Supermartingale)**.** Let $(M_n)_{n\geq 0}$ be adapted to a filtration $(F_n)_{n\geq 0}$ and suppose,

$$\mathbb{E}[|M_n|] < \infty \quad \forall n \geq 0.$$

The process is called a,

(i) *Martingale* if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \quad \text{almost surely for every } n \geq 0.$$

(ii) *Submartingale* if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \geq M_n \quad \text{almost surely for every } n \geq 0.$$

(iii) *Supermartingale* if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \leq M_n \quad \text{almost surely for every } n \geq 0.$$

Because M_n is $\mathcal{F}_n$-measurable, the martingale condition is equivalent to,

$$\begin{aligned}\mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n] &= \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] - M_n \\ &= 0.\end{aligned}$$

Thus, a martingale has zero conditional expected increment. A submartingale has nonnegative conditional expected increments, while a supermartingale has nonpositive conditional expected increments. These conditions concern conditional averages and do not require individual paths to be constant, increasing, or decreasing.

For the symmetric Simple Random Walk (SRW) above, independence of ξ_{n+1} from $\mathcal{F}_n$ gives,

$$\begin{aligned}\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[S_n + \xi_{n+1} \mid \mathcal{F}_n] \\ &= S_n + \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] \\ &= S_n + \mathbb{E}[\xi_{n+1}], \quad \xi_{n+1} \perp \mathcal{F}_n \\ &= S_n.\end{aligned}$$

Since S_n is adapted and $|S_n| \leq n$, it is integrable. Therefore, the simple SRW is a martingale.

A nonlinear function of a martingale need not remain a martingale. The following example shows how an exponential transformation can be normalized to recover the martingale property.

Example 3.3 (An Exponential martingale)**.** Let $(S_n)_{n\geq 0}$ for $S_0 = 0$ be a simple SRW adapted to $(\mathcal{F}_n)_{n\geq 0}$ with $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ where the increments are independent Rademacher random variables and $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Fix $\theta \in \mathbb{R}$ and define,

$$M_n := \frac{e^{\theta S_n}}{(\cosh \theta)^n}, \quad \text{where } \cosh \theta := \frac{e^{\theta} + e^{-\theta}}{2}, \quad n \geq 0.$$

We show that $(M_n)_{n\geq 0}$ is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$.

Proof. We verify adaptedness, integrability, and the conditional expectation identity.

First, S_n is $\mathcal{F}_n$-measurable and $x \mapsto e^{\theta x}$ is continuous. Since $(\cosh\theta)^n$ is a strictly positive deterministic constant, M_n is $\mathcal{F}_n$-measurable. Thus, the process is adapted.

Next, $M_n > 0$ so $\mathbb{E}[|M_n|] = \mathbb{E}[M_n]$. For each increment,

$$\mathbb{E}[e^{\theta\xi_i}] = \frac{1}{2}e^{\theta} + \frac{1}{2}e^{-\theta} = \cosh\theta.$$

By independence,

$$\begin{aligned}\mathbb{E}[e^{\theta S_n}] &= \mathbb{E}\left[e^{\theta\sum_{i=1}^n \xi_i}\right] = \mathbb{E}\left[\prod_{i=1}^n e^{\theta\xi_i}\right] \\ &= \prod_{i=1}^n \mathbb{E}[e^{\theta\xi_i}] \\ &= (\cosh\theta)^n.\end{aligned}$$

Therefore,

$$\mathbb{E}[|M_n|] = \frac{1}{(\cosh\theta)^n}\mathbb{E}[e^{\theta S_n}] = 1 < \infty.$$

This also holds at $n = 0$ since $M_0 = 1$. Finally using $S_{n+1} = S_n + \xi_{n+1}$,

$$\begin{aligned}\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}\left[\frac{e^{\theta(S_n+\xi_{n+1})}}{(\cosh\theta)^{n+1}}\middle|\mathcal{F}_n\right] \\ &= \frac{e^{\theta S_n}}{(\cosh\theta)^{n+1}}\mathbb{E}[e^{\theta S_{n+1}} \mid \mathcal{F}_n] \\ &= \frac{e^{\theta S_n}}{(\cosh\theta)^{n+1}}\mathbb{E}[e^{\theta\xi_{n+1}}] \\ &= \frac{e^{\theta S_n}}{(\cosh\theta)^{n+1}}\cosh\theta \\ &= \frac{e^{\theta S_n}}{(\cosh\theta)^n} \\ &= M_n.\end{aligned}$$

The second equality takes out the $\mathcal{F}_n$-measurable factor. The third uses independence of ξ_{n+1} from $\mathcal{F}_n$. This proves the martingale property.

□

The normalization has a simple interpretation,

$$M_{n+1} = M_n\frac{e^{\theta\xi_{n+1}}}{\cosh\theta}.$$

Given $\mathcal{F}_n$, the current value M_n is known, while the next multiplier has conditional expectation one. Dividing by $(\cosh\theta)^n$ precisely compensates for the expected multiplicative growth of $e^{\theta S_n}$.

Example 3.4 (Square of a Simple Random Walk)**.** Let $(S_n)_{n\geq 0}$ with $S_0 = 0$ be a symmetric SRW with filtration $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ for $n \geq 1$.

We will show that $(S_n^2)_{n\geq 0}$ is a submartingale and that $(-S_n^2)_{n\geq 0}$ is a supermartingale.

First, let $X_n := S_n^2$. Because S_n is $\mathcal{F}_n$-measurable, so is S_n^2. Moreover, $|S_n| \leq n$ giving,

$$\mathbb{E}[|X_n|] = \mathbb{E}[S_n^2] \leq n^2 < \infty.$$

Thus, $(X_n)_{n\geq 0}$ is adapted and integrable at every time. To verify the submartingale condition, recall that we can decompose $S_{n+1} = S_n + \xi_{n+1}$. Squaring both sides,

$$\begin{aligned} X_{n+1} &= S_{n+1}^2 \\ &= (S_n + \xi_{n+1})^2 \\ &= S_n^2 + 2S_n\xi_{n+1} + \xi_{n+1}^2. \end{aligned}$$

The next increment is independent of $\mathcal{F}_n$ so,

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[\xi_{n+1}] = 0.$$

$\xi_{n+1}^2 = 1$ almost surely therefore,

$$\begin{aligned} \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[S_n^2 + 2S_n\xi_{n+1} + \xi_{n+1}^2 \mid \mathcal{F}_n] \\ &= \mathbb{E}[S_n^2 \mid \mathcal{F}_n] + 2\mathbb{E}[S_n\xi_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[\xi_{n+1}^2 \mid \mathcal{F}_n] \\ &= S_n^2 + 2S_n\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] + 1 \\ &= S_n^2 + 2S_n(0) + 1 \\ &= S_n^2 + 1 \\ &= X_n + 1 \geq X_n. \end{aligned}$$

Here, we take S_n outside the conditional expectation because it is $\mathcal{F}_n$-measurable.

Hence $(S_n^2)_{n\geq 0}$ is a submartingale.

Although the random walk itself has zero conditional expected increment, its squared distance from zero has conditional expected increment one $\mathbb{E}[S_{n+1}^2 - S_n^2 \mid \mathcal{F}_n] = 1$. This does not mean that the squared distance increases along every path. A step toward zero can decrease it. The statement concerns its average change given the current information.

Now define $Y_n := -S_n^2$. This process is also adapted and integrable.

By linearity of conditional expectation,

$$\begin{aligned} \mathbb{E}[Y_{n+1} \mid \mathcal{F}] &= \mathbb{E}[-S_{n+1}^2 \mid \mathcal{F}_n] \\ &= -\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] \\ &= -(S_n^2 + 1) \\ &= -S_n^2 - 1 = Y_n - 1 \leq Y_n. \end{aligned}$$

Therefore, $(-S_n^2)_{n\geq 0}$ is a supermartingale.

The calculation above also shows how to remove the expected growth. Since S_n^2 gains one unit in conditional expectation at each step, define $M_n := S_n^2 - n$.

This process is adapted and integrable and,

$$\begin{aligned}\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] \\ &= S_n^2 + 1 - (n+1) \\ &= S_n^2 - n \\ &= M_n.\end{aligned}$$

Thus, subtracting the accumulated expected growth produces the martingale $(S_n^2 - n)_{n\geq 0}$.

A predictable betting strategy allows the size of each bet to depend on previous outcomes. For a fair game, however, changing the stake using past information does not change the conditional expected gain on the next round. Consider the Doubling Betting Strategy example.

Example 3.5 (Doubling Betting Strategy)**.** Let $\xi_1, \xi_2, \ldots$ be Rademacher random variables such that,

$$\xi_n = \begin{cases} +1, & \text{win on round } n, \\ -1, & \text{loss on round } n. \end{cases}$$

Define the underlying fair-game process by $M_0 = 0$ with $M_n = \sum_{i=1}^n \xi_i$ with filtration,

$$\mathcal{F}_0 = \{\emptyset, \Omega\}, \quad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n).$$

Then $M_n - M_{n-1} = \xi_n$ and $\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0$. The *doubling strategy* starts with a bet of one unit. After each loss, the next bet is doubled. After each win, the next bet returns to one unit.

Thus,

$$H_1 = 1, \quad H_{n+1} = \begin{cases} 2H_n, & \xi_n = -1, \\ 1, & \xi_n = 1. \end{cases}$$

$(H_n)_{n\geq 1}$ is the amount bet on round n. That is, the stake H_{n+1} depends only on outcomes observed through round n so it is $\mathcal{F}_n$-measurable. Equivalently, H_n is $\mathcal{F}_{n-1}$-measurable since the bet on round n is chosen before observing ξ_n. Therefore, $(H_n)_{n\geq 1}$ is predictable.

Starting from zero, define the cumulative net gain by,

$$X_0 = 0, \qquad X_n = \sum_{i=1}^n H_i(M_i - M_{i-1}) = \sum_{i=1}^n H_i \xi_i.$$

On round i, the gain is H_i if the bet wins and $-H_i$ if it loses. In particular,

$$X_{n+1} = X_n + H_{n+1}\xi_{n+1}.$$

Negative values of X_n represent cumulative losses.

Example 3.6 (Two Losses Followed by a Win)**.** Suppose the first three outcomes are,

$$\xi_1 = -1, \quad \xi_2 = -1, \quad \xi_3 = +1.$$

The doubling strategy produces the following sequence,

Round n	Stake H_n	Outcome ξ_n	Round gain $H_n\xi_n$	Cumulative gain X_n	Next stake H_{n+1}
0	–	–	–	0	1
1	1	-1	-1	-1	2
2	2	-1	-2	-3	4
3	4	$+1$	$+4$	$+1$	1

After two losses followed by a win, the cumulative gain is,

$$-1 - 2 + 4 = 1.$$

More generally, a run of k losses followed by a win produces a net gain of,

$$-\sum_{j=0}^{k-1} 2^j + 2^k = -(2^k - 1) + 2^k = 1,$$

where the sum is zero if $k = 0$. Although every completed cycle produces one unit of profit, the stakes and accumulated losses grow rapidly during a losing streak. We now verify that the cumulative gain process is nevertheless a martingale.

Claim. The process $(X_n)_{n\geq 0}$ defined above is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$.

Proof. We check adaptedness, integrability, and the conditional expectation identity.

First, note that H_i is $\mathcal{F}_i$-measurable and ξ_i is $\mathcal{F}_i$-measurable. Since $\mathcal{F}_{i-1} \subseteq \mathcal{F}_i \subseteq \mathcal{F}_n$ for $i \leq n$, each product $H_i\xi_i$ is $\mathcal{F}_n$-measurable. Hence,

$$X_n = \sum_{i=1}^{n} H_i\xi_i$$

is $\mathcal{F}_n$-measurable, so the process is adapted. Next, the largest possible stake on round i occurs when all preceding rounds are losses. Consequently, $0 < H_i \leq 2^{i-1}$. Since $|\xi_i| = 1$,

$$\begin{aligned} |X_n| &= \left|\sum_{i=1}^{n} H_i\xi_i\right| \leq \sum_{i=1}^{n} H_i\xi_i \\ &= \sum_{i=1}^{n} H_i \\ &\leq \sum_{i=1}^{n} 2^{i-1} \\ &= 2^n - 1. \end{aligned}$$

Thus, for every fixed n, $\mathbb{E}[|X_n|] \leq 2^n - 1 < \infty$.

Finally, using the one-step update decomposition,

$$\begin{aligned}\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[X_n + H_{n+1}\xi_{n+1} \mid \mathcal{F}_n] \\ &= \mathbb{E}[X_n \mid \mathcal{F}_n] + \mathbb{E}[H_{n+1}\xi_{n+1} \mid \mathcal{F}_n] \\ &= X_n + H_{n+1}E[\xi_{n+1} \mid \mathcal{F}_n] \\ &= X_n + H_{n+1}\left(\frac{1}{2} - \frac{1}{2}\right) \\ &= X_n.\end{aligned}$$

Note that H_{n+1} is taken outside because it is $\mathcal{F}_n$-measurable. Independence of ξ_{n+1} from $\mathcal{F}_n$ then gives its conditional mean zero. Hence $(X_n)_{n\geq 0}$ is a martingale.

Fundamentally, the core computation of the conditional expectation is,

$$\mathbb{E}[X_{n+1} - X_n \mid \mathcal{F}_n] = H_{n+1}\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0.$$

The stake is already known when conditioning on $\mathcal{F}_n$ while the next outcome remains a fair random increment. Predictability preserves this fairness, provided the resulting gains are integrable. In particular, at every fixed time,

$$\mathbb{E}[X_n] = \mathbb{E}[X_0] = 0.$$

This is consistent with the profit from completed cycles since an unfinished losing streak can leave a large negative balance. Stopping at a random winning time is a different question from evaluating the gain at a fixed time, and requires additional care when we study stopping times.

A useful way to construct a martingale is to begin with a single integrable random variable and update its conditional expectation as information becomes available. The tower property ensures that these successive estimates form a martingale.

Definition 3.4 (*Doob* martingale)**.** Let $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\geq 0}, \mathbb{P})$ be a filtered probability space, and let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. The process,

$$M_n := \mathbb{E}[X_n \mid \mathcal{F}_n], \quad n \geq 0$$

is called the *Doob martingale* associated with X.

Proof. By definition of conditional expectation, M_n is $\mathcal{F}_n$-measurable. Thus, the process is adapted. Each M_n is also integrable. Indeed, conditional monotonicity gives,

$$|\mathbb{E}[X \mid \mathcal{F}_n]| \leq \mathbb{E}[|X| \mid \mathcal{F}_n] \quad \text{almost surely}$$

Therefore by monotonicity we may write,

$$\begin{aligned}\mathbb{E}[|M_n|] &= \mathbb{E}[|\mathbb{E}[X \mid \mathcal{F}_n]|] \\ &\leq \mathbb{E}[\mathbb{E}[|X| \mid \mathcal{F}_n]] \\ &= \mathbb{E}[|X|] < \infty.\end{aligned}$$

Finally, since $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$, the tower property gives,

$$\begin{aligned}\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] \\ &= \mathbb{E}[X \mid \mathcal{F}_n] \\ &= M_n.\end{aligned}$$

Hence, $(M_n)_{n\geq 0}$ is a martingale. □

Remark 3.3 (Updating an estimate). Suppose we interpret X as a random quantity of interest and $\mathcal{F}_n$ as the information available at n. Then $M_n = \mathbb{E}[X \mid \mathcal{F}_n]$ is the conditional estimate of X based on that information.

If $X \in \mathcal{L}^2$, it is also the best approximation to X in mean-squared error among all square-integrable, $\mathcal{F}_n$-measurable random variables.

As new information arrives, this estimate may increase or decrease. However,

$$\mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n] = 0.$$

Thus, given the current information, the next revision has conditional mean zero.

Every martingale on a fixed finite time horizon $0, \ldots, N$ has this form,

$$M_n = \mathbb{E}[M_N \mid \mathcal{F}_n], \quad 0 \leq n \leq N.$$

We obtain the above by repeatedly applying the tower property. Representing an infinite-horizon martingale using a single integrable terminal random variable requires additional conditions.

3.3 Martingale Differences

The changes between successive martingale values are often easier to study than the process itself. Their defining feature is that each change has conditional mean zero given the preceding information. We provide the definition of a *Martingale Difference* next.

Definition 3.5 (Martingale Difference Sequence)**.** A sequence $(D_n)_{n\geq 1}$ is a *martingale difference sequence* with respect to $(\mathcal{F}_n)_{n\geq 0}$ if for every $n \geq 1$,

1. D_n is $\mathcal{F}_n$-measurable,
2. $\mathbb{E}[|D_n|] < \infty$,
3. $\mathbb{E}[D_n \mid \mathcal{F}_{n-1}] = 0$ almost surely.

The variable D_n represents a new increment observed at time n. Its conditional mean is zero given all information available at time $n-1$.

Proposition 3.1 (Martingales and Their Differences)**.** *Let $(M_n)_{n\geq 0}$ be an adapted process such that $M_n \in \mathcal{L}^1$ for every $n \geq 0$. Define,*

$$D_n := M_n - M_{n-1}, \quad n \geq 1.$$

Then $(M_n)_{n\geq 0}$ is a martingale if and only if $(D_n)_{n\geq 1}$ is a martingale difference sequence.

Proof. First, each D_n is $\mathcal{F}_n$-measurable since M_n is $\mathcal{F}_n$-measurable while M_{n-1} is $\mathcal{F}_{n-1}$-measurable and $\mathcal{F}_{n-1} \subseteq \mathcal{F}_n$. Also, integrability is satisfied because,

$$\begin{aligned}\mathbb{E}[|D_n|] &= \mathbb{E}[|M_n - M_{n-1}|] \\ &\leq \mathbb{E}[|M_n|] + \mathbb{E}[|M_{n-1}|] \\ &< \infty.\end{aligned}$$

Thus, the differences are adapted and integrable.

Suppose first that $(M_n)_{n\geq 0}$ is a martingale. Because M_{n-1} is $\mathcal{F}_{n-1}$-measurable,

$$\begin{aligned}\mathbb{E}[D_n \mid \mathcal{F}_{n-1}] &= \mathbb{E}[M_n - M_{n-1} \mid \mathcal{F}_{n-1}] \\ &= \mathbb{E}[M_n \mid \mathcal{F}_{n-1}] - \mathbb{E}[M_{n-1} \mid \mathcal{F}_{n-1}] \\ &= M_{n-1} - M_{n-1} \\ &= 0.\end{aligned}$$

Therefore, $(D_n)_{n\geq 1}$ is a martingale difference sequence.

Conversely, suppose $(D_n)_{n\geq 1}$ is a martingale difference sequence. Since,

$$M_n = M_{n-1} + D_n$$

we obtain,

$$\begin{aligned}\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] &= \mathbb{E}[M_{n-1} + D_n \mid \mathcal{F}_{n-1}] \\ &= \mathbb{E}[M_{n-1} \mid \mathcal{F}_{n-1}] + \mathbb{E}[D_n \mid \mathcal{F}_{n-1}] \\ &= M_{n-1} + 0 \\ &= M_{n-1}.\end{aligned}$$

Together with adaptedness and integrability, this proves that $(M_n)_{n\geq 0}$ is a martingale.

□

Consequently, a martingale can be written as its initial value plus its successive differences,

$$M_n = M_0 + \sum_{i=1}^{n} D_i.$$

Conversely, starting with an integrable, $\mathcal{F}_0$-measurable M_0 and a martingale difference sequence, this formula precisely constructs a martingale.

Remark 3.4 (Conditional Mean Zero does not imply independence). The condition $\mathbb{E}[D_n \mid \mathcal{F}_{n-1}] = 0$ does not state that D_n is independent of $\mathcal{F}_{n-1}$. Adaptedness and the inclusion $\mathcal{F}_{n-1} \subseteq \mathcal{F}_n$ do not imply independence either.

For example, suppose that $\xi_1, \xi_2, \ldots \overset{i.i.d.}{\sim}$ Rademacher with their natural filtration and define,

$$D_1 = \xi_1, \qquad D_n = (2 + \xi_{n-1})\xi_n, \quad n \geq 2.$$

The coefficient $2 + \xi_{n-1}$ is known at time $n - 1$ hence,

$$\mathbb{E}[D_n \mid \mathcal{F}_{n-1}] = (2 + \xi_{n-1})\mathbb{E}[\xi_n \mid \mathcal{F}_{n-1}] = 0.$$

Nevertheless,

$$|D_n| = 2 + \xi_{n-1} = \begin{cases} 3, & \xi_{n=1} = 1, \\ 1, & \xi_{n-1} = -1. \end{cases}$$

The past therefore determines the magnitude of the next increment, even though its conditional mean remains zero. A martingale difference may depend on past information through its distribution while still satisfying the required conditional mean-zero property.

Example 3.7 (Geometric Martingale)**.** Let $\xi_1, \xi_2, \ldots$ be Rademacher random variables.

Consider the filtration,

$$\mathcal{F}_0 = \{\emptyset, \Omega\}, \quad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n), \quad n \geq 1.$$

Define $M_0 = 1$ and,

$$M_n := \prod_{k=1}^{n} \left(1 + \frac{\xi_k}{2}\right).$$

For example,

$$M_1 = 1 + \frac{\xi_1}{2}, \quad M_2 = \left(1 + \frac{\xi_1}{2}\right)\left(1 + \frac{\xi_2}{2}\right).$$

That is, at each step, the current value is multiplied by either $\frac{3}{2}$ or $\frac{1}{2}$ each with probability $\frac{1}{2}$.

We will show that $(M_n)_{n\geq 0}$ is a martingale by verifying that its increments form a martingale difference sequence. First, M_n depends only on $\xi_1, \ldots, \xi_n$ so it is $\mathcal{F}_n$-measurable.

Moreover,

$$0 < M_n \leq \left(\frac{3}{2}\right)^n \iff \mathbb{E}[|M_n|] \leq \left(\frac{3}{2}\right)^n < \infty.$$

Thus, the process is adapted and integrable at every time. Separating the last factor in the product gives the product decomposition of the adapted process,

$$\begin{aligned} M_n &= \prod_{k=1}^{n} \left(1 + \frac{\xi_k}{2}\right) \\ &= \left[\prod_{k=1}^{n-1} \left(1 + \frac{\xi_k}{2}\right)\right] \left(1 + \frac{\xi_n}{2}\right) = M_{n-1}\left(1 + \frac{\xi_n}{2}\right). \end{aligned}$$

Therefore, the increment $D_n := M_n - M_{n-1}$ satisfies,

$$\begin{aligned} D_n &= M_n - M_{n-1} \\ &= M_{n-1}\left(1 + \frac{\xi_n}{2}\right) - M_{n-1} \\ &= M_{n-1}\left(1 + \frac{\xi_n}{2} - 1\right) \\ &= \frac{M_{n-1}}{2}\xi_n. \end{aligned}$$

Because M_{n-1} is $\mathcal{F}_{n-1}$-measurable and ξ_n is independent of $\mathcal{F}_{n-1}$,

$$\begin{aligned} \mathbb{E}[D_n \mid \mathcal{F}_{n-1}] &= \mathbb{E}\left[\frac{M_{n-1}}{2}\xi_n \middle| \mathcal{F}_{n-1}\right] \\ &= \frac{M_{n-1}}{2}\mathbb{E}[\xi_n \mid \mathcal{F}_n] \\ &= \frac{M_{n-1}}{2}\mathbb{E}[\xi_n] \\ &= \frac{M_{n-1}}{2}(0) \\ &= 0. \end{aligned}$$

Hence, $(D_n)_{n\geq 1}$ is a martingale difference sequence. By *Proposition 3.1*, $(M_n)_{n\geq 0}$ is a martingale. We can also verify the martingale identity directly,

$$\begin{aligned} \mathbb{E}[M_n \mid \mathcal{F}_{n-1}] &= \mathbb{E}\left[M_{n-1}\left(1 + \frac{\xi_n}{2}\right) \middle| \mathcal{F}_{n-1}\right] \\ &= M_{n-1}\mathbb{E}\left[1 + \frac{\xi_n}{2} \middle| \mathcal{F}_{n-1}\right] \\ &= M_{n-1}\left(1 + \frac{1}{2}\underbrace{\mathbb{E}[\xi_n \mid \mathcal{F}_{n-1}]}_{=\mathbb{E}[\xi_n]=0}\right) \\ &= M_{n-1}. \end{aligned}$$

3.4 Convex Transforms of Martingales

Convex functions provide a general way to turn martingales into submartingales. The key is Jensen's inequality, which compares applying a convex function before averaging with applying it after averaging. We state it in the proposition below.

Proposition 3.2 (Jensen's inequality)**.** *Let X be an integrable random variable and let $\phi : \mathbb{R} \to \mathbb{R}$ be convex. If $\phi(X)$ is integrable, then*

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)].$$

Proof. Set $m := \mathbb{E}[X]$. Since ϕ is convex, its graph has a supporting line at m such that there exists a scalar $a \in \mathbb{R}$ such that,

$$\phi(x) \geq \phi(m) + a(x - m), \quad x \in \mathbb{R}.$$

This line agrees with ϕ at m and lies below its graph everywhere.

Substituting X for x and taking expectations,

$$\begin{aligned}\mathbb{E}[\phi(X)] &\geq \mathbb{E}[\phi(m) + a(X - m)] \\ &= \phi(m) + a(\mathbb{E}[X] - m) \\ &= \phi(m) \\ &= \phi(\mathbb{E}[X]).\end{aligned}$$

This proves the inequality.

□

Observe that the inequality also holds with $\mathbb{E}[\phi(X)] = +\infty$. Indeed, the supporting-line bound ensures that the negative part of $\phi(X)$ is integrable, so its expectation is well-defined even when its positive part has infinite expectation.

Remark 3.5 (The Jensen Gap). When both expectations are finite, the quantity,

$$\mathbb{E}[\phi(X)] - \phi(\mathbb{E}[X]) \geq 0.$$

This is called the *Jensen's Gap*. It measures the difference between averaging the transformed values and transforming the averaging. In some settings, this quantity is called a ϕ-entropy, but is *not* the general definition of entropy. If ϕ is strictly convex then,

$$\mathbb{E}[\phi(X)] - \phi(\mathbb{E}[X]) = 0 \iff X = \mathbb{E}[X] \quad \text{almost surely.}$$

A supporting line at $m = \mathbb{E}[X]$ satisfies,

$$\phi(x) - \phi(m) - a(x - m) \geq 0$$

with equality only when $x = m$. Its expected value is precisely the Jensen gap. A nonnegative random variable has expectation zero only if it is zero almost surely, giving the stated equivalence.

Proposition 3.3 (Conditional Jensen's inequality)**.** *Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra and let $\phi : \mathbb{R} \to \mathbb{R}$ be convex. If X and $\phi(X)$ are integrable, then,*

$$\phi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\phi(X) \mid \mathcal{G}] \quad \textit{almost surely.}$$

Proof. We use the concept of supporting lines with determines coefficients so that conditional linearity applies directly. For each rational number q, choose a supporting line,

$$l_q(x) := \phi(q) + a_q(x - q), \quad l_q(x) \leq \phi(x) \quad x \in \mathbb{R}.$$

A finite convex function on $\mathbb{R}$ is continuous and these supporting lines recover its graph,

$$\phi(x) = \sup_{q \in \mathbb{Q}} l_q(x).$$

Indeed, each line lies below ϕ and supporting lines at rational points approaching x have values at x approaching $\phi(x)$. For each fixed q, conditional monotonicity and linearity give us,

$$\begin{aligned}\mathbb{E}[\phi(X) \mid \mathcal{G}] &\geq \mathbb{E}[l_q(X) \mid \mathcal{G}] \\ &= \phi(q) + a_q(\mathbb{E}[X \mid \mathcal{G}] - q) \\ &= l_q(\mathbb{E}[X \mid \mathcal{G}]) \quad \text{almost surely.}\end{aligned}$$

There are only countably many rational numbers so these inequalities hold simultaneously outside a single event of probability zero. Taking the supremum over $q \in \mathbb{Q}$,

$$\begin{aligned}\mathbb{E}[\phi(X) \mid \mathcal{G}] &\geq \sup_{q \in \mathbb{Q}} l_q(\mathbb{E}[X \mid \mathcal{G}]) \\ &= \phi(\mathbb{E}[X \mid \mathcal{G}]).\end{aligned}$$

This proves conditional Jensen's inequality.

□

Example 3.8 (Convex functions of Martingales)**.** Let $(M_n)_{n\geq 0}$ be a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$. Suppose $\phi : \mathbb{R} \to \mathbb{R}$ is convex and,

$$\mathbb{E}[|\phi(M_n)|] < \infty, \quad n \geq 0.$$

Then $(\phi(M_n))_{n\geq 0}$ is a submartingale.

By convexity, ϕ is continuous and hence Borel-measurable. Since M_n is $\mathcal{F}_n$-measurable, $\phi(M_n)$ is $\mathcal{F}_n$-measurable. Thus, the transformed process is adapted, and it is integrable by assumption.

Applying conditional Jensen's inequality,

$$\begin{aligned}\mathbb{E}[\phi(M_{n+1}) \mid \mathcal{F}_n] &\geq \phi(\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]) \\ &= \phi(M_n).\end{aligned}$$

The final equality uses the martingale property, precisely yielding the submartingale condition.

In particular,

(i) Taking $\phi(x) = |x|$ shows that $(|M_n|)_{n\geq 0}$ is a submartingale for every martingale, and
(ii) Taking $\phi(x) = x^2$ shows that $(M_n^2)_{n\geq 0}$ is a submartingale when each M_n is square integrable.

Let $(S_n)_{n\geq 0}$ be a symmetric SRW defined by $\xi_k \overset{i.i.d.}{\sim}$ Rademacher. We already know that $(S_n)_{n\geq 0}$ is a martingale. Since $|S_n| \leq n$, each S_n is square-integrable, so Jensen's inequality immediately implies that $(S_n^2)_{n\geq 0}$ is a submartingale.

Computing its conditional expectation change explicitly using $S_{n+1} = S_n + \xi_{n+1}$,

$$\begin{aligned} S_{n+1}^2 &= (S_n + \xi_{n+1})^2 \\ &= S_n^2 + 2S_n\xi_{n+1} + \xi_{n+1}^2. \end{aligned}$$

Because S_n is $\mathcal{F}_n$-measurable, ξ_{n+1} is independent of $\mathcal{F}_n$ and $\xi_{n+1}^2 = 1$ almost surely,

$$\begin{aligned} \mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] &= S_n^2 + 2S_n\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[\xi_{n+1}^2 \mid \mathcal{F}_n] \\ &= S_n^2 + 2S_n\mathbb{E}[\xi_{n+1}] + 1 \\ &= S_n^1 + 1 \\ &\geq S_n^2. \end{aligned}$$

Here, $\mathbb{E}[\xi_{n+1}^2 \mid \mathcal{F}_n] = 1$ follows directly from $\xi_{n+1}^2 = 1$ almost surely.

The martingale property fixed the conditional mean of the next value, but it does not prevent fluctuations around that mean. Convexity ensure that averaging these fluctuations cannot lower the transformed value below the transformation of the current conditional mean.

3.5 Stopping Times, First, and Last Hitting Times

A stopping rule determines when to stop observing or acting on a process. The time may depend on the observed path, but the decision to stop must use only information already available.

Hence, *Stopping times* formalize this requirement.

Definition 3.6 (Stopping time)**.** Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration. A random variable,

$$\tau : \Omega \longrightarrow \{0, 1, \dots, \} \cup \{+\infty\}$$

is a *stopping time* with respect to $(\mathcal{F}_n)_{n\geq 0}$ if,

$$\{\tau \leq n\} \in \mathcal{F}_n \quad \forall n \geq 0.$$

For each outcome ω, the value $\tau(\omega)$ specifies when the rule stops. For example, $\tau(\omega) = 5$ means we stop at time 5 while $\tau(\omega) = +\infty$ means that we never stop. Hence the event,

$$\{\tau \leq n\} = \{\omega \in \Omega : \tau(\omega) \leq n\}$$

contains the outcomes for which stopping has occurred by time n.

Requiring that this event to belong to $\mathcal{F}_n$ means that the information available at time n determines whether we have already stopped. The condition, however, does not require us to know the eventual stopping time in advance and requires only that we can recognize whether stopping has occurred as observations arrive.

Example 3.9 (The First Heads)**.** Let $\xi_1, \xi_2, \ldots \overset{i.i.d}{\sim}$ Rademacher represent coin flips, where $\xi_n = 1$ represents heads and $\xi_n = -1$ represents tails.

Let $\mathcal{F}_0$ represent the trivial σ-algebra and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$. Define,

$$\tau := \inf\{n \geq 1 : \xi_n = 1\}$$

with convention $\inf \emptyset = +\infty$. Thus, τ is the time of the first heads.

For example,

$$\{\tau \leq 3\} = \{\xi_1 = 1\} \cup \{\xi_2 = 1\} \cup \{\xi_3 = 1\}$$

asks whether at least one of the first three flips was heads, which can be determined from the information in $\mathcal{F}_3$. More generally, for $n \geq 1$,

$$\{\tau \leq n\} = \bigcup_{k=1}^{n} \{\xi_k = 1\}$$

where each event $\{\xi_k = 1\}$ belongs to $\mathcal{F}_k \subseteq \mathcal{F}_n$ so their finite unions belongs to $\mathcal{F}_n$.

Also, it follows that $\{\tau \leq 0\} = \emptyset \in \mathcal{F}_0$. Therefore, τ is a stopping time.

Remark 3.6 (Equivalent descriptions of a Stopping time)*.* In discrete time, the stopping-time condition is equivalent to writing,

$$\{\tau = n\} \in \mathcal{F}_n, \quad \forall n \geq 0.$$

Thus, the available information determines not only whether we have stopped by time n, but also whether we stop exactly at time n. To verify this equivalence, first suppose that τ is a stopping time. Then, for $n \geq 1$,

$$\{\tau = n\} = \{\tau \leq n\} \backslash \{\tau \leq n-1\}.$$

The first event belongs to $\mathcal{F}_n$ and the second belongs to $\mathcal{F}_{n-1} \subseteq \mathcal{F}_n$. Hence, $\{\tau = n\} \in \mathcal{F}_n$.

At $n = 0$,

$$\{\tau = 0\} = \{\tau \leq 0\} \in \mathcal{F}_0.$$

Conversely, suppose that $\{\tau = k\} \in \mathcal{F}_k$ for every $k \geq 0$. Then,

$$\{\tau \leq n\} = \bigcup_{k=0}^{n} \{\tau = k\} \in \mathcal{F}_n$$

because $\mathcal{F}_k \subseteq \mathcal{F}_n$ whenever $k \leq n$. Thus, τ is a stopping time. Similarly,

$$\{\tau \geq n\} \in \mathcal{F}_{n-1}$$

Indeed, $\{\tau \geq n\} = \{\tau \leq n-1\}^c$ meaning we have not stopped before time n. Whether this is true is already known at time $n-1$. In particular, the process,

$$H_n := \mathbb{I}_{\{\tau \geq n\}}, \quad n \geq 1$$

is predictable. It records precisely whether the stop from $n-1$ to time n is taken before or at the stopping time. This observation will be useful when constructing stopped processes.

An important stopping rule is to stop the first time an observed process enters a specific set.

Theorem 3.1 (First Hitting Times are Stopping Times)**.** *Let $(X_n)_{n\geq 0}$ be a real-valued process adapted to $(\mathcal{F}_n)_{n\geq 0}$ and let $B \in \mathcal{B}(\mathbb{R})$. Define the first hitting time of B by,*

$$\tau_B := \inf\{n \geq 0 : X_n \in B\}$$

with $\inf \emptyset = +\infty$. Then τ_B is a stopping time.

The value of τ_B records the first entry into B. For a given outcome ω,

(i) If $X_0(\omega) \in B$, then $\tau(\omega) = 0$
(ii) If $X_0(\omega), X_1(\omega) \notin B$ but $X_2(\omega) \in B$, then $\tau_B(\omega) = 2$
(iii) If $X_n(\omega) \notin B$ for every $n \geq 0$, then $\tau_B(\omega) = +\infty$.

Proof. We show that $\{\tau_B = n\} \in \mathcal{F}_n$ for every $n \geq 0$.

For $n \geq 1$, the first entry occurs at time n precisely when the process is in B at time n and was outside B at every earlier time. Therefore, we can define this event,

$$\{\tau_B = n\} = \{X_n \in B\} \cap \bigcap_{k=0}^{n-1} \{X_k \notin B\}.$$

Since the process is adapted, X_k is $\mathcal{F}_k$-measurable. Because B is Borel, $\{X_k \in B\} \in \mathcal{F}_k$. By closure under complements, we also see that $\{X_k \notin B\} \in \mathcal{F}_k$.

Hence, for every $k \leq n$, $\mathcal{F}_k \subseteq \mathcal{F}_n$. Thus, every event appearing in the intersection belongs to $\mathcal{F}_n$. Since a σ-algebra is closed under finite intersections,

$$\{\tau_B = n\} \in \mathcal{F}_n.$$

At time $n = 0$, $\{\tau_B = 0\} = \{X_0 \in B\} \in \mathcal{F}_0$. By equivalent characterization in *Remark 3.6*, τ_B is therefore a stopping time, as required.

□

Alternatively, we can see the stopping-time directly from,

$$\{\tau \leq n\} = \bigcup_{k=0}^{n} \{X_k \in B\} \in \mathcal{F}_n.$$

To determine whether the process has entered B by time n, we only need its observed values $X_0, \ldots, X_n$. No future observations, hence, are required.

A first hitting time can be recognized as soon as the process reaches the specified set. A last visit may require information about what happens afterward. The following example makes this distinction clear.

Theorem 3.2 (A Last Visit need not be a Stopping Time)**.** *Let $\xi_1, \xi_2, \ldots$ be Rademacher random variables and define,*

$$\mathcal{F}_0 = \{, \Omega\}, \quad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n), \quad n \geq 1.$$

Consider the symmetric SRW started at 1 and absorbed at 0,

$$S_0 = 1, \quad S_n = \begin{cases} S_{n-1} + \xi_n, & S_{n-1} > 0, \\ 0, & S_{n-1} = 0. \end{cases}$$

Define $\tau := \inf\{n \geq 1 : S_n = 0\}$ and let,

$$\sigma := \begin{cases} \tau - 1, & \tau < \infty, \\ +\infty, & \tau = \infty. \end{cases}$$

Then τ is a stopping time, but σ is not.

While the process is positive, it moves up or down by one unit. Once it reaches zero, it stays there. Interpreting S_n as wealth, τ is the bankruptcy time. When bankruptcy occurs, σ is the last time the process was strictly positive.

Proof. The process $(S_n)_{n\geq 0}$ is adapted because S_n is determined by $\xi_1, \ldots, \xi_n$. Since τ is the first hitting time of the Borel set $\{0\}$, *Theorem 3.1* implies that τ is a stopping time.

To show that σ is not a stopping time, it suffices to find a time n for which,

$$\{\sigma = n\} \notin \mathcal{F}_n.$$

Consider $n = 0$. Time 0 is the last positive time exactly when the process reaches zero on its first step. Thus,

$$\begin{aligned} \{\sigma = 0\} &= \{\tau = 1\} \\ &= \{S_1 = 0\} = \{1 + \xi_1 = 0\} = \{\xi_1 = -1\}. \end{aligned}$$

Because the first increment is symmetric, $\mathbb{P}(\sigma = 0) = \mathbb{P}(\xi_1 = -1) = \dfrac{1}{2}$.

However, the trivial σ-algebra $\mathcal{F}_0 = \{\emptyset, \Omega\}$ contains only events with probability zero or one, not $\frac{1}{2}$. Therefore, $\{\sigma = 0\} \notin \mathcal{F}_0$. Hence, σ is not a stopping time.

□

Remark 3.7 (Recognizing a Last Visit requires future information)*.* At time 0, we know that $S_0 = 1$ but we do not yet know whether this is the final positive value before absorption. That precisely depends on the next increment,

$$\{\sigma = 0\} = \{\xi_1 = -1\} \in \mathcal{F}_1, \quad \{\sigma = 0\} \notin \mathcal{F}_0.$$

We can recognize bankruptcy when it occurs but we cannot generally recognize the time immediately before bankruptcy using only the information available at that earlier time.

This example demonstrates that subtracting one from a stopping time need not produce another stopping time. The conclusion depends on the filtration. Here, the first increment is not available at time 0. Although moving a stopping time backward can destroy the stopping-time property, taking the earlier or later of two stopping times preserves it.

For random times σ, τ taking values in $\{0, 1, \dots\} \cup \{+\infty\}$ define,

$$\sigma \wedge \tau := \min\{\sigma, \tau\}, \quad \sigma \vee \tau := \max\{\sigma, \tau\}.$$

These operations are applied separately to each outcome,

$$(\sigma \wedge \tau)(\omega) = \min\{\sigma(\omega), \tau(\omega)\},$$
$$(\sigma \vee \tau)(\omega) = \max\{\sigma(\omega), \tau(\omega)\}.$$

Thus, $\sigma \wedge \tau$ records the earlier time, while $\sigma \vee \tau$ records the later time. For each fixed time n,

$$\{\sigma \wedge \tau \leq n\} = \{\sigma \leq n\} \cup \{\tau \leq n\},$$
$$\{\sigma \vee \tau \leq n\} = \{\sigma \leq n\} \cap \{\tau \leq n\}.$$

The earlier time has occurred by n if at least one of the two times has occurred. The later time has occurred by n if both have occurred. The events in the union need not be disjoint.

Proposition 3.4 (Minimum and Maximum of Stopping Times)**.** *Let σ and τ be stopping times with respect to the same filtration $(\mathcal{F}_n)_{n\geq 0}$. Then both,*

$$\sigma \wedge \tau, \quad \sigma \vee \tau$$

are stopping times.

Proof. Fix $n \geq 0$. Since σ and τ are stopping times,

$$\{\sigma \leq n\} \in \mathcal{F}_n, \quad \{\tau \leq n\} \in \mathcal{F}_n.$$

For the minimum,

$$\{\sigma \wedge \tau \leq n\} = \{\sigma \leq n\} \cup \{\tau \leq n\}.$$

Because $\mathcal{F}_n$ is closed under unions, this event belongs to $\mathcal{F}_n$. Therefore, $\sigma \wedge \tau$ is a stopping time. For the maximum,

$$\{\sigma \vee \tau \leq n\} = \{\sigma \leq n\} \cap \{\tau \leq n\}.$$

Because $\mathcal{F}_n$ is closed under finite intersections, this event also belongs to $\mathcal{F}_n$. Therefore, $\sigma \vee \tau$ is a stopping time. Finally, a deterministic time N is itself a stopping time, since

$$\{N \leq n\} = \begin{cases} \emptyset, & n < N, \\ \Omega, & n \geq N. \end{cases}$$

Both events belong to $\mathcal{F}_n$. Applying the preceding result with $\tau = N$ proves the final claim.

□

3.6 Stopped Processes & Stopped Martingales

A stopping time specifies when to stop. The corresponding stopped process follows the original process up to that time and then holds its value fixed.

Definition 3.7 (Stopped Process)**.** Let $(X_n)_{n\geq 0}$ be adapted to $(F_n)_{n\geq 0}$ and let τ be a stopping time with respect to the same filtration.

The *process stopped* at τ is $(X_{n\wedge\tau})_{n\geq 0}$ where, for each $\omega \in \Omega$,

$$X_{n\wedge\tau}(\omega) := X_{\min\{n\wedge\tau(\omega)\}}(\omega).$$

Equivalently, the stopped process can be written as,

$$X_{n\wedge\tau}(\omega) := \begin{cases} X_n(\omega), & n < \tau(\omega), \\ X_{\tau(\omega)}(\omega), & n \geq \tau(\omega). \end{cases}$$

If $\tau(\omega) + \infty$ then $n \wedge \tau(\omega) = n$ for every finite n, so the process never stops on that outcome. No value X_∞ is needed for this definition to be complete.

Remark 3.8 (Following the process until stopping). Before τ, the stopped process agree with the original process. At time τ, it takes value X_τ and it remains at that value afterward. In particular, the movement into the stopping time is included.

The stopped process is also adapted. Indeed for each $n \geq 0$,

$$X_{n\wedge\tau} = \sum_{k=0}^{n} X_k \mathbb{I}_{\{\tau=k\}} + X_n \mathbb{I}_{\{\tau>n\}}.$$

For $k \leq n$, both X_k and $\mathbb{I}_{\{\tau=k\}}$ are $\mathcal{F}_n$-measurable because they are $\mathcal{F}_k$-measurable and $\mathcal{F}_k \subseteq \mathcal{F}_n$. Also $\{\tau > n\} \in \mathcal{F}_n$. Thus, $X_{n\wedge\tau}$ is $\mathcal{F}_n$-measurable.

Example 3.10 (Stopping at a Boundary with Bounded increments)**.** Let $C_1 > 0$ and $C_2 \geq 0$ be constants. Suppose that $|X_0| \leq C_1$ almost surely and the increments satisfy,

$$|X_{n+1} - X_n| \leq C_2 \quad \text{almost surely } \forall n \geq 0.$$

Define,

$$\tau := \inf\{n \geq 0 : |X_n| \geq C_1\}, \quad \inf \emptyset = +\infty.$$

Since this is the first hitting time of the Borel set $(-\infty, -C] \cup [C_1, \infty)$, τ is hence a stopping time. Then the stopped process satisfies,

$$|X_{n\wedge\tau}| \leq C_1 + C_2 \quad \forall n \geq 0, \quad \text{almost surely.}$$

Proof. Since there are only countably many increment bounds, we may work on an event of probability one where the initial bound and all the increment bounds hold simultaneously. Fix an outcome in this event and a time $n \geq 0$. If $n < \tau$, then stopping has not yet occurred.

By definition of τ,

$$|X_n| < C_1 \implies |X_{n\wedge\tau}| = |X_n| < C_1 \leq C_1 + C_2.$$

Note this case also covers any finite n on a path where $\tau = +\infty$. If $n \geq \tau$ and $\tau \geq 1$, then the stopped process equals X_τ. Because τ is the first time the threshold is reached, $|X_{\tau-1}| < C_1$.

By triangle inequality and increment bound,

$$\begin{aligned} |X_{n\wedge\tau}| &= |X_\tau| \\ &= |X_{\tau-1} + (X_\tau - X_{\tau-1})| \\ &\leq |X_{\tau-1}| + |X_\tau - X_{\tau-1}| \\ &< C_1 + C_2. \end{aligned}$$

Finally, if $\tau = 0$, the process is stopped immediately so,

$$|X_{n\wedge\tau}| = |X_0| \leq C_1 \leq C_1 + C_2.$$

Combining the cases proves the bound.

□

The initial bound matters because the process could already be outside the interval at time 0. Without a restriction on X_0, stopping immediately would leave the potentially large initial value unchanged. Before a later exit, the process has magnitude less than C_1, and the final step adds at most C_2 which is why we precisely include both constants in the bound.

Stopping a martingale preserves its martingale property. That is, before stopping, its increments have conditional mean zero and afterward, its increment are zero. The stopping-time condition ensures that we know whether to include the next increment using only the information currently available. We formalize this claim with the following proposition.

Proposition 3.5 (A Stopped Martingale is a Martingale)**.** *Let $(M_n)_{n\geq 0}$ be a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$ and let τ be a stopping time. Then the stopped process,*

$$(M_{n\wedge\tau})_{n\geq 0}$$

is a martingale with respect to the same filtration.

Likewise, stopping a submartingale produces a submartingale and stopping a supermartingale produces a supermartingale. We verify *Proposition 3.5* in the proof below.

Proof. We verify adaptedness, integrability, and the conditional expectation identity.

For each fixed $n \geq 0$, we write,

$$M_{n\wedge\tau} = \sum_{k=0}^{n} M_k \mathbb{I}_{\{\tau=k\}} + M_n \mathbb{I}_{\{\tau>n\}}.$$

The sum accounts for paths that have now stopped by time n while the final term accounts for paths that have not. For each $k \leq n$, the stopping-time property gives us,

$$\{\tau = k\} \in \mathcal{F}_k \subseteq \mathcal{F}_n.$$

Also, M_k is $\mathcal{F}_k$-measurable and therefore $\mathcal{F}_n$-measurable. Thus, each product term $M_k \mathbb{I}_{\{\tau=k\}}$ is $\mathcal{F}_n$-measurable. Finally, we write the event,

$$\{\tau > n\} = \{\tau \leq n\}^c \in \mathcal{F}_n$$

so the remaining term is also $\mathcal{F}_n$-measurable. Consequently, $M_{n\wedge\tau}$ is $\mathcal{F}_n$-measurable.

At each outcome, $M_{n\wedge\tau}$ equals one of $M_0, \ldots, M_n$. Hence,

$$|M_{n\wedge\tau}| \leq \sum_{k=0}^{n} |M_k| \iff \mathbb{E}[|M_{n\wedge\tau}|] \leq \sum_{k=0}^{n} \mathbb{E}[|M_k|] < \infty.$$

The final sum is finite because each M_k is integrable and there are only finitely many terms.

Finally, suppose we fix $n \geq 0$. Then, there are two possibilities,

(i) If $\tau \leq n$, the process has already stopped so,

$$M_{(n+1)\wedge\tau} - M_{n\wedge\tau} = M_\tau - M_\tau = 0.$$

(ii) If $\tau \geq n+1$, the process takes the next step, including the step at which stopping occurs if $\tau = n+1$. Thus, we may write the increment as

$$M_{(n+1)\wedge\tau} - M_{n\wedge\tau} = M_{n+1} - M_n.$$

Combining these cases, we may write the stopped process increments as,

$$M_{(n+1)\wedge\tau} - M_{n\wedge\tau} = \mathbb{I}_{\{\tau\geq n+1\}}(M_{n+1} - M_n).$$

Because τ is a stopping time, $\{\tau \geq n+1\} = \{\tau \leq n\}^c \in \mathcal{F}_n$. Its indicator is therefore $\mathcal{F}_n$-measurable and can be taken outside the conditional expectation. Hence,

$$\begin{aligned}
\mathbb{E}[M_{(n+1)\wedge\tau} - M_n \mid \mathcal{F}_n] &= \mathbb{E}\left[\mathbb{I}_{\{\tau\geq n+1\}}(M_{n+1} - M_n) \,\middle|\middle|\, \mathcal{F}_n\right] \\
&= \mathbb{I}_{\{\tau\geq n+1\}}\mathbb{E}[M_{n+1} - M_n \mid \mathcal{F}_n] \\
&= \mathbb{I}_{\{\tau\geq n+1\}}\left(\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] - M_n\right) \\
&= \mathbb{I}_{\{\tau\geq n+1\}}\left(M_n - M_n\right) \\
&= 0.
\end{aligned}$$

Since $M_{n\wedge\tau}$ is $\mathcal{F}_n$-measurable this implies,

$$0 = \mathbb{E}[M_{(n+1)\wedge\tau} \mid \mathcal{F}_n] - M_{n\wedge\tau}.$$

Therefore, $\mathbb{E}[M_{(n+1)\wedge\tau} \mid \mathcal{F}_n] = M_{n\wedge\tau}$. Hence, with adaptedness and integrability, the stopped process satisfies the martingale conditional expectation identity, hence it is a martingale.

For a submartingale, the same argument gives,

$$\mathbb{E}[M_{(n+1)\wedge\tau} - M_{n\wedge\tau} \mid \mathcal{F}_n] = \mathbb{I}_{\{\tau\geq n+1\}} (\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] - M_n) \geq 0$$

because both factors on the right are nonnegative. Hence, the stopped process is a submartingale.

For a supermartingale, the conditional expectation increment is nonpositive, so the same expression is at most zero. Thus, the stopped process is a supermartingale.

□

The increment identity also yields the useful representation,

$$M_{n\wedge\tau} = M_0 + \sum_{k=1}^{n} \mathbb{I}_{\{\tau\geq k\}}(M_k - M_{k-1})$$

where the indicator $\mathbb{I}_{\{\tau\geq k\}}$ is $\mathcal{F}_{k-1}$-measurable. Stopping therefore amounts to a predictable rule that keeps each increment up to and including time τ, then discards all subsequent increments.

For every fixed finite n, the proposition implies,

$$\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0].$$

This does not, by itself, imply $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ for an unbounded stopping time. Even when $\tau < \infty$ almost surely, passing the limit as $n \to \infty$ requires additional justification.

The stopped martingale property gives an expectation identity at every fixed finite time. To obtain an identity at the stopping time itself, we must justify passing the limit.

Example 3.11 (Random Walk Stopped at Two Barriers)**.** Let $(S_n)_{n\geq 0}$ be a symmetric SRW with $S_0 = 0$ and,

$$S_n = \sum_{k=1}^{n} \xi_k.$$

Let $\mathcal{F}_0$ be the trivial σ-algebra and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n) = \sigma(S_0, \ldots, S_n)$.

For positive integers define the stopping time,

$$\tau := \inf\{n \geq 0 : S_n \in \{-a, b\}\}.$$

Thus, τ is the first time the walk reaches either the lower barrier $-a$ or the upper barrier b. Since τ is the first hitting time of an adapted process, it is a stopping time. The SRW $(S_n)_{n\geq 0}$ is a martingale, so by *Proposition 3.5* $(S_{n\wedge\tau})_{n\geq 0}$ is also a martingale.

Consequently, for every $n \geq 1$, and repeating the identity

$$\begin{aligned}
\mathbb{E}[S_{n\wedge\tau}] &= \mathbb{E}[\mathbb{E}[S_{n\wedge\tau} \mid \mathcal{F}_{n-1}]] \\
&= \mathbb{E}[S_{(n-1)\wedge\tau}] \\
&= \cdots \\
&= \mathbb{E}[S_{0\wedge\tau}] \\
&= \mathbb{E}[S_0] = 0.
\end{aligned}$$

Hence, $\mathbb{E}[S_{n\wedge\tau}] = 0$ for every finite $n \geq 0$. There is a useful identity involving the squares position. Recall that $(S_n^2 - n)_{n\geq 0}$ is a martingale. Indeed,

$$\begin{aligned}\mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] &= \mathbb{E}[(S_n + \xi_{n+1})^2 \mid \mathcal{F}_n] - (n+1) \\ &= S_n^2 + 2S_n\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[\xi_{n+1}^2 \mid \mathcal{F}_n] - (n+1) \\ &= S_n^2 + 1 - (n+1) \\ &= S_n^2 - n.\end{aligned}$$

Stopping this martingale at τ produces $(S_{n\wedge\tau}^2 - (n \wedge \tau))_{n\geq 0}$. Its initial value is zero,

$$0 = \mathbb{E}[S_{n\wedge\tau}^2 - (n \wedge \tau)] = \mathbb{E}[S_{n\wedge\tau}^2] - \mathbb{E}[n \wedge \tau].$$

Therefore, $\mathbb{E}[S_{n\wedge\tau}] = 0$ and $\mathbb{E}[S_{n\wedge\tau}^2] = \mathbb{E}[n \wedge \tau]$ for $n \geq 0$.

These identities hold without assuming that τ is bounded. For each fixed n, the time $n \wedge \tau$ is automatically bounded by n. The remaining question is whether we may replace $n \wedge \tau$ by τ.

Remark 3.9 (Passing from $n \wedge \tau$ to τ). For a general martingale $(M_n)_{n\geq 0}$ the stopped-martingale property gives $\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0]$ for every finite n. To conclude that $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$, we must address both the existence of the terminal value and convergence of expectations.

If $\tau < \infty$ almost surely, then M_τ is defined on the event $\{\tau < \infty\}$. That is, it may be assigned any fixed value on the null event $\{\tau = \infty\}$. For almost every ω, once $n \geq \tau(\omega)$,

$$M_{n\wedge\tau}(\omega) = M_{\tau(\omega)}(\omega).$$

Hence $M_{n\wedge\tau} \to M_\tau$ almost surely. However, almost-sure convergence does not alone imply convergence of expectations. We still need a justification for,

$$\mathbb{E}\left[\lim_{n\to\infty} M_{n\wedge\tau}\right] = \lim_{n\to\infty} \mathbb{E}[M_{n\wedge\tau}].$$

An integrable bound on the stopped process is one condition that makes this interchange valid. In the two-barrier example, such a bound is available. Because the walk moves in steps of size one and both barriers are integers, it cannot cross either barrier without hitting it.

Thus, $-a \leq S_{n\wedge\tau} \leq b$ for $n \geq 0$ and therefore $|S_{n\wedge\tau}| \leq \max\{a, b\}$. We can also use the squared-martingale identity to establish that stopping occurs almost surely,

$$\mathbb{E}[n \wedge \tau] = \mathbb{E}[S_{n\wedge\tau}^2] \leq \max\{a, b\}^2.$$

Since $n \wedge \tau \to \tau$, monotone convergence gives,

$$\mathbb{E}[\tau] = \lim_{n\to\infty} \mathbb{E}[n \wedge \tau] \leq \max\{a, b\}^2 < \infty.$$

In particular, $\tau < \infty$ almost surely. We now have both almost-sure convergence and a deterministic bound, $S_{n\wedge\tau} \to S_\tau$ almost surely and $|S_{n\wedge\tau}| \leq \max\{a, b\}$. Hence,

$$\mathbb{E}[S_\tau] = \lim_{n\to\infty} \mathbb{E}[S_{n\wedge\tau}] = 0.$$

Similarly, applying dominated convergence to the squares gives,

$$\mathbb{E}[S_\tau^2] = \lim_{n\to\infty} \mathbb{E}[S_{n\wedge\tau}^2] = \mathbb{E}[\tau].$$

Thus the expectation identities do extend to the actual hitting time in this extend.

The justification comes from the bounded stopping positions and the almost-sure finiteness of τ, rather than from the martingale property alone.

To make the distinction explicit, $\tau < \infty$ almost surely means that each path, except on a null event, eventually stops. It does not mean that all paths stop before some common deterministic time. At any finite time n, there may still be paths on which $\tau > n$.

For a general martingale, the values on these increasingly rare events may be sufficiently large to prevent convergence of expectations. The concern is therefore not the null event $\{\tau = \infty\}$, but the contribution from paths that have not yet stopped.

In this example, however, both $S_{n\wedge\tau}$ and S_τ lie in $[-a, b]$ almost surely, and they agree on $\{\tau \le n\}$. Consequently,

$$\mathbb{E}[|S_{n\wedge\tau} - S_\tau|] \le (a+b)\mathbb{P}(\tau > n) \longrightarrow 0.$$

Thus, almost-sure finiteness gives convergence of the stopped values, while the bound ensures convergence of their expectations.

3.7 Optional Stopping Theorem & Failure Cases

The Stopped-martingale property tells us that,

$$\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0]$$

at every fixed finite time n. The *Optional Stopping Theorem* gives conditions under which we can replace $n \wedge \tau$ by the stopping time τ itself.

Theorem 3.3 (Optional Stopping Theorem)**.** *Let $(M_n)_{n\ge0}$ be a martingale with respect to $(\mathcal{F}_n)_{n\ge0}$ and let τ be a stopping time satisfying,*

$$\mathbb{P}(\tau < \infty) = 1.$$

Suppose at least one of the following conditions holds,

(i) There exists a deterministic integer $N < \infty$ such that,

$$\tau \le N \quad \textit{almost surely.}$$

(ii) There exists a deterministic constant $C < \infty$ such that,

$$|M_{n\wedge\tau}| \le \quad \textit{almost surely for every } n \ge 0.$$

Then M_τ is integrable and,

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

Under the same hypotheses, if $(M_n)_{n\geq 0}$ is a submartingale then $\mathbb{E}[M_\tau] \geq \mathbb{E}[M_0]$ and if it is a supermartingale then $\mathbb{E}[M_\tau] \leq \mathbb{E}[M_0]$. We distinguish the two types of bounds.

Remark 3.10 (Two Different Types of Bounds). Condition **(i)** bounds the stopping time. Outside a null event, every path stops by the same fixed time N and values of the process need *not* be bounded under this condition. In condition **(ii)**, we bound the stopped values meaning the stopping time may take arbitrarily large finite values, but the stopped process always remains fixed in the interval $[-C, C]$.

In both cases, $\tau < \infty$ almost surely ensuring that a terminal value M_τ exists almost surely. We may assign it any fixed value, such as zero on the null event $\{\tau = \infty\}$.

Proof. By *Proposition 3.5*, since $(M_n)_{n\geq 0}$ is a martingale, the stopped process $(M_{n\wedge\tau})_{n\geq 0}$ is a martingale as well. In particular, its values are integrable and, for every $n \geq 0$,

$$\mathbb{E}[M_{(n+1)\wedge\tau} \mid \mathcal{F}_n] = M_{n\wedge\tau} \quad \text{almost surely.}$$

We prove the result under each condition separately.

Stopping Time is Bounded. Assume there exists a deterministic integer $N < \infty$ such that,

$$\tau \leq N \quad \text{almost surely.}$$

Then, for almost every ω,

$$N \wedge \tau(\omega) = \min\{N, \tau(\omega)\} = \tau(\omega).$$

Hence, $M_{N\wedge\tau} = M_\tau$ almost surely, so $\mathbb{E}[M_\tau] = \mathbb{E}[M_{N\wedge\tau}]$. Observe this equality need not hold with N replaced by a smaller $n < N$. Some paths may not yet have stopped by time n. All that matter is that the same deterministic N is large enough for almost every path.

First, M_τ is integrable as, on $\{\tau \leq N\}$, it equals one of $M_0, \ldots, M_N$,

$$\mathbb{E}[|M_\tau|] \leq \sum_{k=0}^{N} \mathbb{E}[|M_k|] < \infty.$$

N is finite hence the last sum is finite.

Now, for the martingale conditional expectation identity, for every $k \geq 1$, the martingale property of the stopped process $(M_{n\wedge\tau})_{n\geq 0}$ shows that,

$$\mathbb{E}[M_{k\wedge\tau} \mid \mathcal{F}_{k-1}] = M_{(k-1)\wedge\tau}.$$

Taking expectations and applying *Law of Total Expectation*,

$$\begin{aligned}\mathbb{E}[M_{k\wedge\tau}] &= \mathbb{E}[\mathbb{E}[M_{k\wedge\tau} \mid \mathcal{F}_{k-1}]] \\ &= \mathbb{E}[M_{(k-1)\wedge\tau}].\end{aligned}$$

Each stopped time step equals the expected value at the preceding stopped time step.

If $N \geq 1$, applying this identity successively for $k = N, N-1, \ldots, 1$,

$$\begin{aligned}
\mathbb{E}[M_\tau] &= \mathbb{E}[M_{N\wedge\tau}] \\
&= \mathbb{E}[\mathbb{E}[M_{N\wedge\tau} \mid \mathcal{F}_{N-1}]] \\
&= \mathbb{E}[M_{(N-1)\wedge\tau}] \\
&= \cdots \\
&= \mathbb{E}[M_{1\wedge\tau}] \\
&= \mathbb{E}[\mathbb{E}[M_{1\wedge\tau} \mid \mathcal{F}_0]] \\
&= \mathbb{E}[M_{0\wedge\tau}] \\
&= \mathbb{E}[M_0].
\end{aligned}$$

$\tau \geq 0$ so $0 \wedge \tau = 0$. Hence,

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

Therefore, at a fixed time N, the stopped process equals its terminal value almost surely.

Stopped Process is Uniformly Bounded. Now assume that, for some deterministic $C < \infty$,

$$|M_{n\wedge\tau}| \leq C \quad \text{almost surely } \forall n \geq 0.$$

We do not assume that τ is bounded by a deterministic time. Different paths may therefore stop at arbitrarily large times. Because there are only countably many integer n, there is a single event of probability one on which all the bounds,

$$|M_{n\wedge\tau}| \leq C, \quad n \geq 0$$

hold simultaneously. Intersect this event with $\{\tau < \infty\}$ which also has probability one.

Fix $\omega \in \Omega$ in this intersection. $\tau(\omega)$ is a finite integer hence, there eventually will be a point where $n \geq \tau(\omega)$. From that point onward, $n \wedge \tau(\omega) = \tau(\omega)$ hence,

$$M_{n\wedge\tau}(\omega) = M_{\tau(\omega)}(\omega) = M_\tau(\omega).$$

Thus, along almost every path, the stopped process eventually becomes exactly equal to its terminal value. In particular, we write that $M_{n\wedge\tau} \to M_\tau$ almost surely.

Choosing any $n \geq \tau(\omega)$ we see that M_τ is hence bounded,

$$|M_\tau(\omega)| = |M_{n\wedge\tau}(\omega)| \leq C.$$

Consequently, $|M_\tau| \leq C$ almost surely, with $\mathbb{E}[|M_\tau|] \leq C < \infty$.

Now we justify convergence of expectations. Fix an finite $n \geq 0$.

On the event $\{\tau \leq n\}$, stopping has already occurred so,

$$M_\tau = M_{n\wedge\tau}.$$

Hence, the difference $M_\tau - M_{n\wedge\tau}$ is nonzero only on paths where $M_{n\wedge\tau} \neq M_\tau$ which occurs on $\{\tau > n\}$. Hence, the difference can be written as an indicator,

$$M_\tau - M_{n\wedge\tau} = (M_\tau - M_{n\wedge\tau})\mathbb{I}_{\{\tau>n\}} \quad \text{almost surely.}$$

Taking expectations inside absolute values,

$$\begin{aligned}
|\mathbb{E}[M_\tau] - \mathbb{E}[M_{n\wedge\tau}]| &= |\mathbb{E}[M_\tau - M_{n\wedge\tau}]| \\
&= \left|\mathbb{E}\left[(M_\tau - M_{n\wedge\tau})\,\mathbb{I}_{\{\tau>n\}}\right]\right| \\
&\leq \mathbb{E}\left[|M_\tau - M_{n\wedge\tau}|\mathbb{I}_{\{\tau>n\}}\right], \qquad |\mathbb{E}[X]| \leq \mathbb{E}[|X|] \\
&\leq \mathbb{E}\left[|M_\tau| + |M_{n\wedge\tau}|\mathbb{I}_{\{\tau>n\}}\right], \qquad |X-Y| \leq |X| + |Y| \\
&\leq \mathbb{E}\left[2C\mathbb{I}_{\{\tau>n\}}\right], \qquad |M_\tau| = |M_{n\wedge\tau}| \leq C \\
&= 2C\mathbb{E}\left[\mathbb{I}_{\{\tau>n\}}\right] \\
&= 2C\mathbb{P}(\tau > n), \qquad \text{Expectation of Indicator.}
\end{aligned}$$

For $n \geq 0$, the events $\{\tau > n\}$ form a decreasing sequence,

$$\{\tau > 0\} \subseteq \{\tau > 1\} \subseteq \{\tau > 2\} \subseteq \cdots.$$

Their intersection is,

$$\bigcap_{n=0}^{\infty} \{\tau > n\} = \{\tau = \infty\}.$$

Indeed, a stopping time exceeding every finite integer is precisely when it is infinite. Hence,

$$\lim_{n\to\infty} \mathbb{P}(\tau > n) = \mathbb{P}(\tau = \infty) = 0.$$

Hence we have the bounds,

$$0 \leq |\mathbb{E}[M_\tau] - \mathbb{E}[M_{n\wedge\tau}]| \leq 2C\mathbb{P}(\tau > n) \to 0.$$

Therefore, $M_{n\wedge\tau} \to \mathbb{E}[M_\tau]$. Meanwhile, the stopped process is a martingale, so the repeated-expectation from (i) tells us that $\mathbb{E}[M_{n\wedge\tau}] = \mathbb{E}[M_0]$ for every finite n. Taking the limit,

$$\begin{aligned}
\mathbb{E}[M_\tau] &= \lim_{n\to\infty} \mathbb{E}[M_{n\wedge\tau}] = \lim_{n\to\infty} \mathbb{E}[M_0] \\
&= \mathbb{E}[M_0].
\end{aligned}$$

□

Example 3.12. Let $\xi_1, \xi_2, \ldots$ be independent symmetric Rademacher random variables and define the symmetric SRW with $S_0 = 0$ and $S_n = \sum_{k=1}^n \xi_k$ with natural filtration,

$$\mathcal{F}_n = \sigma(S_0, \ldots, S_n) = \sigma(\xi_1, \ldots, \xi_k), \quad n \geq 1$$

and $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Let a, b be positive integers and define,

$$\tau := \inf\{n \geq 0 : S_n \in \{-a, b\}\}.$$

This is the first time the walk reaches either lower boundary $-a$ or the upper boundary b. Since it is a first hitting time of an adapted process, τ is a stopping time.

We wish to compute the probability of reaching b before $-a$,

$$\mathbb{P}(S_\tau = b).$$

We apply the *Optional Stopping Theorem* to $(S_n)_{n\geq 0}$ so that we can justify,

$$\mathbb{E}[S_\tau] = \mathbb{E}[S_0] = 0.$$

Hence, we must first verify that the stopped process is uniformly bounded and that $\tau < \infty$ almost surely. Then, we can determine the respective probabilities of the two terminal values, $-a$ and b. Before time τ, the walk has not reached either boundary.

Since the walk moves in steps of size one and the boundaries are integers, it cannot pass either boundary without first hitting it. Hence, on $\{\tau > n\}$, $-a < S_n < b$.

If $\tau \leq n$ then,

$$S_{n\wedge\tau} = S_\tau \in \{-a, b\}.$$

Combining the cases when $\{\tau > n\}$ and $\{\tau \leq n\}$,

$$-a \leq S_{n\wedge\tau} \leq b, \quad n \geq 0.$$

In particular, $S_{n\wedge\tau}$ is uniformly bounded by,

$$|S_{n\wedge\tau}| \leq \max\{a, b\} \quad \forall n \geq 0.$$

Thus, the bounded stopped process condition in *Optional Stopping Theorem* is satisfied.

We now show that $\mathbb{P}(\tau < \infty)$, that is the walk reaches a boundary almost surely. Hence, we show that the probability of remaining between the boundaries decreases geometrically when time is divided into blocks of $a + b$-steps.

Suppose the walk has not yet stopped at some time n. Then,

$$-a < S_n < b.$$

Because S_n is integer-valued, its lowest possible position is $-a + 1$. If the next $a + b$ increments are all $+1$-steps, then even from this lowest position, the walk would reach,

$$(-a + 1) + (a + b) = b + 1.$$

Since these steps move upward one unit at a time, the walk must hit b before completing the block. Hence, from every position strictly between the boundaries, a block of $a + b$ consecutive $+1$-increments forces a boundary hit within that block.

Let's formalize along the first block. Define the event,

$$A_1 := \{\xi_1 = \cdots = \xi_{a+b} = +1\}.$$

Increments are independent, so the probability that the first block is all $+1$-increments,

$$\mathbb{P}(A_1) = \mathbb{P}(\{\xi_1 = \cdots = \xi_{a+b} = +1\}) = \left(\frac{1}{2}\right)^{a+b} = 2^{-(a+b)}.$$

This is analogous to a fair coin flip being all heads, if heads is $+1$ with probability $\frac{1}{2}$. Observe that the event that all of the steps of the first $a+b$-block are $+1$ is a subset of the event that the walk hits a boundary in that block. Hence,

$$A_1 = \{\xi_1 = \cdots = \xi_{a+b} = +1\} \subseteq \{\tau \leq a+b\}.$$

Consequently,

$$\begin{aligned}\mathbb{P}(\tau > a+b) &= 1 - \mathbb{P}(\tau \leq a+b) \\ &\leq 1 - \mathbb{P}(A_1) \\ &= 1 - 2^{-(a+b)}.\end{aligned}$$

Let's generalize to some r-th block for $r \geq 1$. Define,

$$A_r := \{\xi_{(r-1)(a+b)+1} = \cdots = \xi_{r(a+b)} = +1\}$$

the event that every increment in the r-th block is a $+1$-increment. If the walk has survived up to the beginning of this block then,

$$\tau > (r-1)(a+b) \implies -a < S_{(r-1)(a+b)} < b$$

otherwise it would have already hit a boundary and thus *not* survived. On this event, the occurrence of A_r forces the walk to hit b by the end of the block, thus the event that the walk survives up to the beginning of the r-th block and the next $a+b$-steps being $+1$-increments is,

$$A_r \cap \{\tau > (r-1)(a+b)\} \subseteq \{\tau \leq r(a+b)\}.$$

Equivalently, surviving through the entire r-th block requires both surviving to its beginning and avoid the event A_r from occurring,

$$\{\tau > r(a+b)\} \subseteq \{\tau > (r-1)(a+b)\} \cap A_r^c.$$

Hence, such an event $\{\tau > (r-1)(a+b)\}$ belongs to $\mathcal{F}_{(r-1)(a+b)}$ while A_r depends only on the next block of independent increments. Therefore, these events are independent and,

$$\begin{aligned}\mathbb{P}(\tau > r(a+b)) &\leq \mathbb{P}(\{\tau > (r-1)(a+b)\} \cap A_r^c) \\ &= \mathbb{P}(\tau > (r-1)(a+b))\mathbb{P}(A_r^c) \\ &= \left(1 - 2^{-(a+b)}\right)\mathbb{P}(\tau > (r-1)(a+b)).\end{aligned}$$

This argument also covers the case where the probability of surviving the beginning of a block is zero. Whenever that survival probability is positive, the same reasoning can be expressed conditionally as,

$$\begin{aligned}\mathbb{P}(\tau \le r(a+b) \mid \tau > (r-1)(a+b)) &= \frac{\mathbb{P}(\tau \le r(a+b))}{\mathbb{P}(\tau > (r-1)(a+b))} \\ &= 1 - \frac{\mathbb{P}(\tau > r(a+b))}{\mathbb{P}(\tau > (r-1)(a+b))} \\ &\ge 1 - \left(1 - 2^{-(a+b)}\right) \\ &\ge 2^{-(a+b)}.\end{aligned}$$

Thus, conditional on reaching a new block without stopping, there is always at least the same positive probability of stopping during that block. Applying this recursive bound repeatedly,

$$\begin{aligned}\mathbb{P}(\tau > r(a+b)) &\le \left(1 - 2^{-(a+b)}\right) \mathbb{P}(\tau > (r-1)(a+b)) \\ &\le \left(1 - 2^{-(a+b)}\right)^2 \mathbb{P}(\tau > (r-1)(a+b)) \\ &\le \cdots \\ &\le \left(1 - 2^{-(a+b)}\right)^r \mathbb{P}(\tau > 0) \\ &= \left(1 - 2^{-(a+b)}\right)^r.\end{aligned}$$

The final equality holds because $S_0 = 0$ lies strictly between $-a$ and b so $\tau > 0$.

Since $0 < 1 - 2^{-(a+b)} < 1$ we have that,

$$\left(1 - 2^{-(a+b)}\right)^r \to 0, \quad r \to \infty.$$

Hence the probability that the walk survives the r-th block $\mathbb{P}(\tau > r(a+b))$ goes to zero. Finally, the events $\{\tau > r(a+b)\}$ decrease as r increases and,

$$\bigcap_{r=1}^{\infty} \{\tau > r(a+b)\} = \{\tau = \infty\}.$$

By continuity of probability from above,

$$\mathbb{P}(\tau = \infty) = \lim_{r \to \infty} \mathbb{P}(\tau > r(a+b)) = 0.$$

Therefore, $\mathbb{P}(\tau < \infty) = 1$. We have verified that τ is a stopping time and,

$$\mathbb{P}(\tau < \infty) = 1, \qquad |S_{n \wedge \tau}| \le \max\{a, b\}, \quad \forall n \ge 0.$$

Since $(S_n)_{n \ge 0}$ is a martingale, hence condition (ii) of the *Optional Stopping Theorem* gives,

$$\mathbb{E}[S_\tau] = \mathbb{E}[S_0] = 0.$$

Finally, we solve for the boundary-hitting probabilities. Let $p := \mathbb{P}(S_\tau = b)$. The walk stops almost surely and cannot overshoot either boundary hence, $S_\tau \in \{-a, b\}$ almost surely. So,

$$\mathbb{P}(S_\tau = b) = p, \qquad \mathbb{P}(S_\tau = -a) = 1 - p.$$

Its terminal expectation is therefore,

$$\begin{aligned}\mathbb{E}[S_\tau] &= b\mathbb{P}(S_\tau = b) + (-a)\mathbb{P}(S_\tau = -a)\\ &= bp - a(1-p).\end{aligned}$$

Since $\mathbb{E}[S_\tau] = \mathbb{E}[S_0] = 0$,

$$\begin{aligned}\mathbb{E}[S_\tau] = 0 &= bp - a(1-p)\\ &= bp - a + ap\\ &= (a+b)p - a.\end{aligned}$$

Hence, solving for both probabilities,

$$p = \frac{a}{a+b}, \qquad 1 - p = 1 - \frac{a}{a+b} = \frac{b}{a+b}.$$

We obtain the *Gambler's Ruin* probabilities,

$$\mathbb{P}(S_\tau = b) = \frac{a}{a+b}, \qquad \mathbb{P}(S_\tau = -a) = \frac{b}{a+b}.$$

The probability of reaching the upper boundary increases as the lower boundary moves farther away. When $a = b$, symmetry gives probability $\frac{1}{2}$ for each boundary.

Almost-sure finiteness of a stopping time does not, by itself however, guarantee that stopping preserves the expected value of a martingale. The following example shows precisely why the additional hypotheses in the Optional Stopping Theorem matter.

Example 3.13 (Almost-Surely finite Stopping is Not Sufficient)**.** Let $(S_n)_{n\geq 0}$ be a symmetric SRW with $S_0 = 0$ and $S_n = \sum_{k=1}^n \xi_k$. Let $\mathcal{F}_n = \sigma(S_0, \ldots, S_n)$ and define,

$$\tau := \inf\{n \geq 0 : S_n = 1\}.$$

Hence, τ is the first time the walk reaches $+1$. Since it is a first hitting time, it is a stopping time. We will show that $\mathbb{P}(\tau < \infty) = 1$ but nevertheless,

$$\mathbb{E}[S_\tau] = 1 \neq 0 = \mathbb{E}[S_0].$$

This is not a contradiction of the Optional Stopping Theorem since neither of its two sufficient conditions is satisfied. Fix a positive integer a, and introduce define the stopping time,

$$\sigma_a := \inf\{n \geq 0 : S_n \in \{-a, 1\}\}.$$

This stops the walk up reaching either $-a$ or $+1$. By *Gambler's Ruin*, $\sigma_a < \infty$ almost surely,

$$\mathbb{P}(S_{\sigma_a} = 1) = \frac{a}{a+1}, \qquad \mathbb{P}(S_{\sigma_a} = -a) = \frac{1}{a+1}.$$

If $S_{\sigma_a} = 1$, then the walk has reached $+1$ before reaching $-a$. In particular, it has reached $+1$ at a finite time hence $\tau = \sigma_a$ on this event. Therefore,

$$\{S_{\sigma_a} = 1\} \subseteq \{\tau < \infty\}.$$

Consequently,

$$\mathbb{P}(\tau < \infty) \geq \mathbb{P}(S_{\sigma_a} = 1) = \frac{a}{a+1}.$$

However, there may be a path that first reaches $-a$ giving $S_{\sigma_a} = -a$ then later reaches $+1$. The auxiliary stopping rule does not prevent the original random walk from continuing.

Since the lower bound holds for every positive integer,

$$1 \geq \mathbb{P}(\tau < \infty) \geq \frac{a}{a+1}.$$

Letting $a \to \infty$, $1 \geq \mathbb{P}(\tau < \infty) \geq 1$ so $\mathbb{P}(\tau < \infty) = 1$.

By definition, τ is the first time the walk reaches $+1$. Since this time is finite almost surely, $S_\tau = 1$ almost surely. Thus, S_τ is integrable and $\mathbb{E}[S_\tau] = 1$. However, $S_0 = 0$ so,

$$\mathbb{E}[S_\tau] = 1 \neq 0 = \mathbb{E}[S_0].$$

The terminal value is therefore particular simple. The strategy of waiting until the walk reaches $+1$ ends with a gain of one unit almost surely. We must examine why the *Optional Stopping Theorem* does not preserve the initial expectation here.

Let $N \geq 1$ be any fixed integer and consider,

$$A_N := \{\xi_1 = \cdots = \xi_N = -1\}.$$

On this event, $S_k = -k$ for $1 \leq k \leq N$. Hence, the walk has not reached $+1$ by time N so,

$$A_N \subseteq \{\tau > N\}.$$

By independence,

$$\mathbb{P}(\tau > N) \geq \mathbb{P}(A_N) = 2^{-N} > 0.$$

Thus for every finite N, there is a positive probability that stopping occurs after N hence no deterministic N satisfies $\tau \leq N$ almost surely.

Therefore, condition (i) of the *Optional Stopping Theorem* fails.

Suppose we propose a finite constant C as a bound for all stopped values. Choose an integer $N > C$. On the same event A_N, we have that $\tau > N$,

$$S_{N \wedge \tau} = S_N = -N.$$

Therefore,

$$\mathbb{P}(|S_{N \wedge \tau}| > C) \geq \mathbb{P}(A_N) = 2^{-N} > 0.$$

Since this argument works for every finite C, there is no constant satisfying,

$$|S_{n\wedge\tau}| \leq C \quad \text{almost surely } \forall n.$$

Although the walk stops when it reaches $+1$, there is no lower barrier preventing arbitrarily large losses before stopping. Hence condition (ii) also fails.

For each deterministic n, the stopping time $n\wedge\tau$ is bounded by n. Therefore, the boundary-time case of the *Optional Stopping Theorem* applies,

$$\mathbb{E}[S_{n\wedge\tau}] = \mathbb{E}[S_0] = 0, \quad n \geq 0.$$

Equivalently, it follows because $(S_{n\wedge\tau})_{n\geq 0}$ is a martingale. On the other hand, $\tau < \infty$ almost surely. For almost every $\omega \in \Omega$, eventually $n \geq \tau(\omega)$,

$$S_{n\wedge\tau}(\omega) = S_\tau(\omega) = 1.$$

Hence, $S_{n\wedge\tau} \longrightarrow 1$ almost surely. Despite this convergence,

$$\lim_{n\to\infty} \mathbb{E}[S_{n\wedge\tau}] = 0 \neq 1 = \mathbb{E}\left[\lim_{n\to\infty} S_{n\wedge\tau}\right].$$

The failure occurs when passing the limit in the expectations, not when stopping the process at a fixed finite time. The contribution responsible for such a failure can be computed directly.

On the event $\{\tau \leq n\}$, the stopped value is 1 whereas on $\{\tau > n\}$, it is S_n. Therefore,

$$S_{n\wedge\tau} = \mathbb{I}_{\{\tau\leq n\}} + S_n\mathbb{I}_{\{\tau>n\}}.$$

Taking expectations and noting $\mathbb{E}[S_{n\wedge\tau}] = \mathbb{E}[S_0] = 0$,

$$\begin{aligned}\mathbb{E}[S_{n\wedge\tau}] = 0 &= \mathbb{E}[\mathbb{I}_{\{\tau\leq n\}}] + \mathbb{E}[S_n\mathbb{I}_{\{\tau>n\}}] \\ &= \mathbb{P}(\tau \leq n) + \mathbb{E}[S_n\mathbb{I}_{\{\tau>n\}}].\end{aligned}$$

Re-arranging,

$$\mathbb{E}[S_n\mathbb{I}_{\{\tau>n\}}] = -\mathbb{P}(\tau \leq n) \longrightarrow -1.$$

Although $\mathbb{P}(\tau > n) \to 0$, the expected contribution from paths that have not yet stopped does not tend to zero. Those increasingly rare paths carry sufficiently large negative values to keep the total expectation equal to zero.

Return to σ_a. Its terminal distribution is characterized as,

$$S_{\sigma_a} = \begin{cases} 1, & \text{w.p.} \quad \frac{a}{a+1}, \\ -a, & \text{w.p.} \quad \frac{1}{a+1}. \end{cases}$$

For each fixed a, the stopped process is bounded between $-a$ and 1 so Optional Stopping is valid. Hence, its expected value,

$$\begin{aligned}\mathbb{E}[S_{\sigma_a}] &= 1 \cdot \frac{a}{a+1} + (-a) \cdot \frac{1}{a+1} \\ &= \frac{a}{a+1} - \frac{a}{a+1} \\ &= 0.\end{aligned}$$

As $a \to \infty$, the probability of the negative outcome tends to zero,

$$\mathbb{P}(S_{\sigma_a} = -a) = \frac{1}{a+1} \to 0.$$

But the magnitude of that loss grows to infinity, and its contribution to the expectation satisfies,

$$(-a)\mathbb{P}(S_{\sigma_a} = -a) = -\frac{a}{a+1} \to -1.$$

Meanwhile, the positive contribution satisfies,

$$1 \cdot \mathbb{P}(S_{\sigma_a} = 1) = \frac{a}{a+1} \to 1.$$

These two contributions cancel for every a. In fact, $S_{\sigma_a} \to 1$ almost surely as $a \to \infty$. To see this, fix a path with $\tau < \infty$. Before reaching $+1$, this path visits only finitely many positions,

$$S_0, S_1, \ldots, S_\tau.$$

For sufficiently large a, the boundary $-a$ lies below all these positions. On that path, the walk then reach $+1$ before $-a$ so $\sigma_a = \tau$ and $S_{\sigma_a} = 1$ for all sufficiently large a.

The stopped values in the example above converge almost surely to 1 while their expectations remain zero. Almost-sure convergence controls what eventually happens along each typical path, but it does not itself control the expected contribution of rare, increasingly large losses.

The doubling strategy provides an explicit example in which the terminal gain is always $+1$, but the expected gain at every fixed finite time is zero. The distinction comes from the large losses that can accumulate before the first win.

Example 3.14 (Doubling Strategy Stopped at its First Win)**.** Let $\xi_1, \xi_2, \ldots$ be independent symmetric Rademacher random variables where $+1$ represents a win and -1 represents a loss.

Define,

$$M_0 = 0, \quad M_n = \sum_{k=1}^{n} \xi_k, \quad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$$

with $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Recall the doubling strategy,

$$H_1 = 1, \quad H_{n+1} = \begin{cases} 2H_n, & \xi_n = -1, \\ 1, & \xi_n = 1. \end{cases}$$

Also, define its cumulative net gain as,

$$X_0 = 0, \quad X_n = \sum_{k=1}^{n} H_k(M_k - M_{k-1}) = \sum_{k=1}^{n} H_k \xi_k.$$

H_n represents the stake chosen before observing ξ_n so the strategy is predictable. As established previously, $(X_n)_{n\geq 0}$ is a martingale. Define the stopping time,

$$\tau := \inf\{n \geq 0 : X_n = 1\}.$$

Since X_n is adapted, τ is a stopping time.

We will show that τ is exactly the time of the first win.

Before the First win. If the first r bets are losses, then their stakes are,

$$2^0, 2^1, \ldots, 2^{r-1}.$$

Hence, the cumulative gain across the first r bets is,

$$X_r = -\sum_{k=1}^{r} 2^{k-1} = -(2^r - 1) = 1 - 2^r.$$

In particular, the strategy has not yet reached $+1$.

At the First win. If the first r bets lose and the next bet wins, the winning stake is 2^r hence,

$$\begin{aligned} X_{r+1} &= X_r + 2^r \\ &= -(2^r - 1) + 2^r \\ &= 1. \end{aligned}$$

Thus, the first win recovers all preceding losses and produces a net gain of $+1$. Consequently, $\tau := \inf\{n \geq 1 : \xi_n = 1\}$. For every $n \geq 1$, the strategy has not yet stopped by the time n exactly when the first n bets are all losses,

$$\{\tau > n\} = \{\xi_1 = \cdots = \xi_n = -1\}.$$

By independence, $\mathbb{P}(\tau > n) = 2^{-n}$. Therefore,

$$\begin{aligned} \mathbb{P}(\tau = \infty) &= \lim_{n\to\infty} \mathbb{P}(\tau > n) \\ &= \lim_{n\to\infty} 2^{-n} \\ &= 0. \end{aligned}$$

Hence, $\tau < \infty$ almost surely. By the definition τ, $X_\tau = 1$ almost surely so $\mathbb{E}[X_\tau] = 1$.

However, $(X_{n\wedge\tau})_{n\geq 0}$ is a stopped martingale. At every fixed time,

$$\mathbb{E}[X_{n\wedge\tau}] = \mathbb{E}[X_\tau] = 0.$$

The stopped gain at time n has only two possible values.

(i) $\tau \leq n$ implies the first has already occurred, and the stopped gain is 1,
(ii) $\tau > n$ implies all n bets are losses, and cumulative gain is $1 - 2^n$.

Hence,

$$X_{n\wedge\tau} = \begin{cases} 1, & \tau \leq n, \\ 1 - 2^n, & \tau > n. \end{cases}$$

Since $\mathbb{P}(\tau > n) = 2^{-n}$ its distribution is,

$$X_{n\wedge\tau} = \begin{cases} 1, & p = 1 - 2^{-n}, \\ 1 - 2^n, & p = 2^{-n}. \end{cases}$$

Computing this expectation explicitly,

$$\begin{aligned} \mathbb{E}[X_{n\wedge\tau}] &= 1 \cdot (1 - 2^{-n}) + (1 - 2^n) \cdot 2^{-n} \\ &= 1 - 2^{-n} + 2^{-n} - 1 \\ &= 0. \end{aligned}$$

This agrees with the stopped-martingale property.

However, notice that for every fixed integer $N \geq 1$,

$$\mathbb{P}(\tau > N) = 2^{-N} > 0$$

so there is always a positive probability that the first N bets are all losses and the strategy has not yet stopped. Hence, no deterministic $N < \infty$ satisfies $\tau \leq N$ almost surely.

There is, however, no conflict with $\tau < \infty$ almost surely. Each path stops eventually outside a null event, but there is no single finite deadline by which almost every path has stopped.

Fix any proposed finite bound C. Choose an integer r large enough that $2^r - 1 > C$. On the event that the first r bets are losses, we have $\tau > r$ so,

$$X_{r\wedge\tau} = X_r = -(2^r - 1).$$

Consequently,

$$\mathbb{P}(|X_{r\wedge\tau}| > C) \geq \mathbb{P}(\tau > r) = 2^{-r} > 0.$$

Therefore, no deterministic constant C satisfies $|X_{n\wedge\tau}| \leq C$ almost surely for every $n \geq 0$. $X_\tau = 1$ almost surely, but before stopping the process can take values,

$$X_r = -(2^r - 1).$$

Hence the terminal value is bounded, but the entire stopped process is not.

Now, since $\tau < \infty$ almost surely, for almost every ω, eventually $n \geq \tau(\omega)$ so,

$$X_{n\wedge\tau}(\omega) = X_\tau(\omega) = 1.$$

Therefore $X_{n\wedge\tau} \to 1$ almost surely. Nevertheless,

$$\lim_{n\to\infty} \mathbb{E}[X_{n\wedge\tau}] = 0 \neq 1 = \mathbb{E}\left[\lim_{n\to\infty} X_{n\wedge\tau}\right].$$

The contributions from paths that have already stopped is,

$$1 \cdot \mathbb{P}(\tau \leq n) = 1 - 2^{-n} \to 1.$$

Yet, the contribution from paths that have not yet stopped is,

$$\begin{aligned}(1-2^n)\mathbb{P}(\tau > n) &= (1-2^n)2^{-n} \\ &= 2^{-n} - 1 \to -1.\end{aligned}$$

Although the probability of an unfinished losing streak tends to zero, its loss grows sufficiently quickly that its expected contribution approaches -1. The positive and negative contributions cancel at every finite time. Indeed, we can write the stopped gain as,

$$X_{n\wedge\tau} = 1 - 2^n \mathbb{I}_{\{\tau > n\}}.$$

Hence, for almost every path, the indicator eventually becomes zero. But the expected absolute difference from the terminal value remains

$$\begin{aligned}\mathbb{E}[|X_{n\wedge\tau} - X_\tau|] &= \mathbb{E}[2^n \mathbb{I}_{\{\tau>n\}}] \\ &= 2^n \mathbb{P}(\tau > n) \\ &= 2^n 2^{-n} = 1.\end{aligned}$$

Thus the process converges almost surely to its terminal gain, while its expected absolute error does not decrease at all. Increasing rare but increasingly large losses prevent the passage of the limit through expectation.

3.8 Martingale Convergence

Upcrossings measure how often a sequence makes a complete upward passage between two fixed levels. They provide a way to detect persistent oscillation, that is, if a sequence fails to converge, it must repeatedly cross some interval. We formalize this definition.

Definition 3.8 (Upcrossings)**.** Let $(X_n)_{n\geq 0}$ be a deterministic real-valued sequence, and fix $a < b$. An *upcrossing* of $[a, b]$ is a passage from a value at or below a to a later value at or above b.

To count successive upcrossings, set $\tau_0 = 0$ and define recursively for $j \geq 1$,

$$\begin{aligned}\sigma_j &:= \inf\{n \geq \tau_{j-1} : X_n \leq a\}, \\ \tau_j &:= \inf\{n \geq \sigma_j : X_n \geq b\}.\end{aligned}$$

The convention $\inf \emptyset = +\infty$ allows us to define when a time in this construction is infinite so that all subsequent times are also set equal to $+\infty$.

Here, σ_j is the starting time of the j-th attempted upcrossing, and τ_j is its completion time. Whenever the relevant times are finite, $a < b$ ensures that,

$$\sigma_1 < \tau_1 < \sigma_2 < \tau_2 < \cdots .$$

Indeed, a value cannot be both at or below a and at or above b.

The *Number of Upcrossings by time* N is,

$$U_N(a, b) := \#\{j \geq 1 : \tau_j \leq N\}.$$

An upcrossing that has started but has not been completed by time N is not included. For fixed $a < b$, the count $U_N(a, b)$ is non-decreasing in N.

Thus, its limit exists in $\{0, 1, 2, \dots\} \cup \{+\infty\}$ and we write this as,

$$U_\infty(a, b) = \lim_{N\to\infty} U_N(a, b).$$

To interpret this construction, suppose X_n is a stock price. Buy one share when the price first reaches or falls below a, and sell it when the price subsequently reaches or exceeds b.

Then we repeat this rule.

Each completed trade correspond to one upcrossing and its gain satisfies,

$$X_{\tau_j} - X_{\sigma_j} \geq b - a.$$

Thus, $U_N(a, b)$ counts completed trades by time N, each producing a gain of at least $b - a$. Any position still held at time N is unfinished and must be treated separately when computing the total gain. The following result explains why upcrossings are useful for proving convergence.

Lemma 3.1 (Finite Upcrossings Imply Convergence)**.** *Let $(X_n)_{n\geq 0}$ be a deterministic real-valued sequence. Suppose that, for every pair of rational numbers $a < b$,*

$$\lim_{N\to\infty} U_N(a, b) < \infty.$$

Then, $(X_n)_{n\geq 0}$ converges to a limit in the extended real line $\mathbb{R} \cup \{-\infty, +\infty\}$.

Proof. Define,

$$l := \lim_{n\to\infty} \inf X_n, \qquad u := \lim_{n\to\infty} \sup X_n$$

where $l, u \in \mathbb{R} \cup \{-\infty, +\infty\}$. Recall that,

$$l = \lim_{N\to\infty} \inf_{n\geq N} X_n, \qquad u := \lim_{N\to\infty} \sup_{n\geq N} X_n$$

describe the limiting lower and upper values of the tails of the sequence. The sequence converges in the extended real line precisely when $l = u$.

Suppose, for a contradiction, that the sequence does not converge. Then, $l < u$.

By density of rational numbers, we can choose rational $a < b$ such that,

$$l < a < b < u.$$

Observe this is also possible if $l = -\infty$ and $u = +\infty$. We first show that the sequence visits values at or below a infinitely often. Otherwise, there would be some N_0 such that,

$$X_n > a \quad \forall n \geq N_0$$

implying that $\lim_{n\to\infty} \inf X_n \geq a$ contradicting $l < a$.

Hence, there are arbitrarily large times at which $X_n \leq a$.

Similarly, the sequence visits values at or above b infinitely often. Otherwise, for some N_1,

$$X_n < b \quad \forall n \geq N_1$$

which would imply $\lim_{n\to\infty} \sup X_n \leq b$ contradicting $b < u$.

Thus, there are arbitrarily large times at which $X_n \geq b$.

We now use these two facts to construct infinitely many completed upcrossings. Because the sequence visits $(-\infty, a]$ infinitely often, the first starting time,

$$\sigma_1 = \inf\{n \geq 0 : X_n \leq a\}$$

is finite. Since it also visits $[b, \infty)$ at arbitrarily large times, there is a visit σ_1. Thus,

$$\tau_1 := \inf\{n \geq \sigma_1 : X_n \geq b\} < \infty.$$

This completes the first upcrossing. After τ_1, there is another visit to $(-\infty, a]$ followed by another visit to $[b, \infty)$. Thus σ_2 and τ_2 are finite as well.

Repeating this argument, every σ_j and τ_j is finite with,

$$\sigma_1 < \tau_1 < \sigma_2 < \tau_2 < \cdots .$$

Consequently for every positive integer j, $U_N(a, b) \geq j$ whenever $N \geq \tau_j$. Since j is arbitrary,

$$U_N(a, b) \to +\infty \quad \text{as } N \to \infty.$$

This contradicts the assumption that $U_\infty(a, b) < \infty$ for every rational $a < b$. Therefore,

$$l = u$$

and the sequence converges in the extended real line.

□

Remark 3.11 (Oscillation and Rational Levels). For a concrete illustration suppose,

$$X_n = (-1)^n + z_n, \quad z_n \to z.$$

Then,

$$X_{2k} \longrightarrow z + 1,$$
$$X_{2k+1} \longrightarrow z - 1.$$

Choosing rational levels,

$$z - 1 < a < b < z + 1$$

where the odd terms eventually lie below a while the even terms lie above b. The sequence therefore completes infinitely many upcrossings of $[a, b]$.

Rational levels suffice because any two distinct limiting lower and upper values contain two rational numbers between them. Their countability will also be important for random processes. If each rational interval has finitely many upcrossings almost surely, we can intersect these countably many probability-one events and apply the lemma path by path.

Lemma 3.1 allows infinite limits. For example, $X_n = n$ has at most one upcrossing of each fixed interval and converges to $+\infty$. Additional bounds are needed to conclude that the limit is finite.

The deterministic upcrossing lemma shows that persistent oscillation forces infinitely many upcrossings of some rational interval. To apply this to a stochastic process, we need a bound on how many upcrossings can occur.

The upcrossing inequality supplies such a bound in expectation.

Lemma 3.2 (Upcrossing Inequality)**.** *Let $(X_n)_{n\geq 0}$ be a submartingale with respect to $(\mathcal{F}_n)_{n\geq 0}$ and fix positive integers $a < b$. Write,*

$$x^+ := \max\{x, 0\}$$

and let $U_N(a, b)$ denote the number of completed upcrossings of $[a, b]$ by time N.

Then, for every integer $N \geq 0$,

$$(b - a)\mathbb{E}[U_N(a, b)] \leq \mathbb{E}[(X_N - a)^+] - \mathbb{E}[(X_0 - a)^+].$$

Proof. We construct a predictable strategy that holds during each attempted upcrossing. Applied to a suitable transformation on X_n, each completed upcrossing earns at least $b - a$ while an unfinished upcrossing contributes a nonnegative amount. Define,

$$Y_n := (X_n - a)^+ = \begin{cases} X_n - a, & X_n \geq a, \\ 0, & X_n \leq a. \end{cases}$$

Note that the function $\phi(x) := (x - a)^+$ is convex, nondecreasing, and continuous.

Therefore, Y_n is $\mathcal{F}_n$-measurable. Also,

$$0 \leq Y_n \leq |X_n| + |a|$$

so Y_n is integrable. By conditional Jensen's inequality,

$$\begin{aligned} \mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n], && X_n \text{ is submartingale} \\ &\geq \phi(\mathbb{E}[X_{n+1} \mid \mathcal{F}_n]), && \phi \text{ is nondecreasing} \\ &\geq \phi(X_n) \\ &= Y_n. \end{aligned}$$

Hence $(Y_n)_{n\geq 0}$ is a nonnegative submartingale. This transformation is useful as,

$$X_n \leq a \implies Y_n = 0, \qquad X_n \geq b \implies Y_n \geq b - a.$$

Set $\tau_0 = 0$ and define recursively,

$$\begin{aligned}\sigma_j &:= \inf\{n \geq \tau_{j-1} : X_n \leq a\},\\ \tau_j &:= \inf\{n \geq \sigma_j : X_n \geq b\}, \quad j \geq 1.\end{aligned}$$

As before, $\inf = +\infty$ and all subsequent times are infinite once the construction cannot continue. These times are stopping times. To verify this recursively, suppose τ_{j-1} is a stopping time. Then,

$$\{\sigma \leq n\} = \bigcup_{m=0}^{n} \left(\{\tau_{j-1} = m\} \cap \bigcup_{r=m}^{n} \{X_r \leq a\} \right).$$

This is precisely the event that X_n is below or at a. That is, the previous upcrossing finishes at m and at some time r from m to n, $X_r \leq a$. Every event on the right belongs to $\mathcal{F}_n$, hence $\{\sigma_j \leq n\} \in \mathcal{F}_n$. Likewise we can define,

$$\{\tau_j \leq n\} = \bigcup_{m=0}^{n} \left(\{\sigma_j = m\} \cap \bigcup_{r=m}^{n} \{X_r \geq b\} \right) \in \mathcal{F}_n,$$

the event where the current upcrossing has started at some time $m \leq n$ and at some time r between m and n, $X_r \geq b$. Starting from the deterministic stopping time $\tau_0 = 0$, this proves the claim. For $k \geq 1$ define the analogous trading rule over $k-1$ to k,

$$H_k := \sum_{j=1}^{\infty} \mathbb{I}_{\{\sigma_j < k \leq \tau_j\}}$$

which tells us whether we hold one share over $[k-1, k]$. That is,

(i) Buy 1 share at σ_j when $X \leq a$,
(ii) Hold through the steps $\sigma_j < k \leq \tau_j$
(iii) Sell at τ_j when $X_j \geq b$, then hold nothing till the next buying time σ_{j+1}.

The inequalities $\sigma_j < k \leq \tau_j$ ensure that we start collecting increments after the buying time σ_j and include the increment that completes the upcrossing at τ_j.

Since we buy at σ_j, the first increment we earn is,

$$X_{\sigma_j+1} - X_{\sigma_j}.$$

Similarly, since we sell at τ_j the last increment we earn is $X_{\tau_j} - X_{\tau_{j-1}}$.

To check predictability observe that,

$$\{\sigma_j < k \leq \tau_j\} = \{\sigma_j \leq k-1\} \cap \{\tau_j \leq k-1\}^c$$

since $\{\sigma_j < k\} = \{\sigma_j \leq k-1\}$ and $\{\tau_j \geq k\} = \{\tau_j \leq k-1\}^c$.

Hence, since both events are $\mathcal{F}_{k-1}$-measurable, $\{\sigma_j < k \leq \tau_j\}$ is $\mathcal{F}_{k-1}$-measurable. Hence, the indicator is $\mathcal{F}_{k-1}$-measurable. In other words, before observing the increment $Y_k - Y_{k-1}$ we already know whether we are holding a share. Hence H_k is $\mathcal{F}_{k-1}$-measurable.

Consider a completed upcrossing (σ_j, τ_j). At its starting time,

$$X_{\sigma_j} \leq a \implies Y_{\sigma_j} = (X_{\sigma_j} - a)^+ = 0.$$

At its completion time,

$$X_{\tau_j} \geq b \implies Y_{\tau_j} = (X_{\tau_j} - a)^+ \geq b - a.$$

Therefore, the gain accumulated during this upcrossing is,

$$\begin{aligned}\sum_{k=\sigma_j+1}^{\tau_j} (Y_k - Y_{k-1}) &= Y_{\tau_j} - Y_{\sigma_j} \\ &= Y_{\tau_j} \\ &\geq b - a.\end{aligned}$$

Recall that the first increment we receive $Y_{\sigma_j+1} - Y_{\sigma_j}$ is $[\sigma_j + 1, \sigma_j]$ hence why the sum begins at $k = \sigma_j + 1$ buying at time σ_j. There are $U_N(a, b)$ completed upcrossings by time N.

Summing their gains gives,

$$(b - a)U_N(a, b) \leq \sum_{j=1}^{U_N(a,b)} (Y_{\tau_j} - Y_{\sigma_j}).$$

We must also account for a possible unfinished upcrossing. Suppose its starting time, if it has begun, is $\sigma_{U_N(a,b)+1}$. If $\sigma_{U_N(a,b)+1} \leq N$, then the next upcrossing has started but has not finished by N. Since its starting value satisfies $Y_{\sigma_{U_N(a,b)+1}} = 0$, its contribution is,

$$\sum_{k=\sigma_{U_N(a,b)+1}}^{N} (Y_k - Y_{k-1}) = Y_N - Y_{\sigma_{U_N(a,b)+1}} = Y_N \geq 0.$$

If it starts exactly at N, the sum is empty and $Y_N = 0$. If,

$$\sigma_{U_N(a,b)+1} > N,$$

the next upcrossing has not started, so there is no unfinished contribution. Thus, in all cases, the unfinished contribution can be written as,

$$Y_N \mathbb{I}_{\{\sigma_{U_N(a,b)+1} \leq N\}}$$

which is nonnegative. This expression also avoid evaluating Y at a potentially infinite starting time. Recall H_k selects precisely the increments earned during the holding intervals.

Consequently,

$$\begin{aligned}\sum_{k=1}^{N} H_k(Y_k - Y_{k-1}) &= \sum_{j=1}^{U_N(a,b)} \sum_{k=\sigma_j+1}^{\tau_j} (Y_k - Y_{k-1}) + Y_N \mathbb{I}_{\{\sigma_{U_N(a,b)+1} \le N\}} \\ &= \sum_{j=1}^{U_N(a,b)} (Y_{\tau_j} - Y_{\sigma_j}) + Y_N \mathbb{I}_{\{\sigma_{U_N(a,b)+1} \le N\}} \\ &\ge \sum_{j=1}^{U_N(a,b)} (Y_{\tau_j} - Y_{\sigma_j}) \\ &\ge (b-a) U_N(a,b).\end{aligned}$$

Hence, we have the pathwise lower bound,

$$(b-a) U_N(a,b) \le \sum_{k=1}^{N} H_k(Y_k - Y_{k-1}).$$

Next, we need an upper bound on the expected sum on the right. Hence, we examine the increments during which we do not hold. Since $H_k \in \{0,1\}$ is $\mathcal{F}_{k-1}$-measurable, the variable $1 - H_k$ is also $\mathcal{F}_{k-1}$-measurable, bounded, and nonnegative.

Also, because Y is a submartingale,

$$\begin{aligned}\mathbb{E}[Y_k - Y_{k-1} \mid \mathcal{F}_{k-1}] &= \mathbb{E}[Y_k \mid \mathcal{F}_{k-1}] - Y_{k-1} \\ &\ge 0.\end{aligned}$$

Taking out the known factor $1 - H_k$,

$$\begin{aligned}&\mathbb{E}[(1 - H_k)(Y_k - Y_{k-1}) \mid \mathcal{F}_{k-1}] \\ &\quad = (1 - H_k)\mathbb{E}[Y_k - Y_{k-1} \mid \mathcal{F}_{k-1}] \\ &\quad \ge 0.\end{aligned}$$

The final inequality holds because both factors are nonnegative. All these expectations are well-defined since boundedness of $1 - H_k$ gives,

$$|(1 - H_k)(Y_k - Y_{k-1})| \le |Y_k| + |Y_{k-1}|$$

and both Y_k and Y_{k-1} are integrable. Taking ordinary expectations and by law of expectation,

$$\begin{aligned}\mathbb{E}[(1 - H_k)(Y_k - Y_{k-1})] &= \mathbb{E}[\mathbb{E}[(1 - H_k)(Y_k - Y_{k-1}) \mid \mathcal{F}_{k-1}]] \\ &\ge 0.\end{aligned}$$

Therefore, summing over the finitely many times $k = 1, \ldots, N$,

$$\begin{aligned}\mathbb{E}\left[\sum_{k=1}^{N} (1 - H_k)(Y_k - Y_{k-1})\right] &= \sum_{k=1}^{N} \mathbb{E}[(1 - H_k)(Y_k - Y_{k-1})] \\ &\ge 0.\end{aligned}$$

Thus, the increments omitted by the strategy have a nonnegative total contribution to expectation. Splitting each increment into its held and unheld portions,

$$Y_k - Y_{k-1} = H_k(Y_k - Y_{k-1}) + (1 - H_k)(Y_k - Y_{k-1}).$$

Summing over $k = 1, \ldots, N$ and re-arranging,

$$\begin{aligned}\sum_{k=1}^{N} H_k(Y_k - Y_{k-1}) &= \sum_{k=1}^{N}(Y_k - Y_{k-1}) - \sum_{k=1}^{N}(1 - H_k)(Y_k - Y_{k-1}) \\ &= Y_N - Y_0 - \sum_{k=1}^{N}(1 - H_k)(Y_k - Y_{k-1}).\end{aligned}$$

The first sum telescopes,

$$(Y_1 - Y_0) + (Y_2 - Y_1) + \cdots + (Y_N - Y_{N-1}) = Y_N - Y_0.$$

Taking expectations,

$$\begin{aligned}\mathbb{E}\left[\sum_{k=1}^{N} H_k(Y_k - Y_{k-1})\right] &= \mathbb{E}[Y_N - Y_0] - \mathbb{E}\left[\sum_{k=1}^{N}(1 - H_k)(Y_k - Y_{k-1})\right] \\ &\leq \mathbb{E}[Y_N - Y_0] \\ &= \mathbb{E}[Y_N] - \mathbb{E}[Y_0].\end{aligned}$$

Taking expectations over the pathwise lower bound and combining the two inequalities,

$$\begin{aligned}(b - a)U_N(a, b) &\leq \mathbb{E}\left[\sum_{k=1}^{N} H_k(Y_k - Y_{k-1})\right] \\ &\leq \mathbb{E}[Y_N] - \mathbb{E}[Y_0] \\ &= \mathbb{E}[(X_N - a)^+] - \mathbb{E}[(X_0 - a)^+].\end{aligned}$$

This proves the upcrossing inequality.

□

The upcrossing inequality controls persistent oscillation. If the expected positive parts of a submartingale remain bounded, then every fixed interval has only finitely many upcrossings almost surely. The deterministic upcrossing lemma then gives convergence, and a further argument rules out infinite limits.

Theorem 3.4 (Martingale Convergence Theorem)**.** *Let $(X_n)_{n\geq 0}$ be a submartingale with respect to $(\mathcal{F}_n)_{n\geq 0}$. Suppose,*

$$\sup_{n\geq 0} \mathbb{E}[X_n^+] < \infty.$$

Then there exists a finite, integrable random variable X such that,

$$X_n \xrightarrow{a.s.} X.$$

Proof. First we show that almost every sample path has finitely many upcrossings of every rational interval. This implies convergence in the extended line.

Then we show that the limit cannot be $+\infty$ or $-\infty$.

Suppose we fix numbers $a < b$. By the upcrossing inequality,

$$\begin{aligned}(b-a)\mathbb{E}[U_N(a,b)] &\leq \mathbb{E}[(X_N - a)^+] - \underbrace{\mathbb{E}[(X_0 - a)^+]}_{\geq 0} \\ &\leq \mathbb{E}[(X_N - a)^+].\end{aligned}$$

Therefore, $(b-a)\mathbb{E}[U_N(a,b)] \leq \mathbb{E}[(X_N - a)^+]$. We next show that pointwise, $(X_N - a)^+ \leq X_N^+ + |a|$. Consider when $X_N - a \leq 0$, then,

$$(X_N - a)^+ = \max\{X_N - a, 0\} = 0 \leq X_N^+ + |a|.$$

Otherwise, when $X_N - a > 0$, then using $-a \leq |a|$, $X_N \leq X_N^+$ so,

$$\begin{aligned}(X_N - a)^+ &= \max\{X_N - a, 0\} = X_N - a \\ &\leq X_N + |a| \\ &\leq X_N^+ + |a|.\end{aligned}$$

Hence, in both cases, $(X_N - a)^+ \leq X_N^+ + |a|$. Taking expectations,

$$\begin{aligned}\mathbb{E}[(X_N - a)^+] &\leq \mathbb{E}[X_N^+ + |a|] \\ &= \mathbb{E}[X_N^+] + |a| \leq \sup_{n\geq 0} \mathbb{E}[X_n^+] + |a|.\end{aligned}$$

Combining these inequalities we obtain,

$$(b-a)\mathbb{E}[U_N(a,b)] \leq \sup_{k\geq 0} \mathbb{E}[X_k^+] + |a|.$$

Because $b - a > 0$, division preserves the inequality,

$$\mathbb{E}U_N(a,b) \leq \frac{\sup_{n\geq 0} \mathbb{E}[X_n^+] + |a|}{b-a}, \qquad \forall N.$$

Hence, the right-hand side is finite and does *not* depend on N. For any positive integer r, we wish to show that this fixed interval has finitely many upcrossing almost surely.

Markov's inequality gives, for $r > 0$,

$$\begin{aligned}\mathbb{P}(U_N(a,b) \geq r) &\leq \frac{\mathbb{E}[U_N(a,b)]}{r} \\ &\leq \frac{1}{r}\frac{\sup_{n\geq 0}\mathbb{E}[X_n^+] + |a|}{b-a} = \frac{\sup_{n\geq 0}\mathbb{E}[X_n^+] + |a|}{r(b-a)}.\end{aligned}$$

As time increases, completed upcrossings are never lost, that is $U_N(a,b) \leq U_{N+1}(a,b)$. Consequently, the event $\{U_N(a,b) \geq r\} \subseteq \{U_{N+1}(a,b) \geq r\}$. Moreover,

$$\left\{\sup_{N\geq 0} U_N(a,b) \geq r\right\} = \bigcup_{N\geq 0} \{U_N(a,b) \geq r\}.$$

Indeed, because the counts are integers, having at least r upcrossings in total means that at least r have been completed by some finite time N. By continuity of probability,

$$\begin{aligned}\mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) \geq r\right) &= \mathbb{P}\left(\bigcup_{N\geq 0} \{U_N(a,b) \geq r\}\right) \\ &= \lim_{N\to\infty} \mathbb{P}(U_N(a,b) \geq r) \leq \frac{sup_{n\geq 0}\mathbb{E}[X_n^+] + |a|}{r(b-a)}.\end{aligned}$$

If infinitely many upcrossings occur, then at least r upcrossings occur for every positive integer r, hence we may write the null event of infinite upcrossings as,

$$\left\{\sup_{N\geq 0} U_N(a,b) = \infty\right\} \subseteq \left\{\sup_{N\geq 0} U_N(a,b) \geq r\right\}.$$

Hence, we obtain a bound on the probability on such an event,

$$\begin{aligned}0 &\leq \mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) = \infty\right) \\ &\leq \mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) \geq r\right) \leq \frac{\sup_{n\geq 0}\mathbb{E}[X_n^+] + |a|}{r(b-a)} \to 0, \qquad r \to \infty.\end{aligned}$$

The numerator is finite while $r(b-a) \to \infty$ as $r \to \infty$ hence,

$$\mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) = \infty\right) = 0.$$

Hence, by complements $\mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) < \infty\right) = 1$. Thus, for every fixed rational pair $a < b$, we have proved that the probability that the process makes infinitely many upcrossings on the interval $[a,b]$ is zero. It remains for us to show that this conclusions holds for all rational intervals simultaneously. Consider the event,

$$\bigcup_{a<b} \left\{\sup_{N\geq 0} U_N(a,b) = \infty\right\}$$

i.e., the event that at least one rational interval has infinitely many upcrossings. Observe that there are only countably many rational pairs, hence by countable subadditivity of probability,

$$0 \leq \mathbb{P}\left(\bigcup_{a<b}\left\{\sup_{N\geq 0} U_N(a,b) = \infty\right\}\right) \leq \sum_{a<b} \mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) = \infty\right) = 0.$$

The complementary event is that every rational interval has finitely many upcrossings. Therefore,

$$\begin{aligned}
&\mathbb{P}\left(\bigcap_{a<b}\left\{\sup_{N\geq 0} U_N(a,b) < \infty\right\}\right) \\
&\qquad = 1 - \mathbb{P}\left(\bigcup_{a<b}\left\{\sup_{N\geq 0} U_N(a,b) = \infty\right\}\right) \\
&\qquad = 1 - 0 = 1.
\end{aligned}$$

Hence, outside an event of probability zero, the deterministic sequence $X_0(\omega), X_1(\omega), X_2(\omega), \ldots$ has finitely many upcrossings of every rational interval. By *Finite Upcrossings Lemma*, each such sequence converges in $\mathbb{R} \cup \{-\infty, +\infty\}$. Hence, X_n converges almost surely to an extended real-valued limit. It remains for us to rule out infinite limits.

For every real number $x \in \mathbb{R}$, $|x| = 2x^+ - x$. To verify this,

(i) If $x \leq 0$, then $x^+ = 0$ so,

$$2x^+ - x = 2(0) - x = -x = |x|.$$

(ii) If $x > 0$, then $x^+ = x$ so,

$$2x^+ - x = 2x - x = x = |x|.$$

Therefore, $|X_n| = 2X_n^+ - X_n$. Taking expectations and using linearity,

$$\mathbb{E}|X_n| = 2\mathbb{E}X_n^+ - \mathbb{E}X_n.$$

Because $(X_n)_{n\geq 0}$ is a submartingale, $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n$. Taking expectations,

$$\begin{aligned}
\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] &\geq X_n \\
\mathbb{E}[\mathbb{E}[X_{n+1} \mid \mathcal{F}_n]] &\geq \mathbb{E}[X_n], && \text{total expectation} \\
\mathbb{E}[X_{n+1}] &\geq \mathbb{E}[X_n], && \text{repeated-iteration} \\
&\geq \mathbb{E}[X_{n-1}] \\
&\geq \cdots \\
&\geq \mathbb{E}[X_0].
\end{aligned}$$

Multiplying by -1 reverses the inequality so $-\mathbb{E}[X_n] \leq -\mathbb{E}[X_0]$.

Also by definition of the supremum, $\mathbb{E}[X_n^+] \leq \sup_{k\geq 0} \mathbb{E}[X_k^+]$. Combining these bounds we obtain,

$$\begin{aligned}\mathbb{E}|X_n| &= 2\mathbb{E}X_n^+ - \mathbb{E}X_n \\ &\leq 2\mathbb{E}X_n^+ - \mathbb{E}X_0 \\ &\leq 2\sup_{k\geq 0}\mathbb{E}[X_k^+] - \mathbb{E}X_0 < \infty \quad \forall n.\end{aligned}$$

The bound is finite because the supremum is finite by hypothesis and X_0 is integrable by definition of a submartingale. We wish to now rule out convergence to either infinity.

Fix $R > 0$. Applying Markov's inequality to the nonnegative random variable $|X_n|$,

$$\begin{aligned}\mathbb{P}(|X_n| \geq R) &\leq \frac{\mathbb{E}|X_n|}{R} \\ &\leq \frac{2\sup_{k\geq 0}\mathbb{E}X_k^+ - \mathbb{E}X_0}{R}.\end{aligned}$$

If $X_n(\omega) \to +\infty$, then eventually $X_n(\omega) \geq R$ so eventually $|X_n(\omega)| \geq R$. Otherwise, if $X_n(\omega) \to -\infty$, then eventually $X_n(\omega) \leq -R$ so again eventually $|X_n(\omega)| \geq R$.

In either case, there exists some time m such that $|X_n(\omega)| \geq R$ for every $n \geq m$.

In event notation,

$$\{X_n \to +\infty\} \cup \{X_n \to -\infty\} \subseteq \bigcup_{m\geq 0}\bigcap_{n\geq m}\{|X_n| \geq R\}.$$

For each fixed m,

$$\bigcap_{n\geq m}\{|X_n| \geq R\} \subseteq \{|X_m| \geq R\}$$

because the intersection requires the inequality at every time $n \geq m$ including $n = m$.

Consequently,

$$\mathbb{P}\left(\bigcap_{n\geq m}\{|X_n| \geq R\}\right) \leq \mathbb{P}(|X_m| \geq R) \leq \frac{2\sup_{k\geq 0}\mathbb{E}X_k^+ - \mathbb{E}X_0}{R}.$$

Furthermore we may write, $\bigcap_{n\geq m}\{|X_n| \geq R\} \subseteq \bigcap_{n\geq m+1}\{|X_n| \geq R\}$. The right-hand event imposes fewer requirements as it no longer requires the inequality at m. Hence these events increase as m increases. By continuity of probability from below,

$$\begin{aligned}\mathbb{P}\left(\bigcup_{m\geq 0}\bigcap_{n\geq m}\{|X_n| \geq R\}\right) &= \lim_{m\to\infty}\mathbb{P}\left(\bigcap_{n\geq m}\{|X_n| \geq R\}\right) \\ &\leq \frac{2\sup_{k\geq 0}\mathbb{E}X_k^+ - \mathbb{E}X_0}{R} \longrightarrow 0, \qquad R \to \infty.\end{aligned}$$

Hence, with the earlier event inclusion,

$$0 \leq \mathbb{P}(\{X_n \to +\infty\} \cup \{X_n \to -\infty\}) \leq \frac{2\sup_{k\geq 0}\mathbb{E}X_k^+ - \mathbb{E}X_0}{R} \longrightarrow 0.$$

This holds for every $R > 0$ since the numerator is a fixed finite number. We have proved that X_n converges almost surely in the extended real line, and that the probability of either infinite limit is zero. Therefore,

$$\mathbb{P}\left(\lim_{n\to\infty} X_n \text{ exists and belongs to } \mathbb{R}\right) = 1.$$

Define X equal to this limit wherever it exists and is finite, and set $X = 0$ on the exceptional null event. This is a real-valued random variable and,

$$X_n \xrightarrow{a.s.} X.$$

□

Two useful cases follow directly from the *Martingale Convergence Theorem.*

Corollary 3.1. *Let $(X_n)_{n\geq 0}$ be a process satisfying either of the following conditions.*

(i) $(X_n)_{n\geq 0}$ is a martingale and

$$\sup_{n\geq 0} \mathbb{E}[|X_n|] < \infty.$$

(ii) $(X_n)_{n\geq 0}$ is a nonnegative supermartingale

$$X_n \geq 0 \quad \textit{almost surely } \forall n \geq 0.$$

Then there exists a finite random variable X such that,

$$X_n \longrightarrow X \quad \textit{almost surely.}$$

Proof. We verify the hypotheses of *Corollary 3.1* in each case.

(i) Martingale with Uniformly Bounded expected absolute values.

Since $(X_n)_{n\geq 0}$ is a martingale,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$$

so it is, in particular, a submartingale. Also $0 \leq X_n^+ \leq |X_n|$.

Taking expectations then the supremum over n,

$$\begin{aligned} \mathbb{E}[X_n^+] &\leq \mathbb{E}[|X_n|], \\ \sup_{n\geq 0} \mathbb{E}[X_n^+] &\leq \sup_{n\geq 0} \mathbb{E}[|X_n|] < \infty. \end{aligned}$$

Thus, $(X_n)_{n\geq 0}$ satisfies the hypotheses of the *Martingale Convergence Theorem.* Consequently, there exists a finite random variable X such that,

$$X_n \xrightarrow{a.s.} X \quad \text{almost surely.}$$

(ii) A Nonnegative Supermartingale.

Define $Y_n := -X_n$. Because X_n is adapted and integrable, so is Y_n.

Moreover, the supermartingale property gives,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n.$$

Multiplying by -1 reverses the inequality so by conditional inequality,

$$\begin{aligned}\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[-X_{n+1} \mid \mathcal{F}_n] \\ &= -\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \\ &\geq -X_n = Y_n.\end{aligned}$$

Therefore, $(Y_n)_{n\geq 0}$ is a submartingale. Since $X_n \geq 0$ almost surely,

$$Y_n = -X_n \leq 0 \quad \text{almost surely.}$$

Its positive part is therefore zero,

$$Y_n^+ = (-X_n)^+ = 0 \quad \text{almost surely.}$$

Hence, $\sup_{n\geq 0} \mathbb{E}[Y_n^+] = 0 < \infty$. Applying the *Martingale Convergence Theorem* to $(Y_n)_{n\geq 0}$, there exists a finite random variable Y such that,

$$Y_n \xrightarrow{a.s.} X.$$

Negating both sides,

$$X_n = -Y_n \xrightarrow{a.s.} -Y.$$

Setting $X := -Y$ gives the required finite limit. Finally, because there are countably many times, the inequalities $X_n \geq 0$ hold simultaneously on an event of probability one. On that event, whenever the sequence converges, its limit is nonnegative. Hence,

$$X \geq 0 \quad \text{almost surely.}$$

This completes the proof.

□

The convergence theorem guarantees a finite almost-sure limit under suitable bounds. Identifying that limit is a separate question. In the first example, the process can converge only by reaching its absorbing state. In the second, the proportion of red balls converges, but the argument does not identify its limiting distribution.

Example 3.15 (A Simple Random Walk Absorbed at Zero)**.** Let $\xi_1, \xi_2, \dots$ be independent symmetric Rademacher random variables and define the associated trivial σ-algebra $\mathcal{F}_0$ and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$. Let $S_0 = 1$ and,

$$S_n = S_{n-1} + \mathbb{I}_{\{S_{n-1}>0\}}\xi_n, \quad n \geq 1.$$

When $S_{n-1} > 0$, the next step is $S_n = S_{n-1} + \xi_n$ so the process moves up or down by one. When $S_{n-1} = 0$, the indicator is zero, giving

$$S_n = S_{n-1} = 0.$$

Thus, zero is an absorbing state. Since the process starts at 1 and takes unit steps while positive, it cannot become negative. We first verify that $(S_n)_{n\geq 0}$ is a martingale.

S_n is determined by $\xi_1, \dots, \xi_n$ so it is $\mathcal{F}_n$-measurable. Also,

$$0 \leq S_n \leq n + 1$$

so therefore,

$$\mathbb{E}[|S_n|] \leq n + 1 < \infty.$$

Hence the process is adapted and integrable at every time.

Since S_{n-1} and $\mathbb{I}_{\{S_{n-1}>0\}}$ are $\mathcal{F}_{n-1}$-measurable

$$\begin{aligned}
\mathbb{E}[S_n \mid \mathcal{F}_{n-1}] &= \mathbb{E}\left[S_{n-1} + \mathbb{I}_{\{S_{n-1}>0\}}\xi_n \mid \mathcal{F}_{n-1}\right] \\
&= S_{n-1} + \mathbb{I}_{\{S_{n-1}>0\}}\mathbb{E}[\xi_n \mid \mathcal{F}_{n-1}] \\
&= S_{n-1} + \mathbb{I}_{\{S_{n-1}>0\}}\mathbb{E}[\xi_n], \qquad \xi_n \perp \mathcal{F}_{n-1} \\
&= S_{n-1}.
\end{aligned}$$

Hence, $(S_n)_{n\geq 0}$ is a nonnegative martingale and, in particular, a nonnegative supermartingale. By *Corollary 3.1* there exists a finite random variable S_∞ such that,

$$S_n \xrightarrow{a.s.} S_\infty.$$

Since $S_n \geq 0$, we also have that $S_\infty \geq 0$ almost surely. We now identify the limit.

Fix a sample path on which S_n converges to a finite value.

By *Martingale Convergence Theorem*, there exists N such that,

$$|S_n - S_\infty| < \frac{1}{4} \quad \forall n \geq N$$

where $1/4$ is an arbitrary ε. For any $m, n \geq N$,

$$\begin{aligned}
|S_n - S_\infty| &\leq |S_n - S_\infty| + |S_m - S_\infty| \\
&< \frac{1}{4} + \frac{1}{4} \\
&= \frac{1}{2}.
\end{aligned}$$

But $S_n - S_m$ is an integer and the only integer with absolute value less than $\frac{1}{2}$ is zero so,

$$S_n = S_m \quad \forall m, n \geq N.$$

Therefore, a convergent integer-valued sequence must eventually be constant. That eventual constant cannot be positive. If $S_N > 0$, then the recursion requires,

$$S_{N+1} - S_N = \xi_{N+1} \in \{-1, 1\}$$

contradicting eventual constancy. Hence, the eventual constant is zero. We conclude,

$$S_n \xrightarrow{a.s.} 0.$$

Moreover, the process reaches zero in finite time almost surely and remains there afterward,

$$\mathbb{P}(\inf\{n \geq 0 : S_n = 0\} < \infty) = 1.$$

Although the almost-sure limit is zero, the martingale property gives,

$$\mathbb{E}[S_n] = \mathbb{E}[S_0] = 1 \quad \forall n.$$

Hence, again, almost-sure convergence does not imply convergence of expectations.

Example 3.16 (Polya's Urn)**.** An urn initially contains one red ball and one blue ball. At each turn, draw a ball uniformly at random, return it to the urn, and add one additional ball of the same color. Let $\mathcal{F}_n$ contain the information revealed by the first n draws and let R_n be the number of red balls after those draws. Since the urn begins with two balls and gains one ball per draw, its total number of balls at time n is $n + 2$.

Define the proportion of red balls by,

$$M_n := \frac{R_n}{n+2}, \quad n \geq 0.$$

In particular, $R_0 = 1$ and $M_0 = \frac{1}{2}$. We show that $(M_n)_{n\geq 0}$ is a martingale and therefore converges almost surely. The first n draws determine R_n, so R_n and M_n are $\mathcal{F}_n$-measurable.

Moreover, $0 \leq M_n \leq 1$ so $\mathbb{E}[|M_n|] \leq 1$. Hence, the process is adapted and integrable. Conditional on $\mathcal{F}_n$, the current composition of the urn is known. Therefore,

$$\mathbb{P}(\text{red on draw } n+1 \mid \mathcal{F}_n) = \frac{R_n}{n+2} = M_n,$$
$$\mathbb{P}(\text{blue on draw } n+1 \mid \mathcal{F}_n) = 1 - M_n.$$

If the next ball is red, an additional red ball is added so,

$$R_{n+1} = R_n + 1, \qquad M_{n+1} = \frac{R_n + 1}{n+3}.$$

Otherwise, if the next ball is blue, the number of red balls remains unchanged so,

$$R_{n+1} = R_n, \qquad M_{n+1}\frac{R_n}{n+3}.$$

Averaging these two possible proportions using their conditional probabilities,

$$\begin{aligned}
\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= M_n \cdot \frac{R_n + 1}{n+3} + (1 - M_n)\frac{R_n}{n+3} \\
&= \frac{M_n(R_n+1) + (1-M_n)R_n}{n+3} \\
&= \frac{M_nR_n + M_n + R_n - M_nR_n}{n+3} \\
&= \frac{R_n + M_n}{n+3} = \frac{R_n + \frac{R_n}{n+2}}{n+3} \\
&= \frac{R_n(n+2) + R_n}{n+3} \\
&= \frac{\frac{R_n(n+2)+R_n}{n+2}}{n+3} = \frac{R_n(n+3)}{(n+2)(n+3)} \\
&= \frac{R_n}{n+2} \\
&= M_n.
\end{aligned}$$

Hence, $(M_n)_{n\geq 0}$ is a martingale. Because $0 \leq M_n \leq 1$, $\sup_{n\geq 0} \mathbb{E}[|M_n|] \leq 1 < \infty$. By *Corollary 3.1*, we have a finite random variable M_∞ such that,

$$M_n \xrightarrow{a.s.} M_\infty.$$

Since $M_n \in [0,1]$, its limit also satisfies $0 \leq M_\infty \leq 1$ almost surely.

Thus, the proportion of red balls converges along almost every sequence of draws. The martingale property says that, given the current composition, the expected proportion after the next draw equals the current proportion. However, it does not make a statement that says the proportion is unchanged after each draw, or that its almost-sure limit must equal its initial value $\frac{1}{2}$.

3.9 Backward Martingales & Convergence

Definition 3.9 (Backward Filtration and Backward Martingale)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A family of sub-σ-algebras $(\mathcal{F}_n)_{n\leq 0}$ is a *backward filtration* if,

$$\cdots \subseteq \mathcal{F}_{-2} \subseteq \mathcal{F}_{-1} \subseteq \mathcal{F}_0 \subseteq \mathcal{F}.$$

A process $(M_n)_{n\leq 0}$ is a *backward martingale* with respect to this filtration if,

(i) M_n is $\mathcal{F}_n$-measurable for every $n \leq 0$,
(ii) $\mathbb{E}[|M_n|] < \infty$ for every $n \leq 0$, and
(iii) for every $n \leq -1$,

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \quad \text{almost surely.}$$

A backward martingale is indexed by $0, -1, -2, \ldots$. As the index moves toward $-\infty$, the available information becomes smaller. Its values are successive conditional expectations based on this decreasing information.

Remark 3.12 (Direction of Information). The variable M_{n+1} is measurable with respect to the larger information set $\mathcal{F}_{n+1}$. Conditioning it on the smaller information set $\mathcal{F}_n$ recovers M_n. Repeated application of the tower property gives,

$$M_n = \mathbb{E}[M_0 \mid \mathcal{F}_n], \quad n \leq 0.$$

For example,

$$\begin{aligned} M_{-2} &= \mathbb{E}[M_{-1} \mid \mathcal{F}_{-2}] \\ &= \mathbb{E}[\mathbb{E}[M_0 \mid \mathcal{F}_{-1}] \mid \mathcal{F}_{-2}] \\ &= \mathbb{E}[M_0 \mid \mathcal{F}_{-2}]. \end{aligned}$$

Thus, moving toward $-\infty$ means estimating the same integrable variable M_0 using progressively less information, hence the name, *Backward* martingale.

Theorem 3.5 (Backward Martingale Convergence)**.** *Let $(M_n)_{n\leq 0}$ be a backward martingale with respect to $(\mathcal{F}_n)_{n\leq 0}$. Then there exists a finite, integrable random variable $M_{-\infty}$ such that,*

$$M_n \to M_{-\infty} \quad \textit{almost surely as } n \to \infty.$$

Proof. We follow the upcrossing argument used for forward martingale convergence. First we obtain a uniform bound on the expected absolute values. Then we control upcrossings over finite blocks and use this to establish convergence as the left endpoint moves toward $-\infty$.

Since

$$M_n = \mathbb{E}[M_{n+1} \mid \mathcal{F}_n]$$

conditional Jensen's inequality for the absolute-value function gives,

$$|M_n| = |\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]| \leq \mathbb{E}[|M_{n+1}| \mid \mathcal{F}_n].$$

Taking expectations of this inequality we obtain,

$$\mathbb{E}[|M_n|] \leq \mathbb{E}[\mathbb{E}[|M_{n+1}| \mid \mathcal{F}_n]] = \mathbb{E}[|M_{n+1}|].$$

Iterating from n to 0,

$$\mathbb{E}[|M_n|] \leq \mathbb{E}[|M_{n+1}|] \leq \cdots \leq \mathbb{E}[|M_0|].$$

Set $C := \mathbb{E}[|M_0|] < \infty$. Therefore, $\sup_{n\leq 0} \mathbb{E}[|M_n|] \leq C$. This bounds the expectations of the absolute values however. does not assert that $|M_n| \leq C$ almost surely.

Suppose we fix rational numbers $a < b$. For each integer $N \geq 0$, let $U_N(a, b)$ count the completed upcrossings of $[a, b]$ in the block $M_{-N}, M_{-N+1}, \ldots, M_0$ read from left to right in increasing time order. Each such finite block is a martingale with respect to,

$$\mathcal{F}_{-N} \subseteq \mathcal{F}_{-N+1} \subseteq \cdots \subseteq \mathcal{F}_0.$$

It therefore is a submartingale so the upcrossing inequality applies,

$$\begin{aligned}(b-a)\mathbb{E}[U_N(a,b)] &\leq \mathbb{E}[(M_0-a)^+] - \mathbb{E}[(M_{-N}-a)^+] \\ &\leq \mathbb{E}[(M_0-a)^+] \\ &\leq \mathbb{E}[|M_0|] + |a| \\ &= C + |a|.\end{aligned}$$

Hence, re-arranging we obtain $\mathbb{E}[U_N(a,b)] \leq \dfrac{C+|a|}{b-a}$ for $N \geq 0$. Hence, we see that the bound is independent of how far into the past the block extends.

Increasing N adds observations to the beginning of the block that is,

$$(M_{-N}, \ldots, M_0) \longrightarrow (M_{-N+1}, M_{-N}, \ldots, M_0).$$

Every collection of completed upcrossings in the shorter block is still available in the longer block. The usual construction, which always starts and completes each upcrossing as early as possible, counts the maximum number of successive completed upcrossings.

Consequently, $U_N(a,b) \leq U_{N+1}(a,b)$ and so $U_N(a,b) \to \sup_{N\geq 0} U_N(a,b)$.

Hence, by monotone convergence theorem,

$$\mathbb{E}\left[\sup_{N\geq 0} U_N(a,b)\right] = \lim_{N\to\infty} \mathbb{E}[U_N(a,b)] \leq \frac{C+|a|}{b-a} < \infty.$$

For any $r > 0$, Markov's inequality gives,

$$\mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) \geq r\right) \leq \frac{\mathbb{E}[\sup_{N\geq 0} U_N(a,b)]}{r} \leq \frac{C+|a|}{r(b-a)}.$$

Letting $r \to \infty$,

$$0 \leq \mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) = \infty\right) \leq \frac{C+|a|}{r(b-a)} \longrightarrow 0.$$

Hence, by complements, for this fixed rational interval,

$$\mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) < \infty\right) = 1.$$

There are countably many rational pairs $a < b$ hence by countable subadditivity,

$$\mathbb{P}\left(\bigcup_{a<b}\left\{\sup_{N\geq 0} U_N(a,b) = \infty\right\}\right) \leq \sum_{a<b} \mathbb{P}\left(\sup_{N\geq 0} U_N(a,b) = \infty\right) = 0.$$

Again, by complements,

$$\mathbb{P}\left(\bigcap_{a<b}\left\{\sup_{N\geq 0} U_N(a,b) < \infty\right\}\right) = 1.$$

We must pay attention to the direction of time. The upcrossings above are counted in the order $-N, -N+1, \ldots, 0$ while the desired convergence concerns $M_0, M_{-1}, M_{-2}, \ldots$.

Suppose we fix a sample path for which the upcrossing counts are bounded for every rational interval. Suppose for a contradiction that M_{-N} does not converge in the extended real line as $N \to \infty$. Then,

$$\liminf_{N\to\infty} M_{-N} < \limsup_{N\to\infty} M_{-N}.$$

Choose rational numbers $a < b$ strictly between these two values,

$$\liminf_{N\to\infty} M_{-N} < a < b < \limsup_{N\to\infty} M_{-N}.$$

As in the finite upcrossing argument, this implies that there are arbitrarily large N for which $M_{-N} < a$ and arbitrarily large N for which $M_{-N} \geq b$.

Let $j \geq 1$ be arbitrary. We can choose,

$$0 \leq N_1 < N_2 < \cdots < N_{2j}$$

such that,

$$\begin{aligned} M_{-N_1} \geq 1, \quad & M_{-N_2} \leq a, \\ M_{-N_3} \geq b, \quad & M_{-N_4} \leq a, \\ \vdots \quad & \vdots \\ M_{-N_{2j-1}} \geq b, \quad & M_{-N_{2j}} \leq a. \end{aligned}$$

Each new index exists because both types of visits occur arbitrarily far into the past. Now, read these observations in increasing time order,

$$-N_{2j} < -N_{2j-1} < \cdots < -N_2 < -N_1.$$

In this order, the values alternate from at or below a to at or above b,

$$M_{-N_{2j}} \leq a, \quad , M_{-N_{2j-1}} \geq b, \quad \cdots \quad , M_{-N_2} \leq a, \quad M_{-N_1} \geq b.$$

Hence, the block $M_{-N_{2j}}, \ldots, M_0$ contains at least j completed upcrossings of $[a, b]$. Therefore, $U_{N_{2j}}(a, b) \geq j$. Since j was arbitrary, $\sup_{N\geq 0} U_N(a, b) = \infty$ contradicting our choice of sample path. Therefore, almost surely, M_n converges as $n \to -\infty$ to a limit in,

$$\mathbb{R} \cup \{-\infty, +\infty\}.$$

For every $R > 0$ and every integer $N \geq 0$, we can apply Markov's inequality to obtain a bound of the form $\frac{C}{R}$ for $\mathbb{P}(|M_{-n}| \geq R)$. Hence if $M_{-N} \to +\infty$ or $M_{-N} \to -\infty$ then eventually $|M_{-N}| \geq R$ for every subsequent R. We use $\sup_{n\leq 0} \mathbb{E}|M_n| \leq C$ to rule out convergence to $\pm\infty$.

We have redacted this portion as the claims are equivalent to those in the proof of the *Martingale Convergence Theorem*. Hence, the limit is finite almost surely. Therefore,

$$M_n \xrightarrow{a.s.} M_{-\infty} \quad n \to -\infty.$$

□

Sample averages can be viewed as a backward martingale when we retain their aggregate value but discard the information about the individual observations making up that average.

Example 3.17 (Sample Averages as a Backward Martingale)**.** Let $X_1, X_2, \dots$ be an infinite sequence of independent, identically distributed, integrable random variables. Define,

$$S_n := \frac{1}{n}\sum_{i=1}^{n} X_i, \quad n \geq 1$$

and set $M_{-n} := S_n$ with $\mathcal{F}_{-n} := \sigma(S_n, X_{n+1}, X_{n+2}, \dots)$.

The information $\mathcal{F}_{-n}$ consists of the average of the first n observations together with every subsequent observation but does not generally reveal the first n observations individually.

We show that $\mathcal{F}_{-(n+1)} \subseteq \mathcal{F}_{-n}$ and that $\mathbb{E}[X_1 \mid \mathcal{F}_{-n}] = S_n$. The backward-martingale property will then follow from the tower property. For every $n \geq 1$,

$$\begin{aligned} S_{n+1} &= \frac{1}{n+1}\sum_{i=1}^{n+1} X_i \\ &= \frac{\sum_{i=1}^{n} X_i + X_{n+1}}{n+1} \\ &= \frac{nS_n + X_{n+1}}{n+1}. \end{aligned}$$

Both S_n and X_{n+1} are F_{-n}-measurable so S_{n+1} is also $\mathcal{F}_{-n}$-measurable. Moreover, every variable in the list $X_{n+2}, X_{n+3}, \dots$ is already $\mathcal{F}_{-n}$-measurable.

Moreover, every variable in the list $X_{n+2}, X_{n+3}, \dots$ is already $\mathcal{F}_{-n}$-measurable. Thus, all the variables generating $\mathcal{F}_{-(n+1)} = \sigma(S_{n+1}, X_{n+2}, X_{n+3}, \dots)$ are measurable with respect to $\mathcal{F}_{-n}$.

Therefore, $\mathcal{F}_{-(n+1)} \subseteq \mathcal{F}_{-n}$. Consequently,

$$\cdots \subseteq \mathcal{F}_{-3} \subseteq \mathcal{F}_{-2} \subseteq \mathcal{F}_{-1}.$$

Moving from $\mathcal{F}_{-n}$ to $\mathcal{F}_{-(n+1)}$, we combine S_n and X_{n+1} into the new average S_{n+1} We retain this combined average but no longer retain X_{n+1} separately.

By construction, S_n is $\mathcal{F}_{-n}$-measurable and $\mathbb{E}[|S_n|] < \infty$. Hence, S_n satisfies the measurability and integrability requirements for being a conditional expectation.

Fix $A \in \mathcal{F}_{-n}$. We must show that,

$$\mathbb{E}[X_1 \mathbb{I}_A] = \mathbb{E}[S_n \mathbb{I}_A].$$

Permuting $X_1, \dots, X_n$ leaves their average unchanged. Moreover, it leaves every later observation $X_{n+1}, X_{n+2}, \dots$ unchanged. Hence, such a permutation leaves unchanged the information generating $\mathcal{F}_{-n}$ and therefore whether the event A occurs.

Because the observations are i.i.d., exchanging X_1 and X_i does not change their joint distribution with the remaining observations. Under this exchange, $\mathbb{I}_A$ remains unchanged while X_1 becomes X_i. Consequently, $\mathbb{E}[X_1 \mathbb{I}_A] = \mathbb{E}[X_i \mathbb{I}_A]$ for $i = 1, \dots, n$.

All expectations are finite and since each of these n expectations is precisely $\mathbb{E}[X_1\mathbb{I}_A]$, their average equals that same quantity hence,

$$\begin{aligned}\mathbb{E}[X_1\mathbb{I}_A] &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i\mathbb{I}_A] \\ &= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}X_i\mathbb{I}_A\right] \\ &= \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right)\mathbb{I}_A\right] = \mathbb{E}[S_n\mathbb{I}_A].\end{aligned}$$

This holds for every $A \in \mathcal{F}_{-n}$. Together with measurability and integrability, it proves that $S_n = \mathbb{E}[X_1 \mid \mathcal{F}_{-n}]$ almost surely. By the same symmetry argument,

$$\mathbb{E}[X_i \mid \mathcal{F}_{-n}] = S_n, \quad i = 1, \ldots, n.$$

Since $\mathcal{F}_{-(n+1)} \subseteq \mathcal{F}_{-n}$, the tower property gives,

$$\begin{aligned}\mathbb{E}[M_{-n} \mid \mathcal{F}_{-(n+1)}] &= \mathbb{E}[S_n \mid \mathcal{F}_{-(n+1)}] \\ &= \mathbb{E}[\mathbb{E}[X_1 \mid \mathcal{F}_{-n}] \mid \mathcal{F}_{-(n+1)}] \\ &= \mathbb{E}[X_1 \mid \mathcal{F}_{-(n+1)}] = S_{n+1} = M_{-(n+1)}.\end{aligned}$$

Thus, $(M_{-n})_{n\geq 1}$ is a backward martingale. This process is indexed by $-1, -2, \ldots$. Thus, to match the definition with a time-zero endpoint, we may set $M_0 := X_1$ and $\mathcal{F}_0 := \mathcal{F}_{-1}$. Since $M_{-1} = S_1 = X_1$, the additional conditional expectation identity holds automatically.

By the *Backward Martingale Convergence Theorem*, there exists a finite, integrable random variable $M_{-\infty}$ such that,

$$S_n = \frac{1}{n}\sum_{i=1}^{n}X_i \xrightarrow{a.s.} M_{-\infty} \quad n \to \infty.$$

3.10 Strong Law of Large Numbers

The backward-martingale argument shows that averages of i.i.d. integrable random variables converge almost surely to some finite random variable. To prove the *Strong Law of Large Numbers*, we must identify that limit as their common mean.

We will first explain how the Weak Law identifies the limit when the variance is finite. We then use truncation to remove the finite-variance assumption.

Theorem 3.6 (Strong Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be independent, identically distributed random variables satisfying,*

$$\mathbb{E}[|X_1|] < \infty, \qquad \mu := \mathbb{E}[X_1].$$

Define the sample averages,

$$S_n := \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad n \geq 1.$$

Then, $S_n \xrightarrow{a.s.} \mu$.

No assumption that $\mathbb{E}[X_1^2] < \infty$ is required. Before proving the theorem in full, consider the simpler case in which the second moment is finite. That is, suppose additionally that,

$$\mathbb{E}[X_1^2] < \infty, \qquad \mathrm{Var}(X_1) = \sigma^2.$$

By *Backward Martingale Convergence Theorem*, there exists a finite random variable S_∞ such that $S_n \xrightarrow{a.s.} S_\infty$. Meanwhile, the Weak Law of Large Numbers gives,

$$S_n \xrightarrow{\mathbb{P}} \mu.$$

We show that $S_\infty = \mu$ almost surely. Fix $\varepsilon > 0$. By triangle inequality,

$$|S_\infty - \mu| \leq |S_\infty - S_n| + |S_n - \mu|.$$

If both terms on the right are at most $\varepsilon/2$, their sum is at most ε. Therefore,

$$\{|S_\infty - \mu| > \varepsilon\} \subseteq \left\{|S_\infty - S_n| > \frac{\varepsilon}{2}\right\} \cup \left\{|S_n - \mu| > \frac{\varepsilon}{2}\right\}.$$

Taking probabilities,

$$\mathbb{P}(|S_\infty - \mu| > \varepsilon) \leq \mathbb{P}\left(|S_\infty - S_n| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(|S_n - \mu| > \frac{\varepsilon}{2}\right).$$

The first probability tends to zero because almost-sure convergence implies convergence in probability. The second tends to zero by the Weak Law. Hence,

$$\mathbb{P}(|S_\infty - \mu| > \varepsilon) = 0.$$

Since this holds for every $\varepsilon > 0$,

$$\begin{aligned}\mathbb{P}(S_\infty \neq \mu) &= \mathbb{P}\left(\bigcup_{m=1}^{\infty}\left\{|S_\infty - \mu| > \frac{1}{m}\right\}\right)\\ &\leq \sum_{m=1}^{\infty}\mathbb{P}\left(|S_\infty - \mu| > \frac{1}{m}\right)\\ &= 0.\end{aligned}$$

Thus, $S_\infty = \mu$ almost surely, proving the Strong Law under the additional finite-variance assumption. For the general integrable case, we will first establish convergence in probability without assuming a finite variance.

The following inequality will control the averages of bounded approximations.

Lemma 3.3 (Cauchy-Schwarz Inequality)**.** *If $X, Y \in \mathcal{L}^2$ then,*

$$|\mathbb{E}[XY]| \leq \left(\mathbb{E}[X^2]\mathbb{E}[Y^2]\right)^{1/2}.$$

Proof. First, XY is integrable because,

$$2|XY| \leq X^2 + Y^2.$$

Therefore,

$$\mathbb{E}[|XY|] \leq \frac{1}{2}\mathbb{E}[X^2] + \frac{1}{2}\mathbb{E}[Y^2] < \infty.$$

For every $t \in \mathbb{R}$, the square $(X - tY)^2$ is nonnegative. Expanding and taking expectations,

$$\begin{aligned}0 &\leq \mathbb{E}[(X - tY)^2]\\ &= \mathbb{E}[X^2 - 2tXY + t^2Y^2]\\ &= \mathbb{E}[X^2] - 2t\mathbb{E}[XY] + t^2 + t^2\mathbb{E}[Y^2].\end{aligned}$$

If $\mathbb{E}[Y^2] = 0$ then $Y = 0$ almost surely. So $\mathbb{E}[XY] = 0$ and the desired inequality holds.

Suppose instead that $\mathbb{E}[Y^2] > 0$ and choose,

$$t = \frac{\mathbb{E}[XY]}{\mathbb{E}[Y^2]}.$$

Substituting,

$$\begin{aligned}0 &\leq \mathbb{E}[X^2] - 2\frac{\mathbb{E}[XY]}{\mathbb{E}[Y^2]}\mathbb{E}[XY] + \left(\frac{\mathbb{E}[XY]}{\mathbb{E}[Y^2]}\right)^2\mathbb{E}[Y^2]\\ &= \mathbb{E}[X^2] - \frac{2(\mathbb{E}[XY])^2}{\mathbb{E}[Y^2]} + \frac{(\mathbb{E}[XY])^2}{\mathbb{E}[Y^2]}\\ &= \mathbb{E}[X^2] - \frac{(\mathbb{E}[XY])^2}{\mathbb{E}[Y^2]}.\end{aligned}$$

Re-arranging,

$$\frac{(\mathbb{E}[XY])^2}{\mathbb{E}[Y^2]} \leq \mathbb{E}[X^2].$$

Multiplying by $\mathbb{E}[Y^2] > 0$,

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

Taking square roots proves the result.

□

Remark 3.13 (Inner-Product Form). In $\mathcal{L}^2$, the inner product and norm are,

$$\langle X, Y\rangle = \mathbb{E}[XY], \qquad \|X\|_2 = (\mathbb{E}[X^2])^{1/2}.$$

Thus, Cauchy-Schwarz can be written as,

$$|\langle X, Y\rangle| \leq \|X\|_2 \|Y\|_2.$$

The same inequality holds in any inner-product space.

In particular, applying it to $|X|$ and the constant random variable 1 gives,

$$\mathbb{E}[|X|] \leq (\mathbb{E}[X^2])^{1/2}.$$

Proof of SLLN. Recall that,

$$M_{-n} := S_n, \qquad \mathcal{F}_{-n} := \sigma(S_n, X_{n+1}, X_{n+2}, \dots)$$

define a backward martingale. Therefore, the *Backward Martingale Convergence Theorem* gives a finite random variable S_∞ such that $S_n \xrightarrow{a.s.} S_\infty$ almost surely.

It remains for us to identify S_∞. For $K > 0$ define,

$$X_i^{(K)} := (X_i \wedge K) \vee (-K) = \begin{cases} -K, & X_i < -K, \\ X_i, & -K \leq X_i \leq K, \\ K, & X_i > K. \end{cases}$$

This replaces values outside $[-K, K]$ by the nearest endpoint. Define the corresponding average,

$$S_n^{(K)} := \frac{1}{n}\sum_{i=1}^{n} X_i^{(K)}.$$

Since the same Borel-measurable function is applied separately to each X_i, the truncated variables $X_i^{(K)}$ remain independent and identically distributed. Also, $|X_i^{(K)}| \leq K$ so,

$$\mathbb{E}[(X_i^{(K)})^2] \leq K^2 < \infty.$$

Thus, the truncated observations have finite variance.

By linearity of expectation and identical distributions,

$$\begin{aligned}\mathbb{E}[S_n^{(K)}] &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i^{(K)}] \\ &= \frac{1}{n} n\mathbb{E}[X_1^{(K)}] = \mathbb{E}[X_1^{(K)}].\end{aligned}$$

Similarly, by independence,

$$\begin{aligned}\mathrm{Var}(S_n^{(K)}) &= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n}X_i^{(K)}\right) \\ &= \frac{1}{n^2}\,\mathrm{Var}\left(\sum_{i=1}^{n}X_i^{(K)}\right) \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i^{(K)}) \\ &= \frac{1}{n^2} n\,\mathrm{Var}(X_1^{(K)}) = \frac{\mathrm{Var}(X_1^{(K)})}{n}.\end{aligned}$$

Applying *Cauchy-Schwarz* to the absolute deviation and the constant 1,

$$\begin{aligned}\mathbb{E}\left[\left|S_n^{(K)} - \mathbb{E}[X_1^{(K)}]\right|\right] &\le \left(\mathbb{E}\left[\left(S_n^{(K)} - \mathbb{E}[X_1^{(K)}]\right)^2\right]\right)^{1/2} \\ &= \sqrt{\mathrm{Var}(S_n^{(K)})} = \frac{\sqrt{\mathrm{Var}(X_1^{(K)})}}{\sqrt{n}}.\end{aligned}$$

Therefore, for every fixed K, as $n \to \infty$,

$$\mathbb{E}\left[\left|S_n^{(K)} - \mathbb{E}[X_1^{(K)}]\right|\right] \longrightarrow 0.$$

We wish to control the error caused by truncation. Observe that the difference between the original and truncated averages satisfies,

$$\begin{aligned}\mathbb{E}[|S_n - S_n^{(K)}] &= \mathbb{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n}(X_i - X_i^{(K)})\right|\right] \\ &\le \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[|X_i - X_i^{(K)}|], \qquad |\mathbb{E}[X]| \le \mathbb{E}[|X|] \\ &= \frac{1}{n} n\mathbb{E}[|X_1 - X_1^{(K)}|], \qquad \text{by i.i.d.} \\ &= \mathbb{E}[|X_1 - X_1^{(K)}|].\end{aligned}$$

The difference between the two means is bounded by the same quantity,

$$\begin{aligned}\left|\mu - \mathbb{E}[X_1^{(K)}]\right| &= \left|\mathbb{E}[X_1] - \mathbb{E}[X_1^{(K)}]\right| \\ &= \left|\mathbb{E}[X_1 - X_1^{(K)}]\right| \leq \mathbb{E}\left[|X_1 - X_1^{(K)}|\right].\end{aligned}$$

To show that this error vanishes, observe that $X_1 = X_1^{(K)}$ whenever $|X_1| \leq K$.

When $|X_1| > K$,

$$|X_1 - X_1^{(K)}| = |X_1| - K \leq |X_1|.$$

Hence, we can write this absolute difference as an indicator over $|X_1| > K$,

$$|X_1 - X_1^{(K)}| \leq |X_1|\mathbb{I}_{\{|X_1|>K\}}.$$

Taking expectations over this difference,

$$\mathbb{E}\left[|X_1 - X_1^{(K)}|\right] \leq \mathbb{E}[|X_1|\mathbb{I}_{\{|X_1|>K\}}].$$

Because X_1 is integrable, the expectation in its tail tends to zero. Explicitly, $|X_1|\mathbb{I}_{\{|X_1|\leq K\}} \to |X_1|$ as $K \to \infty$. Hence, $\mathbb{E}[|X_1|\mathbb{I}_{\{|X_1|\leq K\}}] \longrightarrow \mathbb{E}[|X_1|]$. Decomposing into two events,

$$\mathbb{E}[|X_1|\mathbb{I}_{\{|X_1|>K\}}] = \mathbb{E}[|X_1|] - \mathbb{E}[|X_1|\mathbb{I}_{\{|X_1|\geq K\}}] \longrightarrow 0.$$

Consequently,

$$\mathbb{E}\left[|X_1 - X_1^{(K)}|\right] \longrightarrow 0, \qquad K \to \infty.$$

We now show convergence of the sample averages in $\mathcal{L}^1$. Compose the error as,

$$S_n - \mu = (S_n - S_n^{(K)}) + \left(S_n^{(K)} - \mathbb{E}[X_1^{(K)}]\right) + \left(\mathbb{E}[X_1^{(K)}] - \mu\right).$$

By the triangle inequality,

$$\begin{aligned}\mathbb{E}[|S_n - \mu|] &\leq \mathbb{E}[|S_n - S_n^{(K)}|] \\ &\quad + \mathbb{E}\left[\left|S_n^{(K)} - \mathbb{E}[X_1^{(K)}]\right|\right] + \left|\mathbb{E}[X_1^{(K)}] - \mu\right| \\ &\leq \mathbb{E}[|X_1 - X_1^{(K)}|] + \frac{\sqrt{\mathrm{Var}(X_1^{(K)})}}{\sqrt{n}} + \mathbb{E}[|X_1 - X_1^{(K)}] \\ &= \frac{\sqrt{\mathrm{Var}(X_1^{(K)})}}{\sqrt{n}} + 2\mathbb{E}[|X_1 - X_1^{(K)}|].\end{aligned}$$

First hold K fixed and let $n \to \infty$. The variance in the numerator is then fixed and finite so,

$$0 \leq \lim_{n\to\infty} \sup \mathbb{E}[|S_n - \mu|] \leq 2\mathbb{E}[|X_1 - X_1^{(K)}|].$$

This inequality holds for every $K > 0$.

Let $K \to \infty$. The right-hand side tends to zero giving $\mathbb{E}[|S_n - \mu|] \to 0$. Thus, $S_n \to \mu$ in $\mathcal{L}^1$. The order of these limits matters as, for fixed K, averaging controls the bounded observations. Afterward, increasing K removes the truncation error.

Finally, we obtain convergence in probability. For every $\varepsilon > 0$, Markov's inequality gives,

$$\mathbb{P}(|S_n - \mu| > \varepsilon) \leq \frac{\mathbb{E}[|S_n - \mu|]}{\varepsilon} \longrightarrow 0.$$

Therefore, $S_n \xrightarrow{\mathbb{P}} \mu$. This establishes the Weak Law under only the assumption $\mathbb{E}[|X_1|] < \infty$. Recall that we previously established that $S_n \xrightarrow{a.s.} S_\infty$ so $S_n \to S_\infty$ in probability as well.

For every $\varepsilon > 0$,

$$\begin{aligned}\mathbb{P}(|S_\infty - \mu| > \varepsilon) &\leq \mathbb{P}\left(|S_\infty - S_n| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(|S_n - \mu| > \frac{\varepsilon}{2}\right) \\ &\longrightarrow 0.\end{aligned}$$

The left-hand side does not depend on n so it must equal zero, $\mathbb{P}(|S_\infty - \mu| > \varepsilon) = 0$.

Thus,

$$\begin{aligned}\mathbb{P}(S_\infty \neq \mu) &= \mathbb{P}\left(\bigcup_{m=1}^{\infty}\left\{|S_\infty - \mu| > \frac{1}{m}\right\}\right) \\ &\leq \sum_{m=1}^{\infty}\mathbb{P}\left(|S_\infty - \mu| > \frac{1}{m}\right) \\ &= 0.\end{aligned}$$

Hence, $S_\infty = \mu$ almost surely and therefore,

$$S_n := \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} \mu.$$

This proves the *Strong Law of Large Numbers*.

□

Remark 3.14. If $X_n \xrightarrow{\mathcal{L}^1}$ then X_n is a Doob martingale,

$$X_n = \mathbb{E}[X_\infty \mid \mathcal{F}_n] = \operatorname{proj}_{\mathcal{L}^2(\mathcal{F}_n)} X_\infty.$$

4 Markov Chains

4.1 Introduction to Markov Theory

A general stochastic process,

$$X_0, X_1, X_2, \ldots$$

may have a next-step distribution that depends on its entire observed history. For example, predicting X_{n+1} might require knowing all of $X_0, \ldots, X_n$.

A *Markov Chain* has a particular simplifying property where once the current state X_n is known, the earlier observations provide no additional information about the distribution of X_{n+1}.

This does not, however, imply that the process has no memory. The current state may summarize information accumulated over many previous steps. The modelling task is to choose a state that contains the information needed to predict the next step.

Example 4.1 (Sequential Language Models)**.** Consider a simplified model that generates a sequence of tokens from a finite vocabulary D. A token may represent a word, part of a word, or punctuation. For each $n \geq 1$, suppose a function,

$$F_n : D^n \longrightarrow \mathcal{P}(D)$$

assigns a probability distribution to every possible history of n tokens. Here $\mathcal{P}(D)$ denotes the set of probability measures on D.

Given the observed history $X_0, \ldots, X_{n-1}$, the model produces the distribution,

$$\mu_n = F_n(X_0, \ldots, X_{n-1})$$

and samples the next token according to that distribution. More precisely,

$$\mathbb{P}(X_n \in B \mid X_0, \ldots, X_{n-1}) = F_n(X_0, \ldots, X_{n-1})(B), \quad B \subseteq D.$$

The distribution of the next token may depend on information from many positions in the history. There is generally no reason for the most recent token alone to contain all the relevant information. Hence, we use a model with a simpler dependence structure.

A *Markov Model* imposes that the current state is sufficient to determine the next-step distribution. We first state some definitions that will be useful.

Definition 4.1 (State Space and Discrete-Time Stochastic Processes)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $E \subseteq \mathbb{R}^d$ be a Borel set. Equip E with the Borel σ-algebra,

$$B(E) := \{E \cap B : B \in \mathcal{B}(\mathbb{R}^d)\}.$$

A discrete-time stochastic process with state space E is a sequence,

$$(X_n)_{n \geq 0}, \qquad X_n : \Omega \longrightarrow E$$

i.e., of measurable maps. The elements of E are called *states*.

For each outcome $\omega \in \Omega$, the sequence,

$$X_0(\omega), X_1(\omega), X_2(\omega), \ldots$$

is a sample path of the process. The natural filtration of such a process is,

$$\mathcal{F}_n^X := \sigma(X_0, \ldots, X_n), \qquad n \geq 0.$$

Thus, $\mathcal{F}_n$ contains information revealed by observing the process up to time n.

Remark 4.1 (The Borel Sets of E). The elements of $\mathcal{B}(E)$ are subsets of E obtained by intersecting E with Borel subsets of $\mathbb{R}^d$. They are *not* Borel sets that contain E.

For example, if $E = [0, \infty)$ then,

$$[1, 3], \quad (2, \infty), \quad \{0\}$$

are all elements of $\mathcal{B}(E)$.

Definition 4.2 (Countable State Space)**.** A state space is *countable* if it is finite or countably infinite. In this case, we equip E with the discrete σ-algebra,

$$2^E := \{B : B \subseteq E\}.$$

If E is a countable subset of $\mathbb{R}^d$, then $\mathcal{B}(E) = 2^E$.

Indeed, every singleton set is Borel, and every subset of E is a finite or countable union of singletons. Consequently, on a countable state space, every subset is an event in the state space, and probabilities can be computed by summing over individual states.

4.2 Markov Property & Transition Kernels

We first describe the *Markov property* without assuming the transition rule is the same at every timestep. Thus, consider the following definition.

Definition 4.3 (Markov Property)**.** Let $(X_n)_{n \geq 0}$ be an E-valued stochastic process with the natural filtration,

$$\mathcal{F}_n^X := \sigma(X_0, \ldots, X_n).$$

The process has the *Markov property* if, for every $n \geq 0$ and $B \in \mathcal{B}(E)$,

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n^X) = \mathbb{P}(X_{n+1} \in B \mid \sigma(X_n)) \quad \text{almost surely.}$$

A process with this property is called a *Markov chain*.

Recall that conditional probability is conditional expectation of an indicator,

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n^X) = \mathbb{E}\left[\mathbb{I}_{\{X_{n+1} \in\}} \middle| \mathcal{F}_n^X\right].$$

On the left-hand side, we condition on the entire observed history. On the right-hand side, we condition *only* on the current state. The *Markov property* states that these should give us the same prediction for every possible target set B.

Remark 4.2 (Conditional Dependence vs. Independence). The *Markov property* does not state that X_{n+1} is independent of X_n or that the observations are independent.

For intuition, tomorrow's weather may depend strongly on today's weather. A Markov weather model, thus, assumes that once today's weather is known, observations from earlier days do not change the next-day distribution.

Definition 4.4 (Markov Property on a Countable State Space)**.** If E is countable, the Markov property can be written using ordinary conditional probabilities.

For every $n \geq 0$ and states $x_0, \ldots, x_n, y \in E$ such that,

$$\mathbb{P}(X_0 = X_0, \ldots, X_n = x_n) > 0$$

we require that,

$$\mathbb{P}(X_{n+1} = y \mid X_0 = x_0, \ldots, X_n = x_n) = \mathbb{P}(X_{n+1} = y \mid X_n = x_n).$$

The positivity assumption ensures that the conditional probability on the left is defined by the usual ratio formula. It also implies that $\mathbb{P}(X_n = x_n) > 0$.

For a general state space, individual states may have probability zero. This is precisely why the formulation using conditional expectations and σ-algebras is more general.

To specify a Markov chain, we need a rule that associates each current state with a probability distribution for the next state. A transition kernel provides exactly this information.

Definition 4.5 (Transition Kernel)**.** A *transition kernel* on E is a function,

$$P : E \times \mathcal{B}(E) \longrightarrow [0, 1],$$

satisfying the following conditions,

(i) For every fixed $x \in E$ the map,

$$B \mapsto P(x, B)$$

is a probability measure on $(E, \mathcal{B}(E))$.

(ii) For every fixed $B \in \mathcal{B}(E)$ the map,

$$x \mapsto P(x, B)$$

is $\mathcal{B}(E)$-measurable.

Hence, we use P for the transition kernel and $\mathbb{P}$ for the probability measure on the underlying state space. Then quantity $P(x, B)$ represents the probability of moving into the set B when the current state is x. For a fixed current state x, the first condition,

$$P(x, E) = 1, \qquad P(x, \emptyset) = 0$$

and, for pairwise disjoint sets $B_1, B_2, \cdots \in \mathcal{B}(E)$,

$$P\left(x, \bigcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} P(x, B_j).$$

Hence, $P(x, \cdot)$ is a complete probability law for the next state.

If $Y \sim P(x, \cdot)$ then,

$$\mathbb{P}(Y \in B) = P(x, B).$$

By (ii), we ensure that we can replace the fixed state x with a random current state X_n. In particular, $P(X_n, B)$ must be a well-defined random variable.

Remark 4.3. Suppose we fix $B \in \mathcal{B}(E)$ and write,

$$f_B(x) := P(x, B).$$

Then, $f_B : E \longrightarrow [0, 1]$ is measurable by the kernel definition.

If $X : \Omega \to E$ is a random variable, the composition,

$$\Omega \xrightarrow{X} E \xrightarrow{f_B} [0, 1]$$

is the random variable $f_B(X) = \mathbb{P}(X, B)$. We verify its measurability. Let $C \in \mathcal{B}(\mathbb{R})$,

$$\begin{aligned} \{\omega : P(X(\omega) \in B) \in C\} &= \{\omega : f_B(X(\omega)) \in C\} \\ &= \{\omega : X(\omega) \in f_B^{-1}(C)\} \\ &= X^{-1}(f_B^{-1}(C)). \end{aligned}$$

Since f_B is measurable, $f_B^{-1}(C) \in \mathcal{B}(E)$. Moreover, since X is measurable,

$$X^{-1}(f_B^{-1}(C)) \in \mathcal{F}.$$

Therefore, probabilities involving $P(X, B)$ are well-defined. If X is $\mathcal{F}_n$-measurable, the same argument shows that $P(X, B)$ is $\mathcal{F}_n$-measurable. Hence, a transition kernel is an arbitrary function satisfying probability-measure and measurability requirements.

4.3 Time-Homogeneous Markov Chains

Definition 4.6 (Time-Homogeneous Markov Chain)**.** Let $(X_n)_{n \geq 0}$ be an E-valued process with natural filtration $\mathcal{F}_n^X := \sigma(X_0, \ldots, X_n)$. The process is a *Time-Homogeneous* Markov chain with transition Kernel P if for every $n \geq 0$ and $B \in \mathcal{B}(E)$,

$$\mathbb{E}\left[\mathbb{I}_{\{X_{n+1} \in B\}} \middle| \mathcal{F}_n^X\right] = \mathbb{P}(X_n, B) \quad \text{almost surely.}$$

Equivalently,

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n^X) = P(X_n, B) \quad \text{almost surely.}$$

The same kernel P is used at every time, hence being time-homogeneous. If the transition rule depends on time, we instead write that,

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n^X) = P_n(X_n, B)$$

where the kernels P_n differ.

Remark 4.4 (Recovering the Markov Property). The kernel identity implies that conditioning on the current state gives the same answer as conditioning on the entire history.

Indeed because $\sigma(X_n) \subseteq \mathcal{F}_n^X$ the tower property gives,

$$\begin{aligned}
\mathbb{E}\left[\mathbb{I}_{\{X_{n+1}\in B\}} \middle| \sigma(X_n)\right] &= \mathbb{E}\left[\mathbb{E}\left[\mathbb{I}_{\{X_{n+1}\in B\}} \middle| \mathcal{F}_n^X\right] \middle| \sigma(X_n)\right] \\
&= \mathbb{E}[P(X_n, B) \mid \sigma(X_n)] \\
&= P(X_n, B), \qquad P(X_n, B) \in \sigma(X_n).
\end{aligned}$$

Hence,

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n^X) = \mathbb{P}(X_{n+1} \in B \mid \sigma(X_n)).$$

Time-homogeneity does not imply that X_n has the same distribution for every n. It simply implies that the rule used to move from one state to the next does not change with time.

The kernel formulation gives a precise meaning to the conditional law of the next state.

Definition 4.7 (Conditional Law)**.** Let $X : \Omega \to E$ be a random variable and let $\mathcal{G} \subseteq \mathcal{F}$.

A *conditional law* of X given $\mathcal{G}$ is a map,

$$\mu : \Omega \times \mathcal{B}(E) \longrightarrow [0, 1]$$

such that,

(i) For each $\omega \in \Omega$ the map $B \mapsto \mu(x, B)$ is a probability measure on E,
(ii) For each $B \in \mathcal{B}(E)$, the map $\omega \mapsto \mu(\omega, B)$ is $\mathcal{G}$-measurable, and
(iii) For every $B \in \mathcal{B}(E)$

$$\mu(\cdot, B) = \mathbb{E}\left[\mathbb{I}_{\{X\in B\}} \middle| \mathcal{G}\right] \quad \text{almost surely.}$$

We denote such a conditional law by $\mathcal{L}(X \mid \mathcal{G})$. Unlike the ordinary law $\mathcal{L}(X \mid \mathcal{G})$, a conditional law depends on the information observed. It is therefore a random probability measure where different outcomes may produce different probability distributions.

For a time-homogeneous Markov chain,

$$\mathcal{L}(X_{n+1} \mid \mathcal{F}_n^X) = P(X_n, \cdot).$$

At an outcome ω, this is the probability measure,

$$B \mapsto P(X_n(\omega), B).$$

Remark 4.5 (Conditional Laws and Test functions). A conditional law can also be used to compute conditional expectations. For every bounded Borel-measurable function $\phi : E \to \mathbb{R}$,

$$\mathbb{E}[\phi(X) \mid \mathcal{G}](\omega) = \int_E \phi(x)\mu(\omega, dx) \quad \text{almost surely.}$$

Taking $\phi = \mathbb{I}_B$ recovers the defining conditional probability,

$$\int_E \mathbb{I}_B(x)\mu(\omega, dx) = \mu(\omega, B).$$

For a Markov chain, the corresponding formula is,

$$\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n^X] = \int_E \phi(y)P(X_n, dy).$$

The right-hand side depends on the observed history only through X_n. Consequently,

$$\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n^X] = \mathbb{E}[\phi(X_{n+1}) \mid \sigma(X_n)].$$

In particular, this holds for bounded continuous test functions.

These describe the same conditional-law comparison through integrals, as in our earlier discussion of laws and convergence in distribution.

Example 4.2 (An Autoregressive Process of Order One)**.** Let $X_0 \sim \mu_0$ and $\xi_1, \xi_2, \ldots \overset{i.i.d.}{\sim} \mathcal{N}(0, 1)$, and assume that the entire sequence of increments is independent of X_0.

Fix a constant $a \in \mathbb{R}$, and define,

$$X_{n+1} = aX_n + \xi_{n+1}, \quad n \geq 0.$$

This is an autoregressive process is order one, abbreviated $AR(1)$.

Let $\mathcal{F}_n^X := \sigma(X_0, \ldots, X_n)$. Every observation up to time n is determined by $X_0, \xi_1, \ldots, \xi_n$ hence,

$$\mathcal{F}_n^X := \sigma(X_0, \xi_1, \ldots, \xi_n).$$

Therefore, $\xi_{n+1} \perp \mathcal{F}_n^X$. Given we are in the current state $X_n = x$, the next state has the form,

$$ax = \xi_{n+1}.$$

Since $\xi_{n+1} \sim \mathcal{N}(0, 1)$, its distribution is $\mathcal{N}(ax, 1)$. Hence, the transition kernel is,

$$P(x, B) = \int_B \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(y - ax)^2}{2}\right) dy, \quad B \in \mathcal{B}(\mathbb{R}).$$

For fixed x, this is a Gaussian probability measure. For fixed B, the displayed integral is a measurable function of x, so it defines a transition kernel. Conditioning on the observed history, X_n is already known while ξ_{n+1} retains its standard normal distribution.

Therefore,

$$\begin{aligned}\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n^X) &= \mathbb{P}(aX_n + \xi_{n+1} \in B \mid \mathcal{F}_n^X) \\ &= \int_B \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(y - aX_n)^2}{2}\right) dy \\ &= P(X_n \in B).\end{aligned}$$

Hence, $(X_n)_{n\geq 0}$ is a time-homogeneous Markov chain. The current state determines the centre aX_n of the next-step Gaussian distribution. The new noise has mean zero and variance one.

On finite or countable spaces, transition kernels can be represented by matrices.

4.4 Transition Laws and Matrices

Definition 4.8 (Stochastic Matrix)**.** Let $E = \{x_i : i \in I\}$ where I is finite or countably infinite. A *stochastic matrix* on E is an array,

$$P = (P_{ij})_{i,j\in I}$$

such that,

$$P_{ij} \geq 0, \qquad \sum_{j\in I} P_{ij} = 1 \qquad \forall i \in I.$$

Each row, hence, is a probability distribution over the possible next states.

Definition 4.9 (Transition Matrix)**.** A *Stochastic Matrix* P is a transition matrix for a time-homogeneous Markov chain if,

$$\mathbb{P}(X_{n+1} = x_j \mid \mathcal{F}_n^X) = P_{ij} \quad \text{on } \{X_n = x_i\}$$

almost surely, for every $n \geq 0$, and $i, j \in I$.

Note that whenever $\mathbb{P}(X_n = x_i) > 0$ this gives,

$$P_{ij} = \mathbb{P}(X_{n+1} = x_j \mid X_n = x_i)$$

The row index i identifies the current state and the column index j identifies the next state.

Remark 4.6 (Matrices and Kernels describe the Same Rule)*.* Given a transition kernel on a countable state space, we can define,

$$P_{ij} := P(x_i, \{x_j\}).$$

By countable additivity,

$$\begin{aligned}\sum_{j\in I} P_{ij} &= \sum_{j\in I} P(x_i, \{x_j\}) \\ &= P(x_i, E) = 1.\end{aligned}$$

Conversely, a stochastic matrix determines a kernel through,

$$P(x_i, B) = \sum_{\{j; x_j \in B\}} P_{ij}, \quad B \subseteq E.$$

For a fixed x_i, this is a probability measure. For fixed B, measurability in x_i is automatic because the state space has discrete σ-algebra 2^E. Thus, matrices describe transitions to individual states, while kernels describe transitions to arbitrary measurable sets of states.

Remark 4.7 (States with Zero Probability). If $\mathbb{P}(X_n = x_i) = 0$ the ratio definition of,

$$\mathbb{P}(X_{n+1} = x_j \mid X_n = x_i)$$

does not determine a transition probability at that time.

A transition matrix nevertheless specifies a row for every state. These rows describe the intended dynamics if the chain is placed in that state. A row may also be determined by the process at another time when that state has positive probability.

4.5 Transition Operator & Chain Joint Distribution

A *Time-homogeneous* Markov chain is specified by two components: its initial distribution, which describes where the process starts, and its transition kernel, which describes how it moves from its current state to the next state.

Throughout, let $(X_n)_{n\geq 0}$ be a time-homogeneous Markov chain on E, with transition kernel P, and natural filtration $\mathcal{F}_n := \sigma(X_0, \ldots, X_n)$. Thus, for every $B \in \mathcal{B}(E)$,

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n) = P(X_n, B) \quad \text{almost surely.}$$

Definition 4.10 (Initial distribution)**.** The *initial distribution* of the Markov chain is the probability measure,

$$\mu_0 := \mathcal{L}(X_0) \quad \text{on } (E, \mathcal{B}(E)).$$

Equivalently, for all $B \in \mathcal{B}(E)$,

$$\mu_0(B) = \mathbb{P}(X_0 \in B).$$

If the chain starts at a deterministic state $x \in E$, then,

$$X_0 = x \quad \text{almost surely,} \quad \mu_0 = \delta_x.$$

The initial distribution and the transition kernel have different roles. The measure μ_0 determines the starting state, while $P(x, \cdot)$ determines the next-state distribution whenever the current state is x. Heuristically, we may construct a factorization of a joint density.

Suppose, temporarily, that the initial distribution and all transition laws have densities,

$$\mu_{0:n} = \mathcal{L}(X_0, \ldots, X_n)$$

and denote its joint density by $p(x_0, \ldots, x_n)$.

The usual factorization of the joint density gives us,

$$\begin{aligned} p(x_0, \ldots, x_n) &= p(x_n \mid x_0, \ldots, x_{n-1}) p(x_0, \ldots, x_{n-1}) \\ &= p(x_n \mid x_{n-1}) p(x_0, \ldots, x_{n-1}) \\ &= p(x_n \mid x_{n-1}) p(x_{n-1} \mid x_{n-2}) p(x_0, \ldots, x_{n-1}) \\ &= \cdots \\ &= p(x_0) \prod_{k=1}^{n} p(x_k \mid x_{k-1}). \end{aligned}$$

Once x_{n-1} is known, the earlier states do not change the conditional distribution of X_n, hence the *Markov property*. Thus, the joint density is obtained by multiplying the initial density by the successive transition densities. However, transition laws need not have densities.

Hence, we want a formulation using probability measures,

$$\mu_0(dx_0), \quad P(x_0, dx_1), \quad P(x_1, dx_2), \quad \ldots.$$

The corresponding iterated-integral formula will give a precise version of this factorization.

Recall that, for each fixed $x \in E$, $B \mapsto P(x, B)$ is a probability measure. The notation $P(x, dy)$ therefore means that we integrate with respect to this measure in the next-state variable y.

In particular,

$$P(x, B) = \int_B P(x, dy).$$

Here, x is the current state, and y ranges over possible next states. The notation $P(x, dy)$ therefore denotes a probability measure but does not assume the existence of a density.

If a transition density does exist, then we may write,

$$P(x, dy) = p(y \mid x) dy \iff P(x, B) = \int_B p(y \mid x) dy.$$

Remark 4.8 (The Countable-State Case). Suppose that $E = \{x_i : i \geq 1\}$.

The transition matrix has entries,

$$P_{ij} := P(x_i, \{x_j\}), \qquad P_{ij} \geq 0, \quad \sum_{j \geq 1} P_{ij} = 1.$$

Then, for any $B \subseteq E$,

$$P(x_i, B) = \sum_{j : x_j \in B} P_{ij}.$$

Equivalently, the transition measure from x_i is.

$$P(x_i, dy) = \sum_{j \geq 1} P_{ij} \delta_{x_j}(dy).$$

Indeed, integrating over B,

$$\int_B P(x_i, dy) = \sum_{j\geq 1} P_{ij}\delta_{x_j}(B) = \sum_{j:x_j\in B} P_{ij} = P(x_i, B).$$

Thus, a transition kernel generalizes a transition matrix to general state spaces. It accommodates discrete laws, laws with densities, mixtures, and other probability measures.

A transition matrix can be applied to a vector by matrix multiplication. To extend this idea to a general state space, we regard a function $\phi : E \to \mathbb{R}$ as assigning one numerical value to each state. On a finite state space, these values form a vector. On a general state space, the function itself plays the corresponding role.

Definition 4.11 (Transition Operator)**.** Let P be a transition kernel on E. For a bounded, measurable function $\phi : E \to \mathbb{R}$, define the *transition operator* as,

$$(P\phi)(x) := \int_E \phi(y)P(x, dy), \qquad x \in E.$$

The operator takes the function ϕ and produces a new function $P\phi$.

At the current state x, the value $(P\phi)(x)$ is the average of ϕ evaluated at the next state, with the next state distributed according to $P(x, \cdot)$.

Example 4.3 (Matrix Multiplication)**.** On a countable state space,

$$(P\phi)(x_i) = \int_E \phi(y)P(x_i, dy) = \sum_{j\geq 1} P_{ij} \int_E \phi(y)\delta_{x_j}(dy) = \sum_{j\geq 1} P_{ij}\phi(x_j).$$

This is precisely the i-th entry of the matrix-vector product $P\phi$. The integral in the general-state space definition therefore replaces the sum in ordinary matrix multiplication.

You may recall from STAT 333 that P_{ij} in our stochastic matrix representation the probability of going from state i to j in one-step. We can show that the transition operator, under a countable finite state space, say $E = \{e_1, \ldots, e_m\}$ is simply matrix multiplication.

Suppose additionally that $\phi : E \to \mathbb{R}$ is a test function. Then, the transition operator is,

$$\begin{aligned}(P\phi)(x) &= \int_E \phi(y)P(x, dy) \\ &= \sum_{k=1}^{m} \phi(e_k)P(x, \{e_k\}).\end{aligned}$$

Define the transition probabilities,

$$p_{ij} := P(e_i, \{e_j\}) = \mathbb{P}(X_{n+1} = e_j \mid X_n = e_i).$$

Then, for each starting state e_i,

$$(P\phi)(e_i) = \sum_{j=1}^{m} p_{ij}\phi(e_j).$$

Expanding the matrix expected test-function values with the transition matrix P,

$$\begin{bmatrix} (P\phi)(e_1) \\ (P\phi)(e_2) \\ \vdots \\ (P\phi)(e_m) \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix} \begin{bmatrix} \phi(e_1) \\ \phi(e_2) \\ \vdots \\ \phi(e_m) \end{bmatrix}.$$

Hence, $P\phi = P\phi$ where the left-hand side denotes the transition operator applied to the function and the right-hand side is the ordinary matrix-vector multiplication using the vector of function values. Each row of P describes the probabilities of moving from a particular starting state to every possible next state. Applying the transition operator m times gives,

$$(P^m)(e_i) = \mathbb{E}[\phi(X_{n+m}) \mid X_n = e_i]$$

corresponding to the matrix multiplication $P^m\phi$ as expected from the Chapman-Kolmogorov identity. The result can be interpreted as a column vector of weighted sums,

$$P\phi = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix} \begin{bmatrix} \phi(e_1) \\ \phi(e_2) \\ \vdots \\ \phi(e_m) \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^{m} p_{1j}\phi(e_j) \\ \sum_{j=1}^{m} p_{2j}\phi(e_j) \\ \vdots \\ \sum_{j=1}^{m} p_{mj}\phi(e_j) \end{bmatrix}.$$

Example 4.4 (A Transition Law with Point Masses and a Density)**.** Suppose that the state set $E \subseteq \mathbb{R}$ and, for a fixed current state x, the transition measure has the mixed density form,

$$P(x, dy) = \sum_{j \geq 1} w_j(x)\delta_{x_j}(dy) + f(x, y)dy, \qquad w_j(x) \geq 0, \quad f(x, y) \geq 0.$$

And, the mixed law satisfies the sum/integrates to 1 property,

$$\sum_{j \geq 1} w_j(x) + \int_E f(x, y)dy = 1.$$

Then, the corresponding *transition operator* is,

$$\begin{aligned} (P\phi)(x) &= \int_E \phi(y)P(x, dy) \\ &= \sum_{j \geq 1} w_j(x)\phi(x_j) + \int_E \phi(y)f(x, y)dx \end{aligned}$$

if the corresponding density exists $P(x, dy) = f(x, y)dx$ possibly $p(x \mid y)dx$.

Observe the first term averages the possible point masses, while the second averages over the part describes by a density. Both are included in the single notation,

$$\int_E \phi(y) P(x, dy).$$

Proposition 4.1 (Basic Properties of the Transition Operator)**.** *Let P be a transition kernel. Then P maps bounded measurable functions to bounded measurable functions. Moreover,*

1. *P is linear,*
$$P(c\phi + d\psi) = cP\phi + dP\psi.$$
2. *P preserves order,*
$$\phi \leq \psi \implies P\phi \leq P\psi.$$
3. *P preserves the constant function 1,*
$$P1 = 1.$$
4. *If $|\phi| \leq C$ then,*
$$|P\phi| \leq C.$$
5. *For every Borel set $B \in \mathcal{B}(E)$,*
$$(P\mathbb{I}_B)(x) = P(x, B).$$

In order to prove *Proposition 4.1* we must introduce the concept of approximating simple functions. Suppose that $(E, \mathcal{B}(E))$ is a measurable state space. A *Simple function*, then, is a measurable function ϕ $E \to \mathbb{R}$ taking only finitely many values, which can be expressed as,

$$\phi(x) = \sum_{i=1}^{m} w_i \mathbb{I}_{B_i}(x), \qquad w_i \in \mathbb{R}, \quad B_i \in \mathcal{B}(E).$$

Each indicator function selects whether x belongs to the corresponding Borel set while w_i assigns its weight. Indeed, if the sets $B_1, \ldots, B_m$ form a measurable partition of E then,

$$\phi(x) = w_i, \quad x \in B_i.$$

Hence, a simple function is constant on each reach of a finite measurable partition.

Suppose that we now take $m \to \infty$. Hence, we make note of the following theorem.

Theorem 4.1 (Simple Function Approximation)**.** *Let $\phi : E \to [0, \infty]$ be a nonnegative Borel-measurable function. Then there exists a sequence of nonnegative simple functions $(\phi_n)_{n \geq 1}$,*

$$0 \leq \phi_n(x) \to \phi(x) \qquad \forall x \in E.$$

In particular, each ϕ_n has a finite representation of the form,

$$\phi_n(x) = \sum_{i=1}^{m_n} w_{n,i} \mathbb{I}_{B_{n,i}}(x)$$

and hence the convergence,

$$\phi(x) = \lim_{n\to\infty} \sum_{i=1}^{m_n} w_{n,i} \mathbb{I}_{B_{n,i}}(x).$$

That is, we construct a sequence of finite simple functions that converges to $\phi(X)$ rather than merely replacing m by infinitely in an arbitrary sum.

Hence, for a simple function under a transition operator,

$$\begin{aligned}(P\phi)(x) &= \int_E \left(\sum_{i=1}^{m} w_i \mathbb{I}_{B_i}(y) \right) P(x, dy) \\ &= \sum_{i=1}^{m} w_i \int_E \mathbb{I}_{B_i}(y) P(x, dy) \\ &= \sum_{i=1}^{m} w_i P(x, B).\end{aligned}$$

Hence, the transition operator computes a weighted average of the values of ϕ over the possible next states. For nonnegative measurable ϕ, take $\phi_n \to \phi$.

By Monotone Convergence Theorem,

$$\begin{aligned}(P\phi)(x) &= \int_E \phi(y) P(x, dy) \\ &= \int_E \lim_{n\to\infty} \phi_n(y) P(x, dy) \\ &= \lim_{n\to\infty} \int_E \phi_n(y) P(x, dy) \\ &= \lim_{n\to\infty} \sum_{i=1}^{m_n} w_{n,i} P(x, B_{n,i}).\end{aligned}$$

This is the fundamental construction of the *Lebesgue integral* as we first define integration for indicators and extend it to finite simple functions by linearity, then to nonnegative measurable functions through monotone limits. Finally, we obtain the integral under integrable signed functions using positive and negative parts.

Proof. Linearity and order preservation follow from the corresponding properties of integration. Because $P(x, \cdots)$ is a probability measure for every $x \in E$ across all $B \in \mathcal{B}(E)$,

$$(P1)(x) = \int_B 1 \cdot P(x, dy) = P(x, E) = 1.$$

Similarly, if our test function is bounded in magnitude $|\phi| \leq C$ then,

$$\begin{aligned} |(P\phi)(x)| &= \left| \int_E \phi(y) P(x, dy) \right| \\ &\leq \int_E |\phi(y)| P(x, dy) \\ &\leq C \int_E 1 P(x, dy) \\ &= C. \end{aligned}$$

Hence, we see that $P\phi$ is bounded. For an indicator function, that is we let $\phi = \mathbb{I}_B$,

$$\begin{aligned} (P\mathbb{I}_B)(x) &= \int_E \mathbb{I}_B(x) P(x, dy) \\ &= \int_B P(x, dy) \\ &= P(x, B). \end{aligned}$$

This is measurable as a function of x by the definition of the transition kernel.

Consequently, if ϕ is a measurable simple function, that is we can express ϕ as the sum of Borel sets $B_i \in \mathcal{B}(E)$ weighted by corresponding w_i's,

$$\phi = \sum_{i=1}^m w_i \mathbb{I}_{B_i} \implies (P\phi)(x) = \sum_{i=1}^m w_i P(x, B_i)$$

is also measurable. Finally, a bounded measurable function can be approximated uniformly by measurable simple functions. Explicitly, suppose we define,

$$\phi_m(y) := 2^{-m} \lfloor 2^m \phi(y) \rfloor.$$

Since ϕ is bounded, ϕ_m takes finitely many values. Its level sets are measurable and $0 \leq \phi(y) - \phi_m(y) < 2^{-m}$. Therefore, its absolute difference is,

$$\|(P\phi)(x) - (P\phi_m)(x)| \leq \int_E |\phi(y) - \phi_m(y)| P(x, dy) \leq 2^{-m}.$$

The result follows letting $m \to \infty$. Hence, $P\phi_m \to P\phi$ pointwise and each $P\phi_m$ is measurable.

□

We defined the Markov property using indicators of events. The transition operator lets us express the same property for any bounded measurable function of the next state.

Proposition 4.2 (Conditional Expectation through the Transition Operator)**.** *For every bounded and measurable test function $\phi : E \to \mathbb{R}$,*

$$\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] = (P\phi)(X_n) \quad \textit{almost surely.}$$

Proof. We first verify the identity for indicators, then for simple functions, then extend to bounded measurable functions. The base case for $\phi = \mathbb{I}_B$, the Markov property is,

$$\begin{aligned}\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] &= \mathbb{E}[\mathbb{I}_{\{X_{n+1}\in B\}} \mid \mathcal{F}_n] \\ &= P(X_n, B) \\ &= (P\mathbb{I}_B)(X_n).\end{aligned}$$

Now suppose that we can write,

$$\phi = \sum_{i=1}^{m} w_i \mathbb{I}_{B_i}.$$

By linearity of conditional expectation,

$$\begin{aligned}\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] &= \mathbb{E}\left[\sum_{i=1}^{m} w_i \mathbb{I}_{\{X_{n+1}\in B_i\}} \middle| \mathcal{F}_n\right] \\ &= \sum_{i=1}^{m} w_i \mathbb{E}\left[\mathbb{I}_{\{X_{n+1}\in B\}} \middle| \mathcal{F}_n\right] \\ &= \sum_{i=1}^{m} w_i P(X_n, B_i) \\ &= \int_E \phi(y) P(X_n, dy) = (P\phi)(X_n).\end{aligned}$$

For general bounded, measurable test functions ϕ, we use simple functions,

$$\phi_m(y) = 2^{-m}\lfloor 2^m \phi(y) \rfloor, \qquad |\phi - \phi_m| \le 2^{-m}.$$

By monotonicity of conditional expectation, we obtain,

$$\|\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] - \mathbb{E}[\phi_m(X_{n+1}) \mid \mathcal{F}_n]| \le 2^{-m} \quad \text{almost surely.}$$

Also, previously we saw, $|(P\phi)(X_n) - (P\phi_m)(X_n)| \le 2^{-m}$ so combining these bounds,

$$\begin{aligned}&\|\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] - (P\phi)(X_n)| \\ &\quad \le |\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] - \mathbb{E}[\phi_m(X_{n+1}) \mid \mathcal{F}_n]| \\ &\qquad + |\mathbb{E}[\phi_m(X_{n+1}) \mid \mathcal{F}_n] - (P\phi_m)(X_n)| \\ &\qquad + |(P\phi_m)(X_n) - (P\phi)(X_n)| \\ &\quad \le 2^{-m} + 0 + 2^{-m} = 2^{1-m} \qquad \text{almost surely.}\end{aligned}$$

Since there are only countably many m, these bounds hold simultaneously outside on null event. Letting $m \to \infty$ proves the result. Hence, it follows that,

$$\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] = (P\phi)(X_n) \quad \text{almost surely.}$$

□

Remark 4.9 (Starting from a Fixed State). If $X_0 = x$ almost surely, then

$$\begin{aligned}\mathbb{E}[\phi(X_1)] &= \mathbb{E}[\mathbb{E}[\phi(X_1) \mid \mathcal{F}_0]] \\ &= \mathbb{E}[(P\phi)(X_0)] \\ &= (P\phi)(x) \\ &= \int_E \phi(y) P(x, dy).\end{aligned}$$

Thus, $\mathcal{L}(X_1) = P(x, \cdot)$.

Notice that the integral averages the quantity $\phi(y)$ over the possible next states y. What it measures depends on the choice of test function ϕ,

(i) If $\phi = \mathbb{I}_B$, it gives the probability of moving into B.
(ii) If $E \subseteq \mathbb{R}$ and $\phi(y) = y$, it gives the expected next state, given that the integral is finite.
(iii) If $\phi(y) = y - x$ it gives the expected increment from x again subject to integrability.
(iv) If $\phi(y) = |y - x|$, it gives the expected size of that increment.

Note that the last three functions need not be bounded so their expectations require the corresponding integrability conditions. Hence, we can now predict more than one step ahead. The tower property shows why repeated transitions correspond to composition operators.

Remark 4.10 (Two-step Prediction). Let $\phi : E \to \mathbb{R}$ be bounded and measurable. Since,

$$F_n \subseteq \mathcal{F}_{n+1}$$

we obtain,

$$\begin{aligned}\mathbb{E}[\phi(X_{n+2}) \mid \mathcal{F}_n] &= \mathbb{E}[\mathbb{E}[\phi(X_{n+2}) \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] \\ &= \mathbb{E}[(P\phi)(X_{n+1}) \mid \mathcal{F}_n] \\ &= (P(P\phi))(X_n).\end{aligned}$$

The last equality applies the one-step formula to the bounded measurable function $P\phi$.

For a fixed current state $X_0 = x$,

$$\begin{aligned}(P(P\phi))(x) &= \int_E (P\phi)(y) P(x, dy) \\ &= \int_E \left[\int_E \phi(z) P(y, dz) \right] P(x, dy).\end{aligned}$$

The inner integral averages ϕ over the second transition starting from y. The outer integral then averages over all possible intermediate states y reached from x. Thus, two-step prediction consists of averaging over the second step and then averaging over the first.

Definition 4.12 (Iterated Transition Operators)**.** Let P_0 be the identity operator,

$$P_0\phi = \phi.$$

For $n \geq 1$ define,

$$P_n := \underbrace{P \circ \cdots \circ P}_{n \text{ times}}.$$

Equivalently, $P_{n+1}\phi = P(P_n\phi)$. Following the matrix notation, we may write,

$$P_nP_m\phi := P_n(P_m\phi).$$

Associativity of composition gives,

$$P_{n+m} = P_nP_m = P_mP_n.$$

Proposition 4.3 (Prediction over Several Steps)**.** *For every $n, m \geq 0$ and every bounded measurable test function $\phi : E \to \mathbb{R}$,*

$$\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] = (P_m\phi)(X_n) \quad \textit{almost surely.}$$

Proof. For $m = 0$, $\phi(X_n)$ is $\mathcal{F}_n$-measurable, so

$$\mathbb{E}[\phi(X_n) \mid \mathcal{F}_n] = \phi(X_n) = (P_0\phi)(X_n) = (P\phi)(X_n).$$

Suppose the result holds for m. By the *Tower Property* of conditional expectation,

$$\begin{aligned}
\mathbb{E}[\phi(X_{n+m+1}) \mid \mathcal{F}_n] &= \mathbb{E}[\mathbb{E}[(\phi(X_{n+m+1})) \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] \\
&= \mathbb{E}[(P_m\phi)(X_{n+1}) \mid \mathcal{F}_n], \qquad \text{starting from } n+1 \\
&= (P(P_m\phi))\,(X_n) \\
&= (P_{m+1}\phi)(X_n).
\end{aligned}$$

By the induction hypothesis of starting at time $n+1$, and the one-step formula, this completes the induction and proves the claim.

□

Remark 4.11 (Multi-step Transition Probabilities). Using the same notation for the associated kernels, we can write the m-step transition kernel as,

$$P_m(x, B) := (P_m\mathbb{I}_B)(x).$$

Then, the m step transition kernel becomes, conditioning on $\mathcal{F}_n$,

$$\mathbb{P}(X_{n+m} \in B \mid \mathcal{F}_n) = P_m(X_n, B).$$

Starting with the two-step move,

$$P_2(x, B) = \int_E P(y, B)P(x, dy)$$

we reach B after two steps if the chain moves first from x to an intermediate state y then moves from y into B. The operator composition identity gives,

$$P_{n+m}(x, B) = \int_E P_m(y, B)P_n(x, dy).$$

Thus, to move from current state x into some state in B in $n + m$ steps, the chain must move from state x to some intermediate state y in n steps then y into B in m steps.

This is precisely the *Chapman-Kolmogorov identity*. Refer to *Appendix* for full expansion.

On a countable state space this becomes,

$$(P_{n+m})_{ij} = \sum_{l \geq 1} (P_n)_{il}(P_m)_{lj}$$

which is the ordinary matrix multiplication summing over every possible intermediate state.

Remark 4.12 (Test Functions and Laws). If $\mu_n := \mathcal{L}(X_n)$ then,

$$\mathbb{E}[\phi(X_n)] = \int_E \phi(y)\mu_n(dy).$$

Combining this with the multi-step formula,

$$\begin{aligned} \mathbb{E}[\phi(X_n)] &= \mathbb{E}[(P_n\phi)(X_0)] \\ &= \int_E (P_n\phi)(x)\mu_0(dx). \end{aligned}$$

In particular, when $X_0 = x$ almost surely,

$$\mathbb{E}[\phi(X_n)] = (P_n\phi)(x)$$

which connects transition operators to our earlier use of test functions. For real-valued random variables where $X_n \xrightarrow{d} X_\infty$, means

$$\int_{\mathbb{R}} \phi d\mu_n \longrightarrow \int_{\mathbb{R}} \phi d\mu_\infty \qquad \forall \phi \in \mathcal{C}_b(\mathbb{R}).$$

For probability measures, integration against test functions plays the corresponding role.

Returning to the notion of joint distribution of a given process, the density factorization at the beginning has an exact counterpart that does not require densities.

Proposition 4.4 (Finite-Dimensional Distributions of a Markov Chains)**.** *Let $(X_n)_{n\geq 0}$ be a time-homogeneous Markov Chain with initial distribution μ_0 and transition kernel P. Then, for every $n \geq 0$ and $B_0, \ldots, B_n \in \mathcal{B}(E)$,*

$$\begin{aligned}&\mathbb{P}(X_0 \in B_0, \ldots, X_n \in B_n)\\ &\qquad \int_{B_0} \mu_0(dx_0) \int_{B_1} P(x_0, dx_1) \cdots \int_{B_n} P(x_{n-1}, dx_n).\end{aligned}$$

Proof. For $n = 0$, the statement is the definition of the initial distribution,

$$\mathbb{P}(X_0 \in B_0) = \mu_0(B_0) = \int_{B_0} \mu_0(dx_0).$$

For $n \geq 1$, write the probability as the expectation of a product of indicators,

$$\mathbb{P}(X_0 \in B_0, \ldots, X_n \in B_n) = \mathbb{E}\left[\prod_{k=0}^{n} \mathbb{I}_{\{X_k \in B_k\}}\right].$$

Then for $k \leq n-1$, the random variable $\mathbb{I}_{\{X_k \in B_k\}}$ is $\mathcal{F}_{n-1}$-measurable. Therefore conditioning on the filtration $\mathcal{F}_{n-1}$ and applying the proposition for *Taking Out what is Known*,

$$\begin{aligned}&\mathbb{E}\left[\prod_{k=0}^{n} \mathbb{I}_{\{X_k \in B_k\}}\right]\\ &\qquad = \mathbb{E}\left[\mathbb{E}\left[\prod_{k=0}^{n} \mathbb{I}_{\{X_k \in B_k\}} \middle| \mathcal{F}_{n-1}\right]\right]\\ &\qquad = \mathbb{E}\left[\left(\prod_{k=0}^{n-1} \mathbb{I}_{\{X_k \in B_k\}}\right) \mathbb{E}[\mathbb{I}_{\{X_n \in B_n\}} \mid \mathcal{F}_{n-1}]\right]\\ &\qquad = \mathbb{E}\left[\left(\prod_{k=0}^{n-1} \mathbb{I}_{\{X_k \in B_k\}}\right) P(X_{n-1}, B_n)\right]\\ &\qquad = \mathbb{E}\left[\left(\prod_{k=0}^{n-1} \mathbb{I}_{\{X_k \in B_k\}}\right) \int_{B_n} P(X_{n-1}, dx_n)\right].\end{aligned}$$

Hence, we have integrated out the last state X_n. If $n \geq 2$, we next condition on $\mathcal{F}_{n-2}$. The relevant bounded measurable function of X_{n-1} is,

$$x_{n-1} \mapsto \mathbb{I}_{B_{n-1}}(x_{n-1}) \int_{B_n} P(x_{n-1}, dx_n).$$

Applying the transition operator identity to this function,

$$\begin{aligned}
&\mathbb{E}\left[\mathbb{I}_{\{X_{n-1}\in B_{n-1}\}}\int_{B_n} P(X_{n-1}, dx_n)\middle| F_{n-2}\right] \\
&\qquad = \int_E \mathbb{I}_{B_{n-1}}(x_{n-1})\left[\int_{B_n} P(x_{n-1}, dx_n)\right] P(X_{n-2}, dx_{n-1}) \\
&\qquad = \int_{B_{n-1}} P(X_{n-2}, dx_{n-1}) \int_{B_n} P(x_{n-1}, dx_n).
\end{aligned}$$

Substituting,

$$\begin{aligned}
&\mathbb{P}(X_0 \in B_0, \ldots, X_n \in B_n) \\
&\qquad = \mathbb{E}\left[\left(\prod_{k=0}^{n-2} \mathbb{I}_{\{X_k\in B_k\}}\right)\int_{B_{n-1}} P(X_{n-2}, dx_{n-1}) \int_{B_n} P(x_{n-1}, dx_n)\right].
\end{aligned}$$

Each further conditioning step works in the exact same way where, if we suppose that we've reached an expression of the expectation with form,

$$\mathbb{E}\left[\left(\prod_{k=0}^{r} \mathbb{I}_{\{X_k\in B_k\}}\right)\int_{B_{r+1}} P(X_r, dx_{r+1}) \cdots \int_{B_n} P(x_{n-1}, dx_n)\right], \qquad 1 \leq r \leq n-1.$$

The product of the indicators with $k \leq r-1$ is $\mathcal{F}_{r-1}$-measurable. The remaining expression is a bounded measurable function of X_r taking values in $[0, 1]$.

Conditioning on $\mathcal{F}_{r-1}$ therefore replaces it by,

$$\int_{B_r} P(X_{r-1}, dx_r) \int_{B_{r+1}} P(x_r, dx_{r+1}) \cdots \int_{B_n} P(x_{n-1}, dx_n).$$

Hence it follows,

$$\begin{aligned}
&\mathbb{E}\left[\left(\prod_{k=0}^{r} \mathbb{I}_{\{X_k\in B_k\}}\right)\int_{B_{r+1}} P(X_r, dx_{r+1}) \cdots \int_{B_n} P(x_{n-1}, dx_n)\right] \\
&\qquad = \mathbb{E}\left[\left(\prod_{k=0}^{r-1} \mathbb{I}_{\{X_k\in B_k\}}\right)\int_{B_r} P(X_{r-1}, dx_r) \cdots \int_{B_n} P(x_{n-1}, dx_n)\right].
\end{aligned}$$

Repeating this step until only X_0 remains gives,

$$\begin{aligned}
&\mathbb{P}(X_0 \in B_0, \ldots, X_n \in B_n) \\
&\qquad = \mathbb{E}\left[\mathbb{I}_{\{X_0\in B_0\}}\int_{B_1} P(X_0, dx_1) \cdots \int_{B_n} P(x_{n-1}, dx_n)\right].
\end{aligned}$$

Finally, X_0 has law μ_0 so integrating with respect to that law gives,

$$\begin{aligned}
&\mathbb{P}(X_0 \in B_0, \ldots, X_n \in B_n) \\
&\quad = \int_E \mathbb{I}_{B_0}(x) \left[\int_{B_1} P(x_0, dx_1) \cdots \int_{B_n} P(x_{n-1}, dx_n) \right] \mu_0(dx_0) \\
&\quad = \int_{B_0} \mu_0(dx_0) \int_{B_1} P(x_0, dx_1) \cdots \int_{B_n} P(x_{n-1}, dx_n).
\end{aligned}$$

This proves the formula.

□

Hence, the formula describes the joint law,

$$\mu_{0:n} = \mathcal{L}(X_0, \ldots, X_n)$$

through its probabilities on rectangles $B_0 \times \cdots \times B_n$. Since these rectangles generate the product Borel σ-algebra, they determine the joint probability measure.

As n varies, we obtain every finite-dimensional distribution of the chain. Thus, the initial distribution μ_0 and transition kernel P determine the process law on the σ-algebra generated by its coordinates. It is common to summarize the iterated-integral formula using the notation,

$$\mu_{0:n}(dx_0, \ldots, dx_n) = \mu_0(dx_0) P(x_0, dx_1) \cdots P(x_{n-1}, dx_n)$$

which gives a shorthand for successive integration against kernels.

It is wrong to interpret this as the product of independent marginal distributions as the measure governing x_k depends on the preceding state x_{k-1}. When densities do exist, we may write,

$$p(x_0, \ldots, x_n) = p(x_0) \prod_{k=1}^{n} p(x_k \mid x_{k-1})$$

recovering the heuristic factorization with which we began.

4.6 Finite-to-Continuous State Space Duality

The purpose of this section is to make the notation and interpretation of transition kernels concrete in the finite-state setting. Here, we view probability distributions as row vectors, test functions as column vectors, and the transition kernel as a matrix.

The same matrix acts in two ways. It updates a distribution by multiplication on the right, and it transforms a function by multiplication on the left.

These actions are precisely related through the identity,

$$(\mu P)\phi = \mu(P\phi).$$

We will interpret this identity both algebraically & probabilistically, then connect it to the kernel notation introduced last week. We start with our previous definition of a transition matrix.

Recall that we previous definition a transition matrix on state space $E = \{1, \ldots, m\}$

A matrix $P \in \mathbb{R}^{m \times m}$ is a transition matrix if,

$$P_{ij} \geq 0, \qquad \sum_{j=1}^{m} P_{ij} = 1 \qquad \forall i \in E.$$

The entry P_{ij} represents the probability of moving from state i to state j in one step. For a fixed starting state i, summing over the possible destination states j gives one.

Hence, each row sums to one.

Definition 4.13 (Finite-State Time-Homogeneous Markov Chain)**.** A process $(X_n)_{n \geq 0}$ with values in E is a time-homogeneous *Markov chain* with transition matrix P is,

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i_0, \ldots, X_n = i_n) = P_{i_n, j}$$

for every $n \geq 0$ and $j \in E$ and every history $i_0, \ldots, i_n \in E$ satisfying,

$$\mathbb{P}(X_0 = i_0, \ldots, X_n = i_n) > 0.$$

The right-hand side depends on the history only through the most recent state i_n. The same matrix P is used at every time, which gives time-homogeneity.

The positivity condition is necessary because the conditional probability above is defined by the ratio, $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$. Even on a finite state space, some histories may have probability zero. The ratio formulation above simply imposes no condition on those histories.

Remark 4.13 (Conditioning on the History as a Finite Partition)*.* Let,

$$\mathcal{F}_n := \sigma(X_0, \ldots, X_n).$$

Then, the events $\{X_0 = i_0, \ldots, X_n = i_n\}$ for $(i_0, \ldots, i_n) \in E^{n+1}$ form a finite partition Ω allowing empty cells and generate $\mathcal{F}_n$. Conditioning on $\mathcal{F}_n$ therefore means assigning a conditional expectation on each positive-probability history cell. On the cell,

$$\{X_0 = i_0, \ldots, X_n = i_n\}$$

the *Markov property* gives,

$$\mathbb{E}\left[\mathbb{I}_{\{X_{n+1}=j\}} \middle| \mathcal{F}_n\right] = \mathbb{P}(X_{n+1} = j \mid \mathcal{F}_n) = P_{i_n, j}.$$

Simultaneously, this gives $\mathbb{E}\left[\mathbb{I}_{\{X_{n+1}=j\}} \middle| \mathcal{F}_n\right] = P_{X_n, j}$ almost surely. Here, $P_{X_n, j}$ is a random variable. At an outcome ω, its value is the matrix entry $P_{X_n(\omega), j}$.

Observe that this is the finite-state version of the kernel identity,

$$\mathbb{E}\left[\mathbb{I}_{\{X_{n+1} \in B\}} \middle| \mathcal{F}_n\right] = \mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n) = P(X_n, B).$$

Indeed, on a finite state space $E = \{1, \ldots, m\}$,

$$P(i, \{j\}) = P_{ij}, \qquad P(i, B) = \sum_{j \in B} P_{ij}.$$

Hence, the finite-history definition agrees with the conditional-expectation definition used for general state spaces. We next distinguish two kinds of objects on which the transition matrix acts which are distributions and state spaces.

Definition 4.14 (Distributions as Row Vectors)**.** A probability distribution μ on E is represented by the row vector,

$$\mu = \begin{pmatrix} \mu_1 & \cdots & \mu_m \end{pmatrix}, \qquad \mu_i := \mu(\{i\})$$

where,

$$\mu_i \geq 0, \qquad \sum_{i=1}^{m} \mu_i = 1.$$

For a *Markov chain* we may write,

$$\mu_n := \mathcal{L}(X_n), \qquad \mu_{n,i} := \mathbb{P}(X_n = i)$$

which means that our row vector becomes,

$$\mu_n = \begin{pmatrix} \mu_{n,1} & \cdots & \mu_{n,m} \end{pmatrix}$$

which is precisely a $1 \times m$ row vector. The notation μ_n describes the law at time n.

Definition 4.15 (Functions as Column vectors)**.** A function $\phi : E \to \mathbb{R}$ is represented by the column vector,

$$\phi = \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_m \end{pmatrix}, \qquad \phi_j := \phi(j).$$

Every function on a finite state space is measurable and bounded. Moreover,

$$\phi(i) = \sum_{j=1}^{m} \phi_j \mathbb{I}_{\{j\}}(i).$$

Hence, for a fixed i, exactly one indicator in this sum equals one which is the indicator corresponding to $j = i$. Therefore, the sum equals exactly $\phi_i = \phi(i)$. Notice that $\mathbb{I}_{\{1\}}, \ldots, \mathbb{I}_{\{m\}}$ form a basis for the vector space of real-valued functions on E where the values $\phi_1, \ldots, \phi_m$ are coefficients of ϕ in this basis.

Interestingly, pairing a distribution with a function gives a function that is analogous to integration of a function against a distribution.

Definition 4.16 (Pairing a Distribution with a Function)**.** For a probability distribution μ and a function $\phi : E \to \mathbb{R}$ we can define,

$$\mu\phi := \sum_{j=1}^{m} \mu_j \phi_j.$$

Notice that this is equivalent to writing,

$$\mu\phi = \int_E \phi(x)\mu(dx).$$

If $X_n \sim \mu_n$ then,

$$\begin{aligned}
\mathbb{E}\phi(X_n) &= \sum_{j=1}^{m} \phi(x_j)\mathbb{P}(X_n = j) \\
&= \sum_{j=1}^{m} \phi_j \mu_{n,j} \\
&= \begin{pmatrix} \mu_{n,1} & \cdots & \mu_{n,m} \end{pmatrix} \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_m \end{pmatrix} \\
&= \mu_n \phi.
\end{aligned}$$

We may denote this pairing as an inner product $\langle \mu_n, \phi \rangle$. In finite dimensions, this possess the equivalent algebraic form as a dot product as well representing the integration of a function against a distribution. Notice, choosing $\phi = \mathbb{I}_B$ gives,

$$\mu\mathbb{I}_B = \sum_{j \in B} \mu_j = \mu(B).$$

Thus, event probabilities are particular examples of this pairing.

Lemma 4.1 (Probability Distributions are Characterized by Test Functions)**.** *Let μ and ν be probability distributions on E. Then,*

$$\mu = \nu \iff \mu\phi = \nu\phi \quad \forall \phi : E \to \mathbb{R}.$$

Proof. Suppose first that $\mu = \nu$. Then, $\mu_j = \nu_j$ for every $j \in E$.

Consequently, for every $\phi : E \to \mathbb{R}$,

$$\mu\phi = \sum_{j=1}^{m} \mu_j \phi_j = \sum_{j=1}^{m} \nu_j \phi_j = \nu\phi.$$

Conversely, suppose that $\mu\phi = \nu\phi$ for every $\phi : E \to \mathbb{R}$.

Fixing $i \in E$ and $\phi = \mathbb{I}_{\{i\}}$ then,

$$\begin{aligned}\mu_i &= \sum_{j=1}^{m} \mu_j \mathbb{I}_{\{i\}}(j) = \mu \mathbb{I}_{\{i\}} \\ &= \nu \mathbb{I}_{\{i\}} \\ &= \sum_{j=1}^{m} \nu_j \mathbb{I}_{\{i\}}(j) \\ &= \nu_i.\end{aligned}$$

Therefore, $\mu_i = \nu_i$ for every i. For any $B \subseteq E$,

$$\mu(B) = \sum_{i \in B} \mu_i = \sum_{i \in B} \nu_i = \nu(B).$$

Hence, it follows that $\mu = \nu$.

□

Note that if $\mu \neq \nu$, there is some $B \subseteq E$ such that,

$$\mu(B) \neq \mu(B).$$

Taking $\phi = \mathbb{I}_B$ gives,

$$\mu\phi = \mu(B) \neq \nu(B) = \nu\phi.$$

Hence, equality of expectations for one particular function generally does not imply equality of distributions. Equality for every test function does. On a finite state space, it is enough to test the m-th indicators on single partition.

Proposition 4.5 (Evolution of the Distribution)**.** *If $\mu_n := \mathcal{L}(X_n)$ then $\mu_{n+1} = \mu_n P$.*

In pointwise coordinates, we can write this as

$$\mu_{n+1,j} = \sum_{i=1}^{n} \mu_{n,i} P_{ij}, \qquad \forall j \in E.$$

Proof. Fix $j \in E$. Applying the *Tower Property* and the *Markov Property*,

$$\begin{aligned}\mu_{n+1,j} &= \mathbb{P}(X_{n+1} = j) \\ &= \mathbb{E}\left[\mathbb{I}_{\{X_{n+1}\}}\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\mathbb{I}_{\{X_{n+1}=j\}} \middle| \mathcal{F}_n\right]\right] \\ &= \mathbb{E}\left[P(X_n, j)\right] \\ &= \sum_{i=1}^{m} P_{ij} \mathbb{P}(X_n = i) \\ &= \sum_{i=1}^{m} \mu_{n,i} P_{ij}.\end{aligned}$$

This is precisely multiplying out the j-th coordinate of $\mu_n P$. Hence, it follows that

$$\mu_{n+1} = \mu_n P.$$

□

Notice that each summand $\mu_{n,i}P_{ij}$ accounts for one possible route to state j. So, if the chain is at i at time n, then moves from i to j. Summing over i accounts for all possible current states.

μ_n is a $1 \times m$ row vector and P is a $m \times m$ matrix so μ_{n+1} is a $1 \times m$ row vector as well.

Hence the resulting vector is a probability distribution. Therefore, right-matrix multiplication of μ_n on our transition matrix P results in a probability distribution.

Since the entries are nonnegative,

$$\begin{aligned}\sum_{j=1}^{m}(\mu_n P)_j &= \sum_{j=1}^{m}\sum_{i=1}^{m}\mu_{n,i}P_{ij} \\ &= \sum_{i=1}^{m}\mu_{n,i}\sum_{j=1}^{m}P_{ij} \\ &= \sum_{i=1}^{m}\mu_{n,i} \\ &= 1.\end{aligned}$$

Now, the second action of P is on functions. Rather than updating the distribution, we compute the expected value of a function after one transition, for each possible current state.

Definition 4.17 (Transition Operator in Finite State-Space)**.** For $\phi : E \to R$ define

$$(P\phi)(i) := \sum_{j=1}^{m} P_{ij}\phi_j, \qquad i \in E.$$

Hence, the map $\phi \mapsto (P\phi)$ is called the transition operator. Here, $P\phi$ is a new function on E. Its value i averages the possible next-state values ϕ_j using the transition probabilities from i.

In matrix form,

$$\underbrace{P}_{m\times m}\underbrace{\phi}_{m\times 1} = \underbrace{P\phi}_{m\times 1}$$

so whenever $\mathbb{P}(X_n = i) > 0$,

$$(P\phi)(i) = \mathbb{E}[\phi(X_{n+1}) \mid X_n = i].$$

Hence, the expression $(P\phi)(i)$ is defined by the matrix even when $\mathbb{P}(X_n = i) = 0$.

Proposition 4.6 (Conditional Expectation as a Transition Operator)**.** *For every function* $\phi : E \to \mathbb{R}$*,*

$$\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] = (P\phi)(X_n) \qquad \textit{almost surely.}$$

Proof. Expanding $\phi(X_{n+1})$ using indicators,

$$\phi(X_{n+1}) = \sum_{j=1}^{m} \phi_j \mathbb{I}_{\{X_{n+1}=j\}}.$$

Taking conditional expectations,

$$\begin{aligned}
\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] &= \mathbb{E}\left[\sum_{j=1}^{m} \phi_j \mathbb{I}_{\{X_{n+1}=j\}} \middle| \mathcal{F}_n\right] \\
&= \sum_{j=1}^{m} \phi_j \mathbb{E}\left[\mathbb{I}_{\{X_{n+1}=j\}} \mid \mathcal{F}_n\right] \\
&= \sum_{j=1}^{m} \phi_j P_{X_n,j}, \qquad \text{by Markov Property} \\
&= (P\phi)(X_n).
\end{aligned}$$

Note that $(P\phi)(i)$ is associated with the fixed state i whereas the second is a random variable obtained by evaluating the function $P\phi$ at the random current state X_n. □

Proposition 4.7 (Duality)**.** *Let* μ *be the probability distribution/law on a finite state space* E*, and let* $\phi : E \to \mathbb{R}$ *with transition matrix* P*. Then,*

$$(\mu P)\phi = \mu(P\phi).$$

Equivalently, in pairing notation using inner products,

$$\langle \mu P, \phi \rangle = \langle \mu, P\phi \rangle.$$

Proof. Expanding the right-hand side of the identity above,

$$\begin{aligned}
(\mu P)\phi = \sum_{j=1}^{m} (\mu P)_j \phi_j &= \sum_{j=1}^{m} \left(\sum_{j=1}^{m} \mu_i P_{ij}\right) \phi_j \\
&= \sum_{j=1}^{m} \sum_{i=1}^{m} \mu_i P_{ij} \phi_j \\
&= \sum_{i=1}^{m} \mu_i \left(\sum_{j=1}^{m} P_{ij} \phi_j\right) \\
&= \sum_{i=1}^{m} \mu_i (P\phi)(i) \\
&= \mu(P\phi).
\end{aligned}$$

By associativity of matrix multiplication, we obtain the identity.

□

On the left, we first update the distribution from μ to μP then average ϕ under the new distribution. On the right, we first calculate the expected next-step value $P\phi$ at every current state, then average those values under the current distribution μ.

If $X_n \sim \mu$, then the tower property expresses this relationship,

$$\begin{aligned}(\mu P)\phi &= \mathbb{E}\phi(X_{n+1}) \\ &= \mathbb{E}[\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n]] \\ &= \mathbb{E}[(P\phi)(X_n)] \\ &= \mu(P\phi).\end{aligned}$$

Hence, the identity,

$$(\mu P)\phi = \mu(P\phi)$$

is precisely the matrix form of the tower property.

Under our row-vector convention for distributions and column-vector convention for functions, $\mu \mapsto \mu P$ moves a distribution forward by one time step, while $\phi \mapsto P\phi$ takes a function evaluated at the next time and produces its conditional expected value as a function of the current state.

As you may notice, repeated transitions correspond to repeated matrix multiplication. The conditional-expectation calculation makes the meaning of these powers explicit.

For two-steps, the tower property gives us

$$\begin{aligned}\mathbb{E}[\phi(X_{n+2}) \mid \mathcal{F}_n] &= \mathbb{E}[\mathbb{E}[\phi(X_{n+2}) \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] \\ &= \mathbb{E}[(P\phi)(X_{n+1}) \mid \mathcal{F}_n] \\ &= (P(P\phi))\,(X_n) \\ &= (P^2\phi)(X_n).\end{aligned}$$

E is finite so $P\phi$ is again a bounded measurable function on E.

Proposition 4.8 (Multi-step Evolution)**.** *For every $n, k \geq 0$ and every function $\phi : E \to \mathbb{R}$,*

$$\mathbb{E}[\phi(X_{n+k}) \mid \mathcal{F}_n] = (P^k\phi)(X_n) \qquad \textit{almost surely.}$$

Under our matrix and vector conventions, this equivalent to writing,

$$\mu_{n+k} = \mu_n P^k.$$

Proof. For $k = 0$, $P^0 = I$. Since $\phi(X_n)$ is $\mathcal{F}_n$-measurable,

$$\mathbb{E}[\phi(X_n) \mid \mathcal{F}_n] = \phi(X_n) = (P^0\phi)(X_n).$$

Suppose the conditional-expectation identity holds for k, at every starting time n. Then,

$$\begin{aligned}\mathbb{E}[\phi(X_{n+k+1}) \mid \mathcal{F}_n] &= \mathbb{E}[\mathbb{E}[\phi(X_{n+k+1}) \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] \\ &= \mathbb{E}[(P^k\phi)(X_{n+1}) \mid \mathcal{F}_n] \\ &= \left(P(P^k\phi)\right)(X_n) \\ &= \left(P^{k+1}\phi\right)(X_n).\end{aligned}$$

Thus, we obtain the first identity. Taking expectations,

$$\begin{aligned}\mu_{n+k}\phi &= \mathbb{E}[\phi(X_{n+k})] \\ &= \mathbb{E}[\mathbb{E}[\phi(X_{n+k}) \mid \mathcal{F}_n]] \\ &= \mathbb{E}[(P^k\phi)(X_n)] \\ &= \mu_n(P^k\phi) \\ &= (\mu_n P^k)\phi.\end{aligned}$$

Since this holds for every function $\phi : E \to \mathbb{R}$, the lemma on characterization by bounded and measurable test functions gives,

$$\mu_{n+k} = \mu_n P^k.$$

□

In particular, observe that, $\mu_n = \mu_0 P^n$. Thus, the distribution at any time is determined by the initial distribution and the transition matrix.

Definition 4.18 (Multi-step Transition Matrices)**.** Define $P_n := P^n$ for $n \geq 0$ with $P_0 = I$. The entry $(P_n)_{ij}$ is the n-step transition probability from i to j.

Indeed, if we take $\phi = \mathbb{I}_{\{j\}}$, the multi-step formula gives,

$$\mathbb{P}(X_{r+n} = j \mid \mathcal{F}_r) = (P^n\mathbb{I}_{\{j\}})(X_r) = (P_n)_{X_r,j}.$$

Consequently, whenever $\mathbb{P}(X_r = i) > 0$,

$$\mathbb{P}(X_{r+n} = j \mid X_r = i) = (P_n)_{ij}.$$

Proposition 4.9 (Semigroup Property and Chapman Kolmogorov)**.** *For every $n, k \geq 0$,*

$$P_{n+k} = P_n P_k.$$

Equivalently, for every state $i, j \in E$,

$$(P_{n+k})_{ij} = \sum_{l=1}^{m} (P_n)_{il}(P_k)_{lj}.$$

Proof. By definition of transition matrix P_n,

$$P_{n+k} = P^{n+k} = P^n P^k = P_n P_k.$$

Taking the (i, j)-th entry gives,

$$(P_{n+k})_{ij} = \sum_{l=1}^{m} (P_n)_{il} (P_k)_{lj}.$$

□

Hence, to move from i to j in $n + k$ steps, the chain must occupy some intermediate state l after the first n steps. For each possible intermediate state l,

$$(P_n)_{il} (P_k)_{lj}$$

is the probability of reaching l after n steps and then reaching j after another k steps. Summing over l accounts for every possible intermediate state.

As an operator identity, the same statement

$$P_{n+k}\phi = P_n(P_k\phi)$$

where the inner operator $P_k\phi$ averages over the final k steps and the outer operation P_n then averages over the first n steps. We obtain our transition determination of the transition kernel.

Remark 4.14 (Matrices to Kernels). The finite-state transition matrix determines the kernel,

$$P(i, dy) = \sum_{j=1}^{m} P_{ij} \delta_j(dy).$$

Therefore, by definition of a transition kernel

$$(P\phi)(i) = \int_E \phi(y) P(i, dy) = \sum_{j=1}^{m} P_{ij} \phi(j).$$

Likewise, a distribution μ is the measure,

$$\mu(dx) = \sum_{i=1}^{m} \mu_i \delta_i(dx)$$

so a one-step update is,

$$\begin{aligned}
(\mu P)(B) &= \int_E P(x, B) \mu(dx) \\
&= \sum_{i=1}^{m} \mu_i P(i, B) = \sum_{i=1}^{m} \mu_i \sum_{j \in B} P_{ij} \\
&= \sum_{j \in B} \sum_{i=1}^{m} \mu_i P_{ij}.
\end{aligned}$$

For $B = \{j\}$ the single state $j \in B$,

$$(\mu P)_j = \sum_{i=1}^{m} \mu_i P_{ij}.$$

Hence, generally, this duality can be expressed as,

$$\int_E \phi(y)(\mu P)(dy) = \int_E \left[\int_E \phi(y) P(x, dy) \right] \mu(dx).$$

which is precisely the integral form of $(\mu P)\phi = \mu(P\phi)$. For a chain $X_n \sim \mu$,

$$\begin{aligned} \int_E \phi(y)(\mu P)(dy) &= E[\phi(X_{n+1})] \\ &= \mathbb{E}[\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n]] \\ &= \mathbb{E}[(P\phi)(X_n)] \\ &= \int_E (P\phi)(x)\mu(dx). \end{aligned}$$

Note here that ϕ is bounded and measurable in the general state-space. The notation $P(x, dy)$ represents a probability measure so none of these identities requires a transition density. Returning to the probability of an entire finite path, we can obtain the finite-state form of the joint-distribution formula established previously.

Proposition 4.10 (Finite-Path Probability Formula)**.** *Let $(X_n)_{n\geq 0}$ be a time-homogeneous Markov chain on E with transition matrix P and initial distribution $\mu_0 = \mathcal{L}(X_0)$. Then,*

$$\mathbb{P}(X_0 = i_0, \ldots, X_n = i_n) = \mu_{0,i_0} \prod_{r=0}^{n-1} P_{i_r, i_{r+1}}, \qquad i_0, \ldots, i_n \in E, \quad n \geq 0.$$

Proof. For $n = 0$,

$$\mathbb{P}(X_0 = i_0) = \mathbb{P}(X_0 = i_0) = \mu_{0,i_0}.$$

This is simply the initial distribution. Then, for $n \geq 1$,

$$\begin{aligned} \mathbb{P}(X_0 = i_0, \ldots, X_n = i_n) &= \mathbb{E}\left[\prod_{r=0}^{n-1} \mathbb{I}_{\{X_r = i_r\}} \right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\prod_{r=0}^{n} \mathbb{I}_{\{X_r = i_r\}} \middle| \mathcal{F}_{r-1} \right] \right], \qquad \mathbb{I}_{\{X_r = i_r\}} \in \mathcal{F}_{n-1}, \quad 0 \leq r \leq n-1 \\ &= \mathbb{E}\left[\left(\prod_{r=0}^{n-1} \mathbb{I}_{\{X_r = i_r\}} \right) \mathbb{E}[\mathbb{I}_{\{X_n = i_n\}} \mid \mathcal{F}_{n-1}] \right] \\ &= \mathbb{E}\left[\left(\prod_{r=0}^{n-1} \mathbb{I}_{\{X_r = i_r\}} \right) P_{X_{n-1}, i_n} \right] \\ &= P_{i_{n-1}, i_n} \mathbb{P}(X_0 = i_0, \ldots, X_n = i_n). \end{aligned}$$

Notice that the product of indicators in the first inequality restricts us to the event where $X_{n-1} = i_{n-1}$. Applying the same distribution computation to the remaining history,

$$\begin{aligned}
&\mathbb{P}(X_0 = i_0, \ldots, X_n = i_n) \\
&\qquad = P_{i_{n-1},i_n} P_{i_{n-2},i_{n-1}} \mathbb{P}(X_0 = i_0, \ldots, X_{n-2} = i_{n-2}) \\
&\qquad = \cdots \\
&\qquad = \mathbb{P}(X_0 = i_0) P_{i_0,i_1} \cdots P_{i_{n-1},i_n} \\
&\qquad = \mu_{0,i_0} \prod_{r=0}^{n-1} P_{i_r,i_{r+1}}.
\end{aligned}$$

Hence, this path formula holds for all paths with probability zero.

□

4.7 Extensions to Continuous State Spaces

We now extend these interpretations to a general state space $E \subseteq \mathbb{R}^d$. The matrix products become integrals against transition kernels, but their probabilistic meanings remain the same.

Before doing so, we introduce a distribution that is unchanged by the transition rule.

Definition 4.19 (Invariant or Stationary Distribution)**.** Let P be a transition matrix on a finite state space $E = \{1, \ldots, m\}$. A probability distribution π on E is called invariant/stationary, if

$$\pi P = \pi.$$

Equivalently, in coordinates this is

$$\pi_j = \sum_{i=1}^{m} \pi_i P_{ij} \qquad j \in E.$$

Because distributions evolve according to $\mu_{n+1} = \mu_n P$, the identity $\pi P = \pi$ says that starting at a step with distribution π produces the same distribution after the transition.

Proposition 4.11 (Preservation of an Invariant Distribution)**.** *If $\pi P = \pi$ and $X_0 \sim \pi$ then*

$$X_n \sim \pi \qquad \forall n \geq 0.$$

Proof. By assumption $\mu_0 = \pi$. Therefore

$$\mu_1 = \mu_0 P = \pi P = \pi.$$

If $\pi_n = \pi$ then,

$$\mu_{n+1} = \mu_n P = \pi P = \pi.$$

The result follows by induction.

□

Remark 4.15 (An Unchanging Distribution does not mean an Unchanging State). The chain may continue moving between states even though its distribution remains π. Stationarity concerns the probabilities assigned to states, not whether a particular sample path stays fixed.

Also, the identity $\pi P = \pi$ does not itself imply that a chain started from another distribution converges to π. Invariance and convergence to an invariant distribution are different questions.

On a finite state space, from our transition matrix P take a fixed row,

$$(P_{i1}, \ldots, P_{im})$$

which is a probability distribution for the next state with entries satisfying $P_{ij} \geq 0$ and $\sum_j P_{ij} = 1$. On a general state space, this row is replaced by the probability measure,

$$P(x, \cdot)$$

where for each current state x, it assigns a probability $P(x, B)$ to every Borel set B of possible next states. In particular, $P(x, B) \geq 0$ and $P(x, E) = 1$. Under countable additivity

$$P\left(x, \bigcup_{r=1}^{\infty} B_r\right) = \sum_{r=1}^{\infty} P(x, B_r)$$

wherever the sets B_r are pairwise disjoint. We require that $x \mapsto P(x, B)$ be a measurable ensuring that $P(X_n, B)$ is a random variable under the measure-theoretic sense.

Since $0 \leq P(X_n, B) \leq 1$, it is also hence integrable. Thus, the Markov identity,

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n) = P(X_n, B) \quad \text{almost surely}$$

sharing meaning on a general state space just as,

$$\mathbb{P}(X_{n+1} = j \mid \mathcal{F}_n) = P_{X_n, j}$$

is meaningful on a finite state space. Importantly however, an uncountable state space does not require the transition measure to have density. Consider the following example.

Example 4.5 (Markov Chain with Discrete Transitions)**.** Let

$$\xi_1, \xi_2, \ldots \overset{i.i.d}{\sim} \text{Bernoulli}(p), \qquad p \in (0, 1)$$

and assume that this sequence is independent of X_0 where $X_0 \in [0, 1]$ almost surely.

Also, define preceding increments,

$$X_{n+1} = \frac{X_n + \xi_{n+1}}{2} \qquad n \geq 0.$$

If $X_n \in [0, 1]$ then,

$$X_{n+1} = \begin{cases} \frac{X_n}{2} & \xi_{n+1} = 0 \\ \frac{X_n + 1}{2} & \xi_{n+1} = 1 \end{cases}.$$

Both values belong on $[0,1]$ therefore the process remains in $E=[0,1]$. Given a fixed current state x, the next state is $\frac{x}{2}$ with probability $1-p$ and $\frac{x+1}{2}$ with probability p.

Its transition kernel is,

$$\begin{aligned} P(x,B) &= (1-p)\delta_B\left(\frac{x}{2}\right) + p\delta_B\left(\frac{x+1}{2}\right) \\ &= (1-p)\mathbb{I}_B\left(\frac{x}{2}\right) + p\mathbb{I}_B\left(\frac{x+1}{2}\right). \end{aligned}$$

For each fixed x, this is a probability measure because it is a weighted sum of two Dirac probability measures with nonnegative weights summing to one. For each $B \in \mathcal{B}(\mathbb{R})$,

$$x \mapsto \mathbb{I}_B\left(\frac{x}{2}\right), \qquad x \mapsto \mathbb{I}_B\left(\frac{x+1}{2}\right)$$

are measurable. Hence $x \mapsto P(x,B)$ is measurable, so P is a transition kernel.

Consider $\mathcal{F}_n := \sigma(X_0,\ldots,X_n)$. Since the observations up to time n are determined by $X_0,\xi_1,\ldots,\xi_n$, the σ-algebra generated by X_n is

$$\mathcal{F}_n \subseteq \sigma(X_0,\xi_1,\ldots,\xi_n).$$

Consequently, by this definition, ξ_{n+1} is independent of $\mathcal{F}_n$. Thus, for every $B \in \mathcal{B}(\mathbb{R})$

$$\begin{aligned} \mathbb{I}_{\{X_{n+1}\in B\}} &= \mathbb{I}_B\left(\frac{X_n+\xi_{n+1}}{2}\right) \\ &= \mathbb{I}_{\{\xi_{n+1}=0\}}\mathbb{I}_B\left(\frac{X_n}{2}\right) + \mathbb{I}_{\{\xi_{n+1}=1\}}\mathbb{I}_B\left(\frac{X_n+1}{2}\right). \end{aligned}$$

Taking expectations conditioning on $\mathcal{F}_n$, factors depending on X_n are already known

$$\begin{aligned} \mathbb{E}\left[\mathbb{I}_{\{X_{n+1}\in B\}}\middle|\mathcal{F}_n\right] &= \mathbb{I}_B\left(\frac{X_n}{2}\right)\mathbb{E}\left[\mathbb{I}_{\{\xi_{n+1}=0\}}\middle|\mathcal{F}_n\right] \\ &\quad + \mathbb{I}_B\left(\frac{X_n+1}{2}\right)\mathbb{E}\left[\mathbb{I}_{\{\xi_{n+1}=1\}}\middle|\mathcal{F}_n\right] \\ &= (1-p)\mathbb{I}_B\left(\frac{X_n}{2}\right) + p\mathbb{I}_B\left(\frac{X_n+1}{2}\right) \\ &= P(X_n,B). \end{aligned}$$

Hence, $(X_n)_{n\geq 0}$ is a time-homogeneous Markov chain. Its associated transition operator is particular explicit, that is, we can write it as,

$$(P\phi)(x) = \int_{[0,1]} \phi(y)P(x,dy) = (1-p)\phi\left(\frac{x}{2}\right) + p\phi\left(\frac{x+1}{2}\right).$$

So, therefore, although the state space is uncountable, computing $P\phi$ here requires only a weighted average of two function values. Now, consider the following measure integration.

Definition 4.20 (Pairing a Measure with a Test Function)**.** Let μ be a probability measure on E and let $\phi : E \to \mathbb{R}$ be a bounded and measurable function. Define,

$$\mu(\phi) := \int_E \phi(x)\mu(dx).$$

We may also write this as $\mu\phi$ or $\langle \mu, \phi \rangle$. If $X \sim \mu$, then obviously it follows that $\mu(\phi) = \mathbb{E}[\phi(X)]$. This is precisely the general state-space version of multiplying a distribution row vector by a function column vector. If μ is a discrete law,

$$\mu = \sum_{i \geq 1} \mu_i \delta_{x_i}$$

so the operator becomes,

$$\begin{aligned}
\mu(\phi) = \int_E \phi(x)\mu(dx) &= \int_E \phi(x) \sum_{i \geq 1} \mu\delta_{x_i} \\
&= \sum_{i \geq 1} \mu_i \int_E \phi(x)\delta_{x_i}(dx) \\
&= \sum_{i \geq 1} \mu_i \phi(x_i).
\end{aligned}$$

The values $\phi(x_i)$ are evaluated at the deterministic support points x_i. If μ has density f with respect to the Lebesgue measure, then $\mu(dx) = f(x)dx$ and,

$$\mu(\phi) = \int_E \phi(x)f(x)dx.$$

Remark 4.16 (Pairing vs. an $\mathcal{L}^2$ Inner Product). The expression,

$$\int_E \phi(x)f(x)dx$$

resembles our $\mathcal{L}^2$ representation in an inner product since it is precisely an $\mathcal{L}^2$ inner product when both ϕ and f belong to $\mathcal{L}^2(E, dx)$. Note that those assumptions are not needed here.

A probability density need not be square-integrable and a bounded function on a set of infinite Lebesgue measure need not be square integrable. Nonetheless, the pairing is finite

$$\begin{aligned}
\int_E |\phi(x)|f(x)dx &\leq \sup_{x \in E} |\phi(x)| \int_E f(x)dx \\
&= \sup_{x \in E} |\phi(x)| \\
&< \infty.
\end{aligned}$$

Hence, the appropriate general interpretation is integration of a bounded measurable function against a probability measure. Consider the continuous analogue of this form of integration.

We now define the analogue of multiplying a distribution row vector by P.

Definition 4.21 (Action of a Transition Kernel on a Distribution)**.** Let μ be a probability measure on E and let P be a transition kernel. Define,

$$(\mu P)(B) := \int_E P(x, B)\mu(dx), \qquad B \in \mathcal{B}(E).$$

For each possible current state x, the quantity $P(x, B)$ is the probability of entering B on the next step. Integrating against the law $\mu(dx)$ averages the transition probability over the current-state distribution.

Proposition 4.12. *The set function μP is a probability measure. If $X_n \sim \mu_n$ then,*

$$\mathcal{L}(X_{n+1}) = \mu_n P.$$

Proof. First note that $(\mu P)(B) \geq 0$ which follows since $P(x, B) \geq 0$.

Also, integrating over the entire state space gives probability 1,

$$\begin{aligned}(\mu P)(E) &= \int_E P(x, E)\mu(dx) \\ &= \int_E 1\mu(dx) \\ &= 1.\end{aligned}$$

Let $B_1, B_2, \ldots$ be pairwise disjoint Borel sets. For each possible current state x, countable additivity of the probability measure $P(x, \cdot)$ gives

$$P\left(x, \bigcup_{r=1}^{\infty} B_r\right) = \sum_{r=1}^{\infty} P(x, B_r).$$

Since the terms are nonnegative, *Monotone Convergence Theorem* yields,

$$\begin{aligned}(\mu P)\left(\bigcup_{r=1}^{\infty} B_r\right) &= \int_E P\left(x, \bigcup_{r=1}^{\infty} B_r\right)\mu(dx) \\ &= \int_E \lim_{N\to\infty} \sum_{r=1}^{N} P(x, B_r)\mu(dx) \\ &= \lim_{N\to\infty} \sum_{r=1}^{N} \int_E P(x, B_r)\mu(dx) \\ &= \sum_{r=1}^{\infty} (\mu P)(B_r).\end{aligned}$$

Hence, μP is a probability measure.

Now, suppose that $X_n \sim \mu_n$. For every $B \in \mathcal{B}(E)$,

$$\begin{aligned}
\mathbb{P}(X_{n+1} \in B) &= \mathbb{E}\left[\mathbb{I}_{\{X_{n+1}\in B\}}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\mathbb{I}_{\{X_{n+1}\in B\}}\middle|\mathcal{F}_n\right]\right] \\
&= \mathbb{E}[P(X_n, B)] \\
&= \int_E P(x, B)\mu_n(dx) \\
&= (\mu_n P)(B).
\end{aligned}$$

Hence, it follows that $\mu_{n+1} = \mu_n P$. □

Observe, that from the formulation above, it is easy to recover the finite state formulation.

Remark 4.17. If $E = \{1, \ldots, m\}$ is a finite state-space then,

$$\mu_n(dx) = \sum_{i=1}^{m} \mu_{n,i}\delta_i(dx).$$

Consequently, "right-multiplying" our law vector,

$$\begin{aligned}
(\mu_n P)(B) &= \int_E P(x, B)\mu_n(dx) \\
&= \sum_{i=1}^{m} \mu_{n,i} P(i, B) \overset{j\in B}{=} \sum_{i=1}^{m} \mu_{n,i} P_{ij} \\
&= \sum_{i=1}^{m} \mu_{n,i} \sum_{j\in B} P_{ij} \\
&= \sum_{j\in B} \sum_{i=1}^{m} \mu_{n,i} P_{ij}.
\end{aligned}$$

For a particular Borel set only containing a specific state, say $B = \{j\}$,

$$(\mu_n, P)_j = \sum_{i=1}^{m} \mu_{n,i} P_{ij}.$$

Hence, integrating against $\mu_n(dx)$ therefore replaces the sum over the current-state index i. If $B \in \mathcal{B}(E)$ contains several destination states, we also sum over those states.

As we have introduced, the action on functions is the transition operator already introduced,

$$(P\phi)(x) = \int_E \phi(y)P(x, dy).$$

x is a fixed current state while we average over all possible next states y. Thus, $P\phi$ is a function of the current state whereas μP is a probability measure on next states.

You may notice that this is analogous to,

$$(P\phi)_i = \sum_j P_{ij}\phi_j.$$

Integration against $P(x, dy)$ replaces the sum over the destination-state index j.

Proposition 4.13 (Conditional Expectations through the Transition Operator)**.** *For every bounded measurable function $\phi : E \to \mathbb{R}$,*

$$\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] = (P\phi)(X_n) \quad \textit{almost surely.}$$

Proof. We extend the *Markov Property* from indicators to simple functions and then to bounded measurable functions. On a finite state space, every function is already a finite sum of indicators on single events. On a general state space, we use an approximating sequence of simple functions.

First, for $\phi = \mathbb{I}_B$,

$$\begin{aligned}\mathbb{E}[\mathbb{I}_B(X_{n+1}) \mid \mathcal{F}_n] = P(X_n, B) &= \int_E \mathbb{I}_B(y) P(X_n, dy) \\ &= (P\mathbb{I}_B)(X_n).\end{aligned}$$

Now, consider the measurable simple function,

$$\phi_N = \sum_{i=1}^{m_N} a_{N,i}\mathbb{I}_{B_{N,i}}.$$

By linearity of expectation,

$$\begin{aligned}\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] &= \mathbb{E}\left[\sum_{i=1}^{m_N} a_{N,i}\mathbb{I}_{B_{N,i}}(X_{n+1}) \middle| \mathcal{F}_n\right] \\ &= \sum_{i=1}^{m_N} a_{N,i}\mathbb{E}[\mathbb{I}_{B_{N,i}}(X_{n+1}) \mid \mathcal{F}_n] = \sum_{i=1}^{m_N} a_{N,i}P(X_n, B_{N,i}) \\ &= \int_E \left(\sum_{i=1}^{m_N} a_{N,i}\mathbb{I}_{B_{N,i}}(y)\right) P(X_n, dy) \\ &= (P\phi_N)(X_n).\end{aligned}$$

The same simple function argument proposed previously, we can choose ϕ_N such that,

$$|\phi_N(x) - \phi(x)| \leq 2^{-N} \qquad \forall x \in E.$$

Using this inequality and showing that,

$$|(P\phi_N)(X_n) - (P\phi)(X_n)| \leq 2^{-N} \longrightarrow 0.$$

and letting $N \to \infty$ proves the proposition.

□

Remark 4.18. In general we approximate ϕ by a sequence,

$$\phi_N = \sum_{i=1}^{m_N} a_{N,i} \mathbb{I}_{B_{N,i}}$$

where the number of terms, coefficients, and sets may all depend on N.

The statement is not that an arbitrary fixed series of indicators converges to ϕ. We choose the simple functions specifically to approximate ϕ. For nonnegative measurable functions, they can be chosen to increase pointwise to ϕ. That is, for bounded functions used here, uniform approximation gives the distinct error bounds above.

Proposition 4.14 (Duality on a General State Space)**.** *Let μ be a probability measure on E and let P be a transition kernel, and $\phi : E \to \mathbb{R}$ be a bounded and measurable. Then,*

$$(\mu P)(\phi) = \mu(P\phi).$$

Equivalently, we can write

$$\int_E \phi(y)(\mu P)(dy) = \int_E \left[\int_E \phi(y) P(x, dy) \right] \mu(dx).$$

Proof. For an indicator $\phi = \mathbb{I}_B$, the identity follows from the definition of μP.

$$\begin{aligned}
(\mu P)(\mathbb{I}_B) &= (\mu P)(B) \\
&= \int_E P(x, B) \mu(dx) \\
&= \int_E (P\mathbb{I}_B)(x) \mu(dx) \\
&= \mu(P\mathbb{I}_B).
\end{aligned}$$

Define the simple function given as $\phi_N = \sum_{i=1}^{m_N} a_{N,i} \mathbb{I}_{B_{N,i}}$ with linearity giving,

$$\begin{aligned}
(\mu P)(\phi_N) &= \sum_{i=1}^{m_N} a_{N,i} (\mu P)(B_{N,i}) \\
&= \sum_{i=1}^{m_N} a_{N,i} \int_E P(x, B_{N,i}) \mu(dx) \\
&= \int_E \left[\sum_{i=1}^{m_N} a_{N,i} P(x, B_{N,i}) \right] \mu(dx) \\
&= \int_E (P\phi_N)(x) \mu(dx) \\
&= \mu(P\phi_N).
\end{aligned}$$

For a bounded measurable function ϕ, choose simple functions satisfying,

$$|\phi - \phi_N| \leq 2^{-N}.$$

Since we know that μP is a probability measure, $|(\mu P)(\phi) - (\mu P)(\phi_N)| \leq 2^{-N}$.

And moreover, we know that $|P\phi - P_{\phi_N}| \leq 2^{-N}$ so,

$$|\mu(P\phi) - \mu(P\phi_N)| \leq 2^{-N}.$$

Combining these estimates,

$$\begin{aligned}|(\mu P)(\phi) - \mu(P\phi)| &\leq |(\mu P)(\phi) - (\mu P)(\phi_N)| \\ &\quad + |(\mu P)(\phi_N) - \mu(P\phi_N)| + |\mu(P\phi_N) - \mu(P\phi)| \\ &\leq 2^{-N} + 0 + 2^{-N}.\end{aligned}$$

Letting $N \to \infty$ proves the identity.

□

We interpret this in the same way as in the finite-state case where we can either evolve the distribution and then average ϕ or first compute the expected next-step value $P\phi$ then average over the current distribution. If $X_n \sim \mu$, this is precisely the tower property given as,

$$\begin{aligned}(\mu P)(\phi) &= \mathbb{E}[\phi(X_{n+1})] \\ &= \mathbb{E}[\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n]] \\ &= \mathbb{E}[(P\phi)(X_n)] \\ &= \mu(P\phi).\end{aligned}$$

Definition 4.22 (Invariant Distribution on a General State Space)**.** A probability measure π on E is invariant for the transition kernel if it satisfies,

$$\pi P = \pi.$$

Observe that, explicitly this formulation is

$$\pi(B) = \int_E P(x, B)\pi(dx) \qquad \forall B \in \mathcal{B}(E).$$

This is precisely the finite-state condition we obtained previously,

$$\pi_j = \sum_{i=1}^{m} \pi_i P_{ij}$$

replacing sums with integration against the current-state distribution.

Proposition 4.15 (Test Function Characterization of Invariance)**.** *A probability measure π is invariant if and only if,*

$$\pi(P\phi) = \pi(\phi)$$

for every bounded, measurable $\phi : E \to \mathbb{R}$.

Proof. If $\pi P = \pi$, then by the duality formula

$$\pi(P\phi) = (\pi P)(\phi) = \pi(\phi).$$

Conversely, suppose that $\pi(P\phi) = \pi(\phi)$ for every bounded measurable ϕ. Then, for any $B \in \mathcal{B}(E)$, we can take $\phi = \mathbb{I}_B$ so that

$$\begin{aligned}(\pi P)(B) &= \int_E P(x, B)\pi(dx) \\ &= \pi(P\mathbb{I}_B) = \pi(\mathbb{I}_B) = \pi(B).\end{aligned}$$

Thus, it follows that $\pi P = \pi$.

□

In fact, invariance can be expressed through distributions, such as $\pi P = \pi$ or through the expectation test-function values given as,

$$\int_E (P\phi)(x)\pi(dx) = \int_E \phi(x)\pi(dx).$$

Definition 4.23 (Stationary Distribution)**.** A probability measure π on E is *stationary*, or *invariant*, for a transition kernel P if,

$$\pi P = \pi.$$

Explicitly, this is precisely

$$\int_E P(x, B)\pi(dx) = \pi(B) \qquad \forall B \in \mathcal{B}(E).$$

If $X_0 \sim \pi$, then by repeated application of this argument

$$\begin{aligned}\mathcal{L}(X_0) &= \pi \\ \mathcal{L}(X_1) &= \pi P = \pi \\ &\vdots \\ \mathcal{L}(X_n) &= \pi_n P = \pi \quad \forall n \geq 0.\end{aligned}$$

Hence, starting with distribution π leaves the distribution unchanged at every subsequent time.

Proposition 4.16 (Testing Stationarity with Functions)**.** *A probability measure π is stationary if and only if,*

$$\pi(P\phi) = \pi(\phi).$$

Proof. If $\pi P = \pi$, then by duality,

$$(\pi P)(\phi) = (\pi P)(\phi) = \pi(P\phi).$$

Conversely, suppose the test function identity holds. Then, for every Borel set $B \in \mathcal{B}(E)$, choose $\phi = \mathbb{I}_B$. Then, by duality again

$$\begin{aligned}(\pi P)(B) &= (\pi P)(\mathbb{I}_B) \\ &= \pi(P\mathbb{I}_B) \\ &= \pi(\mathbb{I}_B) = \pi(B).\end{aligned}$$

Hence it follows that $\pi P = \pi$.

□

This characterization of a stationary distribution is useful when we are averaging a general function under transition rules rather than calculating transition probabilities for arbitrary sets.

Example 4.6 (A Uniform Stationary Distribution)**.** Let,

$$\xi_1, \xi_2, \ldots \overset{i.i.d.}{\sim} \text{Bernoulli}\left(\frac{1}{2}\right), \qquad X_0 \in [0,1]$$

and define,

$$X_{n+1} = \frac{X_n + \xi_{n+1}}{2}.$$

We previously established that, this is a Markov chain on $[0,1]$ with transition kernel,

$$P(x, \cdot) = \frac{1}{2}\delta_{\frac{x}{2}} + \frac{1}{2}\delta_{\frac{x+1}{2}}$$

and transition operator

$$(P\phi)(x) = \frac{1}{2}\phi(x/2) + \frac{1}{2}\phi((x+1)/2).$$

Let λ denote the uniform probability measure on $[0,1]$. This is precisely the Lebesgue measure restricted to the interval $[0,1]$ so,

$$\lambda([a,b]) = b - a, \qquad 0 \le a \le b \le 1, \qquad \lambda(\phi) = \int_0^1 \phi(x)dx.$$

We claim that $\lambda P = \lambda$ for $\phi : [0,1] \to \mathbb{R}$ bounded, measurable functions. Then,

$$\begin{aligned}
\lambda(P\phi) &= \int_0^1 (P\phi)(x)dx \\
&= \frac{1}{2}\int_0^1 \phi(x/2)dx + \frac{1}{2}\int_0^1 \phi((x+1)/2)dx \\
&= \frac{1}{2}\int_0^{1/2} \phi(y)2dy + \frac{1}{2}\int_{1/2}^1 \phi(y)2dy \\
&= \int_0^{1/2} \phi(y)dy + \int_{1/2}^1 \phi(y)dy \\
&= \int_0^1 \phi(y)dy \\
&= \lambda(\phi).
\end{aligned}$$

Because this holds for every bounded, measurable test function ϕ, the test-function characterization gives us $\lambda P = \lambda$. Therefore, λ is a stationary distribution.

Remark 4.19. For each fixed x, the transition measure $P(x, \cdot)$ is concentrated at two points. Nonetheless, averaging these two-point measures over a uniformly distributed current state gives,

$$\int_{[0,1]} P(x, \cdot)\lambda(dx) = \lambda.$$

Thus, a chain can have discrete one-step conditional laws and a stationary distribution with a density. Moreover, in this example,

$$\lambda(\phi) = \int_0^1 \phi(x) \cdot 1 dx = \langle \phi, \mathbf{1} \rangle_{\mathcal{L}^2([0,1],dx)}.$$

Because ϕ is bounded and $[0,1]$ has finite Lebesgue measure, both ϕ and the constant function $\mathbf{1}$ belong to $\mathcal{L}^2([0,1], dx)$. More generally, if $\mu(dx) = f(x)dx$, then

$$\mu(\phi) = \int_E \phi(x)f(x)dx.$$

This is an $\mathcal{L}^2$ inner product when both ϕ and f are square-integrable. Without those additional assumptions, it remains a measure-function pairing, which is sufficient for our computations.

The preceding example verified stationary directly. In the next example, we first identify a candidate stationary distribution by examining the law after many steps, then verify that the candidate is invariant. Consider the following (Bill Nye).

Example 4.7 (Stationary Distribution of an AR(1) Process)**.** Let,

$$\xi_1, \xi_2, \ldots \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$$

independent of X_0 and consider,

$$X_{n+1} = aX_n + \xi_{n+1}, \qquad |a| < 1.$$

First suppose that $X_0 = x$ is deterministic. Expanding the recursion,

$$\begin{aligned}
X_1 &= ax + \xi_1 \\
X_2 &= aX_1 + \xi_2 \\
&= a\left(ax + \xi_1\right) + \xi_2 \\
&= a^2 x + a\xi_1 + \xi_2 \\
X_3 &= aX_2 + \xi_3 \\
&= a(a^x + a\xi_1 + \xi_2) + \xi_3 \\
&= a^3 x + a^2\xi_1 + a\xi_2 + \xi_3 \\
&\vdots \\
X_n &= a^n x + \sum_{k=1}^{n} a^{n-k}\xi_k, \qquad n \geq 0
\end{aligned}$$

which follows by induction. If it holds at time n then

$$\begin{aligned}
X_{n+1} &= aX_n + \xi_{n+1} \\
&= a\left(a^n x + \sum_{k=1}^{n} a^{n-k}\xi_k\right) + \xi_{n+1} \\
&= a^{n+1}x + \sum_{k=1}^{n} a^{n-k+1}\xi_k + \xi_{n+1} \\
&= a^{n+1}x + \sum_{k=1}^{n+1} a^{n-k+1}\xi_k.
\end{aligned}$$

ξ_k are independent Gaussians so their linear combination is Gaussian. Hence,

$$\begin{aligned}
\mathbb{E}[X_n] &= a^n x + \sum_{k=1}^{n} a^{n-k}\underbrace{\mathbb{E}[\xi_k]}_{=0} = a^n x \\
\operatorname{Var}(X_n) &= \operatorname{Var}\left(\sum_{k=1}^{n} a^{n-k}\xi_k\right) = \sum_{k=1}^{n} a^{2(n-k)}\operatorname{Var}(\xi_k) \\
&= \sum_{k=1}^{n} a^{2(n-k)} = \sum_{r=0}^{n-1} a^{2r} = \frac{1-a^{2n}}{1-a^2}.
\end{aligned}$$

Therefore, for $n \geq 1$,

$$X_n \sim \mathcal{N}\left(a^n x, \frac{1-a^{2n}}{1-a^2}\right).$$

Because $|a| < 1$, as $n \to \infty$, $a^n x \to 0$ and $a^{2n} \to 0$ so,

$$\frac{1 - a^{2n}}{1 - a^2} \longrightarrow \frac{1}{1 - a^2}.$$

Hence, we consider the limiting distribution $\pi := \mathcal{N}\left(0, \frac{1}{1-a^2}\right)$. Let Φ be the normal CDF,

$$\mathbb{P}(X_n \leq z) = \Phi\left(\frac{z - a^n x}{\sqrt{(1-a^{2n})/(1-a^2)}}\right) \longrightarrow \Phi\left(\frac{z}{\sqrt{1/(1-a^2)}}\right).$$

Convergence of CDFs is equivalent to stating convergence in distribution. Hence the limiting expression is the CDF of π which is continuous everywhere. Thus,

$$X_n \xrightarrow{d} X_\infty, \qquad X_\infty \sim \mathcal{N}\left(0, \frac{1}{1-a^2}\right).$$

It remains for us to verify that π is stationary. Suppose that,

$$X_0 \sim \pi = \mathcal{N}\left(0, \frac{1}{1-a^2}\right), \qquad X_0 \perp \xi_1.$$

Then, $X_1 = aX_0 + \xi_1$ so

$$\begin{aligned}
\mathbb{E}[X_1] &= a\mathbb{E}[X_0] + \mathbb{E}[\xi_1] \\
&= a \cdot 0 + 0 \\
&= 0, \\
\operatorname{Var}(X_1) &= a^2 \operatorname{Var}(X_0) + \operatorname{Var}(\xi_1) \\
&= \frac{a^2}{1-a^2} + 1 \\
&= \frac{a^2}{1-a^2} + \frac{1-a^2}{1-a^2} \\
&= \frac{1}{1-a^2}.
\end{aligned}$$

Consequently, we see that $X_1 \sim \mathcal{N}\left(0, \frac{1}{1-a^2}\right) = \pi$. Hence, since $\mathcal{L}(X_1) = \pi P$ we obtain $\pi P = \pi$. Explicitly, the stationary probability measure can be written as

$$\pi(B) = \int_B \sqrt{\frac{1-a^2}{2\pi}} \exp\left(-\frac{(1-a^2)y^2}{2}\right) dy \qquad B \in \mathcal{B}(\mathbb{R}).$$

Remark 4.20 (Gaussian Laws are Determined by their Parameters). A Gaussian law is completely determined by its mean and variance. In the stationarity calculation, X_0 and X_1 have exactly the same Gaussian parameters, so they have exactly the same distribution.

In the limiting calculation, the Gaussian parameters of X_n approach those of π. The CDF computation above also shows that this parameter convergence produces convergence in distribution.

$X_0 = x$ implies $\mathcal{L}(X_n)$ converges to x in distribution where as the distributional claim that $X_0 \sim \pi$ implies that $\mathcal{L}(X_n) = \pi$ for every n.

Remark 4.21 (Mean Reversion). The conditional mean satisfies,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = aX_n.$$

Because $|a| < 1$, its magnitude is smaller than that of the current state whenever $X_n \neq 0$,

$$|aX_n| = |a||X_n| < |X_n|.$$

Moreover, it follows

$$\mathbb{E}[X_{n+1} - X_n \mid \mathcal{F}_n] = (a-1)X_n.$$

This conditional expected increment is negative when $X_n > 0$ and positive when $X_n < 0$ so it points toward zero. For $0 < a < 1$, the conditional mean moves toward zero without changing sign. For $-1 < a < 0$, it changes sign while decreasing in magnitude.

The rate at which the effect of the initial condition disappears is controlled by $|a|^n$.

The fresh noise continues to enter at every step. Consequently, the process approaches a nondegenerate stationary distribution rather than settling at zero. The result proved here is convergence in distribution, not almost-sure convergence of the sample paths.

4.8 Discrete-time Generators & Kolmogorov Equations

The transition operator P describes the expected value of a function after one step,

$$(P\phi)(x) = \mathbb{E}_x[\phi(X_1)].$$

In this section, we introduce an operator that describes the *expected change* in that function over one step. This leads to two complementary descriptions of the chain. The *Kolmogorov* equations describe the evolution of distributions and expected function values. A related martingale decomposition separates the actual change along a sample path into its conditional expected change and a martingale remainder.

Throughout, denote $(X_n)_{n\geq 0}$ as the time-homogeneous Markov chain on E with transition kernel P and natural filtration $\mathcal{F}_n := \sigma(X_0, \ldots, X_n)$. We write,

$$P_n = P^n, \qquad P_0 = I$$

where I ise the identity operator on functions.

Definition 4.24 (Discrete-time Generator)**.** The generator associated with P is the operator,

$$L := P - I.$$

For every bounded, measurable test-function $\phi : E \to \mathbb{R}$,

$$(L\phi)(x) := (P\phi)(x) - \phi(x).$$

Equivalently, we can expand and express this as

$$\begin{aligned}(L\phi)(x) &= (P-I)\phi(x)\\ &= (P\phi)(x) - \phi(x)\\ &= \int_E \phi(y)P(x,dy) - \phi(x)\\ &= \int_E \phi(y)P(x,dy) - \int_E \phi(x)P(x,dy)\\ &= \int_E [\phi(y)-\phi(x)]\, P(x,dy).\end{aligned}$$

Hence, we can interpret the *Discrete-time Generator* of a *Markov chain* as, when we fix current state x, integrating computes the average change $\phi(y) - \phi(x)$ over all possible next states y.

Remark 4.22 (Conditional Expected Change). Applying the *Markov Property* gives,

$$\begin{aligned}\mathbb{E}[\phi(X_{n+1}) - \phi(X_n) \mid \mathcal{F}_n] &= \mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] - \phi(X_n)\\ &= (P\phi)(X_n) - \phi(X_n)\\ &= (P-I)\phi(X_n)\\ &= (L\phi)(X_n).\end{aligned}$$

Hence, the transition operator gives the conditional expectation of the next expected value while the generator gives the conditional expected increment. Consider the finite state-space interpretation.

Remark 4.23. On a finite state-space $E = \{1, \ldots, m\}$,

$$L = P - I_m, \qquad L_{ij} = \begin{cases} P_{ij}, & i \neq j,\\ P_{ii} - 1, & i = j.\end{cases}$$

The discrete-time finite state generator is therefore a matrix with rows,

$$\sum_{j=1}^{m} L_{ij} = \sum_{j=1}^{m} -1 = 0.$$

For a test-function $\phi : E \to \mathbb{R}$,

$$\begin{aligned}(L\phi)(i) &= \sum_{j=1}^{m} P_{ij}\phi(j) - \phi(i)\\ &= \sum_{j=1}^{m} P_{ij}\left[\phi(j) - \phi(i)\right].\end{aligned}$$

The generator is not itself a transition matrix since its diagonal entries may be negative and its rows sum to zero rather than one, which violate the condition $P_{ij} \geq 0$.

In discrete time,

$$L = P - I \iff P = I + L.$$

Consequently, specifying (μ_0, L) specifies the same transition model as (μ_0, P) and therefore determines all finite-dimensional distributions of the chain.

A fixed process law need not determine the kernel at states the process never visits. The relevant statement here is that the specified initial distribution and kernel determine the process law.

The generator viewpoint will become particularly useful in continuous time, where a model is often specified through its local evolution even when its transition probabilities are difficult to calculate. The continue-time generator requires a different, infinitesimal definition.

To describe expected values as fucntions of the starting state, we first define $\mathbb{E}_x$.

Definition 4.25 (Expectation from a Fixed Starting State)**.** Write $\mathbb{E}_x$ for the expectation under the Markov chain with transition kernel P and initial distribution,

$$\mu_0 = \delta_x.$$

Thus, $X_0 = x$ almost surely under this starting law. For bounded, measurable $\phi : E \to \mathbb{R}$,

$$\mathbb{E}_x[\phi(X_n)] = \int_E \phi(y) P_n(x, dy) = (P_n\phi)(x).$$

By definition, $\mathbb{E}_x[\phi(X_0)] = \phi(x)$. This definition specifies the starting state directly but does not require the event $\{X_0 = x\}$ to have positive probability under some other initial distribution.

Remark 4.24. For a chain with a possibly random initial state,

$$\mathbb{E}_x[\phi(X_n)] = \mathbb{E}[\phi(X_n) \mid \sigma(X_0)] = \mathbb{E}[\phi(X_n) \mid X_0 = x] = (P_n\phi)(X_0) \quad \text{almost surely.}$$

Therefore, we can interpret $\mathbb{E}_x[\phi(X_n)]$ as the expected value of $\phi(X_n)$ given that the chain starts at x. On a continuous state space, the right-hand side is understood through a conditional law rather than through a ratio involving $\mathbb{P}(X_0 = x)$. Consider the densities heuristic.

Example 4.8 (Conditional Transition Densities)**.** Suppose,

$$\mu_0(dx_0) = f_0(x_0)dx_0, \quad P(x_0, dx_1) = p(x_0, x_1)dx_1.$$

By the joint-law formula,

$$\begin{aligned}\mu_{0:1}(dx_0, dx_1) &= \mu_0(dx_0)P(x_0, dx_1) \\ &= f_0(x_0)p(x_0, x_1)dx_0dx_1 \iff f_{0:1}(x_0, x_1) = f_0(x_0)p(x_0, x_1).\end{aligned}$$

$f_0(x_0) > 0$ so we recover the conditional-density formula,

$$p(x_0, x_1) = \frac{f_{0:1}(x_0, x_1)}{f_0(x_0)}.$$

Hence,

$$\mathbb{E}_x[\phi(X_1)] = \int_E \phi(x_1)p(x, x_1)dx_1 = (P\phi)(x).$$

At points where $f_0(x) = 0$, the ratio does not determine a conditional density. The specified kernel $P(x, \cdot)$ still describes how the chain evolves when started at x.

The semigroup property now gives an evolution equaiton for expected function values. For a fixed bounded, measurable function ϕ, define

$$u_n(x) := \mathbb{E}_x[\phi(X_n)] = (P_n\phi)(x).$$

For each $n \geq 0$, u_n is a function of the starting state x. Since $P_{n+1} = PP_n$,

$$\begin{aligned} u_{n+1}(x) - u_n(x) &:= \mathbb{E}_x[\phi(X_{n+1})] - \mathbb{E}_x[\phi(X_n)] \\ &= P(P_n\phi)(x) - (P_n\phi)(x) \\ &= (P - I)(P_n\phi)(x) \\ &= L(P_n\phi)(x) \\ &= Lu_n(x) \end{aligned}$$

where L is the discrete-time generator defined previously. This, as we will see, is the discrete backward *Kolmogorov* equation. There is a corresponding forward equation for the distributions $\mu_n = \mathcal{L}(X_n)$. First, let us define the action of the generator on a measure.

Definition 4.26 (Action of the Generator on a Measure)**.** For a finite probability measure μ, define

$$\mu L := \mu P - \mu.$$

This is a finite signed measure. For all $B \in \mathcal{B}(E)$,

$$(\mu L)(B) = \int_E P(x, B)\mu(dx) - \mu(B).$$

For a bounded measurable function ϕ, by duality we obtain,

$$\begin{aligned} (\mu L)(\phi) &= (\mu P)(\phi) - \mu(\phi) \\ &= \mu(P\phi) - \mu(\phi) \\ &= \mu(L\phi). \end{aligned}$$

Therefore, $(\mu L)(\phi) = \mu(L\phi)$. Unlike μP, the measure μL is generally not a probability measure. It records the change in probability over one step and its total mass is computed as

$$(\mu L)(E) = 1 - 1 = 0.$$

Theorem 4.2 (Discrete Kolmogorov Equations)**.** *Let $(X_n)_{n\geq 0}$ be a time-homogeneous Markov chain with transition kernel P and generator $L = P - I$.*

For a bounded measurable $\phi : E \to \mathbb{R}$ define,

$$u_n(x) := \mathbb{E}_x[\phi(X_n)] = (P_n\phi)(x).$$

Then,

$$\begin{cases} u_{n+1} - u_n = Lu_n, & u_0 = \phi, & \textit{(Backward Kolmogorov)}, \\ \mu_{n+1} - \mu_n = \mu_n L, & \mu_0 = \mathcal{L}(X_0), & \textit{(Forward Kolmogorov)}. \end{cases}$$

Proof. For the backward equation, let $X_0 = x$ almost surely

$$\begin{aligned} u_{n+1}(x) - u_n(x) &:= \mathbb{E}_x[\phi(X_{n+1})] - \mathbb{E}_x[\phi(X_n)] \\ &= P(P_n\phi)(x) - (P_n\phi)(x) \\ &= (P_{n+1}\phi)(x) - (P_n\phi)(x) \\ &= (P - I)(P_n\phi)(x) \\ &= L(P_n\phi)(x) \\ &= Lu_n, \qquad u_n(x) = (P_n\phi)(x). \end{aligned}$$

The initial condition follows from $P_0 = I$ so $u_0 = P_0\phi = \phi$. For the forward equation, recall that $\mu_{n+1} = \mu_n P$ is the distribution update. Hence, we can write

$$\begin{aligned} \mu_{n+1} - \mu_n &= \mu_n P - \mu_n \\ &= \mu_n(P - I) \\ &= \mu_n L. \end{aligned}$$

This equality is an equality on signed measures. Explicitly, for every Borel set $B \in \mathcal{B}(E)$,

$$\mu_{n+1}(B) - \mu_n(B) = \int_E P(x, B)\mu_n(dx) - \mu_n(B).$$

□

Remark 4.25. The forward equation acts on distributions, that is $\mu_n \mapsto \mu_{n+1}$ describing how probability moves between states. The backward equation acts on functions of the starting state,

$$u_n(x) = \mathbb{E}_x[\phi(X_n)].$$

To compute $u_{n+1}(x)$, take one transition from x then use u_n to evaluate the expected outcome over the remaining n steps, that is

$$u_{n+1}(x) = \int_E u_n(y)P(x, dy).$$

Expanding the first term of the backward equation,

$$\begin{aligned} u_{n+1}(x) &= (P_{n+1}\phi)(x) \\ &= (PP_n\phi)(x) \\ &= (Pu_n)(x) \\ &= \int_E u_n(y)P(x,dy) \\ &= \int_E \left(\int_E \phi(k)P_n(y,dk) \right) P(x,dy) \\ &= \int_E \phi(k) \left(\int_E P_n(y,dk)P(x,dy) \right) \\ &= \int_E \phi(k)P_{n+1}(x,dk). \end{aligned}$$

Hence, the term backward refers to the action on expected future values as functions of the starting state. Both equations can also be solved recursively,

$$u_{n+1} = (I+L)u_n = Pu_n, \qquad \mu_{n+1} = \mu_n(I+L) = \mu_n P.$$

Consequently,

$$u_n = P_n\phi, \qquad \mu_n = \mu_0 P.$$

Recall that a probability measure is stationary precisely when $\pi P = \pi$. Since $L = P - I$, this is equivalent to writing that $\pi L = 0$. In test-function form,

$$\pi(L\phi) = 0 \qquad \forall \phi : E \to \mathbb{R}.$$

Thus, under a stationary distribution, the average one-step change in every bounded measurable test function is zero. Hence, the *Kolmogorov Equations* describe expected evolution. Along an individual sample path, the actual increment need not equal its conditional expectation. THe difference between the two produces a martingale.

Proposition 4.17 (Martingale Associated with the Generator)**.** *Let $\phi : E \to \mathbb{R}$ be bounded and measurable. Define,*

$$M_n^\phi := \phi(X_n) - \phi(X_0) - \sum_{k=0}^{n-1} (L\phi)(X_k), \qquad n \geq 0.$$

Proof. Because ϕ is measurable, $\phi(X_n)$ is $\mathcal{F}_n$-measurable. The function $L\phi = P\phi - \phi = (P-I)\phi$ is also measurable. For $k \leq n-1$, $(L\phi)(X_k)$ is $\mathcal{F}_k$-measurable and therefore $\mathcal{F}_n$-measurable. Hence, M_n^ϕ is $\mathcal{F}_n$ measurable. For integrability, choose $C < \infty$ such that,

$$|\phi(x)| \leq C \qquad \forall x \in E.$$

Since $P(x,\cdot)$ is a probability measure,

$$|(P\phi)(x)| \leq C.$$

Therefore, the discrete generator is also bounded

$$|(L\phi)(x)| = |(P\phi)(x) - \phi(x)| \leq 2C.$$

For each fixed n, the defined remainder

$$\begin{aligned} |M_n^\phi| &= |\phi(X_n)| + |\phi(X_0)| + \sum_{k=0}^{n-1} |(L\phi)(X_k)| \\ &\leq C + C + 2nC \\ &= 2(n+1)C. \end{aligned}$$

We use the martingale difference sequence of prove the martingale condition. Subtracting terms

$$\begin{aligned} M_{n+1}^\phi - M_n^\phi &= \left[\phi(X_{n+1}) - \phi(X_0) - \sum_{k=0}^{n}(L\phi)(X_k)\right] \\ &\quad - \left[\phi(X_n) - \phi(X_0) - \sum_{k=0}^{n-1}(L\phi)(X_k)\right] \\ &= \phi(X_{n+1}) - \phi(X_n) - (L\phi)(X_n). \end{aligned}$$

Taking conditional expectations on $\mathcal{F}_n$,

$$\begin{aligned} \mathbb{E}\left[M_{n+1}^\phi - M_n^\phi \middle| \mathcal{F}_n\right] &= \mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] - \mathbb{E}[\phi(X_n) \mid \mathcal{F}_n] - \mathbb{E}[(L\phi)(X_n) \mid \mathcal{F}_n] \\ &= (P\phi)(X_n) - \phi(X_n) - (L\phi)(X_n) \\ &= (P - I)\phi(X_n) - (L\phi)(X_n) \\ &= (L\phi)(X_n) - (L\phi)(X_n) \\ &= 0. \end{aligned}$$

Since M_n^ϕ is $\mathcal{F}_n$-measurable, it follows $\mathbb{E}[M_{n+1}^\phi \mid \mathcal{F}_n] = M_n^\phi$.

Therefore, we conclude that $(M_n^\phi)_{n\geq 0}$ is a martingale.

□

Re-arranging the martingale remainder definition above,

$$\phi(X_n) = \phi(X_0) + \sum_{k=0}^{n-1}(L\phi)(X_k) + M_n^\phi$$

where at each time k, $(L\phi)(X_k) = \mathbb{E}[\phi(X_{k+1}) - \phi(X_k) \mid \mathcal{F}_k]$. Thus, the sum accumulates the conditional expected increments. The martingale M_n^ϕ therefore accumulates the differences between the realized increments and those predictions.

More precisely, it is the realized change in $\phi(X_n)$ after subtracting the accumulated change that could have been predicted with the transition kernel P. Consider the following expansion.

The martingale M_n^ϕ can be broken down into components,

$$\begin{aligned} M_n^\phi &= \phi(X_n) - \phi(X_0) - \sum_{k=0}^{n-1} (L\phi)(X_k) \\ &= \sum_{k=0}^{n-1} [\phi(X_{k+1}) - \phi(X_k)] - \sum_{k=0}^{n-1} (L\phi)(X_k) \\ &= \sum_{k=0}^{n-1} [\phi(X_{k+1}) - \phi(X_k) - (L\phi)(X_k)] . \end{aligned}$$

Hence, $\phi(X_n) - \phi(X_0)$ are the actual, change in the function along the Markov chain while the sum over $(L\phi)(X_k)$ at times k are the cumulative expected one-step change, evaluated along the states the chain actually visits. M_n^ϕ is therefore the difference between the realized and predictable change and this difference sequence is a martingale.

The accumulated conditional expected changes are themselves random, because they are evaluated along the random states $X_0, \ldots, X_{n-1}$. The decomposition does not assert that the path consists of a deterministic trend plus some independent noise.

Proposition 4.18 (Expected Accumulated Change)**.** *For a fixed starting state $X_0 = x$,*

$$\mathbb{E}_x[\phi(X_n)] = \phi(x) + \sum_{k=0}^{n-1} \mathbb{E}_x[(L\phi)(X_k)] = (P_n\phi)(x).$$

Proof. For $n = 0$, it follows since a martingale starts a zero $\mathbb{E}_x[M_n^\phi] = 0$. Taking expectations,

$$\begin{aligned} \mathbb{E}_x[M_n^\phi] = 0 &= \mathbb{E}_x[\phi(X_n)] - \mathbb{E}_x[\phi(X_0)] - \sum_{k=0}^{n-1} [(L\phi)(X_k)] \\ &= \mathbb{E}_x[\phi(X_n)] - \phi(x) - \sum_{k=0}^{n-1} \mathbb{E}_x[(L\phi)(X_k)]. \end{aligned}$$

Re-arranging proves the first identity. The second equality follows from,

$$\begin{aligned} \mathbb{E}_x[(L\phi)(X_k)] &= \mathbb{E}[(L\phi)(X_k) \mid X_0 = x] \\ &= \int_E (P - I)\phi(X_k) P_k(x, dz) \\ &= \int_E [(P\phi)(X_k) - \phi(X_k)] \, P_k(x, dz) \\ &= \int_E \left(\int_E [\phi(y) - \phi(X_k)] \, P(X_k, dy) \right) P_k(x, dz) \\ &= P_k \int_E [\phi(y) - \phi(x)] \, P(X_k, dy) \\ &= P_k(L\phi)(x). \end{aligned}$$

□

Subtracting this identity at times $n+1$ and n gives,

$$P_{n+1}\phi - P_n\phi = P_nL\phi.$$

Because, $P_n = P^n$ and $L = P - I$,

$$\begin{aligned} P_nL &= P^n(P-I) = P^{n+1} - P^n \\ &= (P-I)P^n \\ &= LP_n. \end{aligned}$$

Therefore,

$$P_{n+1}\phi - P_n\phi = LP_n\phi$$

which recovers the backward *Kolmogorov* equation.

Definition 4.27 (Harmonic, Subharmonic, and Superharmonic Functions)**.** Let $\phi : E \to \mathbb{R}$ be bounded and measurable test functions. ϕ is *harmonic* if,

$$L\phi = 0.$$

Similarly, the inequalities also have definitions:

1. ϕ is *subharmonic* if $L\phi \geq 0$, and
2. ϕ is *superharmonic* if $L\phi \leq 0$.

These are pointwise conditions on the state space E. Equivalently,

$$\begin{aligned} L\phi = 0 &\iff P\phi = \phi, \\ L\phi \geq 0 &\iff P\phi \geq \phi, \\ L\phi \leq 0 &\iff P\phi \leq \phi. \end{aligned}$$

Proposition 4.19 (Harmonicity and Martingales)**.** *If ϕ is harmonic, then $(\phi(X_n))_{n\geq 0}$ is a martingale. Similarly, if ϕ is subharmonic, it is a submartingale, if ϕ is superharmonic, it is a supermartingale. Conversely, each corresponding process property when required for the chain started from every state $x \in E$ implies the respective pointwise condition $L\phi$.*

Proof. The process $\phi(X_n)$ is adapted and integrable because ϕ is bounded and integrable. Moreover, we can decompose the change as

$$\mathbb{E}[\phi(X_{n+1}) \mid \mathcal{F}_n] = (P\phi)(X_n) = \phi(X_n) + (L\phi)(X_n).$$

If $L\phi = 0$, the right-hand side equals $\phi(X_n)$. If $L\phi \geq 0$, it is at least $\phi(X_n)$, otherwise if $L\phi \leq 0$, it as at most $\phi(X_n)$. These are exactly the martingale, submartingale, and supermartingale conditions, respectively. For the converse, suppose $\phi(X_n)$ is a martingale under every fixed starting state $X_0 = x \in E$. At time zero,

$$\mathbb{E}_x[\phi(X_1)] = \phi(x).$$

Therefore,

$$(L\phi)(x) = \mathbb{E}_x[\phi(X_1)] - \phi(x) = 0.$$

Since x is arbitrary, $L\phi = 0$ throughout E. The same argument with inequalities process the converses for submartingales and supermartingales.

□

For one particular initial distribution, the martingale property only implies that,

$$(L\phi)(X_n) = 0 \qquad \text{almost surely } \forall n.$$

However, it does not impose conditions at states that the chain never visits. Requiring the property for every starting state yields the pointwise statement $L\phi = 0$ on all E.

To measure long-term behaviour, it is useful to measure the size of functions relative to a stationary distribution π. This introduces a geometry in which states are weighted according to their stationary probabilities.

Definition 4.28 ($\mathcal{L}^2(\pi)$, Inner Product, and Norm)**.** Let π be a probability measure on E. Define,

$$\mathcal{L}^2(\pi) := \left\{ \phi : E \to \mathbb{R} : \int_E |\phi(x)|^2 \pi(dx) < \infty \right\}$$

where functions equal π-almost everywhere are identified. For $\phi, \psi \in \mathcal{L}^2(\pi)$ define,

$$\langle \phi, \psi \rangle_\pi := \int_E \phi(x)\psi(x)\pi(dx).$$

The corresponding norm under $\mathcal{L}^2(\pi)$ is,

$$\|\pi\|_\pi^2 = \langle \phi, \phi \rangle_\pi = \int_E |\phi(x)|^2 \pi(dx).$$

By *Cauchy-Schwarz* inequality, the inner product define above is finite,

$$\int_E |\phi(x)\psi(x)|\pi(dx) \leq \|\phi\|_\pi \|\psi\|_\pi.$$

Remark 4.26. Consider the finite countable state space $E = \{1, \ldots, m\}$ then,

$$\langle \phi, \psi \rangle_\pi = \sum_{i=1}^m \pi_i \phi_i \psi_i, \qquad \|\phi\|_m^2 = \sum_{i=1}^m \pi_i \phi_i^2.$$

The stationary probabilities act as weights while the differences between functions at state with larger π contribute more to their $\mathcal{L}^2(\pi)$ distance.

If π is uniform on this finite state space, $\pi_i = \frac{1}{m}$ hence,

$$\langle \phi, \psi \rangle_\pi = \frac{1}{m} \sum_{i=1}^m \phi_i \psi_i = \frac{1}{m} \phi^\top \psi.$$

Thus, the $\mathcal{L}^2(\pi)$ inner product is a constant multiple of the usual Euclidean inner product. If some $\pi_i = 0$, the value of the function at state i does not affect its norm which is precisely why $\mathcal{L}^2(\pi)$ identifies functions that agree π-almost everywhere.

Before defining reversibility on $\mathcal{L}^2(\pi)$, we verify that the transition operator and generator act on this space when π is stationary. Consider the following proposition.

Proposition 4.20 (Transition Operator on $\mathcal{L}^2(\pi)$)**.** *Suppose $\pi p = \pi$ is the stationary distribution. Then P maps $\mathcal{L}^2(\pi)$ into itself and,*

$$\|P\phi\|_\pi \leq \|\phi\|_\pi.$$

Consequently, $L = P - I$ also maps $\mathcal{L}^2(\pi)$ into itself with,

$$\|L\phi\|_\pi \leq 2\|\phi\|_\pi.$$

Proof. Stationary implies, for every nonnegative bounded, measurable function, the same integration identity holds for bounded test functions. By *Monotone Approximation* and Simple Approximation Theorem,

$$\int_E \left(\int_E |\phi(x)|^2 P(x, dy) \right) \pi(dx) = \int_E |\phi(x)|^2 (\pi P)(dy) = \int_E |\phi(y)|^2 \pi(dy) < \infty.$$

Thus, the inner integral is finite for π-almost every x. For such x, by *Cauchy-Schwarz*

$$\begin{aligned} |(P\phi)(x)|^2 &= \left| \int_E \phi(y) P(x, dy) \right|^2 \\ &\leq \left(\int_E |\phi(y)|^2 P(x, dy) \right) \left(\int_E 1 \, P(x, dy) \right) \\ &= \int_E |\phi(y)|^2 P(x, dy) \end{aligned}$$

under the probability measure $P(x, \cdot)$. Integrating against π,

$$\begin{aligned} \|P\phi\|_\pi^2 &\leq \int_E \left[\int_E |\phi(y)|^2 P(x, dy) \right] \pi(dx) \\ &= \|\phi\|_\pi^2. \end{aligned}$$

Hence, we obtain the contraction inequality. The same inequality shows that if $\phi = \psi$ π-almost everywhere, then $P\phi = P\psi$ π-almost everywhere. Therefore, P is well-defined on the equivalence classes in $\mathcal{L}^2(\pi)$. Finally, the final equality is obtained by

$$\begin{aligned} \|L\phi\|_\pi &= \|P\phi - \phi\|_\pi \\ &\leq \|P\phi\|_\pi + \|\phi\|_\pi \\ &\leq 2\|\phi\|_\pi. \end{aligned}$$

□

Definition 4.29 (Reversibility)**.** Let π be a stationary distribution. The discrete-time generator L is *reversible* with respect to π if,

$$\langle \phi, L\psi \rangle_\pi = \langle L\phi, \psi \rangle_\pi, \qquad \forall \phi, \psi \in \mathcal{L}^2(\pi).$$

This is analogous to L being self-adjoint on $\mathcal{L}^2(\pi)$. In the present discrete-time setting, L is a bounded operator on the whole space, as established in *Proposition 4.20*.

Proposition 4.21 (Equivalent Reversibility Conditions for P and I)**.** *Since $L = P - I$,*

$$\langle \phi, L\psi \rangle_\pi = \langle L\phi, \psi \rangle_\pi, \quad \forall \phi, \psi \in \mathcal{L}^2(\pi) \iff \langle \phi, P\psi \rangle_\pi = \langle P\phi, \psi \rangle_\pi$$

for all such ϕ, ψ.

Proof. Expanding both sides,

$$\begin{aligned} \langle \phi, L\psi \rangle_\pi &= \langle \phi, P\psi - \psi \rangle_\pi \\ &= \langle \phi, P\psi \rangle_\pi - \langle \phi, \psi \rangle_\pi, \\ \langle L\phi, \psi \rangle_\pi &= \langle P\phi - \phi, \psi \rangle_\pi \\ &= \langle P\phi, \psi \rangle_\pi - \langle \phi, \psi \rangle_\pi. \end{aligned}$$

The common terms cancel, proving the equivalence.

□

Example 4.9 (Uniform Distribution on a Finite State Space)**.** Suppose we have a finite state space with,

$$E = \{1, \ldots, m\}, \qquad \pi_i = \frac{1}{m}.$$

Then, the self-adjoint identity

$$\begin{aligned} \langle \phi, L\psi \rangle_\pi &= \frac{1}{m} \phi^\top L \psi, \\ \langle \phi, \psi \rangle_\pi &= \frac{1}{m} (L\phi)^\top \psi = \frac{1}{m} \phi^\top L^\top \psi. \end{aligned}$$

Therefore, reversibility is equivalent to,

$$\phi^\top (L - L^\top) \psi = 0 \qquad \forall \phi, \psi \in \mathbb{R}^m.$$

Choosing ϕ, ψ to be coordinate vectors shows that every entry of $L - L^\top$ must vanish. Hence, we are left with $L = L^\top$. Since $L = P - I_m$ it follows,

$$\begin{aligned} L = L^\top &\iff P - I_m = P^\top - I_m \\ &\iff P = P^\top. \end{aligned}$$

Hence, reversibility with respect to the uniform distribution exists if and only if $P = P^\top$, that is our transition kernel/matrix is symmetric. For completeness, a symmetric transition matrix does not preserve the uniform distribution. Observe,

$$(\pi P)_j = \frac{1}{m}\sum_{i=1}^{m} P_{ij} = \frac{1}{m}\sum_{i=1}^{m} P_{ji} = \frac{1}{m} = \pi_j.$$

Remark 4.27 (Weighted Symmmetry for a Nonuniform Distribution). For a general finite-state stationary distribution, reversibility is symmetry relative to the weights π_i, rather than ordinary matrix symmetry. Taking $\phi = \mathbb{I}_{\{i\}}$ and $\psi = \mathbb{I}_{\{j\}}$, the reversibility identity for P gives

$$\begin{aligned}\langle \mathbb{I}_{\{i\}}, P\mathbb{I}_{\{j\}}\rangle_\pi &= \pi_i P_{ij},\\ \langle P\mathbb{I}_{\{i\}}, \mathbb{I}_{\{j\}}\rangle_\pi &= \pi_j P_{ji}.\end{aligned}$$

Therefore, $\pi_i P_{ij} = \pi_j P_{ji}$. These are the detailed balance equations which equate the stationary probability of a transition from i to j with that of the reverse transition.

Conversely, if these identities hold then

$$\begin{aligned}\langle \phi, P\psi\rangle_\pi &= \sum_{i=1}^{m}\sum_{j=1}^{m} \pi_i P_{ij}\phi_i\psi_j\\ &= \sum_{i=1}^{m}\sum_{j=1}^{m} \pi_j P_{ji}\phi_i\psi_j\\ &= \sum_{j=1}^{m} \pi_j \left(\sum_{i=1}^{m} P_{ji}\phi_i\right)\psi_j\\ &= \langle P\phi, \psi\rangle_\pi.\end{aligned}$$

Thus, detailed balance is precisely the finite-state form of reversibility. This geometry provides a framework for studying convergence toward stationarity.

Reversibility alone however, does not establish convergence but supplies a symmetry of the transition operator that can be used in this analysis.

4.9 Convergence to a Stationary Distribution

Now, we can use the results of reversibility and the generator $L = P - I$ to study how quickly a Markov chain approaches its stationary distribution. Since π is a stationary distribution, $\pi P = \pi$. Recall, the actions of the inner product and norm,

$$\langle \phi, \psi\rangle_\pi = \int_E \phi(x)\psi(x)\pi(dx), \quad \|\phi\|_\pi^2 = \int_E |\phi(x)|^2\pi(dx).$$

Since π is stationary, P maps $\mathcal{L}^2(\pi)$ into itself. We also denote, $P_n = P^n$ and $L := P - I$, as before. The main result will show that $P_n\phi$ decays geometrically, where ϕ has mean zero under π. We will then explain when this gives convergence of the distributions $\mu_n = \mathcal{L}(X_n)$ to π.

Definition 4.30 (Positive Semidefiniteness of a Transition Operator)**.** The transition operator P is *positive semidefinite* in $\mathcal{L}^2(\pi)$ if,

$$\langle \phi, P\phi \rangle_\pi \geq 0, \qquad \forall \phi \in \mathcal{L}^2(\pi).$$

Together with reversibility,

$$\langle \phi, P\psi \rangle_\pi = \langle P\phi, \psi \rangle_\pi.$$

Positive semidefiniteness is different from the order-preserving property of a transition operator as every transition operator satisfies, $\phi \geq 0 \implies P\phi \geq 0$. However, it need not satisfy $\langle \phi, P\phi \rangle_\pi \geq 0$ for all test functions ϕ that take both positive and negative values.

For a finite state space $E = \{1, \ldots, m\}$, π is uniform, and P is reversible, it is equivalent to express our transition kernel as a symmetric transition matrix $P = P^\top$,

$$\langle \phi, P\phi \rangle_\pi = \frac{1}{m} \phi^\top P \phi.$$

For a symmetric matrix, P is positive semidefinite if and only if $\lambda_{\min}(P) \geq 0$. Hence, the positive semidefiniteness condition excludes negative eigenvalues of P.

To study convergence toward π, we separate constant functions from functions whose averages under π is zero. Write $\mathbf{1}$ for the constant function equal to one. Since $P(x, \cdot)$ is a valid probability measure, over the entire state space E

$$(P\mathbf{1})(x) = \int_E 1 P(x, dy) = 1.$$

Therefore, $P\mathbf{1} = \mathbf{1}$ and $L\mathbf{1} = 0$. Constants are unchanged by the transition operator. Consequently, we cannot expect $P_n\phi \to 0$ for every function ϕ, hence we remove its constant component first. First we define the following subspace.

Definition 4.31 (Mean-zero Subspace)**.** Define,

$$\mathcal{L}^2_0(\pi) := \left\{ \phi \in \mathcal{L}^2(\pi) : \pi(\phi) = 0 \right\}.$$

where $\pi(\phi) = \int_E \phi(x)\pi(dx)$. Because, $\langle \phi, \mathbf{1} \rangle_\pi = \pi(\phi)$,

$$\phi \in \mathcal{L}^2_0(\pi) \iff \phi \perp \mathbf{1} \in \mathcal{L}^2(\pi).$$

From this definition, we can write every $\phi \in \mathcal{L}^2(\pi)$ as

$$\phi = \pi(\phi)\mathbf{1} + (\phi - \pi(\phi)\mathbf{1}).$$

The second term has mean zero,

$$\begin{aligned} \pi(\phi - \pi(\phi)\mathbf{1}) &= \pi(\phi) - \pi(\phi)\pi(\mathbf{1}) \\ &= \pi(\phi) - \pi(\phi) \\ &= 0. \end{aligned}$$

We will abbreviate this centered function $\phi - \pi(\phi)$. A constant test function has the same expectation under every probability distribution, hence it is indistiguishable from one distribution to another. Hence, we can quantify deviations from stationarity using test functions with mean zero under π. $\pi(\phi) = 0$ is of particular interest, rather than simply requiring ϕ be nonconstant.

Definition 4.32 (Spectral Gap)**.** L has a *spectral gap* of at least $\lambda > 0$ in $\mathcal{L}^2(\pi)$ if,

$$-\langle \phi, L\phi \rangle_\pi \geq \lambda \|\phi\|_\pi^2, \qquad \forall \phi \in \mathcal{L}_0^2(\pi).$$

Substituting $L := P - I$,

$$\begin{aligned} -\langle \phi, L\phi \rangle_\pi &= -\langle \phi, (P-I)\phi \rangle_\pi \\ &= -\langle \phi, P\phi - \phi \rangle_\pi \\ &= -\langle \phi, P\phi \rangle_\pi - (-\langle \phi, \phi \rangle_\pi) \\ &= \|\phi\|_\pi^2 - \langle \phi, P\phi \rangle_\pi \geq \lambda \|\phi\|_\pi^2. \end{aligned}$$

Re-arranging the norms, we obtain

$$\begin{aligned} \implies -\langle \phi, P\phi \rangle_\pi &\geq \lambda \|\phi\|_\pi^2 - \|\phi\|_\pi^2 \\ \langle \phi, P\phi \rangle_\pi &\leq (1-\lambda)\|\phi\|_\pi^2. \end{aligned}$$

If P is also positive semidefinite, the two conditions together

$$0 \leq \langle \phi, P\phi \rangle_\pi \leq (1-\lambda)\|\phi\|_\pi^2, \qquad \pi(\phi) = 0.$$

The *Spectral gap* inequality becomes, for $L = L^\top$,

$$-\phi^\top L\phi \geq \lambda \phi^\top \phi, \qquad \phi^\top \mathbf{1} = 0.$$

If $\phi \neq 0$ is a mean-vector eigenvector satisfying $L\phi = \lambda_1 \phi.$, then

$$-\phi^\top L\phi = -\lambda_1 \phi^\top \phi \geq \lambda \phi^\top \phi.$$

Hence, dividing by $\phi^\top \phi > 0$ gives $-\lambda_1 \geq \lambda$. Thus, the eigenvalues of $-L$ on the subspace orthogonal to constants are bounded below by λ. Equivalently, the eigenvalues of $P = I + L$ on that subspace are bounded above $1 - \lambda$.

Under positive semidefiniteness, eigenvalues of P lie in $[0, 1]$. Repeated application of transition matrix P therefore reduces each such eigenvector component geometrically. A positive spectral gap also rules out nonconstant mean-zero functions in the kernel of L.

Indeed, if $L\phi = 0$ and $\pi(\phi) = 0$, then

$$0 = -\langle \phi, L\phi \rangle_\pi \geq \lambda \|\phi\|_\pi^2,$$

so $\phi = 0$ in $\mathcal{L}^2(\pi)$.

Lemma 4.2 (Cauchy-Schwarz for a Positive Semidefinite Operator)**.** *Let P be a reversible and positive semidefinite operator in $\mathcal{L}^2(\pi)$. Then, for every $\phi, \psi \in \mathcal{L}^2(\pi)$,*

$$|\langle \psi, P\phi \rangle_\pi|^2 \leq \langle \psi, P\phi \rangle_\pi \langle \phi, P\phi \rangle_\pi.$$

Proof. For every $t \in \mathbb{R}$, positive semidefiniteness yields,

$$0 \leq \langle \psi - t\phi, P(\psi - t\phi) \rangle_\pi.$$

Expanding the inequality above,

$$\begin{aligned} 0 &\leq \langle \psi, P\psi \rangle_\pi - t\langle \psi, P\phi \rangle_\pi - t\langle \phi, P\psi \rangle_\pi + t^2 \langle \phi, P\phi \rangle_\pi \\ &= \langle \psi, P\psi \rangle_\pi - 2t\langle \psi, P\phi \rangle_\pi + t^2 \langle \phi, P\phi \rangle_\pi, \end{aligned}$$

which follows by reversibility and symmetry of the real inner product. Suppose $\langle \phi, P\phi \rangle_\pi > 0$. Choose, the constant t, to be

$$t = \frac{\langle \psi, P\phi \rangle}{\langle \phi, P\phi \rangle_\pi}.$$

Substituting, we obtain

$$\begin{aligned} 0 &\leq \langle \psi, P\psi \rangle_\pi - 2\frac{\langle \psi, P\phi \rangle_\pi^2}{\langle \phi, P\phi \rangle_\pi} + \frac{\langle \psi, P\phi \rangle_\pi^2}{\langle \phi, P\phi \rangle_\pi} \\ &= \langle \psi, P\psi \rangle_\pi - \frac{\langle \psi, P\phi \rangle_\pi^2}{\langle \phi, P\phi \rangle_\pi^2}. \end{aligned}$$

Multiplying by the positive denominator proves the desired inequality. If instead $\langle \phi, P\phi \rangle_\pi = 0$, then the quadratic inequality reduces to

$$0 \leq \langle \psi, P\psi \rangle_\pi - 2t\langle \psi, P\phi \rangle_\pi, \qquad \forall t \in \mathbb{R}.$$

This can hold for every real t only if $\langle \psi, P\phi \rangle_\pi = 0$. The desired inequality then holds with both sides equal to zero, as required. We obtain the result.

□

Theorem 4.3 (Geometric $\mathcal{L}^2(\pi)$ Decay from a Spectral Gap)**.** *Let P be a transition kernel with stationary distribution π and let $L := P - I$. Suppose that*

(i) L, equivalently P is reversible with respect to π,
(ii) P is positive semidefinite in $\mathcal{L}^2(\pi)$, and
(iii) L has a spectral gap of at least λ where $0 < \lambda < 1$.

Then, for every $\phi \in \mathcal{L}^2(\pi)$ satisfying $\pi(\phi) = 0$ and every $n \geq 0$,

$$\|P_n \phi\|_\pi \leq (1 - \lambda)^n \|\phi\|_n.$$

Proof. First, we show that P preserves mean-zero subspace. By stationarity and duality identity

$$\pi(P\phi) = (\pi P)(\phi) = \pi(\phi).$$

Therefore, if $\pi(\phi) = 0$, then $\pi(P\phi) = 0$. Repeated application yields,

$$\pi(P_n\phi) = 0, \qquad \forall n \geq 0.$$

By positive semidefiniteness and spectral gap,

$$\begin{aligned} 0 \leq \|P\phi\|_\pi &= \langle \phi, P\phi \rangle_\pi \\ &= \langle \phi, \phi + L\phi \rangle_\pi \\ &= \|\phi\|_\pi^2 + \langle \phi, L\phi \rangle_\pi \\ &\leq \|\phi\|_\pi^2 - \lambda \|\phi\|_\pi^2 = (1-\lambda)\|\phi\|_\pi^2. \end{aligned}$$

By *Lemma 4.2*,

$$\begin{aligned} |\langle \psi, P\phi \rangle_\pi| &\leq \sqrt{\langle \psi, P\psi \rangle_\pi \langle \phi, P\phi \rangle_\pi} \\ &\leq \sqrt{(1-\lambda)\|\|\psi\|_\pi^2 (1-\lambda)\|\phi\|_\pi^2} \\ &= (1-\lambda)\|\psi\|_\pi \|\phi\|_\pi. \end{aligned}$$

Choose $\psi = P\phi$. This is a valid mean-zero function as P preserves $\mathcal{L}_0^2(\pi)$. Hence, we obtain

$$\begin{aligned} \|P\phi\|_\pi^2 &= \langle P\phi, P\phi \rangle_\pi \\ &\leq (1-\lambda)\|P\phi\|_\pi \|\phi\|_\pi. \end{aligned}$$

When $\|P\phi\|_\pi = 0$, the one-step estimate is immediate. Otherwise, we divide by $\|P\phi\|_\pi > 0$,

$$\|P\phi\|_\pi \leq (1-\lambda)\|\phi\|_\pi.$$

Setting $u_n := P_n\phi$. We have $u_0 = \phi$ and $\pi(u_n) = 0$, by *Backward Kolmogorov Equation*,

$$\begin{aligned} \|u_n\|_\pi &= \|Pu_{n-1}\|_\pi \\ &\leq (1-\lambda)\|u_{n-1}\|_\pi \\ &\leq (1-\lambda)^2\|u_{n-2}\|_\pi \\ &\cdots \\ &\leq (1-\lambda)^n\|u_0\|_\pi \\ &= (1-\lambda)^n\|\phi\|_n. \end{aligned}$$

Since $u_n := P_n\phi$, this proves the theorem.

□

Alternative Proof. Suppose that $P = P^\top$ and $P\mathbb{I} = \mathbb{I}$ which is equivalent to $\pi P = \pi$. Suppose there exist orthonormal eigenvectors,

$$v_1, \dots, v_m : v_1 = \frac{1}{\sqrt{m}}\mathbb{I}$$

such that

$$Pv_1 = 1 \cdot v_1.$$

Then the spectral gap implies that eigenvalues of P $\mu_1, \dots, \mu_m$ satisfies $\mu_j \leq 1 - \lambda$ for all $j \geq 2$. Then, the test function

$$\phi = \sum_{j=2}^{m} a_j v_j \quad \forall \phi : \mathbb{I}^\top \phi = 0$$

Then, we can express the transition operator as

$$\begin{aligned} \|P\phi\|^2 &= \left\| \sum_{j=2}^{m} a_j P v_j \right\|^2 \\ &= \sum_{j=2}^{m} a_j^2 \mu_j^2 \leq (1-\lambda) \sum_{j=2}^{m} a_j^2 \\ &= (1-\lambda)\|\phi\|^2. \end{aligned}$$

Hence, iterating we obtain

$$\|P_n\phi\|^2 \leq (1-\lambda)^{2n}\|\phi\|^2.$$

□

For a bounded measurable mean-zero function ϕ, recall that $(P_n\phi)(x) = \mathbb{E}_x[\phi(X_n)]$. No density assumption is needed for this identity. The theorem gives,

$$\int_E |\mathbb{E}_x[\phi(X_n)]|^2 \pi(dx) \leq (1-\lambda)^{2n} \int_E |\phi(x)|^2 \pi(dx).$$

Thus, the expected value of the centered observable becomes small in $\mathcal{L}^2(\pi)$ where the starting state x is weighted by π. The estimate is geometric in n, and therefore exponential,

$$(1-\lambda)^n \leq e^{-\lambda n}.$$

A larger lower bound on the spectral gap gives a smaller contraction factor.

Moreover, the spectral gap inequality controls eigenvalues of P from above, away from 1. It does not by itself exclude negative eigenvalues near -1, which can produce oscillations.

For example, consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0, \end{pmatrix} \quad \sim \text{Unif}(\{1, 2\}).$$

For $\phi = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ we have that $P\phi = -\phi$ and $L\phi = -2\phi$. The generator itself satisfies a positive spectral-gap inequality on the mean zero subspace but,

$$P_n\phi = (-1)^n\phi, \quad \|P_n\phi\|_\pi = \|\phi\|_\pi.$$

There is no decay, hence positive semidefiniteness excludes this negative-eigenvalue obstruction. For a test function that is not centered, the constant part remains unchanged while the mean-zero part contracts. Consider the following corollary.

Corollary 4.1 (Convergence of Expected Observables in $\mathcal{L}^2(\pi)$)**.** *Under Theorem 4.3 conditions, for every $\phi \in \mathcal{L}^2(\pi)$,*

$$\|P_n\phi - \pi(\phi)\|_\pi \leq (1-\lambda)^n \|\phi - \pi(\phi)\|_\pi.$$

Proof. Since $P_n\mathbf{1} = \mathbf{1}$,

$$\begin{aligned} P_n(\phi - \pi(\phi)) &= P_n\phi - \pi(\phi)P_n\mathbf{1} \\ &= P_n\phi - \pi(\phi). \end{aligned}$$

The function $\phi - \pi(\phi)$ has mean zero under uniform stationary distribution π.

Applying *Theorem 4.3* gives the result. In particular,

$$P_n\phi \xrightarrow{\mathcal{L}^2(\pi)} \pi(\phi).$$

□

5 Brownian Motion

We now move from discrete time $n = 0, 1, 2, \ldots$ to continuous time $t \geq 0$. A continuous-time process assigns a random variable X_t to every real time $t \geq 0$.

Continuous-time models can describe systems that evolve without a fixed observation interval. They also arise as limits of discrete-time models when the time between successive steps becomes small. Brownian motion is the fundamental example as it describes the limiting fluctuations of a suitably rescaled symmetric walk.

5.1 Properties of Brownian Motion

Consider the scaling of a symmetric random walk that produces this limit.

Example 5.1 (Rescaling a Symmetric Random Walk)**.** Let $\xi_1, \xi_2, \ldots \overset{i.i.d.}{\sim}$ Rademacher,

$$\mathbb{P}(\xi_k = 1) = \mathbb{P}(\xi_k = -1) = \frac{1}{2}.$$

Define symmetric SRW,

$$S_0 = 0, \quad S_n := \sum_{k=1}^{n} \xi_k.$$

Then, $\mathbb{E}[\xi_k] = 0$ and $\mathrm{Var}(\xi_k) = 1$, which gives $\mathbb{E}[S_n] = 0$ and $\mathrm{Var}(S_n) = n$. By CLT,

$$\frac{1}{\sqrt{N}} S_N = \frac{1}{\sqrt{N}} \sum_{k=1}^{N} \xi_k \xrightarrow{d} \mathcal{N}(0, 1).$$

This precisely describes the distribution of the random walk at one rescaled time. To obtain a limiting process, we rescale the entire sequence. Consider the following definition.

Definition 5.1 (Rescaled Random Walk)**.** For each integer $N \geq 1$, define

$$B_t^{(N)} := \frac{1}{\sqrt{N}} S_{\lfloor Nt \rfloor} = \frac{1}{\sqrt{N}} \sum_{k=1}^{\lfloor Nt \rfloor} \xi_k, \quad t \geq 0$$

where $\lfloor x \rfloor := \max\{k \in \mathbb{Z} : k \leq x\}$. An empty sum is defined to be zero. For each $k \geq 0$,

$$\frac{k}{N} \leq t \leq \frac{k+1}{N} \implies B_t^{(N)} = \frac{S_k}{\sqrt{N}}.$$

Thus, $B^{(N)}$ is constant between consecutive times $0, \frac{1}{N}, \frac{2}{N}, \ldots$. At time $\frac{k}{N}$ for $k \geq 1$, its jump is,

$$\frac{S_k - S_{k-1}}{\sqrt{N}} = \frac{\xi_k}{\sqrt{N}} \in \left\{ -\frac{1}{\sqrt{N}}, \frac{1}{\sqrt{N}} \right\}.$$

The time between steps is $\frac{1}{N}$ and the magnitude of each spatial step is $\frac{1}{\sqrt{N}}$.

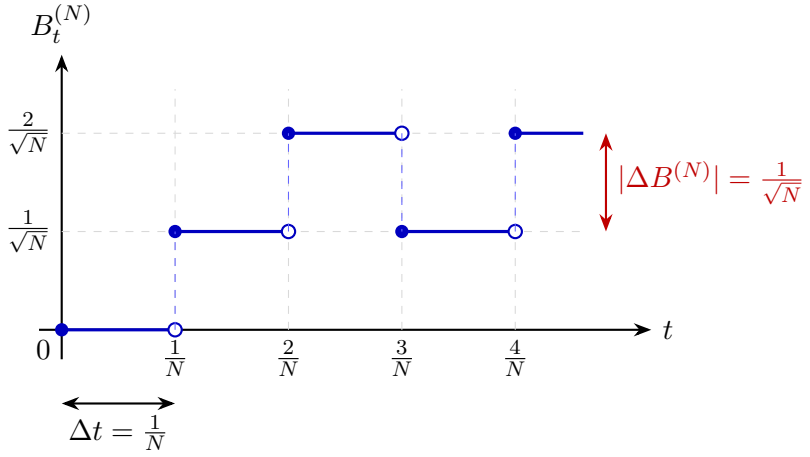


Figure 1: Piecewise-constant sample path of $B_t^{(N)}$.

Remark 5.1 (Continuous Time does not mean Continuous Paths). The process $B_t^{(N)}$ is defined at every real time $t \geq 0$, so it is a continuous-time process. Its sample paths, however, are piecewise constant with jumps. Thus, being indexed by continuous time and having continuous sample paths are different properties. The processes $B^{(N)}$ generally do not take values in $\mathcal{C}([0,T])$.

The factor $\frac{1}{\sqrt{N}}$ balances the increasing number of increments against their decreasing size.

Proposition 5.1 (Mean and Variance of the Rescaled Walk)**.** *For every $N \geq 1$ and $t \geq 0$,*

$$\mathbb{E}[B_t^{(N)}] = 0, \quad \mathrm{Var}(B_t^{(N)}) = \frac{\lfloor Nt \rfloor}{N}.$$

Then, as $N \to \infty$,

$$\mathrm{Var}(B_t^{(N)}) \longrightarrow t.$$

Proof. By linearity,

$$\mathbb{E}[B_t^{(N)}] = \frac{1}{\sqrt{N}} \sum_{k=1}^{\lfloor Nt \rfloor} \mathbb{E}[\xi_k] = 0.$$

By independently and identically distribution ξ_k's,

$$\mathrm{Var}(B_t^{(N)}) = \mathrm{Var}\left(\frac{1}{\sqrt{N}} \sum_{k=1}^{\lfloor Nt \rfloor} \xi_k \right) = \frac{1}{N} \sum_{k=1}^{\lfloor Nt \rfloor} \mathrm{Var}(\xi_k) = \frac{\lfloor Nt \rfloor}{N}.$$

Since $Nt - 1 \leq \lfloor Nt \rfloor \leq Nt$, we have

$$t - \frac{1}{N} \leq \frac{\lfloor Nt \rfloor}{N} \leq t.$$

Therefore, the variance converges to t, as required.

□

At a fixed time $t > 0$, there are approximately Nt independent increments, where each rescaled increment has variance $\frac{1}{N}$. Their total variance is therefore approximately,

$$Nt \cdot \frac{1}{N} = t.$$

The fluctuations remain on a nontrivial scale even though the individual jumps become small.

Remark 5.2. More generally, consider

$$\frac{1}{N^\alpha} S_{\lfloor Nt \rfloor}.$$

Its variance is precisely,

$$\frac{\lfloor Nt \rfloor}{N^{2\alpha}} \sim t N^{1-2\alpha}.$$

For $t > 0$, this tends to zero if $\alpha > 1/2$, remains of order one if $\alpha = 1/2$, and diverges if $\alpha < 1/2$. Thus, the Brownian scaling is proportional to the spatial step size which is the root of its timestep size. Moreover, although N increments could add up to a quantity of order N if they all had the same sign, independent centred increments typically cancel substantially. Their sum has standard deviation $\sqrt{N}$, which is why dividing by $\sqrt{N}$ produces fluctuations of order one.

Remark 5.3 (Law of Large Numbers vs. Central Limit Theorem). For the symmetric random walk, the *Law of Large Numbers* tells us that $\frac{S_N}{N} \to 0$ whereas the *Central Limit Theorem* tells us that,

$$\frac{S_N}{\sqrt{N}} \xrightarrow{d} \mathcal{N}(0, 1).$$

LLN describes the average increment whereas the second describes the scale of the fluctuations around its mean. These are not separate rules for nonzero-mean and zero-mean variables. In general, if $\mathbb{E}[\xi_k] = \mu$, the central limit theorem applies to the centered sum,

$$\frac{1}{\sqrt{N}} \sum_{k=1}^{N} (\xi_k - \mu).$$

Our increments already have mean zero, so no additional centering is needed.

Proposition 5.2 (Convergence at a Fixed Time)**.** *For every fixed $t \geq 0$,*

$$B_t^{(N)} \xrightarrow{d} \mathcal{N}(0, t)$$

where $\mathcal{N}(0, 0)$ denotes the pointmass at zero.

Proof. If $t = 0$, then $B_0^{(N)} = 0$ for every N. Suppose $t > 0$, so $\lfloor Nt \rfloor \to \infty$,

$$B_t^{(N)} = \sqrt{\frac{\lfloor Nt \rfloor}{N}} \left(\frac{1}{\sqrt{\lfloor Nt \rfloor}} \sum_{k=1}^{\lfloor Nt \rfloor} \xi_k \right).$$

The expression in parentheses converges in distribution to a standard normal random variable while, $\sqrt{\frac{\lfloor Nt \rfloor}{N}} \to \sqrt{t}$. To justifying the resulting limit through CDFs, let F_N be the CDF of the expression in parentheses. For $z \in \mathbb{R}$,

$$\mathbb{P}\left(B_t^{(N)} \leq z\right) = F_N\left(\frac{z}{\sqrt{\lfloor Nt \rfloor / N}}\right).$$

The argument tends to $z/\sqrt{t}$. Since,

$$F_N(u) \longrightarrow \Phi(u), \qquad \forall u$$

and Φ is continuous, monotonicity of CDFs gives

$$F_N\left(\frac{z}{\sqrt{\lfloor Nt \rfloor / N}}\right) \longrightarrow \Phi\left(\frac{z}{\sqrt{t}}\right).$$

This is precisely the CDF of $\mathcal{N}(0, t)$.

□

In particular, the limit is standard normal at time $t = 1$. At a general time t, its variance is t.

The fixed-time limits suggest a process with Gaussian distributions whose variance grows linearity with time. To specify Brownian motion, we must also describe how its values at different times relate to one another and require continuity of its sample paths.

Definition 5.2 (Brownian Motion)**.** A real-valued stochastic process $(B_t)_{t \geq 0}$ is a *standard Brownian motion* if it satisfies the following properties:

(i) *Starting at zero.* $B_0 = 0$, almost surely.

(ii) *Independent Increments.* For every finite collection of times $0 = t_0 < t_1 < \cdots < t_m$,

$$B_{t_1} - B_{t_0}, \ldots, B_{t_m} - B_{t_{m-1}},$$

are independent random variables.

(iii) *Gaussian Increments.* For every $0 \leq s < t$,

$$B_t - B_s \sim \mathcal{N}(0, t - s).$$

(iv) *Continuous Sample Paths.*

$$\mathbb{P}\left(t \mapsto B_t \in \mathcal{C}([0, \infty))\right) = 1.$$

The increment distribution depends only on the elapsed time $t - s$ rather than on the starting time s. This is called the stationarity of increments. In paritcular,

$$\mathbb{E}[B_t - B_s] = 0, \quad \mathrm{Var}(B_t - B_s) = t - s.$$

The standard deviation over a time interval of length h is therefore $\sqrt{h}$, matching the random-walk scaling we demonstrated previously.

Remark 5.4. If $0 < s < t$ then,

$$B_t = B_s + (B_t - B_s).$$

The increment $B_t - B_s$ is independent of B_s but B_t contains B_s as a part of its value. COnsequently, B_s and B_t are generally dependent.

Proposition 5.3 (Brownian Motions as a Gaussian Process)**.** *Standard Brownian motion is a centered Gaussian Process with,*

$$\mathbb{E}[B_t] = 0, \qquad \mathrm{Cov}(B_s, B_t) = \min(s, t).$$

Proof. Since $B_0 = 0$ almost surely,

$$B_t = B_t - B_0 \sim \mathcal{N}(0, t).$$

Therefore, $\mathbb{E}[B_t] = 0$ and $\mathrm{Var}(B_t) = t$. We also must verify that every finite collection of values is jointly Gaussian. Fix a finite set of times $0 = t_0 < t_1 < \cdots < t_m$. The increments,

$$B_{t_j} - B_{t_{j-1}}, \qquad j = 1, \ldots, m$$

are independent *Gaussian* random variables. Each process value is a sum of these increments,

$$B_{t_i} = \sum_{j=1}^{i} (B_{t_j} - B_{t_{j-1}}).$$

Thus, we can write the Brownian motion over this finite set of times as,

$$\begin{pmatrix} B_{t_1} \\ B_{t_2} \\ \vdots \\ B_{t_m} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} B_{t_1} - B_{t_0} \\ B_{t_2} - B_{t_1} \\ \vdots \\ B_{t_m} - B_{t_{m-1}} \end{pmatrix}.$$

Hence, a linear transformation of an independent Gaussian vector is jointly Gaussian. Hence, the finite-dimensional distributions of B are Gaussian. To compute the covariance, suppose WLOG that $s \leq t$. Since both means are zero,

$$\begin{aligned} \mathrm{Cov}(B_s, B_t) &= \mathbb{E}[B_s B_t] \\ &= \mathbb{E}[B_s(B_s + (B_t - B_s))] \\ &= \mathbb{E}[B_s^2] + \mathbb{E}[B_s(B_t - B_s)]. \end{aligned}$$

For $0 < s < t$, independent increments imply that $B_s = B_s - B_0$ is independent of the increment $B_t - B_s$. Therefore, it follows

$$\mathbb{E}[B_s(B_t - B_s)] = \mathbb{E}[B_s]\mathbb{E}[B_t - B_s] = 0.$$

When $s = 0$ or $s = t$, we obtain the same result

$$\mathrm{Cov}(B_s, B_t) = \mathbb{E}[B_s^2] = \mathrm{Var}(B_s) + (\mathbb{E}[B_s])^2 = s.$$

Hence, by symmetry,

$$\mathrm{Cov}(B_s, B_s) = \min(s, t).$$

□

Thus, for any finite collection of times $t_1, \ldots, t_m \geq 0$,

$$\begin{pmatrix} B_{t_1} \\ \vdots \\ B_{t_m} \end{pmatrix} \sim \mathcal{N}(0, \Sigma), \qquad \Sigma_{ij} = \min(t_i, t_j).$$

Hence, we can see that B_s and B_t share the increments accumulated up to time $\min(s, t)$ and the variance of the shared part is $\min(s, t)$. We have already shown that the rescaled walk has the correct limiting distribution at each individual time. We now show that it also has the correct limiting joint distributions at any finite collection of times.

5.2 Finite- & Multi-dimensional Brownian Motion

The finite-dimensional convergence result describes the random walk at finitely many observation times. We begin by proving this result using the Central Limit Theorem and independence of increments. We then introduce a space of continuous paths so that we can state convergence of the entire process.

Proposition 5.4 (Finite-Dimensional Convergence to Brownian Motion)**.** *Let $(\xi_k)_{k\geq 1}$ be independent Rademacher random variables, and define*

$$B_t^{(N)} := \frac{1}{\sqrt{N}} \sum_{k=1}^{\lfloor Nt \rfloor} \xi_k, \qquad t \geq 0.$$

Let $(B_t)_{t\geq 0}$ be a standard Brownian motion. Then, for finite collection of times

$$0 \leq t_1 < \cdots < t_m$$

we have that

$$\left(B_{t_1}^{(N)}, \ldots, B_{t_m}^{(N)}\right) \xrightarrow{d} \left(B_{t_1}, \ldots, B_{t_m}\right).$$

Proof. Suppose initially $t_1 > 0$ and set

$$t_0 := 0, \quad N_j := \lfloor t_j \rfloor, \quad \Delta N_j := N_j - N_{j-1}, \quad \Delta t_j := t_j - t_{j-1}.$$

For $j = 1, \ldots, m$ define,

$$\Delta B_{t_j}^{(N)} = B_{t_j}^{(N)} - B_{t_{j-1}}^{(N)}, \qquad \Delta B_{t_j} := B_{t_j} - B_{t_{j-1}}.$$

Expanding the definitions of the scaled random walk increment,

$$\begin{aligned}\Delta B_{t_j}^{(N)} &= \frac{1}{\sqrt{N}}\left(\sum_{k=1}^{N_j}\xi_k - \sum_{k=1}^{N_{j-1}}\xi_k\right)\\ &= \frac{1}{\sqrt{N}}\sum_{k=N_{j-1}+1}^{N_j}\xi_k.\end{aligned}$$

Hence, the j-th increment of Brownian motion $\Delta B_{t_j}^{(N)}$ contains exactly $\Delta N_j := N_j - N_{j-1}$ random walk increments. The process $B^{(N)}$ changes at grid times $\frac{k}{N}$ and every jump has size $\frac{1}{\sqrt{N}}$ or $\frac{-1}{\sqrt{N}}$. The number of such jumps between two observation times is,

$$\Delta N_j = \lfloor Nt_j \rfloor - \lfloor N_{t_{j-1}} \rfloor.$$

Consequently,

$$\frac{\Delta N_j}{N} = \frac{\lfloor Nt_j \rfloor}{N} - \frac{\lfloor Nt_{j-1} \rfloor}{N} \longrightarrow t_j - t_{j-1} = \Delta t_j.$$

Since $\Delta t_j > 0$, then $\Delta N_j \to \infty$. For sufficiently large N, we can rewrite,

$$\Delta B_{t_j}^{(N)} := \sqrt{\frac{\Delta N_j}{N}}\left(\frac{1}{\sqrt{\Delta N_j}}\sum_{k=N_{j-1}+1}^{N_j}\xi_k\right).$$

Notice that, since there are precisely ΔN_j increments in each independent increment of $B_t^{(N)}$ for $j = 1, \ldots, m$, the expression in parentheses is a standardized sum of ΔN_j independent Rademacher random variables, by *Central Limit Theorem*,

$$\frac{1}{\sqrt{\Delta N_j}}\sum_{k=N_{j-1}+1}^{N_j}\xi_k \xrightarrow{d} \mathcal{N}(0,1).$$

Recall that, because of how we've defined our floor,

$$Nt - 1 \leq \lfloor Nt \rfloor \leq Nt \iff \frac{Nt-1}{N} \leq \frac{\lfloor Nt \rfloor}{N} \leq \frac{Nt}{N} \iff t - \frac{1}{N} \leq \frac{\lfloor Nt \rfloor}{N} \leq t.$$

Hence, as $t \to \infty$, $\frac{\lfloor Nt \rfloor}{N} \to t$. Hence, $\sqrt{\frac{\Delta N_j}{N}} \longrightarrow \sqrt{\Delta t_j}$. Therefore, we have shown,

$$\Delta B_{t_j}^{(N)} \xrightarrow{d} \mathcal{N}(0, \Delta t_j).$$

This is precisely the distribution of the Brownian increment ΔB_{t_j} as we defined,

$$\Delta B_{t_j} = B_{t_j} - B_{t_{j-1}} \sim \mathcal{N}(0, \Delta t_j).$$

Next, we consider these increments jointly. For each N, note that the sums of $B_{t_j}^{(N)}$ for $j = 1, \ldots, m$ use disjoint blocks of independent random variables. Hence, the blocks

$$\Delta B_{t_1}^{(N)}, \ldots, \Delta B_{t_m}^{(N)}$$

are independent. Brownian motion also has independent increments. For any $z_1, \ldots, z_m \in \mathbb{R}$,

$$\begin{aligned}
&\mathbb{P}\left(\Delta B_{t_1}^{(N)} \leq z_1, \ldots, \Delta B_{t_m}^{(N)} \leq z_m\right) \\
&= \prod_{j=1}^{m} \mathbb{P}\left(\Delta B_{t_j}^{(N)} \leq z_j\right) \\
&\longrightarrow \prod_{j=1}^{m} \mathbb{P}(\Delta B_{t_j} \leq z_j) \\
&= \mathbb{P}(\Delta B_{t_1} \leq z_1, \ldots, \Delta B_{t_m} \leq z_m).
\end{aligned}$$

The limiting Gaussian joint distribution has continuous CDF, so this establishes,

$$\left(\Delta B_{t_1}^{(N)}, \ldots, \Delta B_{t_m}^{(N)}\right) \xrightarrow{d} (\Delta B_{t_1}, \ldots, \Delta B_{t_m}).$$

Hence, since we have established that each independent increment $\Delta B_{t_j}^{(N)}$ converges in distribution to ΔB_{t_j} of Brownian motion, each block of $B_{t_j}^{(N)}$ for $j = 1, \ldots, m$ is independent, and the product of CDFs of independent $\Delta B_{t_j}^{(N)}$'s converges to the product of CDFs of ΔB_t's, we conclude that the joint distribution converges to the limiting distribution.

□

Finite dimensional convergence compares the processes at any fixed, finite collection of times. To discuss convergence of the entire infinite path requires us to specify a space whose elements are paths, together with a metric on that space.

Recall that convergence in distribution is expressed through bounded continuous test functions. For random paths, these test functions take an entire path as their input. Their continuity therefore depends on how we measure the distance between two paths.

Definition 5.3 (Continuous Path Space and the Uniform metric)**.** Fix $T > 0$. Let,

$$\mathcal{C}([0, T]) := \{f : [0, T] \to \mathbb{R} : f \text{ is continuous}\}.$$

Equip this space with the metric,

$$d(f, g) := \sup_{t \in [0,T]} |f(t) - g(t)|.$$

This is referred to as the *uniform metric*.

Thus, two paths are *close* when their values are close at every time $[0,T]$. In particular, $d(f_n, f) \longrightarrow 0$ implies that f_n converges uniformly to f. The rescaled random walk $B^{(N)}$ has piecewise constant paths with jumps, so its paths do not generally belong to $\mathcal{C}([0,T])$. To obtain continuous paths, we connect successive grid values by straight lines.

Definition 5.4 (Piecewise Linear Interpolation)**.** For $t \geq 0$, let

$$k := \lfloor Nt \rfloor, \quad \lambda := Nt - k.$$

Then $0 \leq \lambda < 1$ and,

$$\frac{k}{N} \leq t < \frac{k+1}{N}$$

Define,

$$\tilde{B}_t^{(N)} := (1-\lambda) B_{k/N}^{(N)} + \lambda B_{k/N}^{(N)}.$$

Substituting definitions of $B_{k/N}^{(N)} = \frac{S_k}{\sqrt{N}}$ and $B_{(k+1)/N}^{(N)} = \frac{S_{k+1}}{N}$ where S_n is the sum of n independent Rademacher random variables, expanding gives us,

$$\begin{aligned}
\tilde{B}_t^{(N)} &:= (1-\lambda) B_{k/N}^{(N)} + \lambda B_{k/N}^{(N)} \\
&= \frac{1}{\sqrt{N}} \left[(1-\lambda) S_k + \lambda S_{k+1} \right] \\
&= \frac{1}{\sqrt{N}} \left[S_k - \lambda S_k + \lambda S_{k+1} \right] \\
&= \frac{1}{\sqrt{N}} \left[S_k + \lambda \xi_{k+1} \right].
\end{aligned}$$

The interpolation agrees with the original rescaled walk at every time and is linear between consecutive grid times. Its paths are therefore continuous, as required.

Theorem 5.1 (Donsker's Theorem)**.** *Let $(\xi_k)_{k \geq 1}$ be i.i.d. random variables satisfying,*

$$\mathbb{E}[\xi_k] = 0, \qquad \mathrm{Var}(\xi_k) = 1.$$

Define,

$$S_0 := 0, \quad S_n := \sum_{k=1}^{n} \xi_k, \quad B_t^{(N)} = \frac{S_{\lfloor Nt \rfloor}}{\sqrt{N}}$$

and let $\tilde{B}^{(N)}$ be its piecewise linear interpolation by Definition 5.4. For every fixed $T > 0$,

$$\left(\tilde{B}_t^{(N)} \right)_{t \in [0,T]} \xrightarrow{d} (B_t)_{t \in [0,T]}$$

as random elements of $\mathcal{C}([0,T])$ equipped with the uniform metric, where B is standard Brownian motion. Equivalently, for every bounded continuous function $\phi : \mathcal{C}([0,T]) \to \mathbb{R}$,

$$\mathbb{E}\left[\phi\left(\tilde{B}^{(N)} \right) \right] \longrightarrow \mathbb{E}\phi(B).$$

Remark 5.5 (Functional Central Limit Theorem). *Donsker's Theorem* is also called the *Functional Central Limit Theorem*. The ordinary CLT gives a Gaussian limit for a normalized sum at one time while Donsker's Theorem gives Brownian motion as the limit of the entire interpolated process. Note that the increments need not be Rademacher. Any i.i.d. increment distribution with mean zero and variance one satisfies the hypotheses and produces the same limiting process. This is the sense in which the Brownian limit is *universal*.

A pointwise limit of continuous functions can be discontinuous. A uniform limit of continuous functions, however, is continuous. Donsker's theorem concerns convergence in distribution on the space $\mathcal{C}([0,T])$ but rather does not assert that the interpolated paths converge uniformly to a Brownian path for each outcome ω.

It is also possible to work directly with the piecewise constant processes $B^{(N)}$. Their paths are right-continuous and have left limits, a property called *cadlag*. Such paths are studied in a different path space, equipped with a topology such as the *Skorohod* J_1 topology.

We now describe the information available as a continuous-time process evolves. The definitiono of a filtration is the same as in discrete time, except that the index ranges over all $t \geq 0$.

Definition 5.5 (Continuous-time Filtration)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *continuous-time filtration* is a family of sub-σ algebras $(\mathcal{F}_t)_{t\geq 0}$ such that,

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}, \qquad 0 \leq s \leq t.$$

The corresponding *natural filtration* of Brownian motion is,

$$\mathcal{F}_t := \sigma(B_s : 0 \leq s \leq t)$$

containing the information revealed by observing the Brownian path up to time t. For a fixed $T > 0$, Brownian motion can be viewed as a random continuous path,

$$B : \Omega \longrightarrow \mathcal{C}([0,T]), \qquad B(\omega)(t) := B_t(\omega).$$

Here, we may define the path arbitrarily on the null event where continuity fails. The cylindrical σ-algebra $\mathcal{F}_{\text{cyl}}$ is generated by sets of the form,

$$\{f \in \mathcal{C}([0,T]) : f(t_1) \in A_1, \ldots, f(t_m) \in A_m\}$$

where $t_1, \ldots, t_m \in [0,T]$ and $A_1, \ldots, A_m \in \mathcal{B}(\mathbb{R})$. Thus, the path is measurable as a map,

$$B : (\Omega, \mathcal{F}) \longrightarrow (\mathcal{C}([0,T]), \mathcal{F}_{\text{cyl}}).$$

That is, of course

$$\mathcal{F}_t = \sigma(B_s : 0 \leq s \leq T) = \sigma\left(B^{-1}(A) : A \in \mathcal{F}_{\text{cyl}}\right).$$

The two descriptions, hence, record the same information where we either observe all the coordinates B_s up to time T, or we observe the path restricted to $[0,T]$.

5.3 Continuous-time Brownian Transition Operators

We next return to the transition kernel notation used for Markov chains. Starting Brownian motion at $x \in \mathbb{R}$ means defining,

$$X_t := x + B_t.$$

Since $B_t \sim \mathcal{N}(0,t)$, then $X_t \sim \mathcal{N}(x,t)$. This determines the distribution of the position, starting at $X_0 = x$, after an elapsed time t, according to Brownian motion.

Definition 5.6 (Brownian Transition Density and Kernel)**.** For $t > 0$, define

$$p_t(x,y) := \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(y-x)^2}{2t}\right), \qquad x, y \in \mathbb{R}.$$

The corresponding transition kernel with respect to density $p_t(x,y)$,

$$P_t(x,A) := \int_A p_t(x,y)dy, \qquad A \in \mathcal{B}(\mathbb{R}).$$

At time zero, we define

$$P_0(x,A) := \mathbb{I}_A(x) = \delta_x(A).$$

$p_t(x,y)$ is used for the density and $P_t(x,A)$ is used for the transition kernel. Thus,

$$P_t(x,dy) = p_t(x,y)dy, \qquad t > 0.$$

For a fixed starting point x, $P_t(x,\cdot)$ is the Gaussian law $\mathcal{N}(x,t)$. As $t \to 0$, this law concentrates at x, giving the pointmass δ_x as its limit, precisely giving our definition of P_0.

Proposition 5.5 (Markov Property of Brownian Motion)**.** *Let,*

$$\mathcal{F}_s := \sigma(B_u : 0 \le u \le s).$$

For every $s, t \ge 0$, and $A \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}(B_{s+t} \in A \mid \mathcal{F}_s) = \mathbb{P}(B_{s+t} \in A \mid \sigma(B_s)) = P_t(B_s, A) \qquad \textit{almost surely.}$$

The σ-algebra $\mathcal{F}_s$ contains the entire observed history up to time s, while $\sigma(B_s)$ contains only the current position. The proposition states both give the same conditional distribution B_{s+t}.

Proof. Suppose that $t > 0$.

We verify that $P_t(B_s, A)$ satisfies the defining properties of conditional expectation,

$$\mathbb{E}[\mathbb{I}_A(B_{s+t}) \mid \mathcal{F}_s].$$

For fixed $A \in \mathcal{B}(\mathbb{R})$, $x \mapsto P_t(x,A)$ is measurable. Since B_s is $\mathcal{F}_s$-measurable, $P_t(B_s, A)$ is also $\mathcal{F}_s$-measurable. Furthermore, since P_t is a valid transition kernel,

$$0 \le P_t(B_s, A) \le 1$$

hence it is integrable. It remains to show that, for every $H \in \mathcal{F}_s$,

$$\mathbb{E}[\mathbb{I}_H \mathbb{I}_A(B_{s+t})] = \mathbb{E}[\mathbb{I}_H P_t(B_s, A)].$$

By independent increments, $B_{s+t} - B_s \sim \mathcal{N}(0, t)$ and this increment is independent of the history $\mathcal{F}_s$. In particular, it is independent of the pair $(\mathbb{I}_H, B_s)$. That is, the future increment is independent of every finite collection of observations before time s and these observations generate $\mathcal{F}_s$. Decomposing over time $s + t$,

$$B_{s+t} = B_s + (B_{s+t} - B_s)$$

and integrating over the independent future increment gives,

$$\begin{aligned}
\mathbb{E}[\mathbb{I}_H \mathbb{I}_A(B_{s+t})] &= \mathbb{E}[\mathbb{I}_H I_A(B_s + (B_{s+t} - B_s))] \\
&= \int_{\mathbb{R}} \mathbb{E}[\mathbb{I}_H \mathbb{I}_A(B_s + z)] p_t(0, z) dz \\
&= \mathbb{E}\left[\mathbb{I}_H \int_{\mathbb{R}} \mathbb{I}_A(B_s + z) p_t(0, z) dz\right].
\end{aligned}$$

Fix $x \in \mathbb{R}$. Substituting $y = x + z$ gives,

$$\begin{aligned}
&\int_{\mathbb{R}} \mathbb{I}_A(x + z) p_t(0, z) dz \\
&\qquad = \int_A p_t(0, y - z) dz \\
&\qquad = \int_A p_t(x, y) dy \\
&\qquad = P_t(x, A).
\end{aligned}$$

Applying this identity at $x = B$, we obtain,

$$\mathbb{E}[\mathbb{I}_H \mathbb{I}_A(B_{s+t})] = \mathbb{E}[\mathbb{I}_H P_t(B_s, A)].$$

By expectations of indicators,

$$\mathbb{P}(B_{s+t} \in A \mid \mathcal{F}_s) = P_t(B_s, A) \quad \text{almost surely.}$$

To condition only on the current position, we use $\sigma(B_s) \subseteq \mathcal{F}_s$. By *Tower Property*

$$\begin{aligned}
&\mathbb{P}(B_{s+t} \in A \mid \sigma(B_s)) \\
&\qquad = \mathbb{E}[\mathbb{E}[\mathbb{I}_A(B_{s+t}) \mid \mathcal{F}_s] \mid \sigma(B_s)] \\
&\qquad = \mathbb{E}[P_t(B_s, A) \mid \sigma(B_s)] \\
&\qquad = P_t(B_s, A).
\end{aligned}$$

$P_t(B_s, A)$ is B_s-measurable. When $t = 0$, both conditional probabilities equal,

$$\mathbb{I}_A(B_s) = P_0(B_s, A),$$

so the result also holds in that case.

□

As in discrete time, the transition kernel also acts on test functions.

Definition 5.7 (Brownian Transition Operator)**.** For a bounded measurable function $\phi : \mathbb{R} \to \mathbb{R}$, define the transition operator,

$$(P_t\phi)(x) := \int_{\mathbb{R}} \phi(y) P_t(x, dy).$$

For $t > 0$,

$$(P_t\phi)(x) = \int_{\mathbb{R}} \phi(y) p_t(x, y) dy = \mathbb{E}[\phi(x + B_t)].$$

At time zero, $(P_0\phi)(x) = \phi(x)$ so $P_0 = I$, the identity operator.

The *Markov property* extends from indicators to bounded measurable test functions,

$$\mathbb{E}[\phi(B_{s+t}) \mid \mathcal{F}_s] = (P_t\phi)(B_s).$$

Indeed, using the independent increment $B_{s+t} - B_s$,

$$\begin{aligned}\mathbb{E}[\phi(B_{s+t}) \mid \mathcal{F}_s] &= \int_{\mathbb{R}} \phi(B_s + z) p_t(0, z) dz \\ &= \int_{\mathbb{R}} \phi(y) p_t(B_s, y) dy \\ &= (P_t\phi)(B_s).\end{aligned}$$

Thus, $(P_t\phi)(x)$ is the expected value of the observable ϕ after an elapsed time t, starting from x.

Proposition 5.6 (Semigroup Property)**.** *The Brownian transition operators satisfy,*

$$P_{t+s} = P_t P_s, \qquad s, t \geq 0.$$

Equivalently, recall that for every bounded measurable ϕ,

$$P_{t+s}\phi = P_t(P_s\phi).$$

Proof. Fix $x \in \mathbb{R}$. The increment,

$$B_{t+s} - B_s$$

is independent fo B_t and has the same distribution as B_s. Therefore,

$$\mathbb{E}[\phi(x + B_t + (B_{t+s} - B_t)) \mid \mathcal{F}_t] = (P_s\phi)(x + B_t).$$

Taking expectations gives,

$$\begin{aligned}(P_tP_s\phi)(x) &= \mathbb{E}[(P_s\phi)(x+B_t)] \\ &= \mathbb{E}[\mathbb{E}[\phi(x+B_t+(B_{t+s}-B_t)) \mid \mathcal{F}_t]] \\ &= \mathbb{E}[\phi(x+B_t+(B_{t+s}-B_t))] \\ &= \mathbb{E}[\phi(x+B_{t+s})] \\ &= (P_{t+s}\phi)(x).\end{aligned}$$

Hence, $P_tP_s = P_{t+s}$.

□

Taking $\phi = \mathbb{I}_A$ gives the corresponding kernel identity,

$$P_{t+s}(x,A) = \int_{\mathbb{R}} P_s(y,A)P_t(x,dy).$$

We can therefore evolve for time $t+s$ by first evolving for time t, then evolving for another time s. Finally, the same transition kernels describe how an initial probability distribution evolves.

Proposition 5.7 (Evolution of the Distribution)**.** *Let X_0 have distribution μ_0, independently of Brownian motion B, and define*

$$X_t := X_0 + B_t.$$

If $\mu_t := \mathcal{L}(X_t)$, then

$$\mu_t = \mu_0 P_t.$$

More explicitly, for every $A \in \mathcal{B}(\mathbb{R})$,

$$\mu_t(A) = \int_{\mathbb{R}} P_t(x,A)\mu_0(dx).$$

Proof. Conditional on $X_0 = x$, the random variable X_t has distribution $P_t(x,\cdot)$. Therefore,

$$\begin{aligned}\mu_t(A) &= \mathbb{P}(X_0 + B_t \in A) \\ &= \mathbb{E}\left[\mathbb{I}_{\{X_0+B_t\in A\}}\right] \\ &= \mathbb{E}[\mathbb{E}[\mathbb{I}_{\{X_0+B_t\in A\}} \mid \sigma(X_0)]] \\ &= \mathbb{E}[\mathbb{P}(X_0+B_t \in A \mid \sigma(X_0))] \\ &= \mathbb{E}[P_t(X_0,A)] = \int_{\mathbb{R}} P_t(x,A)\mu_0(dx) \\ &= (\mu_0 P_t)(A).\end{aligned}$$

For $t > 0$, substituting the Gaussian transition density,

$$\begin{aligned}\mu_t(A) &= \int_{\mathbb{R}} \left(\int_A p_t(x,y)dy\right)\mu_0(dx) \\ &= \int_A \left(\int_{\mathbb{R}} p_t(x,y)\mu_0(dx)\right)dy.\end{aligned}$$

Thus, we average the Gaussian transition law over all possible initial positions, weighted by their probabilities under the initial measure, μ_0, as required.

□

The notation now matches the discrete-time framework,

$$\mu_t = \mu_0 P_t, \quad (P_t\phi)(x) = \mathbb{E}[\phi(x + B_t)], \quad P_{t+s} = P_t P_s.$$

The semigroup describes evolution over arbitrary elapsed times. This is the structure we will use to formulate continuous-time evolution equations.